NATIONAL STANDING ORDERS

FOR COMMUNITY HEALTH OFFICERS

Revised By:

COMMUNITY HEALTH PRACTITIONERS' REGISTRATION BOARD OF NIGERIA
IN COLLABORATION WITH NPHCDA

2024



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USAID FROM THE AMERICAN PEOPLE

2024

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ACRONYMS AND ABBREVIATIONS

ANC	Antenatal Care
BHCPF	Basic Healthcare Provision Fund
CBOs	Community-Based Organizations
CE	Community Engagement
CHEW	Community Health Extension Worker
CHIPS	Community Health Influencers, Promoters and Services
CHPs	Community Health Programmes
CHO	Community Health Officer
DHIS	District Health Information System
HC	Health Clinic
HIS	Health Information System
HRH	Human Resources for Health
CHO	Community Health Extension Worker
LGA	Local Government Area
NPHCDA	National Primary Health Care Development Agency
OIC	Officer In-charge
PHCUOR	Primary Health Care Under One Roof
PHC	Primary Health Care
SBA	Skilled Birth Attendants
SCL	Student-Centered Learning
SMOH	State Ministry of Health
TOR	Terms of Reference
UHC	Universal Health Coverage
VDC	Village Development Community
WDC	Ward Development Community
WHO	World Health Organization

FOREWORD

In today's dynamic healthcare landscape, the quest for quality and uniformity of care at the primary health care level remains paramount. It is with great pleasure that the Community Health Practitioners Registration Board of Nigeria [CHPRBN] presents the 2024 revised National Standing Orders for Community Health Officers (CHOs) It is a comprehensive set of standard operational guidelines meticulously crafted through the collaborative efforts of the reputable regulatory establishment, principally supported by USAID/Nigeria through its Health Workforce Management Activity, the Federal Ministry of Health, the National Primary Health Care Development Agency [NPHCDA], and other Pertinent MDAs.

The significance of the National Standing Orders cannot be overstated. As a guiding framework that links Communities and the national health system, it serves as a cornerstone in ensuring consistency and excellence in healthcare practices across the nation. By providing clear protocols and procedural guidelines, it empowers Community Health Officers to deliver services of the highest standard, thereby enhancing positive Community health outcomes.

Accordingly, the National Standing Orders foster a culture of accountability and professionalism within the Community Health Profession. By adhering to established guidelines, Community Health Officers uphold ethical standards and demonstrate their commitment to providing safe and effective care to all individuals, regardless of their socio-economic status or geographic location.

Furthermore, this document has been painstakingly compiled to serve as a tool for capacity building and continuous improvement. Through regular updates and revisions, it reflects the latest advancements in healthcare and best practices, equipping healthcare providers with the knowledge and skills necessary to adapt to evolving healthcare needs and challenges.

In essence, the National Standing Orders epitomize our collective commitment to advancing healthcare delivery and safeguarding the well-being of our citizens. It is our fervent hope that this document will serve as a catalyst for positive change and contribute significantly to the achievement of national health objectives. Therefore, I strongly endorse the 2024 edition of the National Standing Orders to the Community Health Officers, to provide quality, affordable, and uniformity of care within the national health system. Predominantly, I extend our deepest appreciation to all those who have contributed to this review, recognizing their dedication and expertise in shaping the future of healthcare in Nigeria. It is my ardent hope and trust that Community Health Officers will appreciate the huge efforts put in place for this review by reciprocating with strict adherence to this national document and demonstrating a high sense of responsibility.

Dr. Tunji Alausa
Hon. Minister of state for Health
August,2024

ACKNOWLEDGEMENT

The review and production of the 2024 edition were made possible by USAID/Nigeria through its Health Workforce Management Activity. The regulatory establishment [CHPRBN] reviewed this document through collaborative support from the Federal Ministry of Health, the National Primary Health Care Development Agency [NPHCDA], and other critical stakeholders. The Board acknowledged all the sources of materials used in this publication.

The main objective of the National Standing Orders is to ensure quality, affordable, and uniform care, especially at the Primary Health Care level, provided by Community Health Practitioners. The review committee deserves a high level of commendation for the effort they put into the review process of this important document. Their dedication and collaboration have been instrumental in achieving the overarching objective of ensuring quality, affordable, and uniform care for individuals, families, and communities.

In light of the foregoing and the spirit of collaboration, I am pleased to extend sincere heartfelt appreciation to all contributors to this revised edition. Indeed, your invaluable contributions will undoubtedly benefit countless individuals and communities in Nigeria. Your partnership exemplifies the spirit of collective effort in advancing healthcare delivery and promoting the well-being of our dear citizens.



Dr. Bashir Idris, RCHP, FIPHP [Li idin Jama'a]
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INTRODUCTION

The National Standing Orders serve as a comprehensive set of specific guidelines, arranged by age group, conditions, findings (from history, physical examination, laboratory investigations), clinical judgment, and necessary actions, to standardize care for patients at Primary Health Care (PHC) Facilities. Developed through extensive collaboration between Community Health Practitioners' Registration Board of Nigeria (CHPRBN) and the National Primary Health Care Development Agency (NPHCDA), with support from the United States Agency for International Development (USAID), these comprehensive guidelines equip health workers with standardized procedures and protocols for managing a wide range of health conditions. The objectives of the Standing Orders are to:

- Ensure legal protection for health workers in their primary health care duties.
- Offer a structured approach for history taking and physical examination.
- Aid in the management of less common, easily overlooked, and more serious conditions.
- Maintain a consistently high standard of healthcare.
- Minimize unnecessary, expensive, and time-consuming laboratory investigations.
- Provide a framework for evaluating care and staff performance.

Following the guidelines will better position the Community Health Practitioners at the frontline to provide quality care that is Safe, Timely, Equitable, Effective, Effective, and Person-centered.

WHAT YOU WILL FIND

- Clear and Practical Guidelines: Organized by age groups and specific conditions, the Standing Orders provide step-by-step instructions for history taking, physical examination, treatment plans, and follow-up care. This user-friendly format ensures consistent and high-quality care for all patients.
- A framework to manage common and less frequent conditions. You'll find valuable guidance for even complex or life-threatening situations, along with clear referral pathways which will enhance clinical decision making.

More than just guidelines, these standing orders represent years of field-tested experience and continuous refinement, making them practical and relevant to the environments where Community Health Professionals operate. Each complaint or symptom that a patient may present with is addressed with a concise introduction, providing cues on important aspects, followed by a suggested history, examination, education, further investigation, treatment, and follow-up to ensure optimal patient care. Simplicity, clarity, and ease of use have been prioritized, with more common conditions addressed first and more severe conditions prioritized over moderate and mild ones.

In cases of complicated or life-threatening conditions, the Standing Orders specify referral to the next level of care or the nearest hospital as appropriate. The uniformity of care provided by these guidelines facilitates evaluation, setting a standard for care and allowing for the assessment of overall care adequacy and identification of areas needing improvement.

IMPORTANT NOTE: The Standing orders do not replace clinical judgement and critical thinking. Health workers must still gather essential history and physical examination details to make informed clinical decisions and provide appropriate treatment. These protocols delineate the necessary actions and decisions required to manage a patient's symptoms effectively. The Community Health Practitioner must ensure proper documentation of all inferences made and action taken in the management of the patient. By adhering to these standardized procedures, you can ensure your actions are well-supported and documented, minimizing legal risks and maximizing patient safety.

The effectiveness of these Standing Orders, like the broader primary health care concept, depend on well-trained, committed, and dedicated health workers. These Standing Orders are therefore complemented by the curricula developed for use in Schools of Health Technology and Community Health Officers' Training Institutions. The continued and proper utilization of these Orders, for both TRAINING AND PRACTICE, will ensure the sustenance of a high standard of health care in the community. For the 2024 revision of the Standing Orders, an illustrated version has been created to support the ease of use and improved understanding of the guidelines.

HOW TO USE THIS GUIDE

The Standing Orders are organized by complaints or problems derived by the health workers from the patient's history, an examination, and/or laboratory tests. Follow the steps below in your interaction with the patient:

1. Based on the patient's age group, identify the problems or complaints in the relevant section and refer to the appropriate page.
2. Ask all questions listed in the history box.
3. Perform all examinations listed in the examination box.
4. Record all significant information, including negative findings. For example, if examining the neck for stiffness, note the absence of stiffness as "no stiff neck."
5. Select the box containing all significant findings related to the patient's condition from the list of findings.
6. Follow the actions corresponding to the selected findings and manage the patient's complaints as instructed.

In some cases, it may be necessary to refer to more than one section of the Standing Orders to completely manage all of the patient's issues comprehensively. Take appropriate action based on your findings and record them in the home-based record (PHC Child Health card or green card for children under 5 years and Personal Health Record yellow card for those over 5 years old) or in the relevant patient record platform provided. Review the patient's condition whenever a course of medicine or injection is due to finish and based on the medications' onset of action. This is an excellent way to evaluate your work carefully. Ensure the home-based record card is returned to the mother/caregiver or patient.

ADDITIONAL INSTRUCTIONS:

- Every patient should undergo the complete examination listed under the First Visit at their initial contact with the health facility, except in emergencies, in which case the treatment should be prioritized and the 'first visit' examination should be completed at the

next visit.

- Address the health issues of all family members present in the clinic whenever possible.
- Follow up on all referred cases when patients return to ensure continuity of care and facilitate health worker learning.
- Advise all patients to return to the health facility immediately if their conditions worsen.
- Health education is crucial for patient care. Provide relevant health education to the appropriate patients on topics such as:
  - -Breastfeeding (immediate & exclusive) and complementary feeding where appropriate.
  - -Nutrition for older children & adults.
  - -Newborn care at home.
  - -Child-spacing.
  - -Personal hygiene.
  - -Immunization.
  - -Patient's presenting condition.
  - -STD/HIV/AIDS (or other current health concerns).
  - -Importance of home-based records.

This comprehensive approach ensures that health workers can provide effective and standardized care while continually improving their practice through ongoing evaluation and education.
Collaborative work: When you are in doubt about the care plan for a patient, consult with other healthcare workers at the facility or stabilize the patient and refer appropriately. When you do not have the required resources to manage a patient, promptly refer to the appropriate facility or level of care to avoid delay in the commencement of care for the patient.

FEEDBACK AND SUPPORT CONTACT

We encourage all Community Health Practitioners to actively participate in the continuous improvement of this guide. If you encounter any issues, have suggestions for improvements, or notice any outdated information, please do not hesitate to reach out to us. Your feedback is crucial in maintaining the accuracy and relevance of this essential resource and provide valuable insights for future revisions.

How To Provide Feedback

Email: Send detailed feedback, suggestions, or corrections to info@chprbn.gov.ng
Please include specific references to sections or illustrations where applicable.

Phone: Contact our support team at 08034462362 (Ibrahim Umar) for immediate assistance or to discuss your feedback in detail.

Online Form: Please fill out your structured feedback using the google form via this link

SECTION ONE
THE NEWBORN
(0 – 28 DAYS)



NATIONAL STANDING ORDERS

FOR COMMUNITY HEALTH OFFICERS

1.1 THE NEWBORN

A newborn is defined as a baby aged 0 to 28 days. Properly cared for, they will cry and breathe without assistance. When a newborn cries, it indicates they are breathing, which is generally a positive sign. As the baby's breathing improves, their skin will become pink, a reassuring sign to observe. However, newborns may also exhibit signs of fatigue or weakness due to the stress of birth or underlying illness. Therefore, it is essential to carefully monitor newborns and handle them gently with clean hands. Avoiding slapping the baby on any part of their body is advisable.

Newborns are susceptible to becoming wet and cold easily, so it's crucial to keep them dry and well wrapped. Mothers should be encouraged to bring their children for regular visits: monthly during the first two years of life, every three months until age 5, and as needed thereafter. Additionally, mothers should be encouraged to immunize their babies as soon as possible and continue until they are fully immunized.

1.1.1 IMMEDIATE CARE OF THE NEWBORN

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1. First One Hour	1. Normal Newborn		1.Make sure your hands are thoroughly washed and latex gloves are worn before delivery is taken. 2.As soon as baby is delivered. Baby is crying and is pink in colour, 3.Dry the baby thoroughly with a dry clean cloth. 4.Discard the wet cloth after drying wrap baby with another dry cloth. 5.Ensure the room temperature is at least 25°C -28°C to help keep the baby warm. 6.Keep baby away from direct sunlight 7.Put the baby on the mother's chest and at least for the first 1 hour & initiate breastfeeding within 30 minutes 8.Wipe baby's eyes with clean cotton; apply 0.5% Erythromycin ointment or 1%Teracycline ointment. 9.Delay cord clamping (1-3minutes after birth.) 10.Cut the baby's cord about one hour after birth, and ensure proper cord care: 11.Always wash hands before and after cutting/cleaning the cord. 12.Apply 4% chlorhexidine gel (within 1 hour of birth) on the stump, and ask the mother to continue daily. Dry soiled and wet stump after bathing and apply chlorhexidine gel.

			Physical examination: 17.Conduct a thorough physical examination, paying attention to presence of birth defects 18.check breathing and count the respiratory rate (normal is 40 - 60 breaths/minute) 19.measure the temperature (normal is 36.5 – 37.5°C) 20.Weigh the baby. 21.Document your findings in the baby's notes 22.Delay bathing till 24 hours after birth. 23.Give parents guidance for home care: 24.Register the birth. 25.Practice exclusive breastfeeding for six months. 26.Recognize and manage common breast problems. 27.Practice hand washing. 28.Monitor the newborn closely. If neonatal jaundice or any other danger sign is observed, present baby immediately at the health facility. 29.Use only 4% chlorhexidine (7.1% chlorhexidine Di gluconate)) gel in 25g tube for cord care. 30.Complete immunization schedule as in the child health card. At discharge give the parents the National Pictorial Newborn Discharge Guide Information Leaflet for basic instructions.
	2. Baby born with meconium-stained liquor and unable to breathe	Moderate to Severe condition Baby has asphyxia and unable to breathe (GOLDEN MINUTE)	1.If there is blood or meconium-stained liquor and baby is not crying 2.Put the baby on a ventilation table and immediately commence Neonatal resuscitation (Helping Babies Breathe within a minute.) through the following ways: 3.Position the head and neck to open the airway 4.Clear secretions from the airway using the mucous bulb extractor 5.Dry newborn, stimulate and re-position) 6.Assist the baby to breathe using the bag and mask, making sure you are using the right-sized mask and bag. 7.Ensure nos. 3, 4, 5 and 6 are done within 1 minute for the baby who is unable to breathe at birth. Do not handle the baby upside down, slap the feet or pinch the chest.

			8.If baby fails to breathe, continue to ventilate, and REFER to a higher-level facility for advanced care 9.If baby begins to breathe and cry 10. check if baby is fast breathing (two counts of 60 breaths or more in one minute) and 11.check temperature for high (37.5°C or more) or very low (< 36.5°C). 12.continue with other newborn care steps under "Normal baby" above 13.Continue to care for babies who are crying 14.Do not remove vernix or bath baby until at least 24hours afterbirth 15.Continue to keep baby warm and initiate skin-to-skin contact with mother 16.Encourage mother to initiate breastfeeding within 30minutes of birth (offer help) 17.Do not give artificial teats or pre-lacteal feeds to the newborn (no water. Sugar water or local feeds) 18.Examine newborn 1 hour after delivery of placenta
II. After 1 hourof birth	Within 60-90 minutes of life check examination findings: 1. General appearance -movement of the extremities, hypotonia, colour, respiratory distress, dysmorphic features and obvious malformations. 2. Assess breathing (baby must be calm) 3. Check for fast/ difficult breathing, signs of respiratory distress, breathing pattern, respiratory rate, air entry to the lungs, presence of abnormal sounds in the lungs (listen for grunting) 4.Count the breathes (30-60breaths per minute, repeat the count if elevated] 5. Check for chest in-drawing 6. Check Heart rate per minute (normal rate is between 120 and 160 bpm) 7. Measure Newborn length (normal range is 48-53 cm) 8. Check for movements (normal and symmetrical) 9. Check for jaundice, -yellow palms/ soles/ eyes 10. Presenting parts swelling, bruises Abdomen		1.After the clinical examination of the newborn, ensure DOCUMENTATION of findings: 2.Specifically, record all observed danger signs seen in the newborn: -Not feeding, poor feeding, vomiting -Lethargy -Respiratory distress, chest-in-drawing, nasal flaring -Fast breathing (breathing rate >60 per minute), grunting -High body temperature (38°C) -Convulsions -Any jaundice in first hour of life, or yellow palms and soles at any time AND REFER IMMMEDIATELY. 3.Ensure Referral and follow-up to health facility 4.Communicate/Liaise with health facility 5.Fill out the community referral form (In triplicate) 6.Assist to mobilize for transport (Emergency transport system) 7.Keep newborn in skin-to-skin contact with the mother in KMC position

	11. Skin-feel for warmth or cold, pallor, rash, pallor, plethora, meconium staining, cyanosis, birthmarks, etc. Check for yellow palms/ soles/ eyes 12.Temperature-check Axillary temperature (normal is between 36.5°C and 37.5°C) below36.5°Cishypothermia		8.Express breast milk and feed baby with cup 9.Practice and maintain hygiene 10.Follow-up with the patients 11.Complete examination record in baby's notes, sign and date it. 12.Record all the findings in the newborn's registration books or chart prepared for the purpose. 13.Record any discussion or advice given to parents 14.Record any congenital abnormality in the congenital abnormality registry 15.Classify the newborn into normal or abnormal (specify problems identified) 16.Educate mothers to do the following: a.Ensure baby does not have contact of baby with wet or cold surfaces b.Ensure a room temperature of at least 25°C -28°C to help keep the baby warm but keep the baby away from direct sunlight c.Ensure continuous breast-feeding whenever the baby shows signs of readiness to feed and be sure baby is feeding and sucking well d.Ensure good positioning of the baby during breastfeeding; -Ensure baby can attach well and the mother is comfortable; -Place the baby with the head and body in a straight line; -Baby's body turns toward mother; -Baby's body is close to the mother and baby's abdomen is touching the mother's own; and -Baby's whole body is supported. e.Ensure good attachment of the baby during breastfeeding; -Ensure that baby's chin is touching breast; -Baby's mouth is wide open: Baby's lower lip is turned outward and more areola is visible above than below baby's mouth.

			f.Ensure adequacy of feeds by feeding the baby every 2-3 hours or 8 -12 times a day; baby is getting enough milk per feed through Breasts that is soften with feeding; and Swallowing sounds are heard during feeding g.Prevent breastfeeding problems such as; -Inverted nipples by stimulating nipples before feeding and shaping the breast before attachment; -Breast engorgement (Very full, tight and shiny breasts) by feeding the baby more often and/or express milk prior to attachment -Sore or cracked nipples (usually results from poor attachment or skin infection) by ensuring good attachment and treating by applying drops of breast milk to the skin of the nipple, and washing her breasts with water daily and avoiding soaps, medicated lotions and ointments, and continuing breastfeeding and/or expressing breast milk to feed baby -Mastitis (occurs when there is a blocked duct or from infection -usually affects one breast with a well-defined red, sore and swollen or hardened area) by breastfeeding frequently or expressing breast milk; -Inadequate milk volume by increasing fluid intake and frequency of breastfeeding and by applying warm compress to the breasts, massaging the back, neck, breasts and nipples. h.Always ensure baby is passing stools and urinating i.Ensure baby has received immunizations – BCG, OPV0 and hepatitis B vaccine which should have been given within 24hours of delivery if this has not been done. j.Ensure to commence post-partum family planning k.Visit the health clinic for next immunization visit at 6 weeks

			l.Ensure adequate home care by: a. Registering birth b. Practicing exclusive breast-feeding for six months c. Recognizing and managing common breast problems d. Practicing hand washing e. Monitoring closely for neo-natal jaundice and recognizing other danger signs and visiting the health facility f. Ensure proper cord care: -Always wash hands before and after cleaning the cord. -Apply only 4% chlorhexidine (7.1% chlorhexidine digluco-nate)) gel in 25g tube for cord care daily. -Dry soiled and wet stump after bathing and apply chlor-hexidine gel. -Do not cover the cord with a dressing or diaper, fold diaper below stump. -Do not apply methylated spirit or other medications or substances g. Completing immunization schedule as in the child health card h. Always supporting baby's head and neck and never shake or throw up, whether in play or in frustration.

1.1.2 FIRST POSTPARTUM HOME VISIT (2—3 DAYS AFTER THE BIRTH OF A NEWBORN)

The postnatal period begins immediately after the birth of a newborn, extending to six weeks after delivery. This period is crucial for the survival of the newborn. Proper care must be taken to avoid complications including death. The post-natal care visits days are 3, 5, 7, 9, 14, 21, 28 - If the birth weight of the baby is normal (>2.5kg), at least three home visits are required on days 3, 7 and 14. - If the birth weight of the baby is low.

HISTORY	EXAMINATION/ACTIONS
Ask for the frequency of newborn breastfeeding (infants eat about every 1–3 hours) 1.Ask for the number of times the Newborn wets the diaper in a day (should have about 6 wet diapers a day) 2.Ask for the number of times in a day that the newborn stools (most newborns have 3 or 4 soft bowel movements a day) 3.Ask for number of times the newborn sleeps in a day (A new-born may sleep 14 to 17 hours or more in 24 hours) 4.Ask for the number of hours that the newborn sleeps between feedings (Newborns should not sleep more than 4 hours between feedings until they have good weight gain, usually within the first few weeks).	Perform comprehensive physical examination (PE) within 24 hours of birth, and within 72 hours of life 1.Provide essential newborn care 2.Assess for Danger signs and refer newborn to the health facility if any danger sign is seen/observed 3.Check the baby's weight, length, and head circumference and plot the measurements on a growth chart. 4.Check if Newborns seem hungry 5.Watch as the mother breastfeeds the newborn and offer help with any problems. 6.Let the mother undress the baby and conduct the following: a.an eye exam, b.listening to the baby's heart and feeling pulses if readings are a.abnormal, refer ii.inspecting the umbilical cord If smelly or issuing pus, clean and apply Cord care daily using 4% chlorhexidine gel iii.checking the hips Note and observe abnormalities, and refer iv.Physiologic jaundice

5.Ask which of the actions below the mother has noticed in the child in the first 3-5days of the child's life:
a.Pay attention to faces or bright objects 8–12 inches (20–30 cm) away.
b.Respond to sound — they may quiet down, blink, turn head, startle, or cry.
c.hold arms and legs in a flexed position
d.Move arms and legs equally.
e.Lift head briefly when on stomach (babies should be placed on the stomach only while awake and under supervision).
f.have strong newborn reflexes, such as:
-rooting and sucking: turns toward, then sucks breast/bottle nipple
-grasp: tightly grabs hold of a finger placed within the palm
-straightens arm when the head is turned to that side and bends opposite arm
-throws out arms and legs, then curls them in when startled (startle response)
6.Ask mother if she has commenced Post-partum Family Planning:
7.Ask if the mother is experiencing low energy, fatigue, sleep, or appetite problems (then she may have postnatal blues)
8.Other educational needs

7.Repeat Initial examinations done for the newborn within in the first 60-90 minutes of life as shown in II above
8.Screening for:
-Congenital malformations, syndromes, and associations
-Congenital heart diseases
-Congenital cataracts
-Cryptorchidism
Developmental dysplasia of the hip
9.Discourage harmful traditional care practices and Counsel the mother against these harmful practices including the following:
i.Cord care - apply only CHX and discourage the use of other substances such as toothpaste or Cow dung to the cord.
ii.Colostrum is yellowish-coloured milk and should be given to the newborn because it protects the newborn from infection.
10.Promote Exclusive Breast Feeding - Exclusive breastfeeding means that a baby takes only breast milk with no additional foods or liquids, not even water (as breast milk is predominantly water), except medicine if necessary, for the first 6 months when complementary feeding begins. Baby should be breastfed 8 – 12 times per day and on demand.
11.Tell the mother the advantages of breast milk which include:
-It is the best food for the baby as it digests easily
-It protects newborn from infection
-It promotes bonding between the mother and the newborn
12.Demonstrate good positioning and good attachment and discuss the importance of breastfeeding
13.Promote Immunization: Newborns are particularly vulnerable to infections. Immunization is an important way to protect an infant's health. They prevent illnesses and diseases and failure to immunize may put the infant at risk of serious illness or even death
14.Check immunization status and educate on the importance of completing all the child's immunizations.
15.Immunize newborns according to the immunization schedule -
16.Register birth (if not done already)
17.Provide information on other scheduled visits:
-Second postnatal visit day 3
-Third postnatal visit: day 7
-Fourth visit: between day 14 – 21
-Fifth visit: day 28
-Sixth visit: day 42 (6 weeks)
18.Encourage caregivers to talk, sing, and play with the newborn
19.Educate and counsel mother on Post-partum Family Planning:
Family planning methods - Oral contraceptives (progesterone only.); Education on Long Acting Methods e.g. injectables and self-injectables including DMPA-SC.; Long- acting and permanent methods (IUD/tubal ligation, implants

Provide further Health Education to the Newborn mother on:

A. Routine Care
1.Wash hands before handling the baby and avoid people who may be sick.
2.Keep the diaper below the umbilical cord so the stump can dry.

	4.Use fragrance-free soaps and lotions. 5.Hold the baby and be attentive to their needs. 6.Sing, talk, and read to your baby and look directly to your baby's eyes and smile when breastfeeding. Provide ways for your baby to see, hear, move arms and legs freely, and to 7.Call your Health care worker if your infant has a fever of 100.4°F (38°C),is acting sick, isn't eating, isn't peeing or pooping, looks yellow, or has increasing redness or pus around the umbilical cord or circumcision. 8.Don't give medicine to an infant younger than 2 months old without talking to the health facility 9.Breastfeed your baby 10.If you as a new mother feel intense tired, sad, moody, or anxious, call your Health worker 11.Talk to your health worker if you're worried about your living situation or don't have the things that you need to take care of your baby or have food, or a safe place to live.Your health worker can tell you about community resources or refer you to a social worker. 12.Ensure you give your newborn six basic needs required: security, clothing, enough sleep, exclusive breastfeeding, sensory stimulation, love, and attention **B. Safety** 1.To reduce the risk of sudden infant death syndrome (SIDS): a.Always place your baby to sleep on a firm, flat mattress on their back, and not on the stomach or side, in a crib or bassinet without any crib bumpers, blankets, quilts, pillows, or plush toys. b.Let your baby sleep in your room or crib next to your bed until your baby's first birthday, or for at least 6 months, when the risk of SIDS is highest. c.Avoid overheating by keeping the room temperature comfortable. Dress your baby for room temperature and don't cover your baby's head while they're sleeping.Watch for signs of overheating, such as sweating or feeling hot to the touch. d.Don't let your baby fall asleep on a product that isn't specifically designed for sleeping babies, such as a sitting device (like a car seat), a feeding pillow 2.Don't smoke or use e-cigarettes. Don't let anyone else smoke or vape around your baby. 3.Always put your baby in a rear-facing car seat in the back seat. Never leave your baby alone in a car. 4.While your baby is awake, don't leave your little one unattended, especially on high surfaces or in the bath. 5.Never shake your baby — it can cause bleeding in the brain and even death. Call a friend, relative, or your healthcare provider for help. 6.Avoid sun exposure by keeping your baby covered and, in the shade, when possible.

1.2. NEWBORN WITHOUT ANY COMPLAINT- THE WELL CHILD

A well visit, or well-child visit, is a preventative health appointment with the Child's Health caregiver with the goal of helping well children to stay healthy in their earliest years, so they can grow into healthy adults. The Well Child visits concentrate on the whole body, not on sickness based on the child's age and unique needs. The well-child visit is very important and should always include screening for health problems, growth monitoring, and promotion.

Health education about nutrition, Immunization and protection from illness and other conditions as relevant to the family is given. Mothers should be encouraged to take their children for visits while they are well monthly in the first two years of life and every 3 months thereafter until 5 years old, and once a year until the child reaches 21 years. These visits also provide opportunities to monitor mothers and talk to them about postpartum depression or anxiety.

HISTORY	EXAMINATION/ACTIONS
1.Birth: Was pregnancy and birth normal? a.Where was the child delivered? b.Did the child cry immediately after birth? c.Was the child put to breast within 30 minutes after birth? d.Was the child jaundiced (yellow) at any time? 2.Nutrition: How have you been feeding the child? a.Exclusive Breastfeeding? Supplementary feeding? If not exclusively breast feeding, what else? 3.Social: With whom does the child live? a.Does mother work outside the home? b.Who looks after the child when the mother is away? c.What is your source of water? d.What type of toilet do you use? e.Is your house screened against mosquitoes? f.Do you use mosquito net or Insecticide Treated Net (ITN)? 4.Medical: Has the child had illness in the past? E.g. Yellow eyes (Jaundice), charging eyes, fever etc. a.What treatment was given? Where? Any illness or chronic cough in the family? b.Does the child have any problem now?	1.General appearance: 2.Obvious congenital abnormalities e.g. (mongolism, spina bifida, microcephaly, harelip) 3.Crying, active, limp, floppy. 4.Colour: normal, pale, yellow, blue 5.Weight: weigh and chart. 6.Head Circumference: size, swelling, colour and texture of hair, fontanelle-depressed or bulging, sutures-normal or wide 7.Eyes: discharge (sticky), yellow, red 8. Mouth: ability to suck normally, abnormality (cleft upper palate) 9. Spine: swelling, opening, curvature 10.Chest: breathing normally, respiratory rate (note if 60 per minute or more) 11. Abdomen: condition of cord, umbilical hernia 12. Arms and legs: normal movement, deformity, fingers and toes (for extra or joined webbed fingers) 13. Genitalia: abnormality (check vaginal/penial opening, size of clitoris, both testes in scrotal sac), swelling, congenital hernia. 14.Anus: open or closed. 15.Temperature: check and record. 16.Developmental Milestone 17.At birth sees and responds to voice, 18.Neurological tests (muscle tone, moro, suckling, rooting and grasping reflexes 19.Measure heart rate 20.Take blood pressure 21.Give appropriate childhood vaccines (will vary by age) 22.Give Health education: 23.Handwashing with soap and water (or use a hand sanitizer) before handling your baby. Support baby's head and neck. 24.Never shake baby, whether in play or in frustration. 25.Always fasten baby securely when using a carrier, stroller, or car seat 26.Avoid rough play with newborns, such as jiggling them on the knee or throwing them in the air. 27.Clean around the cord stump with plain water and blot dry until the stump dries up and falls off 28.Feed babies whenever they seem hungry. Baby may show you they're hungry by crying, putting - fingers in their mouth, or making sucking noises. A newborn baby needs to be fed every 2–3 hours 29.Change the position of your baby's head from night to night (first right, then left, and so on). This helps prevent a flat spot from developing on one side of the head 30.Immunization 31.Discouraging application of anything else apart from chlorhexidine gel on the cord 32.Refer for nutrition counseling and food demonstration at 5months. 33.Remind the mother to take home-based records with her any time she takes the baby to the clinic or hospital. 34.Remind mothers about the next Well visit and information to bring for discussion including; 35.Changes in her child's health and development **Changes in social behaviors** 36.Your child's eating habits 37.Any questions or concerns

Health Education

Encourage mothers to:

a.Wash hands with soap and water (or use a hand sanitizer) before handling your baby.
b.Support baby's head and neck.
c.Never shake baby, whether in play or in frustration.
d.Always fasten baby securely when using a carrier, stroller, or car seat
e.Avoid rough play with newborns, such as jiggling them on the knee or throwing them in the air.
f.Clean around the cord stump with plain water and blot dry until the stump dries up and falls off
Feed babies whenever they seem hungry. Baby may show you they're hungry by g.crying, putting fingers in their mouth, or making sucking noises. A newborn baby needs to be fed every 2–3 hours
h.Change the position of your baby's head from night to night (first right, then left, and so on). This helps prevent a flat spot from developing on one side of the head.

Advice on immunization

1. Encourage exclusive breast-feeding for six (6) months.
2. Discourage application of anything else apart from chlorhexidine gel on the cord
3. Refer for nutrition counseling and food demonstration at 5 months.

Remind mother to take home-based records with her anytime she takes baby to the clinic or hospital.

1.3 NEWBORN EMERGENCIES

An ill newborn can quickly become an emergency so all newborns should be treated promptly. Immediate actions must be taken whenever these danger signs: abnormal temperature (Cold or Hot), poor sucking or inability to suckle, breathing problem – fast breathing and severe chest in-drawing, yellowness of the hands and body, poor feeding or not feeding at all, bleeding from the cord or redness around the cord stump, no movement at all or only on stimulation and convulsion, are observed in a newborn. There should be no hesitation in initiating appropriate actions or immediately refer newborn, where necessary to the next higher-level facility.

1.3.1. DIFFICULT BREATHING

Difficult breathing means the baby has to make some effort to breathe. Rapid breathing for a newborn is respiration rate between 40-60 per minute. Difficult breathing in children is an emergency. It may be caused by infection, cold. In babies, chest in-drawing, flaring of nose indicate difficult breathing and a danger sign to look out for.

HISTORY	EXAMINATION/ACTIONS
How old is the baby? 2.What is the problem with the baby? When did it start? 3. Has the baby been ill? Had fever? Or been abnormally sleepy or difficult to wake up? 4. Has the baby stopped feeding well? 5.What do you feed the baby on? 6.Was the baby given any medicine or treatment? 7.What medicines were given to the baby since he was ill? 8.Are you using methylated powder or camphor? 9. 9.Were the baby's eyes yellow at birth? 10.Who takes care of the baby? 11.Was the pregnancy normal? If not, what were the problems? 12.Was the baby born head first or bottom first? 13.Was the delivery normal? 14.Was there any difficulty? 15.Was the baby normal at birth? 16. Did the baby cry immediately after birth? 17. How was the cord cut and dressed?	1.General appearance: a.Skin colour (blue, pale, yellow) b.Spontaneous movement. 2.Chest: Check respiratory rate (note if 60 or above per minute), gasping breath, heartbeat, chest in-drawing. 3.Head: a.swelling, b.anterior fontanelle –for bulging or depression, c.Face –for any obvious abnormalities. 4.Weight: weigh and chart 5.Temperature: Check with low reading or digital thermometer and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
1.Notbreathing or Gasping at birth	-Pale or bluish colour, -Gasping than 7 -Apgar score less than 7 -Limp arms or Legs No heartbeat, no respiration	Severe Condition	
2. Difficult breathing after birth	-Fast breathing or slow breathing. Severe chest in-drawing	Severe Condition Emergency	1.Communicate with parents/ Caregiver: a.The need for referral. b.The referral hospital. c.Available continued care for mother (if needed) at the referral hospital. d.Obtain their consent for referral. 2.Give first dose of antibiotics (e.g. IM. Ceftriaxone 20-30mg/kg) immediately (Stat). REFER immediately to hospital

1.3.2 CONDITIONS AFFECTING HANDS AND FEET

These may be secondary to underlying medical conditions.

HISTORY	EXAMINATION/ACTIONS
1.How old is the baby? 2.What is the problem with the baby? 3.When did it start? 4.Has the baby been ill? Had fever? Or been abnormally sleepy or difficult to wake up? 5.Has the baby stopped feeding well? 6.What do you feed the baby on? 7.Was the baby given any medicine or treatment? 8.What medicines were given to the baby since he was ill? 9.Are you using methylated powder or camphor? 10.Were the baby's eyes yellow at birth? 11.Who takes care of the baby?	1. General appearance a. Skin colour (blue, pale, yellow) b. Spontaneous movement 2. Eyes: jaundice (examine under natural light) 3. Skin: rashes, dryness, jaundice 4. Temperature: check with low reading thermometer and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
Cold hands and feet	-Temperature less than 36.5oC -Cold Stress 36.0 – 36.4oC -Moderate hypothermia 32.0 – 35.9oC Severe hypothermia <32.0oC	Moderate Condition	1.Ensure baby is dry and in clean dry clothing 2.Place baby in skin to skin contact in a pre-warmed blouse/shirt opening at the front with the baby in a nappy, hat and socks 3.Wrap baby in warm blanket 4.Ensure that the room is warm (at least 25oC) and is without drafts 5.If temperature is improving but baby cannot maintain normal temperature, put baby in Kangaroo Mother Care (KMC). 6.If temperature does not improve after one hour of re-warming, REFER immediately to hospital
2.Yellow hands and feet	- Jaundice, No fever	Moderate Condition	REFER immediately to hospital where phototherapy is possible.

Health Education

1.Continue skin to skin contact between mother and baby.
2.Teach mother about Kangaroo Mother Care -KMC).
3.Exclusive breastfeeding.
4.Counsel family members to support mother go through process.

1.3.3 CONDITIONS AFFECTING UMBILICAL CORD

The umbilical cord is very significant in a newborn, any infection around the area can cause Neonatal mortality. Common indications of danger include bleeding from the umbilical cord, redness, pus, and smelly cord. These are conditions affecting umbilical cord which could endanger the baby's life.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? When did it start? 2.Is the baby having fever? Or been abnormally sleepy or difficult to wake? 3.Is umbilical cord smelly or bleeding or just wet? 4.Has the umbilical cord dropped off? What medicine have you given the baby?	1.Check umbilicus for discharge, bleeding, swelling, redness, granulation on umbilicus. 2.Temperature Weight

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
I. Smelly umbilical cord	Discharging (pus), smelling, umbilicus swelling, redness around the umbilicus	Severe Condition	1.Wash with clean water and soap, dry with clean cloth 2.Give Benzyl penicillin 25mg/kg/dose immediately (Stat) 3.IM. Gentamicin 5mg/kg/dose immediately (stat) Refer immediately to hospital.
II. Bleeding umbilical cord	Bleeding umbilicus	Moderate to severe condition	1.Re-tie the cord immediately 2.Give a third tie at the base of the cord 3.Clean with sterile water. Apply chlorhexidine gel 4.Give IM.Vitamin K, 0.5mg (preterm baby), 1.0mg (full term baby) Refer if bleeding persists.
III. Wet umbilicus	•Wet umbilicus Temperature normal (37.50C	Mild condition	1.Clean with sterile water 2.Teach mother to clean with sterile water 3.Expose umbilicus Review daily and Check immunization status
IV. Retained umbilical stump	•Granulation on umbilicus	Mild condition	1.Clean with sterile water 2.Cauterize with blue stone 3.Review daily 4.Administer immunization as necessary Record in home - base card and return to mother

Health Education

1.Encourage mother/caregiver.
2.Teach mother to practice and maintain hygiene
3.Teach mother/caregiver about care of the umbilicus.
4.Counsel mother against harmful traditional practices (DISCOURAGE use of toothpaste, cow dung or any other substance on the cord).

1.3.4 OTHER BLEEDING PROBLEMS

This is bleeding from other parts of the body.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem with the baby? When did it start? 2.Has the baby been ill? Had fever? Or been abnormally sleepy or difficult to wake up? 3.Has the baby stopped feeding well? 4.What do you feed the baby on? 5.Was the baby given any medicine or treatment? 6.What medicines were given to the baby since he was ill? 7.Where is the bleeding coming from? a)Nose, gums b)Rectum c)Ears From Circumcision.	1.Mouth: check for bleeding 2.Ears: check for bleeding 3.Rectum: check for bleeding 4.Nose: check for bleeding 5.Chest: check breathing Sole of feet- pallor

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
I. Baby bleeding from mouth, rectum or other parts of the body	-Bleeding from mouth -Rectum -Pallor Normal breathing	Severe Condition	1.Apply pressure on the site where practicable. REFER to hospital immediately.
II. Bleeding from circumcision	- Bleeding with or without pallor	Moderate condition	1.Put pressure on site for 10 mins. 2.show mother how to maintain pressure (over a period of 1 – 2 hours) 3.if bleeding does not stop Refer If there is pallor REFER.

Health Education

1.Teach mother/caregiver how to apply pressure on bleeding site.
2.Counsel/Reassure mother/caregiver on baby's condition and the need for referral.

1.3.5 CONDITIONS AFFECTING THE HEAD

There are many problems that can affect the head during the delivery process, such as injury to the head, swelling of the head, sunken or bulging fontanelles, cephalohematoma and caput succedaneum.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem with the baby? When did it start? 2.Has the baby been ill? Had fever? Or been abnormally sleepy or difficult to wake up? 3.Has the baby stopped feeding well? 4.What do you feed the baby on? 5.Has the child convulsed? 6.Was the swelling present at birth? 7.Was the head larger than normal at birth or after? What part of the head is involved?	1.General Appearance: Irritable, lethargic, crying excessively, evidence of injury. 2.Head: Swelling, check size(head circumference), Fontanelle(-depressed, bulging) 3.Evidence of attempt to drain swelling. 4.Check for lump. 5.Check wideness of the fontanelle 6.Check if fontanelle is pulsating.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
I. Swelling on the head with Convulsion	-Child irritable, -Lethargic -Convulsing -Excessive crying Evidence of injury	Severe condition Suspect Birth injury	1. REFER immediately to hospital
I. Swelling on the head not present at birth but comes up within about 12-24 hours and continues to increase	-Large swelling on head -Evidence of attempt to drain swelling Baby is blue, or infected swelling	Severe condition	1.IM. Benzyl penicillin25mg/kg immediately (stat) 2.IM. Gentamicin 5mg/kg immediately (stat). REFER immediately to Hospital.
III. Swelling on the side of the head, No other complain	-Lump on the head Baby otherwise well	Moderate condition Suspect birth Injury	1.Do not try to incise or drain. 2.Explain findings to mother. 3.Ask mother not to rub or apply anything. 4.Review in 2days then 1 week If getting bigger, REFER.
iv. Wide opening on front part of head	-Fontanelle too wide Pulsating	Mild condition	1.Explain condition to mother that fontanelle should close at about 18 months 2.Monitor head circumference weekly for 2 months. If head is getting larger, REFER to hospital.

Health Education

1.Counsel and reassure mother/caregiver on baby/condition.
2.Explain to mother/caregiver about open fontanelle.

1.36 CONDITIONS AFFECTING THE EYES

The process of delivery can be a channel for Newborn to develop eye condition when the baby sweeps through the vulva bacterial infection can set in and put the Newborn in danger. These are problems affecting the eyes which if not well managed could lead to impaired vision or blindness.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem with baby's eyes? 2.How long has the baby had the problem? 3.What's the nature of discharge? Blood, pus? 4.Has the baby had it before? 5.Did the problem come suddenly or gradually? 6.Did anything enter the baby's eyes? 7.Did the mother or anyone else put something in the baby's eyes? What was it? Has any medicine been put in baby's eyes? If so what medicine?	1.General Appearance: In pain, Eyes: Redness, pus, sticky eyes

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
i. Discharge from baby's eye(s)	- Sticky eyes Pus in the eyes	Moderate - Severe condition (suspect gonococcus)	1.Wash hands with soap and water before and after treatment. 2.Gently clean eyes with cotton wool soaked in cooled boiled water 6-8 times a day 3.Use separate cotton wool for each eye. 4.Apply 1% tetracycline eye ointment in each eye 8 hourly day for 7days (ensure that the eye are clean before application) 5.Give IM. benzyl penicillin 25mg/kg immediately (stat), 12hourly for 5 days 6.Review in 2 days If no improvement and eye discharge become copious, REFER to hospital.
ii. Small amount of blood in the eye	-Blood-shot in the white part of baby's eyes	Mild condition	1.Explain condition to mother (This is due to pressure at birth) 2.No treatment is needed 3.Follow-up in 2 days then 1 week If mother anxious, REFER to hospital

Health Education

1.Counsel and reassure mother/caregiver.
2.Teach mother/caregiver to practice and maintain hygiene.
3.Discourage mother/caregiver from applying any other substance.

1.3.7 CONDITIONS AFFECTING BABY'S BREAST

These are common problems affecting baby's breast which may be due to the effect of maternal hormones.

HISTORY	EXAMINATION/ACTIONS
1.What's the problem with the baby and when did it start? 2.Is the child crying excessively? 3.Does the child have fever? 4.When did the fever start What treatment have you given at home?	1.General appearance: Ill looking, 2.Breast: check for tenderness, redness, production of little milk, warm breast body temperature: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
i. Swelling of breast in the newborn infant; Excessive crying; Feve	•Swollen, tender, warm breast, Rectal Temp. 38.00C	Moderate to severe condition	1.Give Syrup Paracetamol (120mg/5ml) 2.5ml immediately REFER
ii. Swelling of breast in the newborn infant	•Breast swelling, no redness, not painful, not warm, may produce little milk	Mild condition	1.Explain to mother that condition will disappear by itself 2.Advise her to leave breast alone, squeezing will increase swelling. 3.Follow-up visit until better. If no improvement, REFER

1.38 CONDITIONS AFFECTING THE ANUS

The anus is the terminal end of the alimentary canal. Common problems affecting the anus include fissures, redness and prolapse. Some babies may be born with imperforate anus and may not be able to pass stool.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.Has the child passed stool since birth? Has any medication been given?	1.General appearance: Ill looking, 2.Breast: check for tenderness, redness, production of little milk, warm breast body temperature: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
i. Swelling of breast in the newborn infant; Excessive crying; Fever	•Swollen, tender, warm breast, Rectal Temp. 38.00C	Moderate to severe condition	1.Give Syrup Paracetamol (120mg/5ml) 2.5ml immediately REFER
ii. Swelling of breast in the newborn infant	Breast swelling, no redness, not painful, not warm, may produce little milk	Mild condition	1.Explain to mother that condition will disappear by itself 2.Advise her to leave breast alone, squeezing will increase swelling. 3.Follow-up visit until better. If no improvement, REFER

1.3.8 CONDITIONS AFFECTING THE ANUS

The anus is the terminal end of the alimentary canal. Common problems affecting the anus include fissures, redness and prolapse. Some babies may be born with imperforate anus and may not be able to pass stool.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.Has the child passed stool since birth? Has any medication been given?	1.What is the problem? 2.Has the child passed stool since birth? Has any medication been given?

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
I. Newborn unable to pass stool in the first 48 hours of birth	•Anal opening absence •Confirm condition using rectal thermometer very gently	Severe condition Imperforate anus	1. REFER to hospital immediately
II. Newborn unable to pass stool in the first 48 hours of birth, With or without vomiting	•Anal opening present	Severe condition abdominal obstruction	1. REFER to hospital immediately

Health Education

1.Counsel and reassure mother about baby's condition.

1.3.9 CONDITIONS AFFECTING THE LIMB

These are common condition affecting either the arms and/or legs. Deformities of the arms and legs if present since birth often are not correctable. New deformities or swellings may be due to normal movement of the parts whenever there is history of injury, swelling or deformity.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? When did it begin? 2.Was there an accident or injury? 3.Is the baby able to move limbs? Is there any pain when moving the limbs?	1.What is the problem? When did it begin? 2.Was there an accident or injury? 3.Is the baby able to move limbs? Is there any pain when moving the limbs?1.General appearance: ill looking, crying excessively Limbs: deformity- foot turned inwards, swelling, fracture, tenderness, loss of function, warm to touch(area of tenderness)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
I. Foot is turned inward and cannot be turned outward or upward.	•Hip is out of socket. •Foot is turned inward and cannot be turned outward or upward.	Severe condition	REFER immediately to the hospital.
II. Baby unable to move limb(s] or foot	•Less than normal movement in one or more limbs with or without tenderness	Moderate condition	1. Splint the bad limb (Upper limb) by putting the baby's palm on the chest and strap limbs down. 2. Splint the bad limb. 3. Bandage the bad limb to the good limb. 4. REFER to the hospital for advanced care.
iii.Baby unable to move upper limb pain.	•Fracture of the collar bone. Pain on picking up baby by the armpit.	Moderate condition	1.Apply figure of 8 bandages. 2. Give Paracetamol Suppository 125mg per rectum, if not available, give Syrup Paracetamol(120mg/5ml) 2.5ml immediately (stat). 3. REFER to the Hospital.

Health Education

1.Counsel and Reassure mother/caregiver on baby's condition.
2.Teach mother about accident prevention at home.

1.3.10 CONDITIONS AFFECTING THE SKIN

These are common conditions affecting the skin. It could be due to poor hygiene and lack of safe water for bathing.

HISTORY	EXAMINATION/ACTIONS
1.How long has the postules, abcess, rash been there? 2.Does anyone in the family have a similar rash or skin problem? What is the source of your water supply?	1.General appearance: uncomfortable 2.Skin: (a) location, distribution and size of lesions (b) colour of rash or lesion Temperature, respiratory rate, wheeze: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
I.Abscesses (Boil)	•Abscesses or blisters on the skin with or without fever	Moderate condition	1.Apply warm compress 3 times daily until drained 2.Daily cleaning and dressing 3.If fever is present, Give Paracetamol Suppository 125mg per rectum, if not available, give Syrup Paracetamol(120mg/5ml) 2.5ml 8hourly for 3 days Review in 2 days
II. Pustules, or blisters on the skin	•Pustules	Moderate condition	1.Wash hand with clean water and soap 2.Gently wash off pus and crust with boiled and cooled water and soap 3.Dry the area with clean towel, Paint with GV,Wash hand 4.Advise mother to continue 5.Review in 2 days, if no improvement, REFER to hospital a.Examine mother for skin infection and if present, treat using Adult section of the Standing Orders Encourage personal hygiene.
III.Skin rashes	•Heat rash	Mild condition	1.Advise mother to: a)Avoid heavy clothing. Use light b)Bath child twice daily. c)Calamine lotion after bathing. d)Keep in well ventilated room. Ensure good hygiene practices

Health Education

1.Proper hand washing techniques.
2.Teach mother/caregiver about personal hygiene.

1.3.11 NEONATAL TETANUS

This is a condition characterized by poor or inability to suckle, spasms and stiffness of the body.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem with the baby? 2. How long has the baby had the problem? 3. Has the baby stopped sucking well? 4. Does the baby have fever? Or cold body? 5. Does the baby have stiffness of the body? 6. Is the baby able to open mouth without difficulty? 7. Is the baby's breathing normal? Difficult?	1.General appearance: ill looking, lethargic or unconscious. 2.Mouth: locked jaw. 3.Neck: stiff neck. 4.Body: stiffness, arching of the back, spasms. 5.Limbs: stiffness. Temperature: check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
I. Does not open mouth, cannot feed, "Convulsion	•Locked jaw •Stiff abdomen •Arched back •Stiff limbs Spasms	Severe condition (Suspect neonatal tetanus)	1. REFER immediately to the hospital perform first line management or medication before referral e.g. phenobarbitone 15 to 20mg/kg I.M stat
II. Inability to suckle at breast Fever or cold body Convulsion Difficult breathing	•Warm to touch or temp. •37.5°C and above •Fever or hypothermia •Cold to touch (temp. 35.5°C) Lethargic or unconscious	Severe Condition (Suspect Sepsis)	1.Clear nose if blocked REFER immediately

Health Education

1.Reassure mother/caregiver on baby's condition.

1.3.12 DIARRHOEA AND VOMITING IN NEWBORN

Diarrhoea is passage of 3 or more loose/watery stools in 24 hours.
Vomiting is throwing up of contents of the stomach. Babies under six months often bring up little of their food immediately after being fed (this is normal). A child on exclusive breast-feeding can stool for 5-8 times daily (this is normal).

HISTORY	EXAMINATION/ACTIONS
1.How long has the baby had diarrhea? 2.How many times has the baby passed stool? 3.Is the baby vomiting? How often? Does the baby vomit everything? 4.Is there any other complaint such as fever? 5.Is the baby on exclusive breast feeding? If no, what has the baby being fed with? What is the method of feeding? 6.Has any medication been given? What type? Does the baby take breast eagerly normally or poorly?	1.General appearance- irritable, ill-looking, lethargic, unconscious, alert. 2.Head- anterior fontanelle depressed, 3.Eyes- normal, sunken, tears on crying 4.Mouth- unable to suck/drink, drinking poorly, moist tongue. 5.Chest - check respiratory rate and record. Skin- skin pinch goes back quickly or slowly or very slowly.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
I. Diarrhoea with or without vomiting	•Lethargic or unconscious •Sunken eyes •Skin pinch goes back very slowly Not able to drink or drinking poorly	Severe condition	1.Give fluid, treat as plan C 2.Keep warm and refer immediately to hospital. 3.Encourage to continue breast-feeding. 4.Teach mother how to make Lo-ORS/Zinc 5.For Zinc: Give half tablet (10mg) every day for 10 days 6.For LO-ORS: One half cup (50-100ml) given until diarrhea stops (500ml/day) if mother is not doing exclusive breastfeeding
II. Diarrhoea with or without vomiting	•Restlessness •Irritable •Sunken eyes •Eagerly drinks/feeds Ski pinch goes back slowly	Moderate condition (some dehydration)	7.Treat as plan B Review in 2 days
III. Diarrhoea with or without vomiting	•Not enough signs to classify as some or severe dehydration •Alert •Eyes normal (not sunken) •Tears on crying •Moist mouth/tongue Child drinks normally Skin goes back quickly when pinched	Mild condition	1.Treat as plan A 2.Tell mother to breast-feed frequently and for longer period at each feed (Treatment Plan A). 3.Give Lo-ORS and Zinc in addition to breast milk 4.Teach mother how to make Lo-ORS. 5.Give 50-100ml of ORS after each loose stool in addition to regular fluids. 6.Advise to continue giving extra fluids till diarrhoea stops 7. Teach mother danger signs and ask to bring child to clinic immediately, if any danger sign below is present: Sunken eyes Lethargic or unconscious Drinking poorly or not able to drink. Skin pinch goes back very slowly.

Health Education

1.Advise on regular and complete immunization.
2.Counsel on proper personal hygiene and hand washing.
3.Advise on use of long-lasting insecticide nets (LLINs).
4.Advise on environmental hygiene.
5.Encourage exclusive breast feeding and avoid bottle feeding. Breast feed frequently and longer at each feed.
6.Keep baby warm.
7.Remind mother to take home-based records with her anytime she takes baby to clinic or hospital.

1.3.13 SEVERELY ILL BABY

HISTORY	EXAMINATION/ACTIONS
1.What is wrong with the baby? 2.When did you notice the problem? 3.Has the baby been ill? 4.Is the baby feeding well? 5.Is the baby difficult to wake? 6.What do you feed the baby on? 7.Is the baby cold or hot, how long? 8.Has the baby convulsed before? 9.What medication has been given to the baby for the problem? 10.Was the pregnancy normal? If not, what was the problem? 11.Was the delivery normal? 12.Was the baby normal at birth? Did the baby cry immediately at birth?	1.General appearance: severely ill-looking, unconscious, lethargic. 2.Head: bulging fontanelle. 3.Nose: nasal flaring grunting. 4.Ear: draining pus. 5.Chest: check respiration and record chest in-drawing. 6.Umbilical cord: redness extending to the skin. 7.Skin: many or several skin-pustules. 8.Abdomen: distension. 9.Temperature: check and record. Weight: check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
I. Severely ill baby	•One or a combination of any of the following: a)Warm to touch Rectal temp380C and above. b)Hypothermia (cold to touch below 36.50C. c)Convulsion/Fits. d. Lethargic e)Unconsciousness f)Severe chest-in- drawing g)Fast breathing (60 or more per minute) h)Nasal flaring i)Grunting j)Bulging fontanelle k)Pus draining from ear l)Umbilical redness extending to the skin m)Limited movement n)Abdominal distension o) Weight- not gaining weight	Severe condition Possible serious bacterial infection	1. Give stat dose of Benzyl Penicillin 50,000 I.U. 0.5ml 6 hourly 24 - 48hrs intramuscular antibiotics. 2.Continue to breast feed if baby can suck. 3. If baby cannot suck give expressed breast-milk with cup and spoon or give sugar water 10% sugar water 4.Keep baby warm, if no fever. REFER immediately. If HIV infected mother, after counseling refuse to breastfeed, advise on nutritionally adequate food.

Health Education

1.Encourage exclusive breastfeeding.
2.Keep baby warm.
3.Advise on personal and environmental hygiene.
4.Advise on use of Long-Lasting Insecticidal Nets (LLINs).
5.Advise on regular and complete immunization.
6.Remind mother to take home-based records with her any time she takes baby to clinic or hospital.

1.4 NEWBORN ABNORMALITIES

Sometimes babies are born with some physical abnormalities. The health worker should examine for any physical abnormalities and initiate immediate action to avoid permanent disabilities.

HISTORY	EXAMINATION/ACTIONS
1.How old is the baby? 2.What is wrong with the baby? 3.When did you notice the problem? 4.How old was your pregnancy before delivery? 5.Was the pregnancy normal, if not what was the problem? Is the baby sucking well? 6.Does any other person in the family have this type of problem? What medicine or substances have been given to the baby since s/he was born?	1.General appearance- any obvious abnormality, alert, ill-looking, irritable, excessive crying, sleepy. 2.Head- Swelling, size, head circumference 3.Chest- respiratory rate 4.Eyes- squint 5. Ears- Positioning of the 2 ears 6. Limbs- abnormalities like webbed toes or fingers, extra digit , 7.Mouth- cleft palate, hare lip 8.8 Genitalia- undescended testes, no vaginal perforation/ opening, bleeding, abnormal penile opening 9.Temperature- check and record 10.Weight- weigh and chart 11.Abdomen- distension Skin- Birth marks

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
I. Baby is not able to feed	Roof of mouth is not completely formed	Severe condition	1.REFER immediately to hospital.
II. Mouth Baby's lip is not completely formed Baby is able to feed successfully	Cleft lip	Moderate condition	1.Exclusively feed baby with expressed breast milk with spoon/dropper, starting with breastfeeding before offering the cup 2Feed baby to baby 2-3 hours (wake baby and feed) 3Keep baby longer at the breast, do not interrupt feeding if baby is trying 4Encourage mother to express milk into baby's mouth directly 5Keep baby warm using Kangaroo Mother Care (KMC) method 6Less than 1.5kg REFER immediately to hospital 7Ensure warmth during referral
III. Tongue Tie	Fibrous band holds down the tongue	Mild condition	Explain the finding to the parents. As the baby grows older, the tie may not interfere with speech. REFER to the hospital

IV. Constipation	•Abdominal distention or no finding	Mild / Moderate condition	1Encourage exclusive breast-feeding.
V. Baby not feeding well, Crying always	•No abnormality. •Baby is well Irritable	Mild condition Suspect maternal causes	1Examine mother's breasts and treat if necessary. 2Observe how mother feeds infant and correct if necessary. 3Counsel mother accordingly 4Express mother's breast and feed with cup and spoon if difficulty in sucking. 5Encourage exclusive breast feeding 6Follow-up every 2days Monitor baby weekly including weighing.
VI. More than five fingers or toes	•Extra digits	Mild condition	1.If no bone present(soft) tie tightly with a piece of strong sterile cotton or silk thread 2.Advise family that finger or toe will turn black and fall off in a few days 3.Follow up in 3 days If bone is present in extra digit DO NOT TIE, REFER to hospital immediately
VII. Abnormal appearance of genitalia	•Genitalia abnormal •Undescended testes •Penile opening in wrong position No vaginal perforation	Severe condition	REFER to hospital
VIII. Slight vaginal bleeding in baby girl	•Vaginal bleeding	Mild condition	1. Explain to mother that this is a passing condition, and nothing needs to be done 2. Review in 48 hours, if still bleeding. REFER to hospital.
IX. Birth marks	•Small black or blue patches or red or bluish red	Moderate condition	1. Reassure mother
X. Bleeding from circumcision	• Bleeding with or without pallor	Moderate	1Put pressure on side for 10 minutes 2Show mother how to maintain pressure 1-2 hours. If bleeding does not stop, REFER..

Health Education

1 Explain to mother possible cause of abnormality.
2 Advice on regular and complete immunization.
3 Encourage exclusive breastfeeding.
4 Keep baby warm.
5 Remind mother to take home-based records with her anytime she takes baby to the clinic or hospital.

1.5 ORAL HEALTH

This is care of oral cavity. Examination of the mouth of the newborn is an absolutely essential component of examination of the child in order to detect congenital malformations, swelling, pallor, cyanosis and infections. The areas to inspect are the lips, tongues, gum, palate, cheeks, floor and lining of the mouth.

HISTORY	EXAMINATION/ACTIONS
1.What is wrong with the baby? 2.When did you notice the problem? 3.Has the baby been ill? 4.Is the baby feeding well? 5.Was the baby born with tooth/teeth in the mouth? Or teeth came shortly after birth? Has the baby been given any medication?	1.General appearance- ill-looking or not 2.Mouth- roof of mouth (formed completely 3.Tongue- sore, fissured, smooth, dry, tied- check for band holding tongue, swelling 4.Tooth- presence or absence of white patches 5.Lips- Cleft Temperature- check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
I. Physical abnormalities	•Lump or bone defect at the back •Very big head •Squint eye •Low set ears •Club foot •Hare lips •Webbed toes and or fingers Cleft palate	Moderate or Severe condition	1.Explain findings to mother REFER to hospital
II. Premature/low birth weight	•Weight is 2.5kg or below Baby sucks poorly	Severe Condition	1.Exclusively feed baby with expressed breast milk with spoon/dropper, starting with breastfeeding before offering the cup 2Feed baby to baby 2-3 hours (wake baby and feed) 3Keep baby longer at the breast, do not interrupt feeding if baby is trying 4Encourage mother to express milk into baby's mouth directly 5Keep baby warm using Kangaroo Mother Care (KMC) method 6Less than 1.5kg REFER immediately to hospital 7Ensure warmth during referral
III. Small baby	•Weight is 2.5kg or below Baby can suck	Moderate condition	Explain the finding to the parents. As the baby grows older, the tie may not interfere with speech. REFER to the hospital

IV. Tooth/Teeth in the	Tooth may or may not be mobile	Mild condition	Explain to parents that condition is normal. Reassure that baby is not evil REFER immediately.
V. Soft swelling under the tongue	Swelling contains fluid (Salivary cyst)	Mild condition Suspect salivary cyst	1. REFER to the hospital
VI. White patches in mouth	White patch that rubs Off	Mild condition	No treatment needed. Explain to mother that patch is only milk
VII. White patches in the mouth	White patch that does not rub off on wiping	Mild condition Suspect thrush	Clean mouth with cotton wool soaked in normal saline Give nystatin oral suspension 0.3mls 6 hourly for 7 days 3.Review after 3 days If no improvement, REFER

Health Education

1 Oral hygiene, emphasizing proper utilization of cleaning materials - cotton wool and salt solutions.
2 Nutrition and prevention of tooth decay, encourage exclusive breast feeding
3 Discourage uvulectomy.
4 Remind mother to take home-based records with her anytime she takes baby to the clinic or hospital.
5 Educate the mother on personal and environmental hygiene.
6 Advice on regular and completed immunization.

Remind mother to take home based records with her anytime she takes baby to the clinic or health facility.

SECTION TWO
EARLY CHILDHOOD



NATIONAL STANDING ORDERS
FOR COMMUNITY HEALTH OFFICERS

2.0 EARLY CHILDHOOD

2.1. CHILD'S FIRST VISIT

The child's first visit is a very important one and should always include screening for health problems, growth monitoring and promotion, health education on nutrition, immunization and protection from illness and other conditions that are relevant to child and the family. Mothers should be encouraged to bring their children for visits while they are well monthly in the first 2 years of life, every 3 months thereafter until 5 years, and subsequent visits when necessary.

HISTORY	EXAMINATION/ACTIONS
1. Birth: Were pregnancy and birth normal? Where was the child delivered? Did the child cry immediately after birth? Was the child jaundiced (yellow) at any time? 2. Developmental Milestone: Is the child developing normally? •At about the age of 2 months-sees, responds to voice •At about the age of 4months holds head up •At about the age of 8months sits by self •At about the age of 12months walks alone 3.Immunizations: What immunization has the child had? Does the child have immunization card? 4.Nutrition: How have you been feeding the child? Exclusive breast milk if less than 6 months? Complementary feeding, supplementary feeding and breast milk after 6 months? What food is he/she eating now? If not exclusively breastfeeding, is mother using any family planning method? 5.Social: With whom does the child live? Does mother work outside the home? Who looks after the child when the mother is away? What is your source of water? What type of toilet do you use? Is your house screened against mosquitoes? Do you use Long Lasting Insecticide Net (LLIN) 6.Medical: Has the child had illnesses in the past? E.g. Measles? Mumps? What treatment was given? Where? Any illness or chronic cough in the family? 7. Does the child have any problem now?	Steps for physical examination •Inspection •Percussion •Palpation •Auscultation 1.General appearance: Well-looking, ill-looking 2.Colour: normal, pale, yellow, blue 3.Skin: rashes, sores, bruises 4.Head: size, swelling, colour and texture of hair, fontanelle-depressed or bulging, sutures-normal or wide 5.Eyes: discharge(sticky), yellow, red and swollen eyelids 6.Nose: discharge—blood, mucus, blockage 7.Neck: glands, masses, stiffness 8.Mouth: ability to suck normally, bleeding gums, dental caries, pallor, sores, cracks, redness of the throat (tonsillitis), abnormality (cleft upper palate) 9.Spine: swelling or opening 10.Chest: breathing normally, respiratory rate (note if 60 per minute or more) 11.Abdomen: condition of umbilical cord, umbilical hernia, rigidity, tenderness, mass and distended 12.Arms and legs: normal movement, deformity, fingers and toes (for extra or joined webbed fingers) 13.Genitalia: abnormality (check vaginal/penial opening, size of clitoris, both testis in scrotal sac), swelling, congenital hernia. 14.Anus: Redness or prolapse, irritation and imperforation. 15.Temperature and pulse: check and record. 16.Weight: check and chart 17.Haemoglobin: estimate and record 18.Developmental Milestone a.At about the age of 2 months — sees, responds to voice b.At about the age of 4 months — holds head up c.At about the age of 8 months — sits by self At about the age of 12 months — walks alone

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
i. No complaint (routine visit)	•No abnormality found on checking child health card for: -Immunization status -Growth monitoring Hb (haemoglobin)	Well child	1.Encourage exclusive breast-feeding if under 6 months 2.Explain growth chart, commend mother and advise to report for regular growth monitoring 3. 4 food demonstration. Mothers should attend at least once every month 4.Advise mother to introduce complementary feeding at 6 months and continue breastfeeding till 2 years 5.Give Vitamin A if child has not received Vitamin A in the last 4 months: •2-6 months: 50,000 units •6-11Months: 100,000 units •12months-5years: 200,000 units 6.If not fully immunized for age, advise and immunize appropriately (see appendix) 7.Check haemoglobin every 4 months if child is less than 1 year and every 6 months thereafter. 8.Check mother's DT status and advise appropriately 9.Advise mother on family planning methods. 10.Record findings and management in child's home-based record and give to mother 11.Give appointment appropriate to child's age
Any abnormal findings			Refer to appropriate sub-section in the Standing Orders for instructions on how to manage

Health Education

Encourage the mother on:

1.Exclusive breastfeeding up to 6 months and breastfeeding up to 2 years.
2.Nutrition education.
3.Regular and complete immunization.
4.Regular clinic attendance.
5.Child spacing and good nutrition.
6.Personal and food hygiene.
7.Use of long-lasting insecticide Nets (LLINs).
8.Encourage mother to take home-based records with her anytime she brings child to the clinic or hospital.

2.2. HEALTHY CHILD

The healthy child visit is a very important one for the Community Health Practitioner and should always include screening for health problems, growth monitoring and promotion, health education about nutrition, birth registration, immunization and protection from illness and other conditions as relevant to the family. Mothers should be encouraged to take their children for visits while they are well—two years of life and 3-monthly thereafter until 5 years old. Mothers should equally be taught the importance of keeping clinic appointment(s).

HISTORY	EXAMINATION/ACTIONS
1. Ask the mother for any complaint(s)	1. General Appearance: Healthy looking child, no abnormality found. 2. Check the child's immunization record, growth monitoring chart, and haemoglobin. 3. Look for any congenital abnormality.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGMENT	ACTION
i. No complaint (routine visit)	•Check child health card for: a.Last clinic appointment and current treatment (if any) b.Immunization status c.Growth monitoring Hb (haemoglobin)	Well child	1Encourage exclusive breastfeeding if under 6 months (feeding option if mother is HIV positive) 2Interpret result of child's arm circumference/weight to mother and teach mother how to use arm circumference strip. 3Advise mother to introduce complementary feeding at 6 months 4Refer for food demonstration at 5 months on how to prepare complementary food 5Refer for food demonstration at 5 months. 6Give Vitamin A if child has not received Vitamin A in last 4 months: a.2-6months: 50,000 units b.6-11Months: 100,000 units c.12month-5yrs: 200,000 units 7Advise to report for regular growth monitoring. 8If not fully immunized for age, advise and immunize appropriately (see appendix) 9Check mother's TT status and advise appropriately 10Record findings and management in child's home-based record and give to mother or caregiver Give appointment appropriate to child's age

Health Education

Encourage the mother on;

1. Exclusive breast-feeding up to 6 months (feeding option if mother is HIV-positive) and breastfeeding up to 2years.
2. Regular and complete Immunization.
3. Regular clinic attendance.
4. Child spacing and good nutrition using local food to demonstrate.
5. Personal and food hygiene.
6. The use of Long-Lasting Insecticidal Nets (LLINs).
7. Importance of taking home-based records with her anytime she takes child to the clinic or hospital.

2.3. FEVER

Fever is elevation of body temperature above 37.50C. Fever is a sign of infection or loss of body fluid. Rapid rise of temperature may lead to fits especially in young children of 6 months to five years (febrile convulsion). Some common causes of fever are malaria, tuberculosis, respiratory, gastro-intestinal and other bacterial and viral infections. All patients with a history of fever should be treated and encouraged to drink plenty of water, other fluids and fruits. High fever should be treated as an emergency.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child had the fever? 2.Is there any convulsion? 3.Has the child ever had any convulsion? 4.Is the child having a runny nose? Cough? Or difficulty breathing? 5.Is the child coughing up blood, 6.Is the child vomiting or having diarrhoea? 7.Is the child having bloody diarrhoea? 8.Is the child having skin rashes? 9.Is the child having any ear pains? Sore throat? Headache? Stiffness of the neck? 10.Is the child passing urine normally? Is it painful? Is it more or less frequent? 11.Is the child bleeding inside and outside the body? From the eyes, ears, and nose? 12.Has any analgesic or other medication been given? Is anyone currently ill in the family or compound?	**Steps for physical examination** •Inspection •Percussion •Palpation •Auscultation 1.General appearance: alert, restless, listless, drowsy or difficult to wake, delirious, dehydrated, thin, weak or convulsing. 2.Skin: rashes, septic spots, profuse perspiration. 3.Chest: (a) respiratory rate counted for 1 full minute. 4.Lower chest in drawing/retraction. 5.Wheeze or stridor in a calm child 6.Head: fontanelle—bulging or depressed 7.Nose: discharge–clear or purulent, blood 8.Eyes: pus, blood 9.Ear: tenderness behind the ear, discharge, blood. 10.Throat: sores, pains 11.Neck: stiffness, nodes. 12.Abdomen: tenderness, masses Temperature and pulse: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Fever with any of the following danger signs: a.Vomiting everything b. Unable to drink, or suckle c. Convulsion or history of convulsing d. Stiff Neck	Temp. Above 37.5 0C with any of the following: •Neck stiffness •Change in alertness •Convulsion •Anaemia (lack of blood) •Prostration •Bulging fontanelle •Moderate dehydration Lethargic (weak or no energy)	Very Severe condition Suspect febrile illness	1.Tepid sponge For children being referred with very severe febrile disease 2.Give first dose of intramuscular artesunate OR give first dose of intramuscular Quinine 3.Give first dose of an appropriate antibiotic: Susp. Amoxicillin 125mg/5ml a.Dose: 1month–12months: 2.5ml 8hourly for 5days b.12months–5years: 5ml 8hourly for 5days

			4.Give one dose of Paracetamol 120mg/5ml for high fever (37.5°C or above) a.0.3ml immediately (stat) for 2 months-3years b.5ml immediately (stat) for 3-5years 5.Give breastmilk (30-50ml) to prevent low blood sugar OR sugar water if breastmilk is not available to keep up the sugar level. REFER Immediately to the appropriate health facility; but if referral is not possible: 6.Give first dose of intramuscular artesunate 7.Child should remain lying down for one hour 8.Repeat artesunate injection at 12 and 24 hours later and then once daily x6 days. Artesunate/Age or weight: a.1 month–4 months (4-6kg) - 9.6mg-12mg b.months–12 months (6-10kg) - 14.4mg-21.6mg c.months—2 years (10-12kg) - 24.0mg-26.6mg d.2 years–3 years (12-14kg) - 28.8mg-31.2mg e.years–5 years (14–19kg) - 33.6mg-45.6mg OR 1.Give first dose of intramuscular quinine, child should remain lying down for one hour Repeat quinine injection at 4 and 8 hours later and then every 12 hours until the child is able to take an oral anti-malarial. Do not continue quinine injections for more than 1 week: Age/weight: 150mg/ml(2mlsampoules)300mg/ml(2mlampoules) a.1months–4months(4-6kg) 0.4ml 0.2ml b.4months–12months(6-10kg) 0.6ml 0.3ml c.12months-2years(10-12kg) 0.8ml 0.4ml d.2years–3years(12-14kg) 1.0ml 0.5ml e.3years–5years(14–19kg) 1.2ml 0.6ml **where IM quinine is not available, artesunate suppository can be used as pre-referral drug

			N.B: All cases of very severe febrile illness must be referred to secondary level healthcare facility immediately after giving Pre-referral treatment with a letter describing signs, symptoms and treatment given to the patient at PHC level
Fever without any danger sign	Temp. above 38.50C with no danger sign and stiff neck Malaria test (mRDT) positive	Moderate condition Suspect Uncomplicated Malaria	1.Treat with Oral ACT Anti-malarial Arthemeter/ Lumefantrine (AL 20mg/120mg): Day 1 give 1st dose & 2nd dose 8 hours later then twice daily x2 days Age Weight/ kg Dose 1 month – 3 years 5–14 1 tab twice daily 4-5 years 15-24 2 tabs twice daily OR Artesunate Amodiaquine (AA) both as a fixed dose combination Age Weight/kg Dose: 1st 2nd 3rd 2–11 months <10 25mg/67.5mg 1 tab 1 tab 1 tab 12 months - 5 years 10-18 50mg/135mg 1 tab 1 tab 1 tab 2. Give Syr. Paracetamol 120mg/5ml for high fever a.1 month–3 years: 2.5ml 8hourly for 3 days b.3-5 years: 5ml 8hourly for 3 days 3.Advise the mother to use LLINs for the child 4.Advise mother when to return immediately 5.Follow-up in 2 days if fever persists every day for more than 7 days 6.REFER to the appropriate health facility for 7.Assessment
	Malaria test (mRDT) negative	Mild Condition Suspect Pyrexia of unknown origin (PUO)	1.Give one dose of paracetamol for high fever (37.5°C or above). 2.Advise the mother to use Long Lasting Insecticidal Nets (LLINs) for the child 3.Advise mother when to return immediately. 4.Follow-up in 2 days 5.If fever persists, ASSESS for other causes of fever such as measles, UTI, pneumonia If fever is present every day for more than 7 days, REFER for further assessment

III. Fever, cough with runny nose/sore throat	Runny nose Temp. above 37.50C Red throat No fast breathing -Respiratory rate not more than: 50/min for 2 months–12 months 40/min. For 12 months–5 years)	Mild condition Suspect Upper respiratory tract infection (UTI)	1.Tepid sponge 2.Give Syr. Paracetamol 120mg/5ml a.2 month–2 years: 2.5ml 8hourly for 3 days b.2-5 years: 5ml 8hourly for 3days 3.If child cannot take orally or is vomiting everything, give paracetamol (PCM) suppository Give an appropriate antibiotic Susp. Amoxicillin 125mg/5ml a)2 months–12 months: 2.5ml 8hourly for 5 days b)months–5 years: 5ml 8hourly for 5 days 5.Review in 2 days If no improvement, REFER to the appropriate health facility
IV. Fever, No other findings	Fever with no other findings	Mild Condition Suspect pyrexia of unknown origin (PUO)	1. Give one dose of paracetamol for high fever (37.5OC or above) 2. Advise the mother to use LLINs for the child 3. Advise the mother when to return immediately 4. Follow up in 2 days 5. If fever persists ASSESS for other condition(s)
V. Fever and Measles rash	•Rash with any of the following: -Inability to drink or breastfeed -Lethargic -Stridor (when calm) -Severe malnutrition -Chest in-drawing -Clouding of the cornea -Deep or extensive mouth ulcer Temp. 37.50C or above	Severe condition Suspect complicated measles	1.Give Vitamin A treatment Age 200,000iu 100,000iu a.2-6 months ¼ 1/2 b.6-12 months ½ 1 c.12 months-5 years 1 2 2.Give first dose of an appropriate antibiotic Susp. Amoxicillin 125mg/5ml a.Dose: 2 months–12 months: 2.5ml 8hourly for 5 days b. 12 months–5 years: 5ml 8hourly for 5 day 3.If clouding of the cornea or pus draining from the eye, apply chloramphenicol eye ointment 4.REFER immediately to the appropriate health facility.

VI. Fever with Measles-type rash	•Cough •Measles rash •Pus draining from the eyes or •Mouth ulcers No fast breathing – temperature 37.50C or above	Moderate condition Suspect Measles with eye or mouth complications	1.Give Vitamin A treatment Age 200,000iu 100,000iu 50,000iu M.V Syr a.2–5months ¼ ½ 12.5ml b.6-11months ½ 1 25.0ml c.12months–5years 1 2 45.0ml 1.If pus draining from the eye, treat eye infection with chloramphenicol eye ointment [dosage 2.If mouth ulcers, treat with gentian violet 3.Follow-up in 3 days 4.If no improvement, REFER to the appropriate health facility
VII. Fever with Measles-type rash	•Measles now or within the last 3 months	Mild condition Suspect Measles	1.Give Vitamin A Age 200,000iu 100,000iu 50,000iu MultivitaminSyr a.2 – 6 months ¼ ½ 1 2.5ml b.6 – 11 months ½ 1 2 5.0ml 12 months-5 years 1 2 4 5.0ml
VIII. Fever with facial swelling, sore throat and bleeding from nose, eyes, ears gums	•Temperature 37.50C or above •Headache •Joint and muscle aches •Sore throat •Weakness •Stomach pains •Loss of appetite •Vomiting or cough up blood Bloody	Severe Condition Suspect Haemorrhagic Fever	1.REFER immediately to the appropriate health facility 2.ENSURE UNIVERSAL PRECAUTIONS 3.No specific treatment for the disease is yet available Efforts to help those who are infected include giving either oral rehydration salts/ORS or intravenous fluids.
IX. Fever with chills; feeling like flu, headache, muscle aches	•Temp. 37.50C or above •Dry cough •Shortness of breath •Chest pain when coughing or breathing •Diarrhoea Sore throat; Runny nose	Severe Condition Suspect Lower Respiratory Tract Infection	1.Give Susp.Amoxicillin 125mg/5ml a.2 months–12 months: 2.5ml 8hourly for 5 days b.12 months–5 years: 5ml 8hourly for 5 days 2. REFER immediately to the appropriate health facility.

Health Education

Advise mother/caregiver on:

1.Environmental sanitation and prevention of disease transmission (vector control, drainages).
2.Proper housing condition (ventilation, lighting).
3.Use of long-lasting insecticide nets (LLINs)/materials for malaria control.
4.Safe disposal of animal and human bodies infected with lassa virus, SARS virus and Ebola virus.
5.Wearing protective clothing when around person with the disease.
1.Personal and food hygiene.
2.Encourage fluid intake.
3.Nutrition counseling.
4.Safe water and disease prevention.
5.Importance of immunization in disease prevention and control.
6.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.4. CONVULSION

Convulsion is a jerky movement of the arms and legs and twitching of muscles. It is very common in infants and children and can be caused by high fever from any cause, head injury, tumour or sometimes infections of the brain or spinal cord. Muscle twitching due to tetanus or lockjaw should not be mistaken for convulsion. Tetanus often begins with tightness of the mouth, jaw and muscle twitching.

HISTORY	EXAMINATION/ACTIONS
1.Has the child had a fever? Any accident? Head injury? 2.Did the child take any medicine or any substance accidentally? 3.Has the child had any previous convulsion? With or without fever? 4.Has the child been given any medication? If yes, what? Does any other family member suffer from convulsions?	**Steps for physical examination** •Inspection •Percussion •Palpation •Auscultation 1.General appearance: convulsing, conscious, drowsy, unconscious, smelling of cow's urine 2.Nose: flaring of alae nasi 3.Chest: respiratory rate, lower chest in-drawing 4.Head: swelling, sign of injury, fontanelle for bulging 5.Eyes: pupils for size and reaction to light 6.Neck: stiffness 7.Ears: pus, bulging drums or perforation of drums 8.Temperature and pulse: check and record Weight: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
Convulsion with fever	•Child convulsing now Temperature >37.50C	Severe condition Suspect Febrile Convulsion	1.Lie the child on his or her side to avoid aspiration 2.Do not insert anything in the mouth 3.Keep the airway clear 4.Tepid sponge 5.Give the first dose of i/m Paraldehyde and REFER child to the appropriate health facility urgently: a.Age or Weight ParaldehydeDose:1ml=1gram I.0–2 months (1.5-<4kg) 0.5ml II.2 months–4 months (4-<6kg) 1.0ml III.4 months–12 months (6-<10kg) 1.5 ml IV.12 months–3 years (10-<14kg) 2.0ml V.3years–5years(14-<19kg) 2.5ml VI.After 10 minutes, if convulsion continues, repeat i/m paraldehyde. If convulsion continues 10mins after second paraldehyde, give a third dose

II. History of convulsion more than 24 hours before coming to the clinic	•Not convulsing now •Temperature >37.50C •Any of the following: -Bulging fontanelle -Neck stiffness Signs of head injury	Severe condition Suspect cerebral Meningitis or head injury	1.Tepid sponge 2.Give first dose of paraldehyde Age or Weight ParaldehydeDose:1ml= 1gram 0–2months3 (1.5-<4kg) 0.5ml 2months–4months(4-<6kg) 1.0ml 4months–12months(6-<10kg) 1.5ml 12months–3years(10-<14kg) 2.0ml 3years–5years(14-<19kg) 2.5ml 3. Give parental antibiotics and give oxygen if available 4.Explain findings to the mother 5. REFER immediately to the appropriate health facility
III. History of convulsion	•Not convulsing now •Temperature less than	Moderate condition Suspect febrile illness	1.Conduct RDT and if positive, treat for uncomplicated malaria with ACTs (refer to section on treatment with ACTs) 2.Review in 3 days Follow up

Health Education

Explain to Caregiver the:

1.Causes and prevention of convulsion.

2.Management of a person during convulsion episode.

3.Management of fever especially in children <5 years old.

4.Use of Long-Lasting Insecticidal Nets (LLINs).

5.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.5 COUGH/DIFFICULT BREATHING

Cough is usually a symptom of irritation or blockage of the air passages. It is forceful pushing of air out of the lungs. Difficult breathing is when the child is making extra effort to breathe, is breathing too fast or having noisy breathing. Difficult breathing can be seen by looking for chest in-drawing below the ribs and flaring of the nose as the child breathes in. Lower chest in-drawing and fast breathing indicate pneumonia.

Fast breathing means a respiratory rate of 60 and above per minute for an infant less than 2 months of age; respiratory rate of 50 and above per minute for an infant aged 2-12 months; and for, a respiratory rate of 40 and above per minute For a child 12 months – 5 years The most common causes of difficult breathing are pneumonia, asthma, mucus plug, enlarged adenoids and aspiration of a foreign body or heart diseases. Aspirin or kerosene poisoning can also cause fast breathing.

HISTORY	EXAMINATION/ACTIONS
1.How old is the child? 2.How long has the child had the cough or difficult breathing? 3.Is there wheezing? Any previous episodes? 4.Has the child been vomiting? 5.Has the child convulsed? 6.Is there a fever? For how long? 7.Has the child had measles or immunized for measles? 8.Is the child able to drink or suck?	**Steps for physical examination** •Inspection •Percussion •Palpation •Auscultation 1.General Appearance: alert or abnormally sleepy or difficult to wake, severe malnutrition, respiratory distress. 2.Nose: flaring of alae nasi 3.Chest: a.Respiratory rate (count for 1 full minute in a calm child) b.Subcostal retraction lower chest in-drawing c.Stridor d.Wheeze 4.Eyes: redness, discharge 5.Throat: soreness or redness 6.Feet: swollen 7.Temperature: check and record 8. Document oxygen saturation using a pulse oximeter if available

COMPLAINTS	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1. Cough and/or Difficult breathing with one or more of the following: a.Abnormally sleepy or difficult to wake b.Inability to drink Grunting when calm	•Fast/Difficult breathing. •Chest in-drawing. •Flaring of the nose •Stridor (in a calm child). •Lethargic. •May be unconscious •Severe malnutrition Assessment of fast breathing: -Less than 2 months 60 breaths/min or above -2 months – 12 months, 50 breaths/min or more 12 months-5 years, 40 breaths/min or more Document oxygen saturation using a pulse oximeter if available	Severe Condition Suspect Severe Pneumonia	1.Give first dose of oral Amoxicillin (250mg) dispersible tablets 2. REFER immediately Oxygen saturation is or less than 95% REFER immediately

II. Cough with fast breathing or chest indrawing	•Fast breathing, Chest indrawing	Moderate Condition Suspect Pneumonia	1.Give Susp Amoxicillin 250mg/5ml a.2 months up to12 months – 2.5ml 8hourly for 5 days b.12 months up to 5 years – 5ml 8hourly for 5 days 2.Soothe the throat and relieve the cough with a safe remedy 3.Review in 3 days If no improvement, REFER
III. Cough with difficult breathing with history of force feeding, choking or vomiting just before presentation	•Aspiration of feeds •Difficult breathing •Persistent cough Chest not clear	Severe condition Suspect Poisoning	1.If less than 24 hrs. suck nose, throat and airways with soft-tipped suction tube immediately. a.If respiratory distress continues after suction, REFER immediately 2.If respiratory distress improves, then give oral Susp. Amoxicillin 125mg/5ml for 5 days a.2 months up to12 months – 2.5ml 8hourly for 5 days 12 months up to 5 years – 5ml 8hourly for 5 days 3. Review next day and if no improvement, REFER Note: If sucking is not possible, REFER immediately to the appropriate health facility.
IV. Cough with noisy Breathing	•Difficult breathing Wheezing sound	Severe condition Suspect Asthma	1.Give Aminopline injection IV parentel according age 2.Give Oral Salbutamol 3 times daily (for wheezing) (not stated in the IMCI) Age 2mgtablet 4mgtabletSyrup (2mg/5ml) 12 months – 5 years 1 ½ 5 3.Give Susp Amoxicillin 250mg/5ml a.2 months up to12 months – 2.5ml 8hourly for 5 days b.12 months up to 5 years – 5ml 8hourly for 5 days 4.Review the next day 5.If no improvement, REFER immediately to the appropriate health facility 5.For severe acute wheezing, give inhaled salbutamol for wheezing with a spacer a.Dose: Salbutamol metered dose inhaler (100 microgramme/puff) give 2 puffs b.Repeat up to 3 times every 15 mins NOTES: i.Oral salbutamol is not recommended for treatment of severe acute wheeze. ii.No child under 5 years should be given inhaler without a spacer/nebulizer.

V. Cough with difficult breathing with a whoop	•Peculiar cough with a whoop Blood-shot eyes	Moderate to severe Suspect Whooping cough (pertusis)	1.Give first dose of oral Amoxicillin 250mg 2.If chest in-drawing or fast breathing, REFER immediately 3.If no chest in-drawing or fast breathing, and temp. Is >37.50C: a.Give Syr. Paracetamol 120mg/5ml until fever subsides i.2.5ml 8hourly for 5 days for 2 months – 3 years ii.5ml 8hourly for 5 days for 3 years – 5 years 4.Reviewafter24hours If no improvement, REFER to the appropriate health facility
VI. Cough with measles type rash (red eyes and runny nose)			1.Treat appropritely with parental antibiotics 2. REFER to section on MEASLES TYPE RASH
VII. Cough with runny nose or red throat, fever (Temp. above 37.80C)			1.See section on FEVER See section on THROAT problems
VIII. Cough or cold	•No signs of pneumonia •No fever •No difficulty breathing No chest indrawing	Mild condition (No pneumonia) Suspect common cough cold	1.If coughing for more than 14 days, REFER immediately to the appropriate health facility 2.If less than 14 days use safe remedy such as warm tea and honey and breast feed if still breast feeding. 3.Advise mother to return in one week NOTE: do not give honey to children under one (1) year

Home Care

1.Give extra fluids.
2.Continue feeding/breast feeding.
3.Clear the nose if it interferes with feeding.
4.Do not expose child to cold or chills.
5.Watch out for any of the danger-signs and if any, take the child to the health worker.

Danger signs (2 months - 5years)

1.Fast breathing.
2.Difficult breathing or lower chest in drawing.
3.In-ability to suck/drink.
4.Child becomes more ill.

Health Education

Advise mother/caregiver on:

1.Overcrowding and disease transmission control.

2.Prevention of air pollution (from firewood and kerosene smoke, and frying) and respiratory infection/ problems.

3.Role of adequate nutrition and fluid intake in prevention of infections and quick recovery.

4.Use of Long Lasting Insecticide Nets (LLINs).

5.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.6. DIARRHOEA

Diarrhea is the passage of three or more loose or watery stools in 24 hours. Passing 5 to 8 soft or semi-liquid stools per day is normal in exclusively breast-fed infants. Children with diarrhea may quickly become dehydrated and diarrhea in children often leads to malnutrition. The main danger in diarrheal diseases is dehydration, and it kills children more easily than adults. Replacement of fluids is the most important treatment. Persistent diarrhoea is one that lasts for two weeks or more. HIV/AIDS is a major cause of persistent diarrhoea.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child had diarrhoea? 2.How many times does the child usually pass stool each day? How many now? 3.Has the child passed any stools while in the clinic? (if so, look at it) 4.Are the stools watery? Is there blood or mucus? 5.Is the child vomiting? How often? 6.Has the child been active? 7.Are there other complaints such as cough, fever, draining ear, abdominal pain? 8.What has the child been fed? What is the method of feeding? 9.Has any medication been given? 10.Is anyone else in the household having diarrhoea? Does the child drink normally, eagerly or poorly?	Steps for physical examination •Inspection •Percussion •Palpation •Auscultation 1.General appearance: alert, restless, irritable, thin, dehydrated, lethargic, unconscious 2.Head: anterior fontanelle depressed. 3.Eyes; normal, sunken, very sunken and dry. 4.Mouth and Tongue: moist, dry, very dry 5.Chest: respiratory rate Rapid breathing: 60 or more/minute age less than 2 months. 50 or more/minute age 2-12 months 40 or more/minute age 12 months – 5 years 6.Neck: nodes enlarged throat inflamed 7.Ears: purulent discharge, tenderness behind ear 8.Abdomen: tenderness, distended 9.Skin: skin pinch goes back quickly, slowly or very slowly 10.Weight: weigh and chart 11.Temperature: check and record 12.Offer the child something to drink, and note if the child drinks normally, eagerly or poorly (or is not able to drink) SEE APPENDIX====for chart on assessment of dehydration

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Recurrent bouts of frequent stool with or without weight loss	•Very low weight or growth faltering •Enlarged lymph nodes •Oral thrush •Parotid (salivary gland swelling) •Chronic ear discharges Mother HIV positive	Severe condition Suspect HIV/ AIDS	1.Give Vitamin A <6months - 50,000IU 6-11months - 100,000IU 12months - 5 years - 200,000IU 2.REFER immediately Prepare and give mother LO-ORS/Zinc to give the child slowly on the way.

II. Frequent watery stool with weakness, with or without vomiting.	•Two or more of the following signs: -Lethargic or unconsciousness -Sunken eyes -Drinks poorly or unable to drink Skin pinch goes back very slowly.	Severe condition Suspect dehydration	1.If child has no other severe classification give low osmolarity ORS with dispersible zinc sulphate tablets for severe dehydration (see Plan C) 2.If child has another severe classification, REFER immediately 3.Advise mother to give frequent sips of low osmolarity ORS with dispersible zinc sulphate tablets 4.Advice mother to continue breast-feeding. LO-ORS Age Dosage <2years 50–100ml(1/4to½cup)after each loose stool
III. Frequent stooling for 14 days or more	•Two or more of the following: -Restless or Irritable -Sunken eyes -Child is thirsty and drinks eagerly -Skin pinch goes back slowly -Mouth/tongue dry -Dehydration present Child passes 3 or more watery or loose stools a day	Severe condition Suspect Some Dehydration	LO-ORS Age Dosage <2years 50–100ml(1/4to½cup) after each loose stool 1.See Plan B. Treat SOME dehydration with LO-ORS/ Zinc. 2.If child has another severe classification, REFER immediately Advise mother to give frequent sips of low osmolarity ORS with dispersible zinc sulphate tablets 4.Advice mother to continue breast feeding.
IV. Frequent stooling of less than 14 days	•No dehydration •Well, alert. •Eyes normal. •Tears present. •Mouth/Tongue Moist. •Child drinks normally. •Skin pinch goes back quickly OR Child passes 3 or more watery or loose stools a day	Mild condition No dehydration	1.Give low osmolarity ORS, dispersible zinc sulphate tablets and fluid to treat diarrhoea at home. a.Follow Plan A 2.Review in 2 days 3.REFER if diarrhea persists
V. Frequent stooling of more than 14 days with or without vomiting,	Dehydration present	Severe condition Suspect Persistent Diarrhea	1.Give low Osmolarity ORS and dispersible zinc sulphate tablets and Vitamin A 2.Follow Plan A. If child has another severe classification, REFER immediately

VI. Frequent stooling of more than 14 days with or without vomiting	•No dehydration	Persistent Diarrhoea	1.Give low Osmolarity ORS and dispersible zinc sulphate tablets and Vitamin A Advise mother on feeding child who has persistent diarrhea
VII. Diarrhea with fever, abdominal pain, cough/cold fast breathing, ear pain /sore throat	•Profuse watery diarrhoea (rice water stool) •Weakness Dehydration	Severe condition Suspect Cholera	1. Give IV fluid Ringers solution 2 Darrows solution,normal saline when condition improves continue oral fluid 3.ORS, 7 litres in 4 hours. 4.Give Ciprofloxacin15mg/kg/dose-2 times a day for 3 days 5.Supervise disposal of stool and vomitus 6.Notify appropriate Authority Review Daily
IX. Frequent stool with visible blood.	Blood in stool with or without mucus	Severe condition Suspect dysentry	1.REFER to the appropriate health facility if: a.Child is less than12 months b.Was dehydrated on the first visit c.Has had measles within the last 3 months 1.Give Ciprofloxacin15mg/kg/dose-2 times a day for 3 days Age 500mg tablet 250mg tablet 2months–6months ¼tablet ½ 6months–5 Years ½tablet 1 2.Give LO-ORS and zinc tablets. 3.Review in 3 days. 4.If no improvement, REFER immediately to the appropriate health facility. LO-ORS Age Dosage <2years 50–100ml (1/4to½cup)after each loose stool
X. Diarrhea with severe weight loss	•Weight below lower line on growth chart •Weakness •Dehydration •Pallor •Check MUAC <11.5cm (RED) With good appetite or No appetit	Severe condition: Suspect severe malnutrition (maramus or kwashiorkor)	1.Give Vitamin A a.2-6 months 50,000IU b.6-11 months -100,000 IU c.12 months - 5 years - 200,000IU 2.Give LO-ORS and zinc 3.Give Ready to Use Therapeutic Food (RUTF) if the child has appetite 4.If no appetite refer to Stabilization Care if available If no Outpatient Therapeutic Programme or Stabilization Care counsel mother on adequate nutritious foods (locally available food) 6. Give deworming tablet and food demostration room

Danger Signs

These are signs which a mother should watch for and if present, to return to the health worker IMMEDIATELY.

1.Many watery stools.

2.Vomiting everything.

3.Child becomes more ill.

4.Breast feeding or drinking poorly.

5.Develops fever.

6.Blood in the stool (if not previously present).

Health Education

Advise mother/caregiver on:

1.Personal Hygiene.
2.Environmental sanitation.
3.Exclusive breastfeeding.
4.Complementary feeding if above six months.
5.Use of long lasting insecticide nets.
6.Remind mother to take home-based records with her anytime she takes baby to the clinic or hospital.
7.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.7. VOMITING

Vomiting is the throwing up of the content of the stomach. Babies under six months often bring up little of their food immediately after being fed. This is normal. The common causes of vomiting are gastro-enteritis, meningitis, ear infection, liver disease and abdominal problems.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child been vomiting? How often? 2.Is it associated with eating? 3.Is it projectile? Does it contain blood or greenish fluid? 4.Is there diarrhea? 5.If no diarrhea, when was the last stool? (obstruction) 6.Is the child urinating normally? 7.Does the child have any other signs of illness? Fever, cough, pain in the head, abdominal pain? Has the child been recently dewormed	Steps for physical examination •Inspection •Percussion •Palpation •Auscultation 1.General appearance: ill, thin, jaundiced 2.Head: fontannelle depressed or bulging 3.Eyes: normal sunken, very sunken and dry 4.Neck: stiffness, enlarged and tender nodes 5.Abdomen a.Distension b.Tenderness c.board-like stiffness d.visible intestinal movement 6.Throat: redness, purulent discharge and inflamed tonsils 7.Ears: red drums, purulent discharge, tenderness behind the ear 8.Weight: weigh and chart Temperature, Pulse, Respiration & Blood Pressure: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1.Vomiting with any of the following: a.Blood (brownish vomitus) b.Neck stiffness c.Weakness Forceful vomiting	•Neck stiffness •Abdominal rigidity or •tenderness •Sign of severe dehydration Lethargic	Severe condition. Suspect Meningitis	1.Give Benzyl penicillin double dose stat a.2 months to 11 months (300mg stat) b.12 months to 5 years (600mg stat) 2. Continue management for CSM 3.REFER immediately to the appropriate health facility. If referral is difficult, repeat 6 hourly till you get to hospital.

II. Vomiting, frequent watery stools	•Mild dehydration	Mild condition	1.Follow plan A for diarrhea treatment
III. Vomiting with or without diarrhoea but with fever	•Dehydration Temperature 37.50C or above	Mild condition	1.Follow Plan A for diarrhea and treat fever appropriately
IV. Vomiting for more than 12 hours with no stool or fluids passed	•Baby under 3 months of age •Baby under 2.2kg •Severe dehydration •Severe abdominal pain Abdominal	Severe Condition Suspect Intestinal Obstructio	1.REFER immediately
V. Vomiting for less than 12 hours with normal stool	•Some dehydration •Reduced thirst Reduced activity	Mild Condition Suspect Gastritis	1.Demonstrate to mother how to make and give Lo-ORS 2.Continue feeding 3.Teach mother the causes of vomiting 4.Review next day 5.If better, advise mother to continue feeding If not better, REFER to the appropriate health facility.
VI. Occasional vomiting after feeding, no other complaint	•Child under 6 months Well baby	Mild condition	1.Reassure mother 2.Advise mother to feed baby properly 3.Teach mother to feed baby sitting upright Advise mother to keep baby upright for 10-15mins after feeding

Health Education

Advise mother/caregiver on:

1.Causes of vomiting.
2.Personal, food and environmental hygiene.
3.Adequate fluid intake and prevention of dehydration (excessive loss of body fluids)
4.Frequent feeding with small amounts of food.
5.Use of Long-Lasting Insecticide Nets (LLINs).
6.Importance of taking home-based records on child's next visit to the clinic.

2.8. PALLOR

Pallor is a major sign of anaemia. Anaemia is reduced number of red blood cells or reduced amount of hemoglobin in each red cell (below 10 grams). A child can develop anaemia as a result of not eating adequate nutritious foods, malaria, malnutrition, parasites (hook worm or whipworm), infections and sickle cell disease. Anaemia can lead to death.

HISTORY	EXAMINATION/ACTIONS
1.Is there bleeding from any source? a)Nose, gums b)Coughing or vomiting; c)In the stool or urine; d)Black stools, injury; 2.Does the child have a fever? For how long? 3.Has the child had painful swelling of the fingers, toes, the back of hands and feet? Or bone pains? 4. Does the child have a good appetite? What food does he eat? Does the child reject some foods 5. Does the child crawl, play on the ground or go barefoot? 6. Does the child have abdominal pain? 7. Has the child been swimming or washing in the stream? 8. Does anybody in the family have similar problems? 9. Has any medication been given? Is the mother using camphor?	Steps for physical examination •Inspection •Percussion •Palpation •Auscultation 1.General appearance: tall and thin, malnourished 2.Chest: respiratory rate, lower chest indrawing 3.Heart: rate, murmur 4.Head: bossing of the skull 5.Eyes: jaundice or conjunctival pallor 6.Abdomen: enlarged spleen, distension 7.Limbs: •Swollen and tender joints, feet and/or digits, or back of hands and feet •Nails for pallor 8.Haemoglobin: estimate and record 9.Temperature: check and record Urine: colour, blood, protein

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
•Severe palmar pallor Hb below 7gm/dl	•Severe palmar pallor Hb below 7gm/dl	Severe condition Suspect Severe Anaemia	1. REFER Immediately to the appropriate health facility.
ii. Paleness of the body	•Some palmar pallor •Hb between 7gm/dl and 10gm/dl mRDT positive or negative	Moderate condition Suspect Anaemia	Give appropriate dosage of iron and folic acid (If child has sickle cell disease, give only folic acid) Age (Weight) Iron syrup ferrous 5mg Folic acid tab fumarate 100mg/5ml 2months-4months(4-6kg) 1.5ml 1 4months-12months(6-10kg) 2.5ml 1 12months-3years(10-,14kg) 3.5ml 1 3years-5years(14-19kg) 5ml 1 1.Give oral anti-malarial if mRDT positive 2.Advise mother to put child to sleep under LLIN. 3.Give Albendazole 400mg in the clinic if the child is 1 year or older and has not had any in the last 6 months Review in 14 days
iii. Paleness of the body	No sign of palmar pallor or Hb normal	Mild condition No Anaemia	1.If child is less than 2 years old, assess child's feeding and counsel the mother on feeding recommendations.

IV. Paleness of the body	•Hb5-10gm with enlarged spleen mRDT positive or negative	Severe condition Suspect Anaemia with Splenomegaly	1.Give ACT if mRDT positive, treat for uncomplicated malaria (see dosage in section 2.2) 2.Give Amoxycillin or Ciproxacillinsuspension according age 3.Give Syr. Folic Acid 5ml for 2 weeks (see dosage above) 4.Give Pyrantel Pamoate 1 – 2 tablets Stat 5.Nutritional Counselling 6.Review in two weeks If no improvement, REFER immediately to the appropriate health facility.
V. Anaemia with history of bone pains	•Hb <10gm Swollen hands and finger.	Moderate condition Suspect Sickle Cell Disease	1.Conduct mRDT 2.If positive, treat for uncomplicated malaria 3.Give Ibuprofen syrup first dose (5mg/kg body weight per dose) 4. Give antibiotics according to age 5. REFER immediately to the appropriate health facility.
VI. Anaemia with fever	•Hb <10gm Temp >37.0oC	Moderate Condition Suspect Malaria	1.Treat as 4 above REFER to appropriate section on fever
VII. Sickle Cell Crisis with Bone pains and Abdominal pains	•Jaundice Swollen hands and feet	Severe Condition Suspect Sickle Cell in Crisis	1.Encourage Fluids intake 2.Paracetamol injection 300mg/2ml Stat a.2 months – 2 years 0.5ml (75mg) b.2 years – 5yr 1ml (150mg) REFER immediately to the appropriate health facility.

Health Education

Advise mother/caregiver on:

1.Personal, food and environmental hygiene.
2.Nutritional counseling.
3.Regular and complete Immunization.
4.Use of Long-Lasting Insecticide Nets (LLINs).
5.Regular growth monitoring.
6.Importance of taking home-based records on Child's next visit to the clinic or hospital.

2.9. JAUNDICE

Jaundice is the yellow discolouration of eyes (conjunctiva) and skin due to increase bile pigment in the body. Jaundice is as a result of the buildup of bilirubin - a reddish pigment resulting from haem metabolism in the body. It may or may not be as a result of liver problem. The buildup of this yellow pigment may be due to abnormalities in the formation, transportation, metabolism and excretion. At times the level may not be clinically recognizable until it is high. Jaundice occurs in liver problem, sickle cell disease, haemolysis, amoebiasis and tumour of the pancreas.

HISTORY	EXAMINATION/ACTIONS
1.Is the infant unwell? (sepsis & GIT obstruction can cause jaundice) 2.Is there yellow discolouration of eyes and skin 3.When did you notice the yellowish discolouration of the eyes? At what age? 4.Is there dehydration or poor weight gain? (both exacerbate jaundice) 5.Is there family history of haemolytic disease (ABO/G6PD, spherocytosis, sickle cell) 6.Is there dark urine or pale stools (suggest biliary obstruction) 7.What medication have you given (e.g. herbs)? Birth trauma such as cephalhaematoma, significant bruising (breakdown of haem)	Steps for physical examination •Inspection •Percussion •Palpation •Auscultation 1.General appearance: conscious, drowsy, unconscious, 2.Chest: respiratory rate, lower chest indrawing 3.Head: swelling, sign of injury, fontannelle for bulging 4.Eyes: yellowish 5.Ears: pus, bulging drums or perforation of drums 6.Hands: palms and nail beds for pallor and yellowish discolouration 7.Temperature and pulse: check and record Weight: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I.Yellow eye with or without swelling of hands and feet	•Yellow eyes	Mild condition Suspect Sickle Cell Anaemia	1.Refer for diagnosis (hb genotype test) 2.Routinely follow up and give: a.Syr. Folic acid 5mg daily for life Paludrine 100mg daily
II.Yellow eyes with or without bone pains, abdominal pains	•Yellow eyes •Bone pain Abdominal pain	Severe Condition Suspect Sickle Cell Disease in Crisis	1.Encourage oral fluid intake 2.Paracetamol injection 300mg/2ml a.2 months - 2 years 0.5ml (75mg) b.2 years – 5 yrs. 1ml (150mg) REFER immediately

Health Education

Advise mother/caregiver on:

1.Personal, food and environmental hygiene.
2.Nutritional counseling.
3.Regular and complete Immunization.
4.Regular growth monitoring.
5.Genetic counselling.
6.Use of Long Lasting Insecticide Nets (LLINs).
7.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.10. MALNUTRITION

Malnutrition is a state resulting from a deficiency or excess of one or more essential nutrients. Acute malnutrition is caused by a decrease in food consumption and/or illness resulting in bilateral pitting oedema or sudden weight loss. It is characterized by the presence of bilateral pitting oedema or wasting (low Mid Upper Arm Circumference or low Weight for Height). Acute malnutrition comprises of both severe acute malnutrition (SAM) and moderate acute malnutrition (MAM). All children under 5 years of age attending clinic should be examined and have their weight charted on the growth chart. Failure to gain weight should alert the health worker to take action.

N.B: While taking history fill the OTP (Out Patient Therapeutic Programme) card accordingly. (See Annex for OTP Card)

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? For how long? 2.What is the child's weight at birth? 3.Is the child still breastfeeding? 4.How is the food intake in the last few days? What kind of food has the child been eating? 5.Does the child have good appetite? 6.Does the child have diarrhoea? 7.How many times does the child pass stool in a day? 8.Is the child vomiting? 9.Is the child coughing? 10.Does the child pass adequate urine? 11.Does the child have oedema? (not necessary) 12.Is the child immunized? 13.Has the child been dewormed?	Steps for physical examination •Inspection •Percussion •Palpation •Auscultation 1.General appearance: ill looking, weak, wasted, stunted growth, looking puffy 2.Nutritional conditions as Kwashiorkor and Marasmus 3.Head: colour, texture of hair, bossing of the forehead 4.Face: moon-shaped, oedema 5.Eyes: sunken, discharge, conjunctiva discolouration 6.,Mouth: sores and candida 7.Neck: enlarged lymph nodes, 8.Axilla: enlarged lymph nodes 9.Chest: chest indrawing 10.Skin: scabies, rashes, loss of skin turgor 11.Limbs: oedema of the legs 12.Temperature, pulse and respiration: check and record 13.Weight and height: check and record 14.Mid Upper Arm Circumference (MUAC): measure and record 15.Appetite test: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1. Swollen leg, hand and face with severe weight loss, not breastfeeding	complication: -Intractable vomiting, -Hypothermia <35oC -High Fever >39oC -Chest–indrawing -Anaemia -Severe dehydration Failed appetite test ofeat-ing1/3RUTF–Ready to Use Therapeutic Food-(Plumpy Nuts)	Malnutrition (SAM)	1. Quantity of Water Quantity of Sugar Ratio Teaspoon to Sugar Cube 100ml 10g 2 level teaspoons (2 sugar cubes) 200ml(average cup) 20g 4 level teaspoons (4 sugar cubes) 2.REFER to the appropriate health facility to Stabilization Centre Advise mother to keep child warm on the way to hospital.

II. Swollen leg, hand and face with severe weight loss.	Age 6months and above with Mid Upper Arm Circumference MUAC) <11.5cm or 115mm(RED) or WFH <-3zscores, Bilateral pitting oedema +++ , Failed appetite to eat <1/3 RUTF with any one of the following medical complications: •Intractable vomiting, •Hypothermia <35oC •High Fever >39oC •Chest –indrawing •Anaemia •Severe dehydration •OR No weight gain after 4 visits (1month) to OTP	Severe condition: Suspect Severe Acute Malnutrition (SAM)	1.Give Vit A and treat the child to prevent low blood sugar using 10% sugar water while waiting for transfer to stabilization care Preparation of sugar water (10% dilution) Quantity of Water Quantity of Sugar RatioTeaspoon to Sugar Cube 100ml 10g 2levelteaspoons(2 sugar cubes) Refer to Stabilization Centre
3.Give other Routine Medicines based on history taken Discharge if weight gain on 2 consecutive visits)	More than 6 months and above - MUAC <11.5cm Visible severe wasting (prominent ribs, baggy pant, weak)	Suspect Severe Acute Malnutrition (SAM)	3.Give other Routine Medicines based on history taken Discharge if weight gain on 2 consecutive visits
VII. Child is thin, low Weight for Height (WFH), old man's look	MUAC < 11.5cm No medical complication No weakness Good appetite	Severe condition Severe Acute Malnutrition (SAM)	1.Treat at OTP and give RUTF according to weight (see table 3) After 4 visits to OTP if no improvement REFER to Stabilization Centre 1.Treat at OTP and give RUTF according to weight (see table 3) After 4 visits to OTP if no improvement REFER to Stabilization Centre

Note: Babies who are below 6months and are malnourished are with or without medical complications are treated at Stabilization Centres, Babies above 6 months without medical complications are treated at OTP (Outpatient Therapeutic Programme) and Babies above 6 months with complications (hypoglycemia, high fever, and severe anemia) are treated at the Stabilization Centre.

Correct use of RUTF

1.RUTF is a food and medicine for very thin or swollen children only. Do not share it.

2.Sick children often do not like to eat. Give small regular meals of RUTF and encourage the child to eat often (if possible, eight meals per day). Your child should have the recommended packets per day.

3.For breastfeeding children, continue to breastfeed. Offer breast milk first before every RUTF feed.

4.RUTF is the only food sick and thin/swollen children need to recover during their time in OTP. Always give RUTF before other family foods, such as pap, kunu, ogi, akamu, or other meals.

5.Always offer plenty of clean water or breast milk to drink while eating RUTF. Children will need to drink more water than normal.

6.Use soap and water to wash the child's hands and face before feeding. Keep food packets clean and well stored.

7.Caregivers shall also use soap and water to wash their hands prior to feeding.

8.Sick children get cold quickly. Always keep the child covered and warm.

9.For children with diarrhoea, continue feeding. Give them extra food and water.

10.Return to the health facility whenever the child's condition deteriorates or if the child is not eating sufficiently.

Health Education

Advise mother/caregiver on:

1.Personal, food and environmental hygiene.
2.Adequate nutrition.
3.Regular and complete Immunization.
4.Periodical deworming.
5.Regular growth monitoring.
6.Use of Long-Lasting Insecticide Nets (LLINs).
7.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.11. HIV/AIDS

HIV/AIDS is caused by a virus, which infects human beings. AIDS is the disease condition resulting from low immunity. Babies born to HIV-infected women may become infected during pregnancy, delivery or through breastfeeding. At the same time, many babies who are not breastfed because of fear of HIV transmission may die from diarrhoea, pneumonia or other diseases (not related to HIV) because they are not protected by immune substances found in breast milk. Thus, HIV–infected mothers need to be well guided on how best to feed their babies.

Ensure that more babies survive without becoming infected with HIV. Women can be given antiretroviral to reduce the chances of transmitting the virus to their babies. Babies born to HIV positive women are referred to as HIV exposed babies, a baby can only be said to be HIV positive after diagnostic confirmation.

HISTORY	EXAMINATION/ACTIONS
1.How old is your baby? 2.What immunizations has the child taken 3.Are you breastfeeding your baby? If yes, how many times do you breastfeed? If no, why not? 4.What are you feeding your baby with? 5.Have you introduced complementary/supplementary diet? If yes, when? 6.How many times do you feed your baby per day? 7.How do you feed your baby? 8.Has the child been sick recently? 9.Are you on ARV? 10.Do you give your child ARV?	1.General appearance: ill looking, weak, wasted, stunted growth, 2.Collect blood sample for initial DNA PCR at 4-6 weeks of age (or earliest opportunity thereafter) and 6 weeks after complete cessation of breastfeeding 3.Monitor growth as per child-welfare clinic Growth monitoring using growth chart 4.Weight and Height: check and record 5.Mid Upper Arm Circumference (MUAC): measure and record 6.Physical examination - Head: colour, texture of hair 7.Eyes: sunken, discharge, conjunctiva discolouration 8.Mouth: sores and candida 9.Neck: enlarged lymph nodes 10.Axilla: enlarged lymph nodes 11.Chest: chest indrawing 12.Skin: scabies, rashes, loss of skin turgor 13.Limbs: thin or oedema of the legs Temperature, Pulse and Respiration: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Baby born to HIV- infected mother	HIV–infected woman	HIV exposed baby	1.Give baby Nevirapine daily for first six weeks only and stop 2.Continue baby on cotrimoxazole prophylaxis until baby's status is determined through PCR. 3.Provide information on Early Infant Diagnosis (EID) for HIV exposed baby 4.Collect blood sample [Dry Blood Spot (DBS)] at six weeks and send to higher level for DNA PCR. If positive REFER 5.Advise on routine immunization 6.Encourage exclusive breastfeeding for the first 6 months of life thereafter, introduce adequate complementary foods and continue breastfeeding until 12 months with mother being on ARV Maintain or refill ARV for breastfeeding HIV positive mothers

II.Baby born to HIV- infected mother with malnutrition OR HIV infected infants with Malnutrition	Age < or > 6 months and above with Mid Upper Arm Circum	Severe condition Suspect Severe Acute Malnutrition (SAM)	1.REFER to Malnutrition section 2.REFER to HIV section 3.Give nutritional Counseling
Baby born to HIV-infected mother	HIV-infected woman	Moderate condition HIV-exposed baby	1.Give baby Nevirapine daily for first six weeks only and stop 2.Continue baby on cotrimoxazole prophylaxis until baby's status is determined through PCR 3.Provide information on Early Infant Diagnosis (EID) for HIV-exposed baby 4.Collect blood sample [Dry Blood Spot (DBS)] at six weeks and send to higher level for DNA PCR. If positive REFER 5.Advise on routine immunization 6.Encourage exclusive breastfeeding for the first 6 months of life thereafter, introduce adequate complementary foods and continue breastfeeding until 12 months with mother being on ARV Maintain or refill ARV for breastfeeding HIV positive mothers
			NOTE: In all scenarios, even when mother chose not to breastfeed, her baby should receive daily Nevirapine for the first six week of life.
Baby born to HIV- infected mother with TB co-infection	Baby born to motherwith HIV-TB co-infection	Severe Condition HIV-TB co- infection	1. REFER 2. Baby: Continue INH prophylaxis for the first six months of life Mother: Continue DOT (Director Observation Therapy)

Infant Feeding in the Context of HIV:

Malnutrition is a common condition in HIV infected children and a major contributor to mortality in this population. HIV infection can result in nutritional deficiencies, growth failure and developmental delay. Malnutrition itself results in decreased immunity and greater susceptibility to infections, thus accelerating disease progression. Malnutrition makes HIV infection worse and HIV infection worsens malnutrition.

Health Education

Advise mother/caregiver on:

1.Factors known to increase the risk of mother –to-child transmission and prevention.
2.The continuous use of ART (Nevirapine to babies)
3.Adequate nutrition.
4.Immunization.
5.Periodical deworming.
6.Regular growth monitoring.
7.Use of Long-Lasting Insecticide Nets (LLINs).
8.Personal, food and environmental hygiene.
9.The importance of attending ante-natal care (ANC) in order to provide appropriate care for the mother and foetus.
10.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.12. MEASLES – TYPE RASH

Measles is an acute viral infection characterized by the appearance of rashes in little groups made up of macules (flat spot) and papules (raised spots). It begins with high fever, cough, catarrh, and redness of eyes and mouth. The mouth may later become sore and rashes appear on the whole body after the fourth day of onset. The child may also get other serious illness like pneumonia, diarrhoea, or malnutrition at the same time or soon after the child has had the disease. It is a serious illness because it can easily lead to complications especially in poorly nourished children.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child been ill? 2.Does the child have a fever? 3.Does the child have cough? Is it a barking cough? 4.Is there runny nose? 5.What food is the child taking? Is the child refusing to feed? 6.Is there diarrhoea? Vomiting? (If so, may be force-feeding) 7.Has any medicine or treatment been given? 8.Are there any other children in the family under 2 years who have not been immunized for measles? 9.Are there children in the neighbourhood with the same complaint?	General appearance: alert, abnormally sleepy, or difficult to wake, unconscious, ill- looking or thin 1.Skin: rash, peeling, infected spots, haemorrhagic rashes 2.Chest: a). respiratory rate, lower chest indrawing b). wheeze, stridor 3.Eyes: red, purulent discharge, or sensitive to light (eyes closed), cloudy 4.Ears: bulging, discharging pus or fluid, perforation of eardrum, redness 5.Nose: runny, discharge 6.Mouth: a). sores, Koplik's spots, thrush b). throat-inflamed tonsils, purulent exudates 8. Hydration status: bulging or depressed fontannelle, sunken eyes, dry mouth and skin 9.Weight: weigh and chart 10.Temperature, and Pulse: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
Measles rash with fever and cough	•Rash with any of the following -Inability to drink or suckle - At breast -Lethargic -Stridor (when child is calm) -Severe malnutrition -Chest in-drawing -Clouding of the cornea -Deep or extensive mouth ulcer -Temp 37.5oc or above -Pus drawing from eyes	Severe condition Suspected severe Complicated measles	1.Give Vitamin A treatment: a.2 – 6 months (200,000 IU) ¼ or (100,000IU) ½ or 50,000IU 1 once/day. b.6-12 months (200,000IU) ½ or (100,000IU) 1 or 50,000IU) 2 once/day. c.12mths – 5 yrs (200,000IU) 1 or (100,000IU) 2 or (50,000IU) 4 once/day. 2.Give Multivitamin Syrup: a.2-6 months 2.5ml 8hourly b.6-12 months 5.0ml 8hourly c.12 mths-5yrs. 5.0ml 8hourly 3.Give first dose of an appropriate antibiotic Give Susp Amoxicillin 250mg/5ml c.2 months up to12 months – 2.5ml 8hourly for 5 days d.12 months up to 5 years – 5ml 8hourly for 5 days 4. (If clouding of the cornea or pus draining from the eye, apply chloramphenicol eye ointment REFER to the appropriate health facility.

I. Measles type rash with fever	• Cough • Measles type rash • Red eyes without ulcer • So fast breathing • Koplic spot for white patches	Moderate condition Suspect Measles	1. Give Susp Amoxicillin 250mg/5ml a.2 months up to12 months – 2.5ml 8hourly for 5 days b.12 months up to 5 years – 5ml 8hourly for 5 days 2. Give parental Antibiotics ,Ciprofloxacin according to age 3.Clean eyes with cooled boiled cotton wool 3.If cornea is cloudy or pus from eyes, apply chloramphenicol eye ointment tds x 5 days. 4.Give Vitamin A: a.< 6 months - 50,000IU b.6-ll months - 100,000IU c.12 months – 5 years - 200,000IU 5.Give Syr. Paracetamol 120mg/5ml c.2 months - 2 years : 2.5ml 8hourly for 3 days d.2-5 years: 5ml 8hourly for 3 days 6.Apply Gentian Violet to mouth ulcers 7.Review in 2days If no improvement, REFER to the appropriate health facility
III. Measles rash, with fever	Measles with no complication	Moderate condition Suspect measles	1.Give Vitamin A treatment: a.2 – 6 months (200,000 IU) ¼ or (100,000IU) ½ or 50,000IU 1 once/day. b.6-12 months (200,000IU) ½ or (100,000IU) 1 or 50,000IU) 2 once/day. c.12 mths – 5 yrs (200,000IU) 1 or (100,000IU) 2 or (50,000IU) 4 once/day. 2.Apply Benzoate acid lotion 3.Give Multivitamin Syrup: a.2-6 months 2.5ml 8hourly b.6-12 months 5.0ml 8hourly 12 mths-5yrs. 5.0ml 8hourly

Health Education

Advise mother/caregiver on:

1.Immunization for measles prevention and control.
2.Personal hygiene.
3.Adequate nutrition.
4.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.13. SKIN DISORDERS

Skin disease is the presence of itchy or non-itchy raised lesion on the body. It may be due to poor hygiene, allergy, malnutrition, poor housing condition as well as the environment. Some rashes can easily spread to other members of the family therefore, early referral and correct treatment is important for the welfare of the whole family. Some skin diseases affect the external skin (ringworm) or affect the skin internally (pyoderma). Others affect many parts of the body systems and are generally more important (measles, leprosy, diabetes, HIV/AIDS, Jaundice). Others are scabies, eczema, tinea vesicular, vitiligo, athletes' foot, urticaria and contact dermatitis.

HISTORY	EXAMINATION/ACTIONS
1.How long has the rash been there? 2.Does it itch? Is the child scratching? 3.Are there any other symptoms? Fever, cough, headaches, difficult breathing, sore throat? 4.Has the child eaten any unusual foods? 5.Does anyone else in the family have a similar rash or skin problem? 6.How often do you bathe the child? 7.Where and with whom does the child sleep? 8.What medicine (including injections or home treatment) has the child been given lately? 9.Has the child been passing excessive urine? Has the child been having excessive thirst?	1.General appearance: thin, malnourished, distressed/uncomfortable 2.Skin: a)location, distribution and size of lesions b)colour of rash or lesions c)lesions or rash: flat (macules), raised, pus or fluid-filled, weeping, crusted, ulcerated, scratch marks or d)combination of(a)to(c) 3.Ears: red or discharging 4.Eyes: redness, discolouration of eyelids (dark - due to repeated scratching) 5.Mouth: redness, discolouration of eyelids (dark-due to repeated scratching) 6.Chest: respiratory rate, wheeze 7.Temperature: check and record Urine: sugar, colour.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Itchy skin with rash	Small raised red spots or postular rash between fingers, wrist, buttocks, and ankles	Moderate condition Suspect Scabies	1. Advise bathing daily with medicated soap. 2. Apply Benzyl Benzoate lotion for 3 nights to all parts except face and scalp 3. Examine all members of family and TREAT EVERYONE whether any rash is found or not 4. Advise on the following: a. Boil clothes and beddings b. Dry in the sun c. Iron if possible d. If pyoderma, treat as boil in Findings II 5. If open sores, REFER Immediately to the appropriate health facility. 6. Repeat treatment if no improvement after one week
ii. Boils	• Boils or infected rash	Moderate condition Suspect Boils	1. Give Cotrimoxazole 240mg/5ml a) 2-12 months : 2.5ml 12hourly for 5 days b) 1-5 years : 5ml 12hourly for 5 days 2. Incise and drain any abscess which is fluctuant (ripe). 3. Daily cleaning and dry dressing at clinic. 4. Review in 5 days. 5. If no improvement, REFER to the appropriate health facility.

iii.Pustular Rash	•Crusted lessions •Impetigo •Infected insect bites Small sores	Moderate condition Suspect Pyoderma	1.Give Cotrimoxazole a) 2-12 months : 2.5ml 12hourly for 5 days b) 1-5 years : 5ml 12hourly for 5 days 2.Susp Amoxicillin 250mg/5ml a.2 months up to12 months – 2.5ml 8hourly for 5 days b.12 months up to 5 years – 5ml 8hourly for 5 days 3.Wash with medicate soap twice daily especially crushed lesions. 4.Use Gentian Violet on open sores 5.If fever present, treat as section on FEVER 6.Review in 5 days. If no improvement, REFER to the appropriate health facility.
iv. Itchy skin with red papules which from small blisters	• pustules Crusty rash appears after little or no fever mainly on trunk.	Moderate condition Suspect Chicken Pox	Syr. Paracetamol 120mg/5ml a)2 months - 2 years : 2.5ml 8hourly for 3 days b)2-5 years: 5ml 8hourly for 3 days 1.Calamine lotion – apply locally 2.Gentian Violet, if lesions are infected. 3.Treat Secondary infection with antibiotics Ciprofloxacin according age 4.Encourage fluids. 5.If fever present, conduct mRDT. If positive, treat as appropriate in section on FEVER.
v. Itchy skin with red lesions	Lesions, weals or patches which come and go very quickly	Moderate condition Suspect Hives	1.Ask if child has had it before 2.Ask what medicines have been or are being given. 3.Stop all medicines unless absolutely necessary. 4.Ask if the child ate any unusual food in the past 24 hrs. 5.Treat with Azithromycin according age 6.Give syrup Promethazine 12.5mg stat. 7. Review in 2 days and if no improvement, REFER
vi. White patch with itching	•Skin with well-defined edge •Reddish, with some scaling •Head is slightly itchy Batches of scaling and hair loss.	Mild condition Suspect Ringworm	1.Examine rest of family for similar problem 2.Apply Benzoic acid and salicylic acid (Whitfield ointment) on affected part for a minimum of 6 weeks 3.Give tab Griseofulvin and take as necessary 10mg/kg/day for 6 weeks If no improvement after 6 weeks, REFER to the appropriate health facility.

vii. Scalp scratching with marks	Scratches or marks on scalp	Mild condition Suspect Head lice	1.Examine rest of the family for similar problems. 2.Apply Lorexane lotion to scalp close to roots of hair. 3.Wash and comb hair with fine-tooth comb after 24 hrs. 4.Repeat treatment in 1 week. Note: If infestation is severe, a haircut may be required for effective treatment.
VIII.Itchy skin with rash	•Fine rash all over the body	Mild condition Suspect Heat rash	1.Re-assure mother. 2.Advise to bathe child twice daily with medicated soap. Make sure clothing is light and loose.

Health Education

Advise mother/caregiver on:

1.Personal hygiene (Discourage sharing of underwear and clothes).
2.Importance of washing and ironing of clothes (emphasis on not spreading clothes on grass and other inappropriate places.
3.Adequate nutrition, Role of water in skin infection (Protection and treatment of water sources e.g. ponds with insecticide - Abate).
4.Avoid overcrowding and transmission of skin infections.
5.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.14. OEDEMA

Oedema is the abnormal collection of fluid in the tissues resulting in swelling of the affected part of the body. It could be generalized or localized e.g. one foot or finger. Generalized oedema is most noticeable on the parts of the body which are lowest at that time e.g. legs when standing. Causes include congestive heart failure, severe anaemia, kidney disease, liver disease and kwashiorkor. Oedema may also be the result of a reaction to toxin, drugs, and insect sting. Ascites, which is fluid in the abdominal cavity, may accompany oedema.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child had the swelling? 2.Has the child ever had this swelling before? 3.Has the child been well otherwise? 4.Was the child bitten by an insect? 5.Has any medication been given (for example penicillin, ATS)? 6.Is there itching? 7.Is there cough? 8.Is there blood in stool or urine? 9.Any urinary frequency or pain in micturition? Does the child eat well? What foods?	1. General appearance: pale, lethargic, puffy, distressed. 2. Chest: a) Respiratory rate, difficulty b) Crepitations or decreased breath sounds 3. Heart: murmur, rate 4. Head: a) Hair-colour, texture b) Face-oedema, particularly around eyes 5. Eyes: pallor, jaundice 6. Abdomen: swelling, tenderness, enlarged liver and spleen 7. Limbs: swelling, pitting oedema, thinness 8. Urine: colour, cloudiness, protein, volume, (bilharria ova in the laboratory) 9. Weight: weigh and chart 10. Temperature: check and record 11. Haemoglobin: estimate and record 12. BP (if possible): check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Swelling of both feet and hands, and/or yellowness of eyes	Jaundice, wheezing, stridor or fast breathing	Severe condition Suspect sickle cell disease	1.Encourage oral fluid intake REFER to the appropriate section of sickle cell.
ii. Swelling Legs and / or face.	•Pallor, swollen legs and face •Breathlessness Fast heart beat	Severe condition Suspect Kidney or Heart problem	REFER to the appropriate health facility.
iii.Swelling of limbs and/or face with itching, with or without allergic reaction to toxins drugs, stings or some food	•Swollen legs, urticaria rash Itching	Moderate condition Suspect allergy or Kidney problem	1.If no respiratory distress, give promethazine 0.5ml IM stat 6.25mg 2.If respiratory distress, give Adrenaline test dose 0.01ml then promethazine a)0-5 months: (0.05ml) b)6-11 months: (0.1ml) c)1-2 years: (0.15ml) d)3-5 years: (0.15ml) 3.Calamine lotion: apply locally. 4.Observe for 4-6 hours 5.If condition improves, discharge and review daily 6. If no improvement after 3 days, REFER to the appropriate health facility.

IV. Swelling of feet, arms and/ or face	•Swollen feet (pitting) and/ or face Protein in urine	Moderate condition Suspect Kidney disease	1. Susp. Cotrimoxazole (240mg/5ml) a. 2-12mths- 2.5ml 12hourly for 5day b. 1-5yrs- 5ml 12hourly for 5days 1.Encourage fluid intake 2.Review in 5 days 3.If no improvement, REFER to the appropriate health facility.

Health Education

Advise mother / caregiver on:

1.Management of oedema (postural, application of crepe bandage).
2.Nutritional counseling.
Importance of taking home-based records on child's next visit to the clinic or hospital.

2.15. BURNS AND SCALDS

Burns and scalds are injuries to the skin in which there is coagulative necrosis of the tissue. A burn is caused by dry heat, for example, by an iron heat, fire, chemicals, and electricity. A scald is caused by something wet, such as hot water, hot oil or steam. Burns and scalds can range from being a minor injury to a life-threatening emergency. The chart below tells approximately what the proportion of a child's total body surface is:

S/N	Rule of nines for burns (William, 2023)	Percentage of Body Surface
1.	Head and neck	9%
2.	Chest	9%
3.	Abdomen	9%
4.	Upper back	9 %
5.	Lower back	9%
6.	Right arm	9%
7.	Left arm	9%
8.	Genital area	1%
9.	Left leg	18%
10.	Right leg	18%
	Total	100%

HISTORY	EXAMINATION/ACTIONS
1.How did the child get burnt? And with what? 2.When and where did it happen? 3.Has any medication been given or applied since then? 4.Has the child been urinating since the burn? Has the child been immunized against tetanus? How many times?	1. General appearance: restless, distressed, dehydrated, unconscious 2. Skin: a) Size and area of burns, blisters, leathery skin with no sensitivity to pinprick b) Signs of infections: pus, redness, warmouths 3. Temperature: check and record 4. Pulse: check rate and volume and record 5. Chest: respiratory rate, respiratory distress, wheezing, stridor 6. Check for possible signs of dehydration

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL J UDGEMENT	ACTION
I. Burns	Less than 90%with: - Blisters, - Not lethargic - No involvement of face, hands, groin or joints	Moderate condition	1.Vaseline gauze dressing 2. Phenobarbitone: a) 2-5 months: 7.5mg 12hourly for 3days b)6 months-5 years:15mg 12hourly for 3days 3. Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 4. IM Procaine Penicillin for 5 days or Erythromycin 15=25 mg bd X 5days a) 0-5 months: 75,000IU b) 6-ll months: 150,000IU c) 1-5 years: 300,000IU 5.T.T. 0.5 stat; immunized in last 18 months 6. Rehydrate with salt sugar solution (if dehydrated) 7. Review daily until well 8. Syr.Vitamin C 100mg/5ml a) 2.5ml 8 hourly for 2weeks-2month – 2years b) 5ml 8 hourly for 2weeks3-5 years
III. Burns	Less than 9%; No blisters, not lethargic; No involvement of face, hands, groin or joints	Moderate condition	1.Vaseline gauze dressing or topical wounds dressing with Zinc oxide-based cream 2. Phenobarbitone a) 2-5 months: 7.5mg 12hourly for 3days b)6 months-5 years:15mg 12hourly for 3days 3. Give Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 4.T.T.0.5ml, if not immunized in the last 18 months; Review daily until well. 5.Amoxycillin sml 8hourlyx 5days 6. Syr.Ascorbic acid 50mg 8hourly, l month – 2 years l00mg 3-5 years 7. Do not apply lotions or ointment 8. Do not break blisters or remove any loose skin 9. Remove the cause of the burn
IV. Burns with severe pains, (superficial or deep), burns from fire, flame, hotmetal, sun, high tension, lighting and friction	Redness and swelling of the area of blisters, sign of shock	Moderate to severe	Treat as 3above

V. Scald from boiling water, steam, oil, hot tea and hot liquids	Redness and swelling of the area of blisters, sign of shock	Moderate to severe	Treat as 3above

Health Education

1.Personal hygiene. .
2.Management of burns.
3.Nutritional counseling.
4.Compliance with treatment.
5.Prevention of home accidents.
6.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.16. WOUNDS, VARIOUS BITES, SPRAINS AND FRACTURES

2.16.1 WOUNDS

A wound is any break in the continuity of the skin or mucous membrane. There may also be injury of the underlying tissues such as tendons, large blood vessels or bones. Infections or tetanus can easily start when the skin is broken.

HISTORY	EXAMINATION/ACTIONS
1.How did it happen? 2.When did it happen? 3.Has it bled much? 4.Has anything been put on or in the wound? 5.Can the child move the area? Has the child been immunized with triple antigen or tetanus toxoid? When?	1.General appearance: in pain 2.Signs of shock: cold, clammy pale, fast breathing, rapid pulse rate 3.Wound: size, bleeding, dirt, or foreign body, pus, swelling, redness Temperature: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Injury	•Swelling •Tenderness over area of injury •Unable to move the limb In pains	Severe condition: Suspect blunt injury	1. Immobilize limb (splint) 2. Give Paracetamol (if conscious): Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 3. REFER immediately to the appropriate health facility.
II. Wounds on the Chest and Back from Sharp Knife and Gun shot or any other object, pains in Chest	• Dyspnea, cyanosis of the mouth, nailbeds, and skin • Cough bright red frothy blood if lungs is injured, • Blood-stained fluid from the wound Signs of shock,	Severe Condition Suspect Internal Hemorrhage	1.Clean with available antiseptic lotion 2.Stop bleeding by pressure 3.Gently cover the wound with a sterile Pad 4.Give TT 0.5ml 5.Manage for shock REFER immediately to the appropriate secondary or tertiary health facility.
III. Abdominal Wounds External or Internal Bleeding. Abdominal Pains, Vomiting	• Bleeding from the Wounds, • Part of intestine may be visible or protruding from the wound, Signs of shock	Severe Condition. Suspect Rapture of Internal Organs.	1.Clean with available antiseptic lotion 2.Stop bleeding by pressure 3. If organs are out, repack 4.Gently Cover the wound with a sterile Pad 5.Give TT 0.5ml 6.Manage for shock REFER immediately to the appropriate secondary or tertiary health facility.
IV. Wounds with sign of shock	• More than 4cm • Child is lethargic and not alert Torn tendon (unable to move limb) or fracture	Severe condition Suspect bleeding	1.Apply pressure 2.Give TT 0.5mls immediately 3.Manage shock and the wound 4. REFER Immediately to the appropriate health facility.

V. Wound with bleeding	• Laceration • Bleeding • No torn tendons • Unable to move the limb • Wound more than 40m • No fracture Check HB	Moderate condition	1. Clean with available antiseptic lotion 2. Stop bleeding by pressure 3. Give TT 0.5ml if wound is less than 24 hours 4. Amoxycillin, Erthromycillin ,Aziltromycin Suspension according to age a) 0-5 months: 75,000IU b) 6-ll months: 150,000IU c) 1-5 years: 300,000IU 5. Give Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 6. Review the next day then at 1 week to remove sutures 7. Ask to return in 6 weeks for 2nd dose of T.T 8. If no improvement, REFER to the appropriate health facility.
VI. Bleeding from circumcision	• Bleeding with or without pallor or shock	Moderate condition	1. Apply pressure on side for10 minutes 2. Suture the site of bleeding 3. Show mother how to maintain pressure 1-2 hours 4. If bleeding does not stop, manage the shock 5. Then REFER to the appropriate health facility.
VIII. Wounds with sign of shock	More than 4cm Child is lethargic and not alert Torn tendon (unable to move limb) or fracture	Mild condition	1. Clean with available antiseptic lotion 2. Put pressure on site for 10 minutes, if slowing, observe and maintain pressure for 2hours 3. Give Amoxycillin, Erthromycillin ,Aziltromycin Suspension according to age 4. If bleeding does not stop, 5. Give T.T. 0.5ml, if not immunized 6. Check Hb 7. Give Syrup Paracetamo-l(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 8. Syr. Vitamin C 100mg/5ml a) 2.5ml 8 hourly for 2weeks-2month – 2years b) 5ml 8 hourly for 2weeks3-5 years

Health Education

1. Protection from wounds.
2. Personal hygiene.
3. Importance of compliance with treatment.
4. Nutritional Counseling.
5. Environmental Sanitation.
6. Importance of taking home-based records on child's next visit to the clinic or hospital.

2.16.2 VARIOUS BITES

Bites may be from human, animal (dogs and snakes) and insects (mosquitoes) or stings (scorpion, bees etc.). Children are prone to bites both in the home and outside the home particularly when they go out of school.

HISTORY	EXAMINATION/ACTIONS
1. How did it happen? 2. When did it happen? 3. Has it bled much? 4. Has anything been put on or in the bite? 5. Can the child move the area? 6. Has the child been immunized with triple antigen or tetanus toxoid? When?	1. General appearance: signs of shock a) pale, cold or clammy b) Increased sweating c) Increased respiration and pulse rate d) patient agitated or restless 2. Bite: a) Size and site b) Bleeding c) swelling, redness or warmth around bite 3. Temperature, Pulse and Respiration: check and record 4. Haemoglobin: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Dog bite	• Evidence of bite • Anxious Tenderness at the site of bite with or without bleeding	Severe condition: Suspect rabies	1. Wash wound with copious clean soapy water 2. Advise caregivers to observe dog for 48 hours and report any changes in the dog's condition 3. Give anti rabies injection (if dog found or not) 4. Stop anti – rabies injection after 10 days if dog is alive 5. Give antibiotic of choice 6. Give tetanus toxoid 0.5ml I/M 7. REFER to the appropriate health facility.
II. Snake bite	• Wound at the site of bite Tenderness at the site of bite with or without bleeding	Severe condition Suspect Venom Poisoning	1. Keep client calm and clean the area thoroughly with copious soapy clean water. 2. Give anti- snake venom (read and follow marker's instruction) 3. Give promethazine 6.5mg stat 4. Give vit. K to enhance blood flowing 5. REFER immediately to the appropriate health facility.

III. Scorpion sting	• Evidence of sting (Painful Swelling, itching and may be red)	Moderate condition	1. Plain Lignocaine 5cc stat at the site 2. Give Adrenaline Subcutaneous stat a) 0.05ml for 2 – 5 months b) 0.1ml for 6-11 months c) 0.15ml for 1-5 years if suspecting anaphylactic reactions 3. Give Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 4. Give Syr.Vitamin C 100mg/5ml a) 2.5ml 8 hourly for 2weeks-2month – 2years b) 5ml 8 hourly for 2weeks3-5 years 5. Review within 24 hours 6. If no improvement, REFER to the appropriate health facility.
IV. Human bite	• Wound at the site of the bite • Anxious With or without bleeding	Moderate Condition	1. Clean wound well 2. Give antibiotic of choice 4. Dress wound daily 5. Review after 3 days 6.If no improvement, REFER to the appropriate health facility

Health Education

Advise Mother/caregiver on:

1.Protection from bites/sting.
2.Personal hygiene.
3.Importance of compliance with treatment.
4.Adequate nutrition.
5.Environmental sanitation.
6.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.16.3 SPRAINS, STRAINS AND FRACTURES

A sprain is a stretching or tearing of ligaments - the tough bands of fibrous tissue that connect two bones together in the joints. The most common location for a sprain is in the ankle. Initial treatment includes rest, ice, compression and elevation.

A strain is when a muscle is stretched too much and part of it tears. It is also called a pulled muscle.

A fracture is a sudden break in the continuity of a bone causing deformity, pain, tenderness and loss of power or movement.

HISTORY	EXAMINATION/ACTIONS
1.How did the injury occur? When? 2.Was there any bleeding? 3.Where is the pain? How bad is it? 4.Can the child move/use limb? Is there limitation to movement? Which movement causes greatest pain?	1.General appearance: a) Signs of shock-increased sweating, pallor, restlessness b) Degrees of pain 2. Limb or Joint a) Tenderness or swelling b) Warmouths or redness c)Crepitus (crackling) d) Open wound 3. Temperature: Pulse and Respiration: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Severe pains following injury, inability to move the affected part	Crackling or crepitus with loss of function No open wound	Severe condition Suspect Simple fracture	1. Give Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 2.Apply splint to affected limb 3. Observe for signs of shock 4. REFER immediately to the appropriate health facility.
II. Open wound with inability to move the affected part, following injury	• Exposed bone • Bleeding Pains	Severe condition Suspect compound fracture	1. Clean wound with normal saline 2. Observe for signs of shock 3. Give TT 0.5ml I/M immediately 4. Give antibiotic of choice 5. Give Syrup Paracetamo-l(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days Set an IV line 6. Immobilize, and cover wound 7. REFER immediately to the appropriate health facility.
III. Severe pains around the joints following an injury	• Tenderness of joint and swelling • Loss of function of the affected part • Anxious • In obvious pain With or without bleeding	Moderate Condition Suspect Sprain	1.Apply Cold compress to reduce swelling and pain 2.Apply Ung. Methylsalicylate to the affected area P.R.N 3.Apply Crepe bandage to the affected area 4. Give Syrup Paracetamo-l(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 6. Reviewin 2day 7. Continue management until condition improve

IV. Sudden sharp pains in muscle	• Stiffness, cramps, swellings as result of violent tearing of muscles after fall or accident	Moderate Condition Suspect Strain	1.Apply cold compress 2..Apply Ung Methylsalicylate to the affected area P.R.N 3.Resttheaffected limb 4. Give Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 6. REVIEWIN 3DAYS7, If no improvement, continue management until condition improve

Health Education

Advise Mother/caregiver on:

1.Protection from bites/sting.
2.Prevention of home accident.
3.Personal hygiene.
4.Importance of compliance with treatment.
5.Environmental sanitation.
6.Care of wounds and sores.
7.Nutrition and effect on healing.
8.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.17. DISORDERS AFFECTING THE HEAD

The head contains the brain (with the organs for the sense of hearing, seeing, smelling and tasting). The bones of the head are very strong but sometimes a serious head injury may be caused by a fall or a blow to the head. Some signs of serious head injury are bleeding from the nose, mouth or ears, severe headache or unconsciousness. There also may be vomiting or the client may be confused. Any delay in referring a client with a head injury for proper medical treatment may cause permanent injury or death.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.Has he fallen or hit his head? When? 3.If he has had an injury, has he been drowsy? 4.Has he vomited? How many times? 5.Does he have a headache? 6.Has he convulsed? Is he playing and behaving normally for his age as he usually does?	1.General appearance and behaviour: dull, abnormally, sleepy, or difficult to wake, restless, delirious or unconscious 2.Head: a) Circumference,, shape b)External injuries, sores, or rashes c)Swellings or depressions 3.Neck: stiffness 4. Eyes: haemorrahage, pupils size (equal or unequal), reaction to light, vision, jaundice 5.Gait: steady or unsteady Temperature, Pulse and Respiration: check and record.7. Haemoglobin: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. History of trauma to head, abnormally sleepy or difficult to wake, vomiting	•Pupil abnormal in size or not Reacting to light. Fracture, vomiting •Bleeding from ear or nose Black Eye,	Severe condition Suspect Fracture of base of the skull	1.Maintain a clear airway. 2.Lie patient on the side 3. Give oxygen if available 4. REFER immediately to the appropriate health facility, in that position
II. Trauma to Head from blow, fall or vehicular accident. Nausea and Vomiting. Loss of Memory	• Partial loss of consciousness • Shallow breathing, Face may be pale, cold clammy skin, Rapid or weak pulse,	Severe Condition Suspect Concussion	1.Maintain a clear airway. 2.Lie patient on the side REFER immediately to the appropriate health facility, in that position
III. Trauma to Head, pressure from bone in a depressed fracture, from fall or Vehicular accident	• Noisy breathing, slow and • bounding pulse, Different size of pupils, • Weakness or paralysis of one side of body, Poor response from patient	Severe Condition Suspect Compression	1.Maintain a clear airway. 2.Lie patient on the side REFER immediately to the appropriate health facility, in that position
IV. Headache with any of the following: a.Vomiting b.fever neck stiffness	• Temperature. 37.50C or above • stiff neck With or without positive Mrdt	Severe condition Suspect Cerebral Malaria/ Meningitis	1.If mRDT is positive, treat for uncomplicated malaria with ACT (refer to section on malaria treatment) 2.If mRDT is negative, give Benzyl penicillin stat Manage as CSM condition

V. Bulging forehead	Bossing of forehead (swelling	Severe Condition Suspect Sickle Cell Disease	1.REFER to the appropriate health facility, for sickling test 2.Syr. Folic acid 5ml daily for 2 weeks 3.Conduct mRDT, if positive treat for uncomplicated malaria with ACT (refer to section on malaria treatment) 4.If Hb is<10gm/dl, REFER Review in 2 weeks
VI. Injury to head with a small external swelling	Wound or swelling Normal behaviour No vomiting No drowsiness	Moderate condition	1.Give Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 2.If open wound, a.Clean with saline water and dress wound b.Give ATS then TT 0.5ml I/M stat immediately 4..Amoxycillin 125mg 8hourlyx-5days 5.If no open wound, apply cold compress on swelling 6.review after 24 hours Advise mother to return with the child if worse i.e., becomes drowsy, start vomiting, showing any strange behavior and REFER immediately
VII. Head larger or smaller than normal age	Head appears larger or smaller than normal age	Severe Condition	1.Measure head circumference if above or below normal limits, counsel mother REFER 2.Normal Head Circumference: a.1 month (33 – 37 cm average) b.6 months (41 – 46 cm) 12 months (45 – 50 cm)

Health Education

Advise Mother/caregiver on:

1.Proper care of child.
2.Prevention of head injuries.
3.Prevention of home and road accidents.
4.Importance of protection (use of car seat belt helmet etc.)
5.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.18 NECK DISORDERS

Neck problems include the presence of enlarged nodes and lumps around the neck. There could also be pain or stiffness of the neck. These are usually caused by infection of the ears, the mouth or on the scalp, tuberculosis, leukemia (cancer of the blood), meningitis, and other childhood malignancies. There may also be an injury to the neck following a fall.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.How long has the child had the problem? 3.Is there weight loss or loss of appetite? 4.Is there fever or sore throat? 5.Is there excessive sweating at night? 6.Is there headache? Earache? 7.Is there pain or vomiting? 8.Is the child restless or hyperactive? Has the child convulsed?	1.General appearance: pale, thin, or wasted 2.Head: sores or abscess 3.Mouth: a). condition of teeth a).sores b).tonsils 4.Ears: discharge, red or bulging or perforated ear drum 5.Neck: stiffness, nodes, lumps, glands 6.Chest: stridor Temperature, Pulse and Respiration: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Stiffness of neck with a fall and no fever	•Stiff neck with pain •Normal temperature Abnormal head posture on lying down	Severe condition	1.Restrict movement of head and neck If in extreme pain, give Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 2. REFER immediately
II. Neck pain with Fever, with vomiting, no neck injury	Stiff neck, Temp.37.50c or above, vomiting	Severe condition Suspect Cerebral Meningitis, Influenza meningitis,	1.If Kernig's sign positive, a. Give Benzyl penicillin injection REFER immediately to the appropriate health facility.
iii Dengue fever	Fever(37.5oc and above) ,headache,stiff neck,confusion	Severe condition	Pain reliever,IV fluid ,Close monitoring
IV. Neck pain without Fever	Lump and Nodes present No tenderness Deformities Swelling and Masses	Moderate condition	Give Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days REFER immediately to the appropriate health facility. Facility, in that position
V. Neck pain with Fever	Tender lumps or nodes Temp 37.5^0.c or above of one side of body, Poor response from patient	Moderate condition	1.Conduct mRDT for malaria, if positive treat for uncomplicated malaria with ACT (refer to section on malaria treatment) 2.Review in 2days If no improvement, REFER to the appropriate health facility.

VI. Swelling of both sides of neck (cheeks)	•Enlarged salivary gland behind the jaw under the chin. •Testes may or may not be involved (swollen and/or painful)	Mild condition Suspect mumps	1. Give Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 1.Encourage fluids, Saline mouth wash, explain finding to mother, if male, advise rest and continuous observation of testes 2.Review in 3 days If no improvement, REFER to the appropriate health facility.

Health Education

Advise Mother/caregiver on:

1.Prevention of neck injuries.
2.Complications of mumps in males.
3.Oral hygiene.
4.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.19. EYE DISORDERS

Eye problems are disease conditions affecting the child's eye and are commonly caused by bacteria or viruses, injury or foreign body or lack of enough vitamin A in foods. If these conditions are not prevented or treated early, they may cause blindness. It is therefore important to prevent or treat eye infections.

HISTORY	EXAMINATION/ACTIONS
1.What is the matter with the child's eye? 2.Do you think the child sees? 3.How long has the child had the problem? 4.Has the child had it before? 5.Did the problem come suddenly or gradually? 6.Is there a lot of pain in the eye? 7.Does the child rub the eye frequently? 8.Did anything enter the eye? 9.Has any medication been applied to the eye? If so, what? 10.Does anyone else in the family have this problem? 11.Is the child receiving treatment for any other disease? Does the child take palm oil, carrots, fresh mangoes, pawpaw, and tomatoes?	General appearance: in pains, ill, 1.Skin: rashes, scratch marks 2.Eyes: a)Redness, b)Tearing c)Swelling around the eyes. d)Pus or other discharge e)Pupils : Reaction to light, white pupil f)Protruding eyeballs g)Squint h)White foamy substance at corner of eyes/cloudy or white cornea i)Vision :check if child can follow a bright object, or reach out to receive an object from you without assistance 4.Nose: discharge Temperature, Pulse and Respiration: Measure and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Injury to eyeball with sharp object	Injured white part of eye (s)	Severe condition Suspect Trauma to eye	1.Apply Chloramphenicol eye drops 2.Apply eye pad. REFER immediately to the appropriate health facility.
II. Injury to eyeball as a result of burns, hot water or chemicals	Painful watery eyes (chemical burns to the eye)	Severe condition Suspect conjunctivitis/ keratitis	1.Irrigate the eyes immediately with clean water; 2.Apply Chloramphenicol eye ointment REFER immediately to the appropriate health facility.
III. Abnormalities of the eyes	Drooping of eye lids (ptosis) Eyes protruding (pushing out) Eyes watery or discharging Painful red eyes with or without visual loss Cataract (Growth of white small patches which do not react to light) Poor vision (less than 6/9	Moderate to severe condition	1. REFER immediately to the appropriate health facility
IV. Painful swelling of the eye lids)	Pustule or small abscess on eye lid	Mild condition Suspect Stye	1.Remove offending eyelash if possible 2.Warm compress to eye. 3.Chloramphenicol eye ointment 6hourly for 5days 4.Review in 3 days. If no improvement, REFER to the appropriate health facility.

v. Foreign body in the eye with pain	Watery eyes Restless May see foreign body Impaired vision Eyes may be red.	Moderate condition	1.Irrigate with Saline Apply Chloramphenicol eye drop 2.Cover eye with pad; REFER immediately to the appropriate health facility.
VI. White bubbly (foamy) material at corner of eye or dry eye	White bubbly material seen at corner of eye(s)	Severe condition Suspect Vitamin A Deficiency	1.Vitamin A (daily for two days) a.<6months - 50,000 unit b.6-12months - 100,000 IU c.12months - 5 years - 200,000 IU 2.Encourage eating of coloured (yellow/red) fruits and palm oil; 3.Review weekly until better REFER for proper assessment
VII. Sticky eye(s) or red eye(s); no pain, normal pupils and vision	Sticky (gluey)Eyes No Pain Normal pupils and vision	Sticky (gluey)Eyes No Pain Normal pupils and vision	1.Saline bath to eye 3 times a day. 2.Chloramphenicol eye drop 6hourly for 4 days and ointment at night for 5 days 3.Review in 3 days. If no improvement, REFER to the appropriate health facility.

Health Education

Advise Mother/caregiver on:

1.Personal, food and environmental hygiene.
2.Prevention of eye injuries.
3.Adequate nutrition especially foods rich in Vitamin A
4.Regular eye check-up.
5.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.20. EAR, NOSE AND THROAT DISORDER

Ear Problem

Ear problems, such as middle ear infections, can be detected early by observing common symptoms like ear pain, dullness or redness of the ear drum, and sometimes, perforation and drainage of pus or fluid. It's crucial to examine the bone behind the ear, as tenderness could indicate mastoiditis, a potential complication. Another complication is deafness, which can be tested by clapping your hands behind the child to see if they respond. Early detection empowers healthcare workers and parents to take proactive steps for their child's health.

Nose Disorders

The nose, an organ for breathing and smell, can be affected by diseases or injuries in the throat or sinuses. This can lead to symptoms like watery discharge, mucous blood or pus from the nose. Healthcare workers play a vital role in identifying these issues, especially if there's discharge from only one side of the nose, which could indicate a foreign object. This highlights their value and integral role in the healthcare process.

Throat Disorders

A red throat, often a sign of infection, may be combined with nasal congestion and middle ear infection, especially in young children. It's important to note that tonsils normally enlarge during childhood. However, unless they show signs of difficulty breathing or inflammation, no treatment is needed. This reassurance can give parents a sense of relief and confidence in their child's health.

HISTORY	EXAMINATION/ACTIONS
1.What is wrong with the child? 2.Does the child have hearing problems? 3.Is there any discharge from the ear? 4.Has the child had any ear surgery? 5.Is there any family history of deafness? 6.Does the child have runny nose? 7.Is there any bleeding from the nose? If yes, for how long? 8.Has the child had it before? 9.Is the child able to swallow food or fluids? 10.Has the child received any injection or medication recently? If yes, what is the name? Ototoxic drugs (antibiotics (e.g gentamicin), diuretics, and cytotoxics). 11.Has the child been exposed to loud noises recently? Exposure to noise (e.g. pneumatic drill or shooting) What treatment have you given to the child for this condition?	Nose 1.General appearance: well, ill-looking, pale 2.Nose: watery discharge, mucus, blood, pus, any visible swelling, deformity, or foreign body. 3.Throat: pus or redness Ear 1.Head: tenderness over bone behind the ear (mastoid) 2.Ear: a) Condition of canal, b) Pus or blood, perforated drum c) Foreign body, d). Red or bulging drum 3.Test for hearing: clap 4.Neck: stiffness Throat 1.General appearance: distressed, pale 2.Head: tenderness over bone behind the ear (mastoid) 3.Throat: pus or redness., enlargement of tonsils 4.Throat: General appearance: distressed, pale, 5.Neck: nodes, swellings Ear: discharge, redness or bulging drum 7.Temperature, Pulse and Respiration: check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Ear pain with history of foreign body with bleeding or neck stiffness	•Progressive inflammatory swelling over the Mastoid region •Tender, swelling behind the ear. May or may not be fluctuant	Severe condition Suspect Mastoiditis	1. Susp. Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs 5ml 8hourly for 5days 2. Give Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days REFER immediately to the appropriate health facility.
II. Ear pain with Discharge	•Pus is seen draining from the ear and discharge is reported for less than 14 days •Tenderness behind the ear/on touching the ear	Severe condition Suspect Acute Ear Infection	1.Susp. Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs 5ml 8hourly for 5days 2. Give Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 3.REFER urgently to the appropriate health facility, if no improvement
III. Ear pain with discharge for more than 14 days with or without fever	•Pus is seen draining from the ear and discharge is reported for 14 days or more. Ear pain;Temp > 37.50C	Severe condition Suspect Chronic ear infection	1.Susp. Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs 5ml 8hourly for 5days 2. Give Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 3.Conduct mRDT and if positive, treat for uncomplicated malaria with ACT (refer to section on Malaria fever) 4.Teach mother to wick ear at least 3 times daily until the ear is dry and discourage plugging the ear or using ear drops 5.Review in 3 days If no improvement, REFER to the appropriate health facility.

IV. Swollen Wound in ear	•Boil seen •Tenderness Temp > 37.5^{0}C	Moderate condition Suspect Boil	1.Susp. Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs: 5ml 8hourly for 5days 2. Give Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 3.Review in 3 days If no improvement REFER to the appropriate health facility.
V. Difficulty in Swallowing	•Grey and dirty membrane on tonsil Temp. > 37.5^{0}C	Severe condition Suspect diphtheria	1.Give I/M Paracetamol (300mg/2ml) a.2months -2 years 0.5ml(75mg) b.2years – 5 year 1ml (150mg) REFER immediately to the appropriate health facility.
VI. Pain in the throat, Inability to swallow	•Enlarged red tonsils with pussy exudate with or without tender lymph nodes in the neck	Moderate condition Suspect tonsilitis	1.Wash mouth with Salt water 2.Give I/M Amoxicillin for 3days a.2months–12months 125mg b.12months -5years 250mg 3.Give I/M Paracetamol (300mg/2ml) a.2months -2 years 0.5ml(75mg) b.2years – 5 yrs. 1ml (150mg) Observe for 3days, if no improvement refer
VII. Pain in the throat, Fever, Inability to swallow or refuses to feed	•Sore throat with or without redness •Tonsils not inflamed and no pus Temp. > 37.5^{0}C	Mild condition Suspect Pharyngitis	1.Wash mouth with Salt water 2.Give I/M Amoxicillin stat a.2months–12months 125mg b.12months -5years 250mg 3.Give I/M Paracetamol (300mg/2ml) a.2months -2 years 0.5ml(75mg) b.2years – 5 yrs. 1ml (150mg) 4.REFER immediately to the appropriate health facility. 1.Give I/M Amoxicillin stat a.2months–12months 125mg b.12months -5years 250mg 2. Continue, Susp. Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs: 5ml 8hourly for 5days 3.Give I/M Paracetamol (300mg/2ml) a.2months -2 years 0.5ml(75mg)

			b.2years – 5 yrs. 1ml (150mg) 4. Give Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 5.Syr.Vitamin C 100mg/5ml a) 2.5ml 8 hourly for 2weeks-2month – 2years b) 5ml 8 hourly for 2weeks3-5 years 6.Encourage fluid intake 7. Review in 3 days. If no improvement, REFER to the appropriate health facility.
VIII. Pain on either side of the nose with headache with or without fever	•Pus from nose •tenderness i.Smelly, purulent discharge Temp. > 37.5^{0}C	Moderate condition Suspect acute sinusitis	1.Give Susp. Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs 5ml 8hourly for 5days 2. Give Paracetamol 120mg/ml a.2months-2years 2.5ml 8hourly for 5days b.2-5 years 5ml 8hourly for 5days 3.Syr. Chlorpheniramine (Piriton) 2.5-5ml 12hourly 3 days 4.Syr Ascorbic acid (Vitamin C) tab 100mg/5ml 8hourly for 7 days 5.Review in 3 days. If no improvement, REFER to the appropriate health facility.
IX. Bleeding from Nose	•Blood from the nose Anxious	Moderate condition Suspect Epistaxis	1.Teach mother to pinch side of the nose for 15 minutes 2.Apply Cold Compress to nose for 30 minutes 3.Re-examine 4.If no improvement, REFER to the appropriate health facility. If condition has happened before, REFER immediately to the appropriate health facility.
X. Foreign object or Polyp in the nose •Foreign object seen with or without nasal discharge Moderate condition	Foreign object seen with or without nasal discharge	Moderate condition	1.Try to remove foreign body with blunt forceps If not removable, or if it's a polyp, REFER to the appropriate health facility.
XI. Stuffy nose, with or without Fever, Sneezing with or without watery eyes	•Watery eye •Catarrh Temp. > 37.50C	Moderate condition Suspect Common cold or allergy	1.Promethazine 0.25ml stat then, 2. Give Paracetamol 120mg/ml a.2months-2years 2.5ml 8hourly for 5days b.2-5 years 5ml 8hourly for 5days 2.Syr. Chlorpheniramine (Piriton) 2.5-5ml 12hourly 3 days 3.Syr. Ascorbic acid tab (Vitamin C) 50mg 8hrly for 7 days 4.Review in 3 days If no improvement, REFER to the appropriate health facility.

XII. Stuffy nose or catarrh with or without fever, Sneezing with or without watery eyes, Lack of appetite, Difficulty in breathing	•Running nose and watery eyes Temp. > 37.50C	Mild condition Suspect Upper Respiratory Tract Infection	1.Give Ascorbic acid (Vit. C) 50mg/5ml a.< 1-year 50mg 8hourly for 1 - 2 weeks b.>1 - 3 years 100mg 8hourly for 1 - 2 weeks c.3-5 years 200mg 8hourly for 1 - 2 weeks 2.Paracetamol 120mg/5ml a.2months - 2years 2.5ml 8 hourly for 3days b.2years-5years 5ml 8 hourly for 3 days 3.Suspension Amoxycillin 2.5ml 8 hourly for 3days 3.Conduct mRDT test. If positive refer to section on malaria treatment Continue feeding/breastfeeding 4.Clear the nose if it interferes with breathing 5.Encourage extra fluids a.Keep the child warm 6.Watch out for danger signs (fast or difficult breathing, inability to suck or drink, or child becomes more ill)
XIII. Boil in the nose with or without fever	•Boil can be seen •Temp less than or greater than 37.50C	Mild Condition Suspect Boil	Give Susp. Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs 5ml 8hourly for 5days 2. Give Paracetamol a.2months-2years 2.5ml 8hourly for 5days b.2-5 years 5ml 8hourly for 5days 2.Give Ascorbic acid (Vitamin C) 100mg/5ml a.2,-< 1 year 2.5ml 8hourly for 1 - 2 weeks b.>1 - 3 years 5ml 8hourly for 1 - 2 weeks c.3-5 years 5ml 8hourly for 1 - 2 weeks 3.Warm Compress 4.Review in 3 days. If no improvement, REFER to the appropriate health facility.

XIV. Nasal discharge with or without fever	Running discharge from nose with or without fever	Mild Condition Suspect Acute Sinusitis	1.Give Syr. Paracetamol 120mg/5ml a.2months-2years 2.5ml 8 hourly for 5days b.2-5 years 5ml 8 hourly for 5days 2.Give Syr Ascorbic acid (Vitamin C) 100mg/5ml a.< 1 year 50mg 8hourly for 1 - 2 weeks b.>1 - 3 years 100m 8hourly for 1 - 2 weeks c.3-5 years 200mg 8hourly for 1 - 2 weeks 3. Suspen Amoxycillin 2.5ml 8 hourly for 3days 4 Susp. Prednisolone acording to age 5.Warm Compress 6.Review in 3 days. If no improvement, REFER to the appropriate health facility.
XVI. Ear pain with discharge for more than 14 days with or without fever	•Pus is seen draining from the ear. •Dull ear drum Warm to touch (Temp > 37.50C)	Moderate Condition	1. REFER immediately to the appropriate health facility.

Health Education

Advise Mother/caregiver on:

1.Management of throat problems (gargle and discouragement of uvulectomy).
2.Prevention of exposure to air pollutants (fumes, dust, and pollen).
3.Personal hygiene.
4.Discourage home removal of foreign body.
5.Adequate nutrition.
6.General childcare.
7.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.21. ORAL HEALTH DISORDERS

Oral health is a state of being free from chronic mouth and facial pain, oral and throat cancer, oral sores, birth defects such as cleft lip and palate, periodontal (gum) disease, tooth decay and tooth loss, and other diseases and disorders that affect the oral cavity.

The most common complaint is pain which may or may not be accompanied with swelling inside the mouth or on the face. The condition of the mouth is also a reflection of the client's general health, so it should be inspected as part of physical examination. Early recognition of oral problems with prompt attention or referral will reduce morbidity and mortality arising from otherwise simple disease condition.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.Where is the problem? 3.Is there pain from tongue, palate, mucus membrane, teeth or gum? 4.How long has it been noticed or present? 5.Are you able to eat your regular diet? 6.Does the pain increase with hot or cold drink or food? 7.Does the pain start on its own? 8.Does it keep you awake at night? 9.Is there swelling anywhere? 10.Is there any bleeding from gum or other part of the mouth? 11.Do you notice bad or foul breathe? 12.What do you use for cleaning your teeth? 13.How do you clean your teeth? Do you notice any sore in the mouth?	1.General appearance: ill looking, facial swelling, malnourished. 2.Temperature: Check and record. 3.Mouth: a)Gum: sore, swelling, discharging sinus, bleeding, white patches. b)Tongue sore, coated, redness, fissures, smoothness, dryness. c)Lips: crack d)Chest: lining 4.Teeth: caries, discolouration, tenderness, mobility, missing, plague.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I.Severe pain that keeps patient awake a.Aggravated by chewing With or without headache	Hole in tooth/teeth	Moderate to severe condition Suspect Irreversible pulpitis	1.Syr Paracetamol 5ml 8hourly for 3days 2.Syr Metronidazole 5ml 8hourly for 5days 3.Syr Amoxicillin 5ml 8hourly for 5days REFER to the appropriate health facility.
II. Sharp, short pain triggered by chewing hot/cold drinks Stops as soon as stimulus is removed	Hole on the offending tooth/ teeth	Mild condition Suspect reversible pulpitis	1.Syr Paracetamol 5ml 8hourly for 3days 2.Syr Metronidazole 5ml 8hourly for 5days 3.Syr Amoxicillin 5ml 8hourly for 5days 4.REFER to the appropriate health facility.
III. Gum soreness and bleeding, fever, foul breath	•Ulcers along gingival margins, increased salivation •Temperature greater than 37.50C mRDT negative	Severe condition Suspect Acute Ulcerative Gingivitis (ANUG) Suspect Cancrum oris (Noma)	1.Syr Paracetamol 5ml 8hourly for 3days 2.Syr Metronidazole 5ml 8hourly for 5days 3.Syr Amoxicillin 5ml 8hourly for 5days 4.Oraldene mouth wash 5 days 5. If no improvement, REFER to the appropriate health facility.

IV. Gum soreness, fever, foul breath,	•Mouth ulcers with vesicles on the tongue, mucous membranes •Inflamed gingivae •Temperature greater than 37.50C mRDT negative	Severe condition Suspect viral gingivitis	1.Syr Paracetamol 5ml 8hourly for 3days 2.Syr Metronidazole 5ml 8hourly for 5days 3.Syr Amoxicillin 5ml 8hourly for 5days If no improvement, REFER to the appropriate health facility.
V. White patches in mouth, skin rash, soreness of the gum, not gaining weight	•White patches on tongue, oral mucosa that easily rubs off •Oral ulcers Underweight Failure to thrive	Severe condition Suspect malnutrition and/or HIV/AIDS	1. REFER immediately to the appropriate health facility for screening of HIV if not positive manage malnutrition .
VI. Slow or fast-growing swelling of the jaws	Hard or soft swelling anywhere in the mouth or jaws	Severe condition Suspect Oral Tumour	1. REFER immediately to the appropriate health facility.

Health Education

Advise Mother/caregiver on:

1. Ensure brushing twice daily, in the morning and before bed at night.
2. Use fluoride-containing toothpaste for effective dental care.
3. Replace the toothbrush as soon as the bristles begin to flare or scatter.
4. Schedule regular dental visits at least twice a year for routine cleaning, scaling, and polishing.
5. Avoid consuming sugary foods between meals and opt for mouth-cleansing fruits instead.
6. Maintain a diet rich in adequate nutrition for overall oral health.
7. Cultivate the habit of rinsing the mouth with water after meals.
8. Discourage harmful traditional practices such as uvulectomy.
9. Emphasize the importance of maintaining home-based records during visits to the clinic/hospital.

2.22. CHEST/HEART DISORDERS

A very rapid, slow, or irregular heart rate is always a serious sign. Changes in pulse and respiratory rate are often important signs of heart disease. Blueness of the lips or fingers, clubbing of the fingers and toes with difficult breathing are signs that the heart is not able to do its work well or easily. Congestive heart failure occurs when the heart becomes too weak to pump all the blood around and some of the fluid goes back into the lungs and often into the abdomen and legs. Any history of shortness of breath should make the health worker think of a heart disease.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? For how long? 2.Is there cough? 3.Is there difficulty with feeding due to shortness of breath? 4.Is there shortness of breath at night? 5.Is there chest pain? 6.Is there restlessness? (sign of lack of oxygen) 7.Does the child tire easily or tend to squat much of the time? 8.Have the lips or fingernails ever looked blue? 9.Does the child participate in sports? Is the child able to run around with his/her age mates without difficulty? 10.Any other problem?	1.General appearance: ill-looking, weak, thin, pale 2.Eyes: swollen or puffy 3.Nose: flaring 4.Mouth: blueness of lip 5.Toes/fingers: thickness of toes/fingers, blueness of the fingers 6.Chest: respiratory rate, lower chest in-drawing, grunting 7.Heart: rate and rhythm, murmur 8.Assess child's heart rate: Normal Heart rate 0-1year - 120/min 1-2years - 110/min 2-4 years - 100/min 4-8years - 90/min 9.Limbs: a). fingers blueness, pallor. Clubbing 10.Temperature: check and record 11.Haemoglobin: estimate and record Weight: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I.Chest problem with: a.Shortness of breath, Easily tired, Squats frequently b.Blue lips or fingernails c.Frequent coughing d.Poor weight gain Pain in the chest	•Coughing •Swelling of limbs •Fast heart rate •More comfortable sitting up •Blueness of lips and finger-nails Fatigue. Heart murmur	Severe condition Suspect congenital heart problem	1. REFER immediately to the appropriate health facility, in a comfortable position.
II. Easily tired, squats, Weakness	Slow or fast heart rate, cough, blueness, low Hb	Severe condition	2.REFER immediately to the appropriate health facility, in a comfortable position
III. Easily tired with or without fever or infection	•Increased heart rate Temp 37.5 00C or above	Moderate condition	1. Do an mRDT and if positive, refer to appropriate section on FEVER Give Susp. Amoxicillin 125mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs 5ml 8hourly for 5 days 2. Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3 days 3 .Review in 3 days If no improvement, REFER to the appropriate health facility.

Health Education

Advise Mother/caregiver on:

1.Personal hygiene.
2.Proper care of person with chest/heart problem (keep warm and allow for adequate rest).
3.Adequate nutrition.
4.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.23. ABDOMINAL DISORDERS

The common causes of abdominal pain are gastro-enteritis, malaria, appendicitis, rupture of the intestine, typhoid, sickle cell crisis and obstruction, e.g. strangulated inguinal and umbilical hernias. Constipation, worm infestation may also cause abdominal pain. Abdominal pain may lead to hardness or rigidity of the abdominal wall.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? For how long? 2.Is there pain? If yes, is it continuous or intermittent? 3.Does the pain make the child restless at night? 4.What was the child fed on in the past 24 hours? 5.Is there vomiting? Any blood in the vomitus? 6.Is there diarrhoea? 7.Is there constipation? 8.Has the child passed worms in the stool? 9.Are stools black or is there bright red blood in stools? 10.Any pain on micturition? 11.Any fever? Has any medication been given to the child?	1.General appearance: ill looking, crying(pain), thin, pale 2.Chest: respiratory rate, lower chest indrawing 3.Abdomen: a). swelling (distension), uniform or irregular b)visible peristalsis, bowel sound c)rigidity, tenderness, any palpable mass 4.Temperature and pulse: check and record\ 5.Haemoglobin: estimate and record Genotype: check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I.Abdominal pain; Not passing stool	•Child ill looking, crying and in pain, abdominal rigidity.	Severe condition Suspect Constipation	1. 1. Encourage intake of fruits and vegetables 2. Give liquid paraffin according to age
II.Abdominal pain with swelling on left side of abdomen (spleen) with or without history of injury.	Ill looking, pale, enlarged tender mass with painful swelling on the left upper part of the abdomen	Severe condition Suspect injury to spleen	1. REFER Immediately to the appropriate health facility.
III.Abdominal pain with Constipation;	•Well child; swelling of abdomen increased bowel sounds	Moderate condition	1.Review child's diet and fluids; 2.Advise fruits and vegetables 3.Give liquid paraffin 5 - 10mls daily for 2 days; If no improvement in 2 days, REFER
IV.Abdominal pain with rigidity; vomiting but no diarrhoea	Child in pain and very uncomfortable, tender abdomen	Severe condition Suspect acute abdomen	1. REFER immediately to the appropriate health facility.

V. Abdominal pain with bloody stools; Strains on passing stool	•Pale looking; blood in stools or Tenesmus(painful)	Moderate/Severe condition Suspect; Dysentery, Amoebiasis	1. If Dysentery Give Ciprofloxacin15mg/kg/dose-2 times a day for 3 days Age 500mg tablet 250mg tablet 2mts – 6mths ¼ tablet ½ 6mths – 5 yrs ½ tablet 1 2.If no improvement after 5 days; order stool test; 3.If laboratory confirmation of amoebiasis, give Metronidazole Syr Metronidazole 100mg/5ml a. 2mths-12mths: 2.5ml 8hourly for 5 days b. 12mth-5 yrs 5ml 8hourly for 5days If no improvement, REFER immediately to the appropriate health facility.
VI. Abdominal pain and paleness	•Thin and pale looking (Pallor of conjunctiva and fingernails). No tenderness	Mild/Moderate condition Suspect Sickle Cell disease	1.If Hb is less than 10g/dl but more than 5g/dl: Give Syr Folic Acid 5ml daily for 2 weeks, REFER immediately to the appropriate health facility.
VII. Abdominal pain with history of passing worms,	•Abdomen distended; no tenderness; No constipation	Mild Condition	1.Give Pyrantel pamoate 1-2 tabs stat;or Albendazole ml 200mg 2.Review in 2 weeks If pain persists, REFER to the appropriate health facility.

Health Education

Advise Mother/caregiver on:

1.Food and Personal hygiene.
2.Adequate Nutrition.
3.Genetic counseling.
4.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.24. HERNIA

Hernia is when there is a protrusion of an organ or tissue through a gap or weakness of the body cavity in which it normally lays Hernia can be found in the umbilicus, groin or scrotum.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? For how long? 2.Is there vomiting? How many times? 3.Is there constipation? When was the last stool? 4.Is there any abdominal pain? 5.Has the child had the problem before? Has any medication been given? If yes what?	1.General appearance: pallor, weaknes,distress 2.Respiration: difficult or rapid 3.Abdomen: a). enlarged, distended a).masses, enlarged spleen b).visible intestinal movement d)presence of hernia, tender-ness 4.Groin: presence of scrotal hernia 5.Hydration status: fontannelle, eyes, mouth, tongue, skin 6.Haemoglogin: estimate and record Temperature, pulse and respiratory rate : check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Painful swelling in the groin	•Painful mass in the groin that cannot be reduced	Severe Condition Suspect Obstructed Hernia	. REFER immediately to the appropriate health facility
II. Swelling in Scrotum	•Child in Pain •Scrotum tender and warm Swelling in one or both sides of Scrotum	Severe Condition Suspect strangulated Scrotal hernia	1.Explain condition to mother 2.IM Paracetamol a.2mth-2yrs 125mg Stat b.3-5 yrs 250mg Stat REFER immediately to the appro-priate health facility.
III. Swelling in the groin that disappears on lying down OR painless swelling of the scrotum	•Swelling appears on coughing and it is easily reduced	Moderate condition Suspect Inguinal hernia/ Hydrocele	1.Explain findings to mother REFER immediately to the appro-priate health facility.
IV. Umbilical swelling (Hernia)	•Well child, Not in pain	Mild/Moderate Suspect Umbilical Hernia	1.If reducible, no treatment needed. 2.Explain to mother that it will resolve by 2-3 yrs, but If not reducible at this age, REFER to the appropriate health facility.
V. Painful swelling in the groin	•Painful mass in the groin that cannot be reduced	Severe Condition Suspect Obstructed Hernia	1. REFER immediately to the appropriate health facility.

Health Education

Advise Mother/Caregiver on;

1.Adequate nutrition, prevention of constipation and adequate fibre in diet.
2.Adequate fluid intake.
3.Discourage use of force to reduce hernia.
4.Operation as appropriate.
5.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.25. SEXUAL ASSAULT/ RAPE

Sexual assault is rape or any sexual contact that results from coercion, including seduction of a child through offers of affection or bribe; it is also includes being touched, grabbed, kissed or shown genitals. On the other hand, rape is oral, anal or vaginal penetration that involves threats or force against a person who is unwilling (i.e. non-consenting) or incapacitated (because of cognitive or physical disability or intoxication). Such penetration, whether wanted or not is statutory rape if victims are younger than the age of consent. This may result in pregnancy (if female), transmission of STIs (e.g. HIV, Syphilis, depression, physical injuries, recurrent dreams and nightmares and other emotional problems. Victims of sexual assault/rape should be advised to report the incident to police if not against their wish.

HISTORY	EXAMINATION/ACTIONS
1.Was the baby sexually assaulted (raped)? If yes, where? How? Who did it? 2.How many times? How long did it last? When? 3.Was the baby threatened or forced? 4.Was the child made to drink anything (alcohol or medicine) prior to the assault? 5.Was the baby wearing clothes when the incident occurred? 6.Was the child physically harmed? 7.Was there any penetration of the anus, vagina or mouth? If yes, with penis, finger or object? 8.Was there ejaculation? Where? 9.Any bleeding? Where? 10.Did you bath, change clothes or clean up the child afterwards 11.Has the child been sexually assaulted before? 12.Did you report to anyone else about this? Who? 13.Have you given any treatment? When? Where? 14.Any witness? Any other problem?	1.General appearance: restless, physically exhausted, conscious, depressed, afraid, crying. 2.Check temperature, pulse, respiration, blood pressure and record. 3.Mouth: injuries 4.Skin: injuries 5.Genitalia: bleeding, injuries 6.Anus: injuries 7.VDRL: test and record. 8.Hepatitis B: test and record. HCT: test and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Rape with vaginal or anal bleeding and lacerations	Signs of penetration of with bleeding, lacerations and injuries to other parts of the body.	Severe condition	1.Counsel and reassure mother. 2.Advise mother/caregiver to report the incident to police or appropriate NGO. 2. Report to security agents 3. Decument evidences 4.REFER to the appropriate health facility for further investigations.
II. Rape without vaginal or anal bleeding and lacerations	Signs of penetration without vaginal or anal bleeding and lacerations.	Moderate condition	1.Counsel and reassure mother. 2.Advise mother/caregiver to report the incident to police or appropriate NGO. 3.Give the child Hepatitis B vaccination (if not already immunized) 4.Conduct HIV test. If positive, Treat as finding one above REFER to appropriate Standing Order 5.If negative, commence emergency post exposure prophylaxis within 72 hrs. of rape (if trained). 6.Give: a) IM Ceftriaxone stat. b) Azithromycin orally c) Tab Metronidazole orally. d) Report to security agents

III. Sexual Assault	No signs of penetration of vagina, anus and mouth. No injury to other parts of the body.	Mild condition	1.Counsel and reassure mother. 2.Advise mother/caregiver to report the incident to police or appropriate NGO. Ensure child protection

Health Education

Advise Mother/Caregiver on;

1.Childcare and protection.
2.Appropriate dressing.

2.26. ANAL DISODERS

The anus is the terminal end of the alimentary canal. Common problem affecting the anus include fissures. Redness and prolapsed. An anal fissure is painful li near s or eat margin of anus. In rectal prolapsed; the rectum turns inside out and protrudes out of the anus as a pink mass. It is common in young children who are poorly nourished, have worms and diarrhoea frequently.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? How long have you had the problem? 2.Is there diarrhoea? How many times? 3.Is there constipation? When was the last stool? 4.Is there bright blood in stool? 5.Is there any abdominal pain? 6.Has the child had the problem before? Has any medication been given?	1.General appearance: pale weak, in distress 2.Hydration status: fontanels, eyes, mouth, tongue, skin 3.Anus: prolapsed, redness, fissure 4.Haemoglobin: estimate and record 5.Temperature: check and record Weight: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Protrusion from the anus, not reducible	•Child in distress, •Not dehydrated Rectal prolapse not reducible with or without diarrhoea	Severe condition Suspect Prolapse of rectum	1.Apply warm saline, compress (sitz bath), Dettol ,,anusal sipostory
II. Protrusion from the anus, reducible	•Child in distress Anal prolapse reducible with or without diarrhoea	Moderate condition Suspect reducible Prolapse of rectum	1.Push prolapsed portion back and strap buttocks together. 2.Teach mother how to strap the buttocks 3.If having diarrhoea, Give suspension Cotrimoxazole: a. 2-12mths- 2.5ml 12hourly for 5day b. 1-5yrs- 5ml 12hourly for 5days 4.If anaemic a.Give Syr. Folic Acid 5ml daily for 2 wks. b.Ferrous sulphate (60mg elemental iron) 8hourly for 2 weeks c.LO-ORS Solution 5.Nutrition Counseling 6.Review in 5 days if no improvement, REFER to the appropriate health facility.
III. Redness of anus with or without pain	•Anus red with sore Child uncomfortable	Moderate condition Suspect anal fissure	1.Clean with soap and water 2.Dry thoroughly 3.Apply Zinc Oxide topically 4.Give Paracetamol tablet or PCM suppository if oral medication is not possible. Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 5.Dietary advice 6.Review in 3 days If no improvement, REFER to the appropriate health facility.

Health Education

Advise Mother/Caregiver on;

1.Personal, food and environmental hygiene.
2.Protection of genital area.
3.Early detection and treatment of infection.
4.Discouragement of sharing of sponge, towel, underwear, and clothing.
5.Adequate nutrition.
6.Bowel training.
7.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.27. WORM INFESTATION

Worm infection is the presence of large amount of worm in the gut which causes discomfort to the patient. Most children have worms. Ascaris (round worm) is the commonest. Heavy round worm infestation produces abdominal pain, vomiting, constipation and sometimes obstruction of the intestine. Hookworm infestation is more common in rural areas where heavy infestation causes blood in stool, anaemia, malnutrition. Severe (whip worm) infestation causes bloody diarrhoea, abdominal pain, anaemia and prolapse of the rectum. Thread worms also live in the intestine and cause itching around the anus. Tapeworm is a flat ribbon - like worm that can also be found in the intestine.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? When did it start? 2.Has any worms been passed? If yes, describe 3.Is there diarrhoea? Blood in stool? 4.Is there vomiting? Is the child coughing? Any wheezing? 5.Is there anal itching? 6.Is there abdominal pain and rectal prolapse? 7.Is there any other problem? Has any medication been given? If yes, what?	1.General appearance: thin, pale, 2.Eyes: pallor 3.Chest: wheezing, rhonchi 4.Abdomen: tenderness, distension, visible peristalsis 5.Anus: redness, prolapserectum 6.Temperature Pulse and Respiration: check and record 7.Haemoglobin: estimate and record 8.Weight: weigh and chart Stool examination: for ova of worm (if laboratory available)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I.Passing worms	Passing worms vomiting with abdominal tenderness	Severe Condition	1.Give Albendazole 200mg or Combantrin a.6-11mths ½ tab stat a. 1-2yrs1tabstat b. 3-5yrs: 2tabs 2. Increase fluid intake 3.Give multivitamins 5ml tdsx5days Review in 1 week
Passing worms	Passing round worms with mild abdominal pain May be underweight No other problems	Moderate condition Suspect Ascariasis	Give Pyrantel pamoate stat a.6-11mths ½ tab stat b.1-2yrs1tabstat c.3-5yrs1-2tabsstat Review in one week.
Passing worms with paleness	Passing worms. With distended but not tender abdomen: or Hb less than10 gm. no diarrhoea.	Moderate condition Suspect hookworm infestation	1.Pyrantel pamoate stat a.6-11mths ½ tab b.1-2yrs : 1 tab c. 3-5yrs: 2tabs Give Syr folic acid 5ml daily for 2wks
IV.Loss of weight paleness	Presence of worm in the stool from laboratory result	Moderate condition Suspect tapeworm infestation	Niclosamide 200mg/kg body weight single dose followed by chastic purgative 2hrs later.

Health Education

Advise Mother/Caregiver on;

1.Personal, food hygiene and environmental sanitation.
2.Adequate nutrition.
3.Regular Deworming.
4.Protection of feet.
5.Importance of taking home-based records on child's next visit to the clinic or hospital.

2.28. URINARY AND GENITAL DISODERS

A urinary tract infection, commonly known as UTI is one of the most common urinary problems and is an infection that occurs in the urinary tract. This infection is most often caused by microbes such as fungi, bacteria, and virus and involves the kidneys, ureters, urethra, or bladder. This condition can be very painful, and the consequences can get critical if the infection spreads to the kidneys. Children may present with abnormalities of the genitalia such as undescended testes, imperforate hymen, hydrocele, hypospadias, and hermaphrodite.

Children may also present with painful micturition with or without discharge. Redness or itching of the genitalia may occur in genital infections. Patients may complain of discharge, pain and burning sensation on passing urine and blood in urine may occur as a result of schistosomiasis infection. Other causes of urinary problems are kidney disease and diabetes which can cause frequent micturition.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. How long has it existed? 3. Is there pain on passing urine? Or does the child scream or rub his/legs together before passing urine? If yes, for how long? 4. When last did he urinate? 5. What does the urine look like? Odour? 6. Has the child been bed wetting after a period of full bladder training? 7. Do ants cloud around urine passed by the child? Is the child excessively thirsty?	1. General appearance: puffy, pale, ill -looking 2. Eye: pallor, peri -orbital oedema 3. Mouth: dryness, parched lips 4. Abdomen: tenderness, or masses, oedema 5. Flanks: tap both flank with hand for tenderness 6. Skin: dryness, wrinkle 7. Limbs: hands and feet for oedema, clubbing 8. External genitalia: abnormalities, redness, swelling or pus. In males, look at foreskin to see if is too tight or will not retract. 9. Temperature pulse and respiration: check and record 10. Urine; cloudiness, colour, odour, protein, blood sugar, and microscopy,

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. No urine for more than 12 hours	Looks ill Puffy Restless Anxious Leg crossing Heal sitting	Severe condition Suspect acute renal failure	1. Insert catheter 2. Refer immediately to the appropriate health facility.
II. Excessive urination with history of ants crowding round urine. Excessive thirst	Dry mouth Wrinkled skin Parched lips Sugar in Urine Lethargic Loosing urine by accident	Severe condition Suspect juvenile diabetes	1. Screening for UTI should be done immediately 2. Refer immediately to the appropriate health facility.
III Dark yellow urine with yellow eyes with or without hotness of body	Dark yellow urine Yellow eyes with or without fever	Severe condition Suspect malaria	1. Test for Genotype 2. Syr. Folic acid 5ml dly for 2 weeks 3. Conduct mRDT if positive treat uncomplicated malaria with ACT (Refer to section on malaria treatment) 4. if Mrdt Negative treat think of Weil's disease 5. Encourage fluid and fruits intake 6. Review in 3 days 7. If no improvement, REFER to the appropriate health facility.

IV. Blood in last stream of Urine with or without pain, fever or discharge	Blood in Urine, Schistosoma ova may be detected in urine	Severe condition Suspect Urinary Schistoso-miasis	1. If painful Micturition; 2. Give Susp. Cotrimoxazole (240mg/5ml) a. 2-12mths- 2.5ml 12hourly for 5day b. 1-5yrs- 5ml 12hourly for 5days 3. Encourage fluid intake 4. Review in 3 day, if no improvement, check for Bilharzia ova, if positive, give prezenquentel accordng to the approprite age of the child a. If negative and no improvement, REFER for further investigations.
V. Redness or swelling of gen-italia with or without painful micturition, itching	Redness and swelling ofgenita-lia, scratch marks Bruises	Moderate to severe condition	1. Assess for history of rape or sexual abuse 2. Wash the area with soap and water 3. Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 4. Apply zinc oxide and Mycostain cream or paste to area. 5. Encourage fluids 6. If male, retract foreskin 7. Give Susp. Cotrimoxazole (240mg/5ml) a. 2-12mths- 2.5ml 12hourly for 5day b. 1-5yrs- 5ml 12hourly for 5days 8. Review in 3 days 9. If no improvement REFER to the appropriate health facility. 10. Treatment of constipation with stool softeners is effective in pre-venting recurrent UTI in children without Urinary tract abnormities
VI. Painful urination; No discharge	Painful urination No discharge	Moderate condition Suspect Urinary Tract Infection [UTI]	1. Use STIs Symdromic management approach in the appreprite setion
VII. Pain in the Flank, fever	Tenderness on tapping flank muscle sperm Temp 37.5^{0}c or above	Moderate condition Suspect kidney infection (pyelonephritis0	1. REFER to the appropriate health facility.

VIII. Painful urination with or without frequent, cloudy urine or fever.	Temp – 27.50c protein in urine, painful Micturition	Moderate condition	1. Encourage fluids 2. Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 3. Give Susp. Augmentin Mist pot cit according age 4. Review in 2 days; 5. if no improvement, REFER to the appropriate health facility.
	Dark Urine No other findings	Moderate condition Suspect dehydration	1. Encourage adequate fluid intake 2. Review in 2 days; 3. Rule out sever Malaria by RDT test
X. Painful, urination With or without frequent, cloudy urine or fever	Temp > 37.5^{0}c Protein in Urine; painful Micturition.	Moderate condition Suspects glomerulonephritis	1 .see treatment pylonepritis section 2.If no improvement REFER to the appropriate health facility
XI. Urethra and/or for Vaginal discharge with painful micturition	Urethral/vaginal discharge	Moderate condition	1. Risk assessment to know if child has been raped or abused 2. Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 3. Tab Ciprofloxacin 125mg 4. See syndromic management 5. Cap Doxycycline 25mg 12hourly for 7 days 6. Review in 3days 7. If no improvement, REFER to the appropriate health facility.
XII. Difficulty in passing urine	•Penis blocked by tight foreskin not easily retracted bladder full	Moderate Condition Suspect phimosis constriction	1. Circumcise the child immediately using standard procedure.
XIII.Abnormal Genitalia	Un-descended testis Imperforate hymen Hermaphrodite Large clitoris Hydrocoele; Abnormal (penial opening(s) Hypodyspavisis Epidyspavisis	Mild or Moderate condition Suspect congenital malformation	1. Explain findings to mother and re-assure her; 2. REFER to the appropriate health facility

Health Education

Advise Mother/Caregiver on;

1.Personal hygiene.
2.Drug compliance.
3.Proper parental care and supervision of the child
4.Protection of genital area especially in children.
5.Adequate fluid intake.
6.Importance of taking home -based records on child's next visit to the clinic or hospital.

2.29. ARM AND LEG DISORDERS

Deformities of the arms and legs, if not corrected early could lead to permanent disabilities. The deformities or swellings may be due to bruises, fractures or a dislocation. It is important to test for normal movement of the parts whenever there is a history of injury, swelling or deformity. Swelling of a joint may be due to a sprain of the tendons surrounding the joint. Sickle cell disease may also be a cause of pain in the arms, legs and the hip bone. Tuberculosis can occur in the joints. Polio can seriously affect one or both legs or arms first causing weakness then wasting, or paralysis. Diabetes Mellitus may cause tingling sensation and numbness of the fingers and toes, foot ulcer, that fail to heal and black discoloration of toes and feet (Gangrene).

HISTORY	EXAMINATION/ACTIONS
What is the problem? When did it begin? Was there an accident or injury? If an injury, has the child been able to use the part since the injury? Is there any pain or tingling or numbness? Is it recurrent? Is there fever? Is the child a sickler?	1. General appearance: ill looking, wasted or deformed 2. Limbs (a) deformity (b) Swelling (c) Fracture (d) Tenderness

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Inability to move arm and/ or leg.	Unable to move arm and feet	Severe condition Suspect Polio Paralysis or Sickle cell disorder	1. Note areas of enlargement 2. Inspect the area for discoloration 3. If Polio notify the appropriate PHC authority. 4. REFER immediately to the appropriate health facility and follow up
ii. History of fall from a height, pain, swelling broken bone.	Pain and tenderness Swelling Lower limb hanging Inability to move limb	Severe condition Suspect fracture or spinal injury	1.Immobilize the limb 2. Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 3. Note area of increase warmth (heat) 4. REFER immediately to the appropriate health facility
iii.History of sickle cell disease, severe joint and abdominal pain, swelling on digits, especially of fingers, toes, back of hands and feet	Intense pain Swollen fingers, toes, back of hands and feet. HB below 10 gms Temp > 37.5°C	Severe condition Suspect Sickle cell disease in crises	1. Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 2. Syr. Folic acid 5ml daily 3.See treatment for Malaria 4.REFER immediately to the appropriate health facility.

iv. Swelling of joints	Tenderness of the affected joint Swelling of the joint.	Moderate condition Suspect juvenile Arthritis	1. Syrup Paracetamol(120mg/5ml) a. 2mths - 2yrs: 2.5ml 8hourly for 3days b. 2-5 years: 5ml 8hourly for 3days 2. Ampicillin + Cloxacillin a. 250mg 6hourly for 5 days 2 - 12mths b. 500mg 6 hourly for 5 days 1 - 5yrs 3. REFER to the appropriate health facility for investigations.
v. Deformity of leg with difficulty in walking	Club foot, Bow leg. Walking with difficulty	Moderate condition Suspect birth defect	1. REFER to the appropriate health facility for assessment.
vi. Swelling of fingers, toes, back of hands and fee	Swollen fingers, toes, back of hands and feet HB below 10	Moderate to Severe condition	1. REFER immediately to the appropriate health facility.

Health Education

Advise Mother/Caregiver on;

1. Prevention of accidents.
2. Genetic counseling.
3. Adequate nutrition (Calcium rich food).
4. Importance of taking home base records on child's next visit to the clinic or hospital.

2.30. CHANGES IN ALERTNESS AND BEHAVIOUR

Changes in alertness and behaviour mean that the client does not act normal. He may be drowsy or unconscious or give a history of fainting. Big changes such as unconsciousness or delirium are easy to identify. Small changes such as drowsiness and decreased alertness are not easily noticed. If the client is not alert or is drowsy, it may not be noticed easily. If this happens, the client may not receive treatment and could come back later in more serious conditions. Therefore, careful observation of client is important.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child been like this? 2.Did anything happen to the child? Any injury? 3.Did the child have a fit or convulsion, or is the child clumsy? 4.Has any medication been given? 5.Has the child been ill otherwise? 6.Has the child had fever, diarrhoea, vomiting or ear pain? 7.Has the child been drinking more recently??? 8.Has the child been urinating more frequently? 9.Was there any difficulty with the pregnancy or delivery? 10.Was the child jaundiced at birth or shortly after? Has the child been frothing in the mouth or passing urine when he faints?	1.General appearance: (especially for alertness and irritability), sign of trauma. 2.Chest: respiratory rate, lower chest in-drawing. 3.Neck: stiffness 4.Head: fontanelle for bulging, signs of injury. 5.Ears: pus, redness or perforation of the drum. 6.Limbs: arms and legs for abnormal movements. Temperature: check and record 8.Haemoglobin: estimate and record. Urine: test for protein, sugar, acetone.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Unconscious or difficult to wake, Irritable, abnormal movements of arms and Legs	•Irritable •No stiffness Uncontrollable	Severe condition Suspect Head Injury	1. REFER immediately to the appropriate health facility.
II. Child faints suddenly but is arousal	•Child disoriented •Weak child Acetone in urine	Severe condition Suspect Hypoglycemia	1.Give child sugar or glucose at once and let him rest for 30mins. 2.If no improvement, REFER to the appropriate health facility. 3.If improved, advise on proper feeding. Review next day.
III. Fainted (fit) once or twice with or without fever or frothing from the mouth or family history of epilepsy	•Conscious, normal temperature 37.5°c fits	Moderate condition Suspect epilepsy	1.Explain condition to mother 2.Phenobarbitone 0-5mths 6mths-5yrs a. 1/8tab (3.5mg) 1/4tab (7.5mg) 12hourly 7days 3.Review in 7 days 4.Assess medication, if improved, resupply If no improvement, REFER to the appropriate health facility.

Health Education

Advise Mother/Caregiver on;

1. Prevention of injuries.
2. Compliance with prescribed drug regimen.
3. Avoidance of self-medication.
4. Healthy home environment.
5. Proper parental care and emotional support.
6. Need for therapy as appropriate.
7. Importance of taking home – based records on child's next visit to the clinic or hospital.
8. Improve skills and sense of self efficacy.
9 Promote services for prevention, care and support.
10. Create a demand for information and services.
11. Address stigma.
12 Focus on risk behavior.
13 Health workers should create trust.

2.31. POISONING

A poison is a toxic substance which may enter the body through swallowing, breathing, absorption through the skin or mucous membrane or infection. Poison may be accidental, suicidal, or homicidal. An overdose of a medicinal drug may act as poison e.g. overdose of aspirin. It is common for children under 5 years of age to accidentally ingest medicine not prescribed for them or over dosage of prescribed, drugs; alcohol (gin), kerosene, petrol or corrosive substances e.g. acid, lead or bleaches.

Poisoning can be harmful effects caused by the ingestion, inhalation, absorption, or injection of a toxic substance into the body. Poisons can come in various forms, including chemicals, gases, plants, medications, household products, and venom from certain animals or insects.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.What was the substance taken? How much of it? When was the poison taken? 3.Has any medication been given (modern or traditional)? Has the child vomited? Is there blood in the vomitus?	1.Try to examine poison container 2.General appearance: consciousness, alertness, pallor, cyanosis. 3.Eye: size and reaction of pupils to light. 4.Chest: respiratory rate, difficulty, lower chest indrawing, deep and sighing. 5.Abdomen : tenderness, rigidity. 6.Mouth: burn, bleeding, dribbling, colour of membrane 7.Pulse: check rate, volume and record. Temperature: check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Ingestion of petrol or kerosene or corrosive substance e.g. acid, herbicides, insecticide, rodenticides etc.	Irritable Swelling and mouth burns	Sever condition Suspect lead/Corrosive poisoning	1. Do NOT induce vomiting 2. Give enema e.g. soap 3. Give IM Cloxacillin for 5 days 0.5mths 6-11mths 1-5yrs a. 0.25ml 0.5ml 1ml daily for 5days b. 75,000IU 150,000IU 500,000IU 4. Susp. Mist Magnesium Trisilicate 5ml 8hourly for 5 days 5. Give activated charcoal. 6. Review in 4 hours. 7. If non improvement REFER to the appropriate health facility.
ii. Ingestion of any of the following, gin medicine, not prescribed.	Irritable, Restless, No Mouth burns or swelling May or may not be warm to touch	Moderate condition Suspect poisoning	1. Identify the poison 2. Do No induce vomiting 3. Give milk Drink Trisilicate 5-10 mls 8hourly for 3days 4. Give activated charcoal 5. Review after four hours 6. If no improvement, REFER to the appropriate health facility.
iii. Ingestion of poison with history of convulsion	Irritable, restless and convulsion May or not be warm to touch	Moderate condition	1. IM. Paraldehyde 1-3ml stat 2. Gastric lavage/washout 3. Tab. Phenobarbitone 15mg 12hourly for 3 days 4. Give activated charcoal 5. Review in 3days

Aspirin poison with or without fever or other signs	Temp normal or above 37.5^{0}c Irritable Thirsty Restless	Moderate condition Suspect aspiration	1. Do not induce vomiting 2. Give milk drink or Mist Magnesium Trisilicate. 3. Give plenty of fluids 4. Give activated charcoal 5. Review daily until better

Health Education

Advise Mother/Caregiver on;

1. Prevention of home accidents especially accidental ingestion.
2. Proper storage and labeling of drugs, various liquids and solvents.
3. Keep hot food hot and cold food cold.
4. Don't keep food in the temperature danger zone (i.e. at or below 5 and above 60°C) any longer than necessary.
5. Reheat food to steaming hot before serving (at least 75°C). Cook food properly, heat to at least 75°C.
6. Keep raw and cooked food separate.
7. Keep kitchen and utensils clean.
8. Wash and dry your hands properly.
9. Avoid handling food when you are ill.
10. Heat food to steaming hot before serving (above 75°C).
11. Keep raw food on container that has cover to prevent it from rat contamination e.g urinating and defecating to avoid having Lassa fever.
12. Keep kitchen and utensils clean.
13. Improvement of taking home-based records on child's next visit to the clinic or hospital.

2.32. EMERGENCY CONDITIONS

In children 2 months to 5 years can be life threatening and should be treated promptly to save life of the child. There should be no hesitation in asking for or referring where necessary. Severe acute fever, diarrhoea, malnutrition, child poisoning. conjunctivitis, epilepsy, choking, anaphylaxis are emergency situation and should be treated as soon as possible to save the life of the child.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem with child? 2. Is there fever, vomiting, diarrhoea or abdominal pain? 3. Is there history of convulsion or unconsciousness? 4. Is the history of poisoning? 5. Is the child having difficulty in breathing or chocking? 6. Is there family history of asthma? 7. Is a child allergic to known substances or food? 8. Is there decrease in urine output?	1. General appearance: ill looking, alert, unconscious, convulsing, dehydrated. 2. Temperature: check and record 3. Eye: Sunken 4. Mouth and lips: Dryness 5. Chest: (a) Respiratory rate: check and record (b) Wheezing, crepitation 6. Skin: dryness, dehydration

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Acute fever with temperature above38.5°c and child convulsion	High temperature above 38.50C and convulsing	Severe condition Suspect Acute fever	1. Tepid sponge 2. Give paracetamol suppository 3. Treat Malaria as approprite 4. If no improvement 5. REFER
ii. Acute diarrhea with severed hydration, loose stools, unable to eat, sleep.	Severe dehydration with symptoms as drowsiness, confusion, passing little urine, dry mouth and tongue, sunken eyes, weakness, cold extremities sunken fontanelle in babies / young infants	Severe condition Acute Diarrhoea	1. Give Lo-ORS / Zine 2. Treat as plan C if improvement 2. REFER Immediately to the appropriate health facility.
iii. Child poisoning from medicines, iron pills, cleaning, products, pesticides	Severe nausea, vomiting, body weakness, fever, in babes / young infants	Severe condition Suspect childpoisoning	1. REFER Immediately to the appropriate health facility.
iv. Conjunctivitis with inflammation, redness, and discharge.	Child with inflamed eyes, redness, and discharge	Severe condition	1. Administer appropriate antibiotics suspension 2. Give Chloremphenicol eye drop 3. Review in 3 days if no improvement
v. Severe seizure in children	Unconscious and convulsing	Severe condition Suspect epilepsy	1. Give I dose of appropriate antibiotics 2. REFER
vi. Anaphylactic reaction to food, medications, allergens	Cutaneous manifestations, urticaria, pruritus, angioedema, and flushing. Cardiovascular dizziness, hypotension and syncope GIT: Nausea, vomiting, diarrhea, abdominal	Severe condition Suspect Anaphylaxis	1. Give injectable epinephrine 1:1000, 001mg/kg into the lateral thigh, may repeat every 5-10mins if symptom persist to all children with signs and symptoms suspicious of anaphylaxis before arrival to hospital. 2. Give Hydrocortisone injection stat 3. Provide warmth 4. REFER

vii. Choking with breathlessness	Bluish skin color Difficulty breathing ribs and chest pull in ward Loss of consciousness if block age is not cleared Inability to cry or make much sound Weak ineffective coughing Soft of high – pitched sounds while inhaling.	Severe condition Suspect chocking	1. If your child is not coughing forcefully or does not have a strong cry, follow these steps. 2. Lay the infant facedown. Along your fore arm. Use your thigh or lap for support Hold the infants chest in your hand and jaw with your fingers. Points the infants head downward. Lower than the body. 3. Give up to 5 quick forceful blows between the infant's shoulder blades. Use the palm of your freehand. 4. If the object does not come out of the airway after 5 blows. 5. Turn the infant face up. Use your thigh or lap to for support the head. 6. Place 2 fingers on the middle of his breastbone just below the nipples 7. Give up to 5 quick thrusts down compressing the chest1/3 to ½ the depth of the chest 8. Continue 5 back blows followed by 5 chest 9. Thrusts until the object is dislodged or the infant loses alertness becomes conscious.
viii. Acute asthma attack With wheezing, cough. breathlessness	Worsening asthma condition	Sever condition Difficulty in breathing	1. give 6 puffs of blue reliever inhaler via a spacer 2. If no improvement within 20 minutes repeat 6 puffs 3. If still no improvement within 20 minutes of 2ndlot of 6 puffs 4. REFER

Health Education

Advise Mother/Caregiver on;

1. Prevention of breathing complication.
2. Compliance with prescribed drug regimen.
3. Avoidance of self-medication.
4. Healthy home environment.
5. Proper parental care and emotional support.
7. Importance of taking home – based records on child's next visit to the clinic or hospital.
8. Maintain balance diet.
9. Adhere to personal hygiene.

2.33. HIGH RISK GROUPS (ORPHANS AND VULNERABLE CHILDREN)

An orphan is a child (0 – 17 years) who has lost one or both parents. A child is vulnerable if, because of the circumstances of birth or immediate environment, is prone to abuse or deprivation of basic needs, care and protection and thus disadvantaged relative to his or her peers. (FMW &SD 2008). A vulnerable child is one: with inadequate access to education, health and other social support, has a chronically ill parent, lives in a household with terminally or chronically ill parent(s) or caregiver(s), live outside of family care (lives Overview with extended family, in institution, or on the street), is infected with HIV in addition to the above, a vulnerable child includes, children in need of alternative family care, children who are sexually abused or neglected, children in hard-to-reach areas; children with mental disability; children affected by armed conflict and children in need of legal protection. Most of the current interventions cover OVC aged 6-17 years, leaving out children under five years old.

The general appearance include Growth failure, Signs of malnutrition, Poor personal hygiene, Evidence of poor health care e.g. delayed immunisation, untreated infection etc., Frequent injuries from lack of supervision Behavioural indicators include Self-simulating behaviours (biting, sucking), Infancy lack of social smile and stranger anxiety, Antisocial behaviour (destructiveness, stealing, cruelty to animals or people), Extremes of behaviour (aggressiveness, demanding, over complaints), Suicide attempts., Lags in emotional and intellectual development (especially language).Psychoneurotic reactions (hysteria, obsession, compulsions, phobia, etc.). Apart from the government at various levels, a number of organizations are involved in OVC work in Nigeria. They include international NGOs, mainly USG and Global Fund implementing partners, local NGOs, FBOs and CBOs. Some of the main international NGOs involved in OVC work include the Columbia University International Centre for AIDs Care and Treatment Programs (CU-ICAP), Catholic Relief Services (CRS), Centre for Development and Population Activities (CEDPA), and Winrock international AIM.

HISTORY	EXAMINATION/ACTIONS
1. How old is the child? Who is the child leaving with? 2. Is the child sick or well? 3. If sick, what kind of illness? 4. If well, who is taking care of the child? 5. Is the mother or father a live both or one 6. Is the mother or both parent HIV/AIDs before death? 7. Has the child been sexually abused and is sick as result of that?	1.General appearance: well-nourished, ill, tired, clean, thin, congenital abnormality 2.Immunization status: if card available, check status 3.Weight and Height: weigh, measure, and record 4.Head: lice, ringworm 5.Eyes: discharge, pallor, jaundice, cataract, test of vision 6.Ears: redness, discharge, test of hearing 7.Mouth: odour, caries, missing teeth Temperature: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Baby born to HIV- Infected mother and orphaned	Baby on or not on ARV	Severe condition Suspect HIV	1. REFER to the section on HIV AIDs and treat 2.REFER to approprite treatment centre
2. Children 2 months to 5 years with any of the disease conditions		Moderate to mild condition, suspect any disease condition	1.REFER to appropriate section and treat
iii. Well Orphans and Vulnerable Children(OVC)	• No abnormality	Mild condition	1.Give Psychosocial support care.

Health Education

1.Healthy home environment.
2.Proper parental care and emotional support.
3.Counsel on need for therapy as appropriate.
4.Interview the child.
5.Interview parent, witness, or other significant persons.
6.Physical injuries should be treated appropriately.
7.Psychiatric or social welfare officers maybe involved i.e. co-manage child.
8.If necessary, assist in moving child from home to temporary custody.
9.Counsel family members about abuse and refer if necessary.

2.34 SCHOOL HEALTH SERVICES

The school health services are essential in maintaining and improving the health status of member of the school community and should be available to everyone for as long as they are in the school community. School health services afford the opportunity for personal health care, screening of school food vendor, environmental health care, community mobilization, Advocacy. Personal health care includes health screening for all newcomers, regular health inspection and treatment for minor ailments.

HISTORY	EXAMINATION/ACTIONS
1.Is there any problem? 2.Which immunizations have you had? 3.What type of food do you eat?(ask about main meals and snacks) 4.How many brothers and sister do you have? 5.Whom do you live with?(parents, grandparents, spouse .. other specify)? 6.How many times do you eat in a day? 7.Where do you get your food? 8.What type of work does your fattier do? Mother? You? Spouse? 9.What is your source of water? What type of latrine do you	1. General appearance: well-nourished, ill, tired, clean, thin, congenital abnormality 2. Immunization status: if card available, check status 3. Weight and Height: weigh, measure and record 4.Head: lice, ringworm 5.Eyes: discharge, pallor, jaundice, cataract, test of vision 6.Nose: discharge, blockage, abnormality 7.Ears: redness, discharge, test of hearing 8.Mouth: odour, caries, missing teeth 9.Gum: healthy, bleeding

SECTION THREE

MIDDLE CHILDHOOD

(6 – 12 Years)

3.0 MIDDLE CHILDHOOD

3.1 CHILD FIRST VISIT

This section is for children 6 - 12 years. The health worker should be able to carry out simple sight and hearing tests and to be able to find out whether or not the child has visual or hearing problems that can affect school performance. He must also recognize social problems likely to put the child at risk, must be able to screen for malnutrition. He/she should request for genotype and blood group so that the child/parents know the blood group and genotype as well as counsel if abnormality is detected. He should carry out health and nutrition education; immunize and protect him/her from illnesses and other conditions as relevant to the family. Encourage mother/guardian to take their children for clinic visit even though they are well.

HISTORY	EXAMINATION/ACTIONS
1.Development; Ask Parent/caregiver if child is developing normally. 2.When does the child start crawling, walking, and speaking? 3.Is the child attending school? 4.What is the current class of the child? (Milestone) 5.Family and Social; a)With whom does the child live? b)Do Parent /caregiver work outside home? c)Who looks after child when Parent /caregiver are away? d)What is the source of water? What type of toilet is used? e)Is house screened against mosquito/is insecticide treated net used? f)What illnesses has child had in the past? g)Does child have any problem now? h)What work does the father do? How many Siblings does the child have? i)How many children are there in the family? j)How many wives does the man have? 6.Are there any of these disease conditions in the family: Asthma, sickle cell disease, Diabetes, Hypertension, Tuberculosis, Epilepsy, and other relevant conditions. 7. Immunization; a)Did the child receive all the routine immunization for age? b)Does the child present his/her certificate for completing immunization? 8. Nutrition; a)Was the child exclusively breast-fed? b)What type of winning food was introduced after six month? c)What kind of food is the child currently eating? d)What type of food does the child like most? 9. Medical History; 1.Has the child been hospitalized before? 2.What type of sickness was the child hospitalized for? Has the child convulsed at any time in the past?	1.General appearance: healthy, ill, thin 2.Blood pressure: measure and record 3.Temperature, pulses: check and record 4.Eyes: pallor, jaundice 5.Mouth: conditions of mucous membranes, gums, caries, inflamed throat 6.Neck: swelling, any stiffness, enlarged, thyroid 7. Chest: a) crepitations, rhonchi, any decreased breath sounds b) respiration rate, difficulty 8. Heart: rate, rhythm, murmurs 9. Abdomen: scars, tenderness, masses, hernias 10. Limbs: a) deformation, varicosities, Oedema b) ankles: Oedema 11. Weight: weigh and record 12. Urine: colour, protein, glucose and ova 13. Stool: parasites and ova 14. Haemoglobin: estimate and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Well child	No abnormality	Healthy child	Explain findings to Mother Counsel as appropriate.
II. Any abnormal finding			REFER to appropriate section.

Health Education

Advise Parent/Caregiver on the following:

1. Ensure adequate nutrition.
2. Maintain personal and food hygiene.
3. Practice environmental sanitation.
4. Prioritize hand washing and hygiene practices.
5. Prevent accidents through vigilance and safety measures.
6. Provide proper parental care and attention.
7. Utilize Long Lasting Insecticidal Nets (LLIN) for protection against insects.
8. Highlight the importance of bringing home-based records when taking the baby to the clinic or hospital.

3.2 FEVER

Fever is the elevation of the body temperature. It is body temperature of 37.50C or above. Fever is a sign of infection or loss of body fluid. Rapid rise of temperature leads to fits (Convulsion) especially in young children. Some common causes of fever in children are malaria, tuberculosis, respiratory, gastrointestinal and other bacterial infections and viral infections. All patients with history of fever should be treated and encouraged to drink plenty of water, other fluids and fruits, and should not be overdressed. High fever should be treated as emergency.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child had the fever? Is there any convulsion? 2.Has the child ever had any convulsion? 3.Is the child having a runny nose? Cough? Or difficult breathing? 4.Is the child coughing up blood? 5.Is the child vomiting or having diarrhea? 6.Is the child having bloody diarrhea, 7.Is the child having skin rashes 8.Is the child having any ear pains? Sore throat? Headache? Stiffness of the neck? 9.Is the child passing urine normally? Is it painful? 10.Is it more or less frequent? 11.Is the child bleeding inside and outside the body. from the eyes, ears, and nose 12.Has any analgesic or other medication been given? Is anyone currently ill in the family or compound?	1.General appearance: alert, restless, listless, drowsy or difficult to wake, delirious, dehydrated, thin, weak or convulsing. 2.Skin: rashes, septic spots, profuse perspiration. 3.Chest: (a)Respiratory rate counted for 1 full minute. (b)Lower chest indrawing/retraction. 4.wheeze or stridor in a calm child Nose: discharge – clear or purulent., blood Eyes: pus, blood 5.Ear: tenderness behind the ear, discharge, blood. 6.Throat: sores, pains 7.Neck: stiffness, nodes. Abdomen: tenderness, masses 8.Temperature and pulse: check and record 9..

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Fever with any of the following Convulsion / multiple convulsion Prostration Impaired consciousness Respiratory distress Abnormal bleeding Persistent vomiting	Lethargic or unconscious, warm to Touch and Temp. 37.5^{0}C or above. Pulmonary oedema Jaundice discolored urine not making urine / scanty urine Haemoglobinuria Renal failure Hypoglycemia Severe anaemia (Hb < 5gm/dl) Circulatory collapse	Severe condition Suspect severe malaria	1. mRDT for malaria if positive Give IM Artesunate 2.4-3.0mg/kg body weight stat OR IVF Normal saline 500mls immediately (stat) 2. Give 50% dextrose or glucose water to correct hypoglycaemia Tepid sponge PRN immediately (stat) if convulsing now 3. Give IM. Paraldehyde 5ml stat 4. REFER
II. Fever with neck pain with or without vomiting	Neck stiffness, abdominal rigidity with or without spasms. Warm to touch and Temp. 37.50C or above. Kernig's sign positive	Severe Condition Suspect Cerebral Meningitis	If Rapid Diagnostic Test (RDT) is negative proceed with the following treatment IM. Benzyl Penicillin 3ml (300,000) stat Refer to CSM section Give IM. Paraldehyde 5ml immediately (stat) REFER Appropriately
III. Fever every day for more than 30 days, skin rash, mouth ulcer, weight loss, with or without diarrhoea. Weight loss,	Oral thrush, Skin rash, Lethargy, Pallor.	Severe condition Suspect HIV/AIDS	1 . REFER for HIV screening

IV Fever with rash, cough and runny nose	Generalized rash, Red eyes, runny nose, Warm to touch. Temp. - 37.5^{0}C and above, sticky eyes and mouth ulcer	Moderate condition Suspect Measles.	Clean the mouth with chlorhexidine/ Drug Age Wt.(kg) Dose Artemeter-Lumefantrine AL 6-8yr 15-24 2 tabs at 0hour, 8hours (20/120 mg) later, then, 12hourly for 3days 9-12yr 25-34 3 tabs at 0hour, after 8 hours then, 12hourly for 3days Artesunate Amodiaquine (AA) 7-12yr 100/270mg 1tab daily for 3days Give Tab. Paracetamol 250-500mg 8hourly for 3 days Give Cap. Vitamin A 200,000 I.U. daily for 2 days. Tab. Co-trimoxazole 480 mg 12hourly for 5 days Clean eyes with cotton soaked in cooled boiled water. Chloramphenicol eye ointment 8hourly for 7 days Clean the mouth with Hydrogen Peroxide 1 in 5 part of water for 5 days Apply Gentian Violet to the mouth. Follow up in two days. If no improvement, REFER
V. Fever with ear pain with or without discharge for less than 14 days	Ear discharge; watery or pus or bloody. Warm to touch. Temp. 37.50C or above. Tenderness.	Moderate condition Suspect ear infection	Give Tab. Paracetamol 250-500mg 8hourly for 2 days Clean ears with TCP-soaked cotton wool. Cap. Ampiclox 250mg 6hourly for 5 days OR Amoxicillin 250mg 8hourly for 5 days Tab. Vitamin C 100mg 8hourly for 5 days Review in 5 days If no improvement, REFER immediately. Conduct rapid diagnostic test, if positive treat for uncomplicated malaria Artemeter-Lumefantrine 20/120mg AL 6 – 9yrs, 15-24kg 2 tab at 0hour, after 8 hours, then 12 hourly for days >9-14yr, 25-34kg 3tabs at 0hour, after 8 hours, then 12 hourly for 3 days.

VI. History of fever in the past 24 hours.	Temp. above 37.2^{0}C no other finding	Moderate condition Suspect uncomplicated malaria	Drug Age Wt.(kg) Dose Artemeter-Lumefantrine 20/120mg AL 6 – 9yrs, 15-24kg 2 tab at 0hour, after 8 hours, then 12 hourly for days >9-14yr, 25-34kg 3tabs at 0hour, after 8 hours, then 12 hourly for 3 days. Artesunate Amodiaquin (AA) 7-12yr 100/270mg 1tab daily for 3days Give Tab. Paracetamol 250-500mg 8hourly for 3 days Encourage intake of fluids, fruits and vegetables Review after 3 days If fever persists, Refer to hospital

Health Education

Advise Parent / Caregiver on;

1.Environmental sanitation and prevention of disease transmission.
2.Use of insecticide treated nets/materials in malaria control.
3.Personal and food hygiene.
4.Adequate nutrition.
5.Safe water and disease prevention.
6.Complete treatment and its importance.
7.Importance of taking home-based records to the clinic or hospital.

3.3 CONVULSION

Convulsion is jerky movements of the arms and legs and twitching of the muscles. It may be caused by injury and infection of the meninges, brain and spinal cord. Tetanus or lockjaw must not be mistaken for convulsion. It begins with tightness of mouth, jaw and muscle twitching.

HISTORY	EXAMINATION/ACTIONS
1.Has the child had a fever? 2.Any accident injury? 3.Did the child take any medications or any substance accidentally? 4.Has the child had any previous convulsion? With or without fever? Has the child been given any medication? If yes, what? Does any other family member suffer from convulsions?	1.General appearance: convulsing, conscious, drowsy, unconscious, smelling of cow's urine. 2.Nose: flaring of alae nasi. 3.Chest: respiratory rate, lower chest indrawing Head: swelling, sign of injury. 4.Eyes: pupils for size and reaction to light Neck: stiffness Ears: pus, bulging drums or perforation of drums. Temperature and pulse: check and record Weight: check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. History of convulsion in the past 24hrs with or without history of head injury	Stiffness of neck, Signs of head injury, drowsy, vomiting	Severe condition	Explain findings to mother REFER immediately
II. Convulsion with fever	Convulsing; W Temp. 37.5^{0}C or above	Severe condition Suspect febrile illnesses	Remove from dangerous environment Lie patient on side, clear airway Give IM. Paraldehyde 5ml immediately (stat) If Rapid Diagnostic Test (RDT) is positive give pre-referral treatment for severe malaria (IV/IM Artesunate 2.4-3.0mg/kg immediately (stat) OR Artesunate suppository (rectally) when available. If RDT is negative, Give IM. Amoxicillin 250mg immediately (stat) Encourage intake of Glucose water or sugar in water if patient can take orally. Expose child REFER
III. One or more episodes of convulsion without fever	Convulsing; Temp less than 37.50C	Moderate condition	Give IM. Paraldehyde 5 ml immediately (stat) IM. Phenobarbitone 30mg/ml at night for 1 day Review in 6 hours if there is no improvement REFER

Health Education

Advise Parent / Caregiver on;

1.Causes and prevention of convulsion.
2.Management of fever.
3.Management of a person during convulsion episode.
4.Importance of complying with treatment.
5.Adequate nutrition.
6.Use of LLINs.
7.Importance of taking home-based records on child's next visit to the clinic or hospital.

3.4 COUGH

Cough is a sign of irritation of the throat and wind pipe which comes suddenly. In children, common causes of cough include pneumonia, asthma, mucus blocked nostrils, adenoids, aspiration of foreign body or poison e.g. kerosene. Heart diseases may also present with cough.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child had the cough or difficulty breathing? 2.Is there wheezing? Have there been any previous episodes? Has the child been vomiting? 3.Has the child experienced convulsions? Is there a fever? If so, how long has it been present? Is the child able to drink? Has any medication been administered to the child? If so, what medication?	1.General Appearance: alert or abnormally sleepy or difficult to wake, severe malnutrition, respiratory distress. 2.Nose: flaring of alae nasi 3.Chest: a). respiratory rate (count for 1 full minute in a calm child) subcostal retraction lower chest-indrawing stridor wheeze 4.Eyes: redness, discharge Throat: soreness or redness Feet: swollen 5.Temperature, pulse: check and record Weight: Check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Cough, fever with one of the following: difficult to sleep and fast breathing.	Chest in-drawing, warm to touch. Temp. 37.5^{0}C or above. Respiratory rate of 40 or more per min. Decreased breath sound and crepitation. Clay colour sputum	Severe condition Suspect Pneumonia	1.Give Amoxycillin 250mg 8hourly x2 days 2. Paracetamol syrup 5ml 8hourlx3days stat 3.if no improvement REFER
II. Cough, fever with noisy and difficult breathing	Chest in-drawing. Harsh noise on inspiration or soft noise on expiration. Warm to touch. Temp. 37.5^{0}C or above. 40 breath or more per min	Severe condition Suspect Asthma	Give IM. Amoxicillin 250mg 8hourly x2 days IV. Aminophylline 5ml given slowly over 10 - 20 minutes stat Tab. Salbutamol 2mg 12hourly for 4 days REFER appropriately
III. Cough of 3 wks or more with or without weight loss. Blood in sputum	Loss of weight Cough with or without blood stain sputum	Severe condition Suspect Tuberculosis	Collect sputum for investigation If 2 out of 3 specimens are positive, REFER for treatment Direct Observed Treatment Short course (DOT) see appendix Follow up at home.
IV. Persistent or recurrent cough more than one month, skin rash, mouth ulcer, fever, weight loss, with or without diarrhoea.	Skin rash, marked loss of weight. Oral thrush, warm to touch. Temp. 37.5^{0}C or above, mouth ulcers	Severe condition Suspect HIV/AIDS	1 . REFER to screening centre

V. Cough, fever with fast breathing	Fast breathing. . Temp. 37.50C or above. Reduced breath sound and crepitation. Clay colour sputum	Moderate condition Suspect Pneumonia	Give Cap. Amoxillin 250mg 8hourly for 5 days Give Tab. Paracetamol 250-500mg 8hourly for 2 days Home care. Warm tea and honey. Review after 3 days. If no improvement, REFER
VI. Cough	No fast breathing	Mild condition	1. Give Tabs cotrimoxazole according to age 2. Give paracetamol according age 3 Parent to manage child at home with warm tea, lemon, honey. Review after 2 days.

Health Education

Advise Parent / Caregiver on;

1. Overcrowding and disease transmission.
2. Personal hygiene and prevention of airborne diseases.
3. Air pollution and respirations problem.
4. Recommendations for good ventilation.
5. Role of adequate nutrition and fluid intake in prevention of infections and promotion of quick recovery.
6. Use of LLIN.
7. Importance of taking home-based records to the clinic or hospital.

3.5 DIARRHOEA

Diarrhoea is characterized by the passage of watery stool more than 3 times a day. In children, it stands as one of the leading causes of morbidity and mortality, posing a greater risk than in adults. Hence, fluid replacement stands as the cornerstone of treatment. Causes of diarrhea encompass dysentery, typhoid, cholera, ingestion of poisons, contaminated foods, and HIV/AIDS. Persistent diarrhoea refers to a condition lasting two or more weeks.

HISTORY	EXAMINATION/ACTIONS
1.How long have you been ill? 2.How many stools do you usually pass each day? Now? 3.Have you passed any since coming to the clinic? 4.Are the stools watery? Is there blood or mucus? 5.What is your source of drinking water? 6.What type of toilet facilities do you use? 7.Are you nauseated or have you vomitted? 8.Do you have fever? Abdominal pain? Cough? 9.Are you sometimes constipated? 10.Is there anyone else in the household who is sick? 11.Have you taken any medication or treatment for this? Have you eaten plenty of fresh fruits (e.g. mangoes) or plenty of pepper?	General appearance: alert, restless, irritable, thin, dehydrated, lethargic, unconscious Eyes; normal, sunken, very sunken and dry. Mouth and Tongue: moist, dry, very dry Chest: respiratory rate Neck: nodes enlarged, throat inflamed Ears: purulent discharge, tenderness behind ear Abdomen: tenderness, distended Skin: skin pinch goes back quickly, slowly or very slowly Weight: weigh and chart Temperature: check and record Offer the child something to drink, and note if the child drinks normally, eagerly or poorly (or is not able to drink) SEE APPENDIX ====for chart on assessment of dehydration

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Sudden onset of frequent watery stool	Weak with severe dehydration; skin pinch goes back very slowly; Mouth/ Tongue dry; very sunken and dry eyes;Tears absent	Severe condition Suspect severe dehydration	Start IV infusion, Ringer's Lactate immediately 100ml/kg body weight If no Ringer's Lactate, give normal saline in divided doses as follows:130ml/kg body weight in 30 min. then 70ml/kg body weight in 2½ hours If child is able to drink, give Lo-ORS by mouth while the IV set is up (5ml/kg/hr), re-assess every 1-2 hours Give Dispersible Zinc tablet 20mg daily for 10 days REFER appropriately
II. Frequent stool and vomiting (in area where there is Cholera	Irritable sunken eyes, Mouth and Tongue dry. Tears absent child is thirsty and drinks eagerly, Skin pinch goes back slowly. Some dehydration	Moderate/Severe Suspect Cholera	Explain condition to mother Give IVF as above if severe Give Oral Rehydration Salt (ORS) solution Cap. Amoxicillin 250mg 6hourly for 5 days Tab. Metronidazole 200mg 12hourly for 5 days Tab. Zinc sulphate 20mg daily for 5 days Tab. Metochlopromide 5mg immediately (stat) If no improvement, REFER

III. Frequent stool with visible blood. Some dehydration	Irritable, Mouth and tongue dry. Skin goes back slowly Visible blood in stool	Moderate/severe Suspect dysentery	1. Give Low osmality ORS / Zinc 2. Continue feeding 3. Tab. Co-trimoxazole 480mg 12hourly for 5 days Review in 2 days If there is still blood in stool after 2 days, give Tab. Metronidazole 200mg 8hourly for 5 days Review in 2 days If no blood, complete Metronidazole If blood persists, REFER
IV. Frequent stool with severe weight loss	Miserable looking wasted muscles Weight low for age	Severe condition Suspect severe malnutrition	Give Parent Low-ORS and Zinc to give slowly on the way to hospital Recommed for high protein diet ,micro nutrent Encourage fluid intake
V. Frequent stool, no mucus or blood, no vomiting	Alert, eyes normal, mouth and tongue moist, child drinks normally, skin pinch goes back immediately	Moderate condition Suspect Mild dehydration	Lo-ORS (demonstrate to mother how to prepare and administer, continue feeding frequently). Advise Parent to give more fluids at home. Parent to return if not better in 3 days or if any danger sign appears. Tab. Zinc sulphate 20mg for 10 days
VI. Intermittent diarrhoea for more than one month with or without fever, skin rashes, mouth ulcers,	Emaciated, lethargic, oral thrush, skin rashes, some dehydration	Severe condition Suspect HIV/AIDS	Low-ORS/zinc (demonstrate to Parent how to prepare and administer, continue feeding frequently). REFER for further investigation/ HCT

Health Education

Advise Parent / Caregiver on;

1. Personal, food, and environmental hygiene.
2. Handwashing practices.
3. Ensuring adequate nutrition (including conducting food demonstrations).
4. Preparation of salt-sugar solutions.
5. Ensuring water sanitation.
6. Emphasizing the importance of taking home-based records with her anytime she takes baby to the clinic or hospital.

3.6 VOMITING

Vomiting is the throwing up of the content of the stomach. The common causes of vomiting are gastro-enteritis, typhoid, cholera, meningitis, ear infection, liver disease and abdominal problems. A child who is vomiting should not be given large or heavy foods but small frequent and freshly prepared food. Children vomiting should be cared for urgently through fluid replacement.

HISTORY	EXAMINATION/ACTIONS
1.How long have you been vomiting? 2.How many times a day do you vomit? 3.What is the vomitus like? Projectile (think of obstruction), greenish? With blood? 4.Is there diarrhoea? If not, when was the last stool? 5.Do you have abdominal pain? 6.Do you have any cough, fever or headache? 7.When was your last menstrual period? (if female) 8.What was your last meal? What medication have the child taken?	1.General appearance: ill looking jaundiced, dehydrated, distressed. 2.Mouth: tongue coated, clean. 3.Throat: redness, imflammed tonsils 4.Neck: stiffness. 5.Abdomen: tenderness, rigidity, visible peristalsis, distended. 6.Ear: redness, bulging or perforation of drum, discharge 7.Hydration status: eyes, tongue, mouth, skin 8.Temperature and pulse: check and record Weight: Check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Vomiting more than 3 times daily with any of the following: Blood (brownish, vomitus) Neck stiffness Abdominal pain	Lethargic and ill looking with bloody vomitus or some/severe dehydration or abdominal rigidity or tenderness	Severe condition	Give IM. Amoxicillin 250mg immediately (stat) Give Low-ORS/zinc IM. Metoclopramide 5mg immediately (stat) Refer appropriately
II.Vomiting more than 3 times daily with Abdominal pain	Ill looking, Some dehydration and abdominal rigidity or tenderness	Moderate condition Moderate dehydration	Explain findings to mother Give Low-ORS/zinc IM. Metoclopramide 5mg stat Tab. Hyoscine 10mg daily for 1 day Counsel about giving small but frequent meals. If not improving, REFER
III.Vomiting less than 3 times daily with no other sign	No dehydration Eyes normal Child happy	Mild condition	Give Low-ORS/zinc IM. Metoclopramide 5mg stat Tab. Hyoscine 10mg daily for 1 day Counsel about giving small but frequent meals

Health Education

Advise Parent / Caregiver on;

1.Personal, food and Environmental Hygiene and use of LLINs.
2.Causes of vomiting.
3.Adequate fluid intake and prevention of dehydration (excessive loss of body fluids).
4.Importance of taking home-based records to the clinic or hospital.

3.7 HEADACHE

Headache is a common complaint which may be a symptom of many diseases such as: Malaria, sinus infection, high blood pressure, ear, nose and throat problems, nervous tension, meningitis, brain abscess and typhoid fever. It can come on when there is constipation or inadequate sleep and rest.

HISTORY	EXAMINATION/ACTIONS
1.Where do you feel the headache most? 2.In the front (think of nose, eyes, malaria)? 3.In the back (think of high blood pressure, tension, kidney failure)? 4.On the sides of head (migraine, teeth, ears)? 5.How long have you had it? 6.Do you have fever or chills? Convulsion? 7.Do you have nausea, vomiting or abdominal pain? (typhoid or malaria)? 8.Is headache worse in the morning (Sinuses)? What time of the day is headache worse? 9.Do you have constipation or diarrhea? 10.Are you on birth control-pills? 11.Have you had any head injury? If so, when Do you have any other problem?	1.General appearance: in pain, wasted 2.Temperature and pulse: check and record 3.Blood pressure: measure and record 4.Face: swelling, puffiness 5.Eyes: redness 6.Nose: pus or mucus. Press both sides below the eyes for tenderness 7.Ears: redness or dullness of drums, mastoid pain 8.Mouth: acetone or alcohol on breath 9.Neck: stiffness, nodes 10.Chest: difficulty in breathing, creps, abnormal breath sounds 11.Abdomen: tenderness, distended, rigidity, enlarged spleen, masses 12.Limbs: weakness, paralysis 13.Urine: protein Conduct mRDT,

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Headache with fever	Temp 37.5°C or above	Moderate condition Suspect malaria	Administer paracetamol Tab. 250 – 500mg 8hourly 3days Encourage fluid intake If RDT is negative explore other causes of fever. Commence Caps. Ampiclox 500mg 6hourly x5days Bed rest for patient and observe
II. Headache with any of these conditions:	Palpable nodes in the neck Stuffy nose Tenderness in the ear lobe Temperature 37.5^{0}c degrees	Severe condition Suspect, Sinusitis	If RDT is positive treat for uncomplicated malaria Commence Caps. Ampiclox 500mg 6hourly x5days REFER appropriately
III. Headache, Disturbed sleep, restlessness	Raised blood pressure	Severe condition Suspect childhood hypertension	Tab. Paracetamol Tab 250-500mg 8hourly for 3 days , if RDT is positive Treat uncomplicated Malaria, Encourage bed rest REFER appropriately.
IV. Headache with stuffy nose,, catarrh, hoarse voice	Stuffy nose	Mild condition Suspect upper respiratory tract infection	1.Tab. Paracetamol 250-500mg 8hourly for 3 days 2.Tab. Chlorpheniramine 2mg 12hourly for 3 days 3.Caps. Ampiclox 500mg 6hourly x5days 4.Tab. Vitamin C 100mg 8hourly for 7 days 5. Bed rest, warm tea and honey 6.Encourage fluid intake Treat malaria when positive for RDT 7. Review in 3 days

Health Education

Advise Parent / Caregiver on;

1. Healthy living (exercise, adequate rest, avoidance of stress).
2. Avoidance of animal fat in food in case of raised blood pressure.
3. Avoidance of overcrowding.
4. Personal/food hygiene.
5. Use of LLINs.
6. Importance of taking home-based records to the clinic or hospital.

3.8 PALLOR

Pallor is a major sign of anaemia. Anaemia is reduced number of red blood cells or reduced amount of hemoglobin in each red cell (below 10 grams). A child may develop anaemia as a result of not eating adequate nutritious foods, malaria, malnutrition, parasites (hook worm or whipworm), infections and sickle cell disease. Anaemia can lead to death.

HISTORY	EXAMINATION/ACTIONS
1.Is there bleeding from any source? 2.Nose, gums 3.Coughing or vomiting; 4.In the stool or urine; 5.Black stools, injury; 6.Does the child have a fever? For how long? 7.Has the child had painful swelling of the fingers, toes, the back of hands and feet? Or bone pains? 8.Does the child have a good appetite? What food does he eat? Does the child reject some foods? 9.Does the child go barefoot 10.Does the child have abdominal pain? 11.Has the child been swimming or washing in the stream? 12.Does anybody in the family have similar problems? 13.Has any medication been given? Is the mother using camphor?	1.General appearance: tall and thin, malnourished 2.Chest: respiratory rate, lower chest indrawing 3.Heart: rate, murmur 4.Head: bossing of the skull 5.Eyes: jaundice or conjunctival pallor 6.Abdomen: enlarged spleen, distension 7.Limbs: a). swollen and tender joints, feet and/or digits, or back of hands and feet b).nails for pallor 8.Haemoglobin: estimate and record 9.Temperature: check and record 10.Urine: colour, blood, protein 11.Conduct RDT, if positive treat for uncomplicated malaria with ACT

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I.A pale child with swollen or puffy face. Swollen feet	Nose flaring, Lower chest in-drawing, grunting, fingers clubbing. Hb lower than 10g/dl	Severe condition Suspect Heart failure	1. REFER urgently 2. Place in proper position(cardiac position) 3. Reassure parent/caregiver
II.A pale child with generalized bone pain, joint pains.	Bone tenderness, swollen hands and feet Severe Conjunctival and Palmar pallor Hb lower than 10g/dl	Severe condition Suspect Sickle Cell crisis	1. Encourage fluid intake If RDT is positive treat for uncomplicated malaria with ACT 2. Give Tab. Paracetamol 250-500mg 8hourly for 3days 3. Give Tab. Folic Acid 5mg daily for 2 weeks 4.Give pentazocine injection 30mg stat Advice on taking warm tea REFER
III.Yellow eyes, weakness, breathlessness and body pains	Jaundice of conjunctiva and palmar pallor Hb estimation lower than 10gm/dl	Severe: condition Suspect Sickle cell Anaemia	If RDT is positive, treat for malaria with Artemisinin based combination therapy (ACT) Give Tab. Ibuprofen 200mg 8hourly for 3 days (After food) Give Tab. Folic Acid 5mg daily for 2 weeks Encourage fluid intake REFER
IV. Pale, not growing well with or without yellow eyes. Weakness	Some pallor, bossing of forehead, apparent long limb, thin hair, may be stunted	Severe condition Suspect Sickle Cell Disease	If RDT is positive treat uncomplicated malaria Give Tab. Folic acid 5mg daily for 2 weeks Tab. Paracetamol 500mg 8hourly for 3 days REFER

V. Weight loss, may or not pass worm	Some pallor- mild to moderate. Check Hb, PCV	Moderate : condition Suspect Worm Infestation Nutritional anaemia.	Give Tab. Pyrantel Pamoate 3 tabs stat. Tab. Ferrous Sulphate 200mg (60mg elemental iron) daily for 2 weeks. Tab. Folic acid 5mg daily for 2 weeks Nutrition counseling i.e high protein diet Follow up after 2 weeks

Health Education

Advise Parent / Caregiver on;

1. Nutritional counseling.
2. Genetic counseling.
3. Prevention and care for sickle cell disease and protection of feet.
4. Prevention of malaria (Environmental control).
5. Use of LLINs.
6. Personal, environmental and food hygiene.
7. Avoid extremes of weather and conditions that could trigger crisis.
8. Importance of taking home-based records to the clinic or hospital.

3.9 YELLOWISH EYES (JAUNDICE)

Jaundice is yellowish discoloration of the conjunctiva and skin due to an increase in bile pigmentation. It is commonly due to excessive breakdown of red blood cells (haemolysis) as in sickle cell crisis, severe malaria and liver problems e.g infective hepatitis, obstruction of the bile duct, etc. It is these conditions rather than the Jaundice itself which demands treatment.

HISTORY	EXAMINATION/ACTIONS
1.Is the child unwell? (sepsis & GIT obstruction can cause jaundice) 2.Is there yellowish discolouration of eyes and skin 3.When did you notice the yellowish discolouration of the eyes? 4.Is there dehydration or poor weight gain? (both exacerbate jaundice) 5.Is there family history of haemolytic disease (ABO/G6PD, spherocytosis, sickle cell) 6.Is there dark urine or pale stools (suggest biliary obstruction) 7 What medication have you given (e.g herbs)?	1.General appearance: conscious, drowsy, unconscious, 2.Chest: respiratory rate, lower chest indrawing 3.Head: swelling, sign of injury 4.Eyes: yellowish 5.Ears: pus, bulging drums or perforation of drums 6.Hands: palms and nail beds for pallor and yellowish discolouration 7.Temperature and pulse: check and record 8.Weight: check and record Conduct malaria Rapid Diagnostic Test (mRDT)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I.Yellow eyes, General body weaknesses May or may not have blood in vomit	Yellow eyes May have swelling of the abdomen with rigidity Marked abdominal tenderness Pallor Hb lower than 10g/dl	Severe condition Suspect sickle cell with liver condition	1. REFER immediately
II.Yellow eyes, General body pain, dark yellow urine	Yellow eyes Haemoglobin less than 10gm Dark yellow urine	Severe condition Suspect liver involvement	1. Glucose in water 2. If RDT is positive , give Pre-referral treatment for severe malaria REFER
III.Yellow eyes, Fever, nausea and/or vomiting, abdominal pain	Yellow eyes Temperature 37.50C or above Abdominal tenderness Diarrhoea	Moderate condition Suspect Hepatitis	If RDT is positive treat for uncomplicated malaria with ACT Give Cap. Amoxicilline 250mg 8hourly for 5 days Review in 3 days REFER if no improvement

Health Education

Advise Parent / Caregiver on;

1.Prevention and care for sickle cell disease.
2.Personal, environmental and food hygiene.
3.Use of LLINs.
4.Proper nutrition.
5.Importance of carrying home based records to the clinic.

3.10 NUTRITIONAL DISORDERS

Nutritional disorder is a condition caused by improper diet between what an individual eats and what he requires to maintain health. This can result from eating too much or excess or an inadequate diet such as protein, fat and carbohydrate.

Children with nutritional disorders have higher risk of developing other diseases, and death. Identifying children with nutritional I disorders and treating them can help prevent many severe diseases and deaths. Stunting is common among this age group especially children from families with low income in rural communities and urban slums.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child been losing weight? 2.Has the child been having diarrhea or vomiting (and for how long? 3.Does the child have cough? 4.Does the child have fever? 5.Has the child been eating well? 6.What food is the child eating? (Probe type of food for a 24 hour period) 7.Has the child been ill recently of Measles, Diarrhea, prolonged cough. etc. 8.Has the child recently passed worms? 9.Social History: 10.Does the mother work outside the home? 11.Who takes care of the child? 12.How many other children are in the family? 13.How much do the mother and father earn? 14.Does the child live with the parents? If not, whom does he live with and why? Are the parents separated or divorced?	1.General appearance: listless, irritable, alert, wasted, Oedematous, weak 2.Hair: colour, texture, quantity 3.Skin: peeling, sore, rashes, dark and white patches on legs 4.Hydration status: fontanelle, eyes, mouth, skin 5.Eyes: pallor, cloudy, cornea or foamy patches on white of the eye 6.Ears: discharge, red drums 7.Mouth: sores, thrush, cracked lips 8.Neck: enlarged lymph nodes 9.Chest: fast breathing, lower chest indrawing, wheeze 10.Heart: rate, rhythm and murmurs 11.Abdomen: masses, enlarged spleen 12.Legs: cedemia, muscle wasting. 13.Temperature and pulse: check and record 14.Urine: test for protein and record Weight and height: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Child has poor weight gain with any of the following: Severe wasting Oedema of hands and feet White spot or cloudy cornea Severe pallor	Looks emaciated, oedematous, weak, Muscle wasting White spot on cornea severe pallor	Severe condition Suspect severe malnutrition	1. Refer for nutrition counselling , 2. High protein diet, 3. Dewormed patient. 4. Review in 1 week 5. If no improvement 6.REFER
II. Poor weight gain	not pale, active, growth graph flat	Moderate condition Failure to thrive	1. Give Tab. Pyrantel pamoate tablet 3 stat. 2. Tab. Folic acid 5mg daily for 2 weeks 3. Tab. Vitamin B complex I tab 8hourly for 7 days 4. Tab. Ferrous sulphate 200mg (60mg elemental iron) 8hourly for 7 days 5. Give micronutrient powder if available (see annex for proper use) 6. Nutritional counselling and high nutritional diet 7. Review weekly for one month and monthly subsequently If gaining weight satisfactorily: 8. Explain findings to mother 9. Arrange for: Home visit weekly Discuss with family

III. Child continues to lose weight Failure to gain weight for more than one month, with or without loss of appetite	Weight loss mildly pale, active,	Moderate condition Suspect moderate acute malnutrition	1. Give multivitamin Nutrition counseling 2. Give micro nutrient powder if available (see annex for proper use) 3.Ask of recent illness 4.Give Pyrantel pamoate tablet 3 stat 5. Review weekly for one month If no improvement, REFER
IV. Child's weight is low for age on first child visit. (small for age) Child appears healthy	Alert and appears healthy	Mild condition	1. Explain findings to mother 2. Nutrition counseling 3. Give multivitamin 4. Give micro nutrient powder if available (see annex for proper use) 5. Give Tab. Pyrantel Pamoate 3 tabs stat
V. Persistent loss of weight for 3 months with or without intermittent diarrhoea, Sore mouth, skin rashes, may or may not have cough and night sweat	Child looks emaciated and miserable Oral thrush Skin rashes Some dehydration enlarged lymph nodes	Severe condition Suspect HIV/AIDS or TB	Treat opportunistic infections REFER for further investigation, HCT

Health Education

Advise Parent / Caregiver on;

1.Proper nutrition.
2.Periodic deworming.
3.Personal, food and Environment hygiene.
4.Use of LLINs.
5.Regular clinic attendance and weight measurement.
6.Importance of taking home-based records to the clinic or hospital.

3.11 MEASLES-TYPE RASH

Measles is an acute viral infection characterized by the appearance of rashes in little groups made up of macules (flat spot) and papules (raised spots). It begins with high fever, cough, catarrh, and redness of eyes and mouth. The mouth may later become sore and rashes appear on the whole body after the fourth day of onset. The child may also get other serious illness like pneumonia, diarrhoea, or malnutrition at the same time or soon after the child has had the disease. It is a serious illness because it can easily lead to complications especially in poorly nourished children.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child been ill? Does the child have a fever? 2.Does the child have cough? Is it a barking cough? Is there runny nose? 3.What food is the child taking? Is the child refusing to feed? 4.Is there diarrhoea? Vomiting? (If so, may be force-feeding) Has any medicine or treatment been given? 5.Are there any other children in the family under 2 years who have not been immunized for measles? Are there children in the neighbourhood with the same complaint?	1.General appearance: alert, abnormally sleepy, or difficult to wake, unconscious, ill-looking or thin 2.Skin: rash, peeling, infected spots, haemorrhagic rashes 3.Chest: a. respiratory rate, lower chest indrawing b. wheeze, stridor; 4. Eyes: red, purulent discharge, or sensitive to light (eyes closed), cloudy 5.Ears: bulging, discharging pus or fluid, perforation of eardrum, redness 6.Nose: runny, discharge 7.Mouth: a.sores, Koplik's spots, thrush b.throat-inflamed tonsils, purulent exudates 8.Hydration status: sunken eyes, dry mouth and skin 8.Weight; weigh and chart 9.Temperature, and Pulse: check and record Conduct mRDT Test

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
Measles with white spot on cornea cloudy cornea poor vision	Miserable child, ill looking, eyes closed, Measles rash ++; white spots on cornea, cloudy cornea. Warm to touch Temp 37.5^{0}C or above.	Severe condition: Suspect measles infection complicated by Xerophtalmia Vit. A deficiency	1. Counsell on the importance of going to the hospital (HCF) 2. 2. 2. Give Cap. Vitamin A 200,000 IU/injectable 50,,000unit 6-11 years 3.Calamine lotion when necessary (PRN) 4. Give Tab. Amoxicillin 250mg 8 hourly for 7days 5.Chloramphenicol eye ointment 6 hourly 6High in take of fluid 7REFER if no improvement
II. Measles with tender swelling behind the ear with fever	Ill-looking, , Temp. above 37.5^{0}C, , tender swelling behind the ear	Severe condition Suspect mastoiditis	If mRDT is positive treat for uncomplicated malaria with ACT Give Cap. Amoxicillin 250mg stat Tab. Vitamin C 100mg 8hourly for 1 week Give Tab. Paracetamol 250-500mg stat REFER

III. Measles with any of the following : Lower chest in- drawing Convulsion Inability to drink Difficulty in sleeping, weakness	Very ill looking, irritable, Lower chest in-drawing, Abnormally sleepy Malnourished ,Temp above 37.50C	Severe condition , Suspect complicated Measles Suspect severe malaria	1. If mRDT is positive give pre-referral treatment for malaria 2. IM Artesunate 2.4mg stat OR 3. IM Quinine 10mg/kg stat OR IM Artemeter3.2mg stat or Artesuanate suppository 4. Give oxygen if available 5.REFER

Health Education

Advise Parent / Caregiver on;

1.Proper nutrition.
2.Personal and food hygiene.
3.Other rashes that can be mistaken for Measles
4.Importance of nutritious food.
5.Clean eyes with clean water.
6.Importance of taking home-based records to the clinic or hospital.

3.12 SKIN DISEASES

Skin diseases are conditions that affect the skin. . It may be due to poor hygiene, allergy, malnutrition, poor housing condition as well as the environment. Some rashes can easily spread to other members of the family therefore, early referral and correct treatment is important for the welfare of the whole family. Some skin diseases affect the external skin (ringworm) or affect the skin internally (pyoderma). Others affect many parts of the body systems and are generally more important (measles, leprosy, diabetes, HIV/AIDS, Jaundice). Others are scabies, eczema, tinea vesicular, vitiligo, athletes' foot, urticaria and contact dermatitis.

HISTORY	EXAMINATION/ACTIONS
1.How long has the rash been there? Does it itch? Is the child scratching? 2.Are there any other symptoms? Fever, cough, headaches, difficult breathing, sore throat? 3.Has the child eaten any unusual foods? 4.Does anyone else in the family have a similar rash or skin problem? 5.How often do you bathe the child? 6.Where and with whom does the child sleep? 7.What medicine (including injections or home treatment) has the child been given lately? 8.Has the child been passing excessive urine? Has the child been having excessive thirst?	1.General appearance: thin, malnourished, distressed/uncomfortable. 2.Skin: 3.location, distribution and size of lesions 4.colour of rash or lesions 5.c). lesions or rash: flat (macules), raised, pus or fluid- filled, weeping, crusted, ulcerated, scratch marks or 6.d). combinations of (a)to (c) Ears: red or dark-due to repeated scratching) 7.respiratory rate, wheeze Temperature: check and record Urine; sugar, colour

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Non-tender, boil-like blister. Worm may be seen	Fever,itching ,discomfort,shortness of breath boil-like blister. Worm may be seen	Moderate to Severe condition Suspect Guinea Worm	1.Give Tab. Paracetamol 250-500mg 8hourly for 2 days 2. Clean and dress the wound with Cicatrin powder daily. 3.Tab.Vitamin C 100-200 mg 8hourly for 7 days 4. Gently wind worm on a stick 5. Ensure compliance with Health Education in conjunction with Village Head 6. Give Cap. Amoxicillin 250mg 8hourly for 5 days Since eggs may survive, a second treatment may be necessary if client shows no clinical improvement after 1 week.
II.Very itchy skin rash	Small, raised red spots or pustular rash between fingers, wrists, buttocks and ankles	Moderate condition Suspect Scabies	Bathe daily with medicated soap for 1 week Paint the entire body except the scalp, hair and face with Benzyl Benzoate lotion (25%) at night for 5 days. Additional weekly treatments are warranted only if live mites are present. All household contacts of clients should be promptly treated. Give Cap. Amoxycillin 250mg 8hourly for 5 days Apply Benzyl benzoate when sores are dry Decontamination of articles Advise client to boil/wash clothing, beddings, towels in hot water and iron, used by client and family wash and dry materials in the sun and iron if possible. Review in 1 week.

III. History of painful lump or swelling on any part of the body.	Warm or tender mass	Mild to Moderate Suspect Boil	1. Give Cap. Amoxicillin or Cloxacillin 250mg 6hourly for 5 days. 2. Tab. Paracetamol 250-500mg 8hourly for 3days 3. Warm compress 8hourly until ready to drain 4. Incise and drain abscess when fluctuant 5. Daily cleansing and dry dressing when back in the community
IV. Flat itchy, dry patches on scalp or trunk.	Scaly macule or light spots with raindrop pattern or active borders	Mild condition Suspect Ringworm	1. Examine the rest of the family for similar problems 2. Apply Whitfield Ointment for 4 - 6 weeks 3. If no improvement, give Tab. Griseofulvin 125mg daily for 14 days. REFER
V. Rashes, may or may not have fever, or look ill	Small blister-like rashes mostly on the trunk. May be warm to touch with runny nose. May or may not look ill	Mild to moderate Suspect Chicken pox	1. Give Tab. Paracetamol 250-500mg 8hourly for 3 days 2. Give Cap. Vitamin A 200,000 I.U. daily for 2 days. 3. Apply calamine lotion 4. Encourage fluid intake especially fruit juice. 5. Give Tab. Promethazine 12.5mg stat then, PRN 6. Give Amoxycillin 250mg tdsx5days 7. Review in 3 days 8. If no improvement, REFER
VI. Severe itching with skin changes which disturbs sleep at night	Thickening discolouration of the skin Scratch mark; Skin changes with severe itching. especially at night with or without skin nodules (Onchocera nodules	Moderate condition Suspect Onchocerciasis	1. Give Tab. Ivermectin tablets 2 as a single dose or 3 days (Direct Observation Therapy) 2. Follow up at home.
VII. Itching of the skin followed by raised red lesion which disappears after some time	Raised lesion of the skin; Scratch mark;	Mild condition Suspect Allergy	1. Investigate cause of rash 2. Tab. Chlorpheniramine (Piriton) 2-4mg 12hourly for 3 days Review in 7 days If no improvement, REFER
VIII. Small light patches not raised, with or without itching	Small light patches not raised, not itchy, on the face, back etc.	Mild condition Suspect fungal infection (Tinea vesicolor)	1. Miconazole ointment for 4 - 6 weeks If itchy, give Tab. 2. Chlorpheniramine 2-4mg 12hourly for 3 days 3. Personal hygiene and care of clothing

IX. History of very itchy scalp and pubic area	Lice present	Mild condition Suspect Hair/Pubic lice infestation	1. Examine the rest of the family for similar problem 2. Apply 100% Gamma Lorexane lotion or shampoo 3. Apply to scalp close to roots of hair. 4. Shave pubic hair Wash with antiseptic lotion 5. Repeat treatment in one week

Health Education

Advise Parent / Caregiver on;

1.Personal hygiene; use of disinfectants.
2.Discourage sharing of underwear and clothes.
3.Importance of washing and ironing of clothes (emphasis on not spreading clothes on grass and other inappropriate places).
4.Proper Nutrition.
5.Roles of water in skin infection (protection and treatment of water sources e.g ponds with insecticide - Abate).
6.Overcrowding and transmission of skin infections.
7.Care of beddings.
8.Importance of taking home-based records to the clinic or hospital.

3.13 OEDEMA

Oedema is the abnormal collection of fluid in the tissues resulting in swelling of the affected part of the body. It could be generalized or localised e.g. one foot or finger. Generalized oedema is most noticeable on the parts of the body which are lowest at that time e.g. legs when standing. Causes include congestive heart failure, severe anaemia, kidney and liver diseases and kwashiorkor. Oedema may also be the result of a reaction to toxin, drugs and insect sting. Ascites, which is fluid in the abdominal cavity, may also occur.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child had the swelling? 2.Has the child ever had this swelling before? 3.Is the swelling painful? 4.Was the child bitten by an insect? 5.Is there itching? 6.Is there cough? 7.Is there any problem with urinationn? 8.Does the child eat well? What foods? 9.Has any medication been given If yes, what? Is there any other problem?	1.General appearance: pale, lethargic, puffy, distressed. 2.Head: a). hair-colour, texture 3.b). face-oedema, particularly around eyes 4.Chest: a). difficulty in breathing 5.b). creps or decreased breath sounds Heart: murmur, rate 6.Eyes: pallor, jaundice 7.Abdomen: swelling, tenderness, 8.Limbs: swelling, pitting oedema, 9.Urine: colour, cloudiness, protein, volume, 10.Weight: weigh and chart 11.Temperature, Pulse and Respiration: check and record Haemoglobin: estimate and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Swelling of limbs and/or face with yellowness of eyes	Jaundice, wheezing, stridor or fast breathing	Severe condition Suspect heart failure	1. Place in cardiac position 2. REFER immediately
II. Swelling of legs, arms and/ or face	Swollen feet (pitting) and/or face Protein or blood in urine	Moderate condition Suspect kidney problem	REFER immediately
III. Swelling of limbs and/ or face with itching, with or without allergic reaction to toxins drugs, stings or some food	Swollen legs, urticaria rash Itching	Moderate condition Suspect allergic reaction	1. If no respiratory distress, give 2. IM. Hydrocortisone 50 - 100mg immediately (stat) 3. Observe for 4-6 hours 4. If no improvement, REFER
IV. Swelling of legs and/or face	Pallor, Swollen legs and face, breathlessness Fast heartbeat with or without fever Hb < 10g/dl	Moderate condition Suspect Anemic Heart Failure	1. REFER Immediately

Health Education

Advise Parent / Caregiver on;

1.Management of oedema (postural).
2.Relationship between infection, anaemia and heart failure.
3.Proper nutrition.
4.Environmental sanitation.
5.Importance of taking home-based records to the clinic or hospital..

3.14 BURNS AND SCALDS

Burns are injuries to the body (skin) caused by fire or hot liquid. Burns are dangerous because it could lead to severe pains and shock resulting from loss of large amount of body fluids. The area of body surface affected is important in deciding the degree of a burn. The chart below shows approximately the proportion of total body surface area affected.

S/N	Rule of nines for burns (William, 2023)	Percentage of Body Surface
1.	Head and neck	9%
2.	Chest	9%
3.	Abdomen	9%
4.	Upper back	9 %
5.	Lower back	9%
6.	Right arm	9%
7.	Left arm	9%
8.	Genital area	1%
9.	Left leg	18%
10.	Right leg	18%
	Total	100%

HISTORY	EXAMINATION/ACTIONS
How did the child get burnt? And with what? When did it happen? Has any medication been given or applied since then? Has the child been urinating since the burn? Has the child been immunized against tetanus?	1.General appearance; restless, distressed, dehydrated, unconscious 2.Temperature, Pulse and Respiration: check and record. 3.Chest: respiratory distress, wheezing, stridor 4.Skin: a). size and area of burns, blisters, leathery areas with no sensitivity to pinprick b). signs of infections-pus, redness, warmth

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1. Burns / scalds	Burns of 9% or more with any of the following involved: Face, hands, groin or joints, Lethargic, without sensation or hoarse voice, has not passed urine since burn and sign of shock (weak, thready pulse and cold extremities, low BP	Severe condition Suspect Burns with imminent shock	1. Give IM. T.T. 0.5ml, if not recently immunized REFER immediately. 2. If referral is difficult, give Ringers Lactate I.V. one third of the total amount (500ml) 8 hourly and Darrow's half strength with 5% dextrose 3.Tab. Vitamin C 200mg 8hourly for 7 days 4.Apply Vaseline gauze 5. Give IM. Amoxycillin 250mg 8hourly for 2 days, then convert to oral. DO NOT GIVE ORAL ANALGESIC, If BURNS ABOVE 10%-

II. Burns / scalds	Less than 9%; no blisters, lethargic; no involvement of face, hands, groin or joints	Moderate condition	1. Vaseline gauze dressing 2. Tab. Phenobarbitone 30mg 12 hourly for 3 days 3. Tab. Paracetamol 250-500mg 8 hourly for 3 days 4. IM. T.T. 0.5ml, if not previously immunized in the last 18 months; Review daily until well. 6. Give IM. Amoxycillin 250mg 8hourly x 2 days, then convert to oral. 5. Tab. Vitamin C 100mg 8hourly for 7 days.
III. Burns/ scalds	Less than 9% with blisters, no lethargy, involvement: face, hands, groin or joints	Moderate condition	1. Paint with Gentian violet; Vaseline gauze dressing 2. Tab. Phenobarbitone 30mg 12 hourly for 3 days 3. Tab. Paracetamol 250-500mg 8 hourly for 3 days 4. Cap. Amoxicillin 250mg 8hourly 5 days 5. Tab. Vitamin C 100-200mg 8hourly for 7 days 6. IM. T.T. 0.5ml immediately (stat); if immunized in the last 18 months 7. ORS Liberally 8. Review daily until well

Health Education

Advise Parent / Caregiver on;

1. Prevention of burns/scalding.
2. Personal hygiene.
3. Prevention of burns.
4. Proper nutrition.
5. Compliance with treatment.
6. Importance of taking home-based records to the clinic or hospital.

3.15 WOUNDS, SORES, ULCERS, FRACTURES AND VARIOUS BITES/STINGS

Children can have wounds, sores, ulcers and various bites similar to adults. They are prone to accidents both in the home and outside the home particularly when they go to school.

HISTORY	EXAMINATION/ACTIONS
1.How did it happen? 2.When did it happen? 3.Was there any bleeding? If yes, how much? 4.What have you applied on the wound? 5.Can the child move the area? What immunization has the child taken?	1.General appearance: in pain 2. Signs of shock: cold, clammy, pale, 3.Wound: size, bleeding, dirt, or foreign body, pus, swelling, redness. 4.Temperature, Pulse and Respiration: check and record. 5. Haemoglobin: Check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Injury, Pain	• Wound resulting from injury. • Tenderness over wound or sore area • Evidence of fracture unable to move the limb • Lethargic • Wound more than 4-6cm	Severe condition Suspect Fracture	1. If bleeding, stop by applying pressure Immobilize limb (splint) 2. Give IM. TT 0.5ml, if not immunized 3Give Cap. Amoxicillin 250mg 8 hourly for 5 days 5. Tab. Ibuprofen 200mg 8hourly for 3 days 6. Tab. Vitamin C 100-200 mg 8 hourly for 7days 7. 5. REFER appropriately
II. Wound as a result of injury	•Laceration Bleeding •Unable to move the limb •Wound more than 4cm vNo evidence of fracture	Severe condition Suspect lacerated wound	1. Clean with available antiseptic lotion 2. Stop bleeding by pressure 3. Suture and dress wound 4. Give IM. Tetanus Toxoid (T.T.) 0.5ml, if not immunized, if wound is less than 24 hours, apply sutures, 5. Check Hb 6. Give Cap. Amoxicillin 250mg 8 hourly for 5 days 7. Tab. Paracetamol 250-500mg 8 hourly for 3days 8. Tab. Vitamin C 100-200mg 8hourly for 7 days 9. Review the next day then at 1 week to remove sutures Ask to return in 6 weeks for 2nd dose of T.T.

III. Wound as a result of injury	Laceration Less than 4cm with no sign of shock, No infection	Moderate condition Suspect lacerated wound	1. Clean with available antiseptic lotion 2. Stop bleeding by pressure Give IM. TT 0.5ml, if not immunized, if wound is less than 24 hours, sutures, Check Hb 3. Give Cap. Amoxicillin 250mg 8 hourly for 5 days 4. Tab. Paracetamol 250-500mg 8 hourly for 3days 5. Tab. Vitamin C 100 - 200mg 8 hourly for 7 days Review the next day then at 1 week to remove sutures Ask to return in 6 weeks for 2nd dose of T.T.
IV. Wound as a result of injur	Laceration Bleeding	Mild condition	Put pressure on site for 10 minutes, if slowing, observe and maintain pressure for 1 - 2 hrs. If bleeding does not stop and wound is less than 24hrs, suture If more than 24hrs, Clean with available antiseptic lotion Stop bleeding by applying pressure Give IM. TT 0.5ml, if not immunized, Check Hb IM. Amoxicillin 250mg stat then Cap. 250mg 8hourly for 5 days Tab. Paracetamol 250-500mg 8hourly for 3 days Tab. Vitamin C 100-200mg 8hourly for 7 days Review the next day then at 1 week to remove sutures Ask to return in 6 weeks for 2nd dose of T.T.
V. Bleeding from circumcision	Bleeding	Moderate condition Suspect haemophilia	1. Apply pressure on site for 10mins 2. If slowing down, observe and maintain pressure for 1-2 hrs. 3. If bleeding does not stop, REFER immediately 4. Teach mother how to apply pressure enroute referral facility

VII. Snake bite	Wound at site of bite	Severe condition Suspect venomous snake poisoning	1. Apply crepe bandage above site, but do not use as tourniquet 2. Keep client, calm and clean the area thoroughly with copious soapy water. 3. Give anti-snake venom (read and follow manufacturers instruction) If not available, REFER Give hydrocortizone. 4. Give IM. TT 0.5ml if not immunised. 5. Give Cap. Amoxicillin 250mg 8 hourly for 5 days 6. Tab. Paracetamol 250-500mg 7. 8 hourly for 3days 8. Review in 5 days If there is complications such as bleeding from orifices or Cardiovascular manifestations, . 9.REFER immediately
VIII. Scorpion sting	Evidence of sting (Painful Swelling, itching and may be red)	Moderate condition Scorpion sting	1. Tab. Paracetamol 250-500mg 8 hourly for 3days 2. Lignocaine 5ml stat directly on the site of sting 3. Give Cap. Cloxacillin 250mg 8 hourly for 5days 4. Cold compress 5. Review within 24
IX. Human bite	Wound at the site of the bite	Moderate condition	1. Clean wound well 2. If more than 4cm suture 3. Give Cap. Cloxacillin 250mg 6 hourly for 5days 4. Dress daily 5. Review after 3 days

Health Education

Advise Parent / Caregiver on;

1.Protection from bites/stings.
2.Personal hygiene.
3.Importance of compliance with treatment.
4.Proper nutrition.
5.Environmental sanitation.
6.Importance of taking home-based records to the clinic or hospital.

3.16 HEAD DISORDERS

The head contains the brain, the organs for the senses of hearing, sight, smelling and tasting. The bones of the head are very strong but sometimes a serious head injury may be caused by a fall or a blow to the head. Other head problems include infestation with lice, dandruffs and headaches. Injuries and infections to the head are of great danger because of the large blood vessels and nerves which it contains and the possibility of damage to the brain

Some signs of serious head injury are bleeding from the nose, mouth or ears, severe headache or unconsciousness. There also may be vomiting or the client may be confused. Any delay in referring a client with a head injury for proper medical treatment may cause permanent injury, damage or death.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.Has he fallen or hit his head? When? 3.If he has had an injury, has he been drowsy? 4.Has he vomited? How many times? 5.Does he have a headache? 6.Has he convulsed? Is he playing and behaving normally for his age as he usually does?	1.General appearance and behaviour: dull, abnormally, sleepy, or difficult to wake, restless, delirious or unconscious 2.Head: a). circumference,, shape external injuries, sores, or rashes swellings or depressions 3.Neck: stiffness 4.Eyes: haemorrahage, pupils size (equal or unequal), reaction to light, vision, jaundice 5.Gait: steady or unsteady 6.Temperature, Pulse and Respiration: check and record. Haemoglobin: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. History of trauma to head, abnormally sleepy or difficult to wake ,vomiting	Pupil abnormal size or not reacting to light. Fracture, vomiting Bleeding from ear or nose	evere condition Suspect fracture of the base of the skull	1.Immobilize the patients head and neck Refer
II. Headache with vomiting ,fever , neck stiffness	Warm to touch (Temp. 37.5^{0}C or more) stiff neck Kerning's sign positive	Severe condition Suspect meningitis	1. IM Benzyl Penicillin 3 cc (300,000 unit) 6 horly for 24 hours 2..IV fluid Dextrose 3. Paracetamol 300mg stat 4. Refer
III. .Injury to head with a small external swelling	Injury or swelling normal behaviour Normal pupil, Not vomiting No drowsiness	Moderate condition Suspect moderate head trauma	1. Explain findings to mother 2. Tab. Paracetamol 250-500mg 8 houly for 3 days 3. Apply cold compress 4. Treat local injury with antibiotics Amoxycillin 250 mg 8hourlyx days 5. Review after 48 24 hours 6. Advise mother to return child if worse, becomes drowsy, start vomiting, show any strange behaviour 7. If no improvement; REFER
IV. Bulging forehead	Bossing of forehead	Mild condition Suspect sickle Cell disease	1.Tab. Folic acid 5 mg daily for 2 weeks 2.Tab. Paracetamol 250-500mg 8hourly for 3 days 3.Refer

Health Education

Advise Parent / Caregiver on;

1.Proper child care.
2.Prevention of home and road accidents.
3.Prevention of head injuries.
4.Importance of the use of car seat belt and helmet.
5.Genetic counseling.
6.Importance of taking home-based records to the clinic or hospital.

3.17. NECK DISORDERS

Neck problem is the presence of enlarged nodes and lumps around the neck. There could also be stiffness of neck. These are usually caused by infection of ears, mouth or on the scalp, tuberculosis, leukemia (cancer of the blood), meningitis, other childhood malignancies, swollen neck glands or goiter.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child had the problem? 2.Is there weight loss or loss of appetite? 3.Is there fever or sore throat? 4.Is there excessive sweating at night? 5.Is there headache? Earache? 6.Is there pain or vomiting? 7.Is the child restless or hyperactive? Has the child convulsed?	1.General appearance: pale, thin, or wasted 2.Head: sores or abscess 3.Mouth: a). condition of teeth b). sores c). tonsils 4.Ears: discharge, red or bulging or perforated ear drum 5.Neck: stiffness, nodes, lumps, glands Chest: stridor, Temperature, Pulse and Respiration: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Stiffness of neck with history of a fall and no Fever	Stiff neck	Severe condition	Support the neck IM Paracetamol 300mg stat REFER appropriately
II. Neck pain with Fever, with or without vomiting, no history of neck injury	Stiff neck, Temp.37.50c or above, vomiting, Kerning's sign positive	Severe condition Suspect Meningitis	1. If Kerning's signs is positive, IM Benzyl Penicillin 3 ml (300,000 IU) stat 2...IV fluid Dextrose 3. Paracetamol 300mg stat REFER immediately
III. Neck pain with no Fever	Lump and Nodes present, no tenderness, deformities, swelling,	Moderate condition Suspect tumour	1. REFER
IV. Neck pain with Fever	Tender lumps or nodes Warm to touch (Temp 37.50.c or above) kernig sign positive	Moderate condition	1.Give Cap. Amoxicillin 250mg 8hourly for 5 days 2.Tab. Paracetamol 250-500mg 8 hourly for 3 days 3.Refer if no improvement
V. Swelling of both sides of the neck.	Enlarged salivary gland behind the jaw under the chin. Testicles may or may not be involved (swollen and/or painful)	Mild condition Suspect mumps	1.Tab. Paracetamol 250-500mg for 2 3days 3.Amoxycillin 6hourly x 5 days 4. Encourage fluids, Saline mouth wash, explain finding - to mother, If male, advise on rest 5.Tab. Vitamin C 100 – 200mg 8 hourly for 7days 6. Review in 3 days If no improvement, refer to the hospital.

Health Education

Advise Parent / Caregiver on;

1.Prevention of neck injuries.
2.Prevention of transmission to others.
3.Complications of mumps in males.
4.Oral hygiene.
5.Importance of taking home-based records to the clinic or hospital.

3.18 EYE DISORDERS

Eye problems, especially redness and discharge are commonly caused by injuries, chemicals e.g. soap and infections. The health worker must however be careful not to treat all eye conditions as if they were caused by infection. There are other more serious conditions such as pain and problems with vision which if missed, may lead to blindness e.g. lack of Vitamin A. Visual problems require correction. Cataract occurring in children may be a manifestation of Rubella infection.

HISTORY	EXAMINATION/ACTIONS
1. What is the matter with his eyes? 2. How long has he had the problem? 3. Has he had it before? 4. Did the problem come suddenly or gradually? 5. Is there a great deal of pain in the eye? 6. Does he rub eye or hold it? 7. Did anything enter his eye? 8. Did the mother or anyone else put something in his eye? What was it? 9. Has any medicine been put in his eye? If so, what medicine? 10. Does the child have runny nose and cough? (think of measles) 11. Does the child have severe prolonged bouts of coughing (think of whopping cough) 12. Does anyone else in the family have the problem? 13. Is he receiving treatment for any other disease? 14. Does he take palm oil or eat carrots, fresh mangoes pawpaw tomatoes? Do you think the child sees?	1. General appearance: in pain, ill, thin, wasted. 2. Skin: rashes, scratch marks? 3. Eyes: (a) Redness. where is the redness: (around the edges, in a circle at a colored part, all over or bleeding into white part of eye?). (b) tears (C) swelling around the eye(s) pus or other discharge Pupils: normal or small in size, equal. Protruding squit Cloudiness or whiteness of pupil (s) (d) Test of vision (visual acuity test) 4. Check if the child can follow a bright object, or reach out to receive an object from you without assistance. Nose: discharge.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Injury to eye ball with sharp object	Injured white part of eye(s)	Severe condition	1. Apply eye pad. 2. IM Gentamicin 3mg/kg stat Apply 3. Gentamicin eye drops 4. Referto Eye Clinic
II. Injury to eye ball as a result of burns, Hot water or chemicals	Painful watery eyes	Moderate to Severe condition	1. Rinse eye (s) with saline water immediately; Apply Gentamicin eye drops. Reassure parents/caregivers. Refer immediately to hospital.
III. Complaint of abnormalities of the eyes	Drooping of eye lids (ptosis) Eyes protruding (pushing out) Eyes watery or discharging Painful Red eyes, with or without visual loss Cataract (Growth of white small patches which do not react to light) poor vision (less than 6/9 visual acuity.)	Moderate to severe condition	1. Explain condition to parents/ caregivers 2. Refer to Eye Clinic

IV. Painful swelling of the eye lids	Pustule or small abscess on eye lid	Mild condition	1. Warm compress to eye. 2. Give Cap. Amoxicillin 250mg 8 hourly for 5 days 3. Tab. Paracetamol 250 -500mg 8 hourly for 3days 4. Chloramphenicol eye drops 8 hourly 5. Review in 3 days. 6. If no improvement, refer to Eye Clinic
V. Foreign body in the eye with pain	Watery eyes; Restlessness, May see foreign body, impaired vision, eyes may be red.	Moderate condition	1. Irrigate with Saline 2. Tab. Paracetamol 250-500mg 8 hourly for 3days 3. Give Gentamicin eye drops 4. Cover eye with pad; 5. Refer to Eye Clinic
VI. White bubbly material at corner of eye or dry eye(s)	White bubbly material seen at corner of eye(s)	Mild condition	1. Cap. Vitamin A 200,000 IU for 2 doses 2. Encourage consumption of coloured (yellow/red) fruits, green leafy vegetables and palm oil; 3. . Refer to Eye Clinic
VII. Sticky eye(s) or red eye(s); no pain, normal pupils and vision	Sticky yes; No pain; Normal vision	Mild condition	1. Cap. Vitamin A 200,000 IU for 2 doses 2. Encourage consumption of coloured (yellow/red) fruits, green leafy vegetables and palm oil; 3. . Refer to Eye Clinic
VIII Imflamatory Eye condition	Redness of the eyes Discharge of the eye Swollen eyelids Sensitivity to light Blurred vision	Moderate to severe condition Suspect kerotoconjuctivitis	1. Wash with normal saline water 2. Chlorenphenicol eye drops 2drops every 2 hoursx3 days 3. Antihistamine 5mg odx3 days or 4. Chloramphenicol 2mg odx3 days 5. Prednisolone 2.5mg stat

Health Education

1. Personal, food and environmental hygiene.
2. Prevention of eye injuries.
3. Eye care.
4. Nutrition counseling.
5. Regular eye checkup.
6. Importance of taking home-based records to the clinic or hospital.

3.19 EAR, NOSE AND THROAT DISORDERS

Ear Disorders

Ear problems are diseases affecting the ear and are commonly caused by middle ear infection; usually with pain in the ear, dullness or redness of the ear drum sometimes with perforation and drainage of pus or fluid. It is important to examine the bone behind the ear mastoid bone), because if this is tender to touch, it may mean that the child has mastoiditis. This is a danger sign. The other major complication is deafness which can be tested by clapping your hands behind the child to see if he will turn and look.

Nose Disorders

The nose is the organ for breathing and for the sense of smell. Disease or injury in the throat or sinuses can affect the nose. This could bring about watery discharge, mucous blood or pus from the nose. Also, injury to the nose can affect the throat. The health worker should think of a foreign object if there is discharge from only one side of the nose.

Throat Disorders

A red throat may be caused by infection. It may be combined with nasal congestion and middle ear infection especially in young children. The tonsils normally enlarge during childhood, but unless they show sign of difficult breathing or inflammation, no treatment is needed.

HISTORY	EXAMINATION/ACTIONS
1.Does the child have hearing problems? 2.Is there any discharge from the ear? 3.Has the child had any ear surgery? 4.Is there any family history of deafness?. 5.Does the child have runny nose? 6.Is there any bleeding from the nose? If yes, for how long? 7.Has the child had it before? 8.Is the child able to swallow food or fluids? 9.Has the child received any injection or medication recently? If yes, what is the name? Ototoxic drugs (antibiotics (e.g. gentamicin), diuretics, cytotoxics). 10.Has the child been exposed to loud noises recently? Exposure to noise (e.g. pneumatic drill or shooting). What treatment have you given to the child for this condition?	1.General appearance: well, ill-looking, pale 2.Nose: watery discharge, mucus, blood, pus, any visible swelling, deformity, or foreign 3.Throat: pus or redness 4.General appearance: ill-looking, irritable 5.Head: tenderness over bone behind the ear (mastoid) 6.Ear: a). condition of canal plus or blood, perforated drum 7.foreign body red or bulging drum 8.Test for hearing: clap 9.Neck: stiffness Throat 10.General appearance: distressed, pale, 11.Head: tenderness over bone behind the ear(mastoid) 12.Throat: pus or redness., enlargement of tonsils 13.Neck: nodes, swellings 14.Ear: discharge, redness or bulging drum 6. Temperature, Pulse and Respiration: check and record. Conduct RDT Test.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Ear pain with history of foreign body with bleeding or Neck Stiffness	Bleeding from ear Stiff neck Dull ear drum Warm to touch (Temp > 37.5°C)	Severe condition Suspect Ear problem	1. Give IM. Diclofenac 30 – 50mg immediately (stat) 2. Give First dose of IM. Amoxicillin 250 mg stat 3. Refer appropriately
II. Pain behind the Ear	Tender swelling behind the ear Dull ear drum Warm to touch (Temp > 37.5°C)	Severe condition Suspect mastoiditis	1. Cap. Amoxicillin 250mg stat 2. Tab. Paracetamol 250-500mg 8 hourly for 3days 3. REFER

III. Ear pain with discharge for less than 14 days with or without fever	Pus draining from the Ear Dull ear drum with or without perforation Warm to touch (Temp > 37.50C)	Moderate condition Suspect acute ear infection	(AA) Cap. Amoxicillin 250mg 8hourly for 5days Tab. Paracetamol 250-500mg 8hourly for 3 days Tabs Vitamin C 100mg 8hourly for 5 days Review in 5 days If no improvement, REFER
IV. Ear pain with discharge for more than 14 days with or without fever	Pus is seen draining from the ear. Dull ear drum Warm to touch (Temp > 37.50C)	Moderate condition Suspect Chronic Ear Infection	1. Cap. Amoxicillin 250mg 8hourly for 5days Explain condition to mother 2. Tab. Paracetamol 250-500mg 8 hourly for 3 days 3. Review in 5 days 4. If no improvement, REFER
V. Boil in ear	Boil seen	Moderate condition	1. Cap. Amoxicillin 250mg 8hourly for 5 days 2. Tab. Paracetamol 250-500mg 8 hourly for 3 days 3. Review in 3 days 4. If no improvement, REFER
VI. Difficulty in Swallowing	Grey and dirty membrane on tonsil Warm to touch (Temp. > 37.50C)	Severe condition Suspect tonsillitis	1. IM Amoxycillin 250mg stat, Continue with 2. Give Ibuprofen 50mg odx3days 2. Cap. Amoxicillin 250mg 8 hourly for 5 days 3. Mouth wash with saline warm water 4. Give lozenges bdx3days 5. Review in 5 days 6.If no improvement REFER
VII. Pain in the throat difficulty with swallowing	Enlarged red tonsils with pussy exudate with or without tender lymph nodes in the neck	Moderate condition Suspect Tonsillitis	1. Give Cap. Amoxicillin 250mg 8 hourly for 5 days 2. Tab. Vitamin C 100mg 8hourly for 5 days. 3. Tab. Paracetamol 250-500mg 8 hourly for 3 days 4. Gargle with salt water 4 times daily if able 5. Review in 3 days 6. If no improvement REFER
VIII. Pain on either side of the nose and Headache	Pus from nose tenderness Smelly discharge	Moderate condition Suspect acute sinusities	1. Tab. Co-trimoxazole 480mg 12hourly or Cap. Amoxicillin 250 mg 8hourly for 5 days Tab. Paracetamol 250-500mg 8 hourly for 3 days Tab. Chlorpheniramine (Piriton) 2-4 mg 12hourly for 3 days Tab. Vitamin C 100 – 200mg 7 8 hourly for 7 days 8 Tab Prednisolone 2.5mg stat Review in 3 days If no improvement refer

IX. Bleeding from nose	Evidence of bleeding	Moderate condition Suspect Epistaxis	1. Teach mother to pinch side of the nose intermittently for 15 minutes Apply 2 2. Adrenaline with gauze on to the nose 3. Apply silver nitrate Apply Cold Compress to nose for 30 minutes and re-examine If no improvement, REFER.
X. Stuffy nose, with or without Fever; Sneezing with or without watery eyes, exposure to dust or pollen	Watery eye, Catarrh Warm to touch (Temp. > 37.5°C)	Moderate condition	1. If RDT is positive, treat for uncomplicated malaria 2. Tab. Chlorpheniramine (Piriton) 2 mg 12hourly for 3 days 3. Tab. Paracetamol 250-500mg 8 hourly for 3 days 4 Tab. Vitamin C. 100- 200mg daily 8 hourly for 7 days 5. Review in 3 days
XI. Pain in the throat, Fever, inability to swallow	Sore throat with or without redness Tonsils not inflamed and no pus Warm to touch (Temp. > 37.50C)	Moderate condition	1. If RDT is positive , treat for malaria 2. Give Cap. Ampiclox 250mg 6hourly for 5 days 3. Tab. Paracetamol 250-500mg 8 hourly for 3 days 4. Tab. Vitamin C 100- 200mg 8hourly for 7 days 5 Take fruits
XII. Boil in nose	Boil can be seen	Mild condition	Cap. Ampicillin 250mg 6hourly for 5 days Tab. Paracetamol 250-500mg 8 hourly for 5 days Tab. Vitamin C 100 – 200 mg 8 hourly for 5 days Review in 3 days.,
XIII. Foreign object or Polyp in the nose	Foreign object seen	Moderate condition Foreign object	Try to remove foreign body with blunt forceps If not removable, REFER .

Health Education

Advise Parent / Caregiver on;

1. Prevention of infections.
2. Care of the ear and nose.
3. Management of throat problems (gargle and discourage uvulectomy).
4. Proper nutrition.
5. Personal hygiene.
6. Prevention of exposure to air pollution (fumes, dust and pollen).
7. Discouragement of home removal of foreign body.
8. General childcare.
9. Encourage intake of fruits.
10. Importance of taking home-based records to the clinic or hospital.

3.20 ORAL HEALTH PROBLEMS

Oral health is a state of being free from chronic mouth and facial pain, oral and throat cancer, oral sores, birth defects such as cleft lip and palate, periodontal (gum) disease, tooth decay and tooth loss, and other diseases and disorders that affect the oral cavity.

The most common complaint is pain which may or may not be accompanied with swelling inside the mouth or on the face. The condition of the mouth is also a reflection of the client's general health, so it should be inspected as part of physical examination. Early recognition of oral problems with prompt attention or referral will reduce morbidity and mortality arising from otherwise simple disease condition.

HISTORY	EXAMINATION/ACTIONS
1.Where is the problem? 2.Is there pain from teeth or gum? 3.How long has it been noticed or present? 4.Are you able to eat your regular diet? 5.Does the pain increase with hot or cold drink or food? 6.Does the pain keep you awake at night? 7.Is there any bleeding from gum or other parts of the mouth? 8.Do you notice bad or foul breath? 9.What do you use for cleaning your teeth? 10.How do you clean your teeth? How many times a day? Do you notice any sore in the mouth?	1.General appearance: ill, facial swelling, malnourished emaciated 2.Mouth: a)sore, swelling, discharging, bleeding, white patches b) tongue: sore, coating, redness, fissures, smoothness, dryness lips: cracks corners, cracking and tenderness c) teeth: carries, discolouration, tenderness, mobility, missing, plaque Temperature: check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Slow or fast growing swelling of the jaws with or without pain	Hard or Soft growth in the jaws or soft tissue	Severe condition Suspect Burkitt's lymphoma or other swellings	Need to relief pain , give Tab. Paracetamol 250-500mg 8hourly for 3days Refer to appropriate health facility
II. Ulcers in the mouth. Oral thrush, weight loss, bleeding gums with or without Skin rash	Oral thrush, Bleeding gum	Suspect HIV/AIDS	Refer to appropriate health facility for screening
III. Injury to mouth and /or face	Tooth missing. Tooth broken Laceration, Mobile teeth, bleeding	Moderate to severe condition	Arrest bleeding by applying pressure Tab. Paracetamol 250-500mg immediately (stat) Refer to Dentist
IV. Mouth Ulcer with fever with or without general weakness.	Vesicle on tongue. Mouth Ulcers	Moderate condition: Suspect Oral Herpes	1.Ensure adequate fluid intake. 2.Tab. Paracetamol 250-500mg 8 hourly for 2 days. 3.Cap. Ampicillin 250mg 6hourly 5 days 4.Tab. Metronidazole 200mg for 3 days 5.Tab. Vitamin C 100–200mg 8hourly for 14 days 6.Review in 5 days. If no improvement, refer to Dentist

V. Cracks at the corner of the mouth	Tongue is smooth and red. Stinging sensation in the tongue Whitish cracks at corner of the mouth, tends to be bilateral.	Moderate condition: Suspect Angular Stomatitis, Avitarminosis	1.Tab.Vitamin B Complex 8hourly for 2weeks 2.Tab.Vitamin C 100- 200mg 8hourly for 2weeks 3.Tab. Multivite daily 2 weeks 4.Advise on diet Oral hygiene instruction
VI. Pains in the mouth that keeps patient awake at night with or without jaw swelling	Holes in tooth/teeth (Caries)	Moderate condition Suspect dental carries	1.Tab. Paracetamol 250-500mg 8 hourly for 2 days. Gargle with warm salinet water Cap.Ampicillin 250mg 6hourly for 5 days Tab. Metronidazole 200mg 8hourly 3 days Review in 3 days Refer if no improvement
VIII. Sharp pain in the tooth, Pain elicited by hot/cold foods	Holes in tooth/teeth	Mild condition	1.Gargle with warm saline water 2. Explain condition Refer to Dentist for filling
IX. Overlaping teeth	Teeth not well arranged	Mild condition Malocclusion	Refer to Dentist

Health Education

1.Oral hygiene, emphasizing proper utilization of cleaning materials e.g. brush, chewing stick etc.
2.Nutrition counseling.
3.Discouragement of uvulectomy.
4.Prevention of accident.
5.Importance of taking home-based records to the clinic or hospital.

3.21 CHEST/HEART DISORDERS

A very rapid, low or irregular heart rate is always a serious sign. Blueness of the lips or fingers, clubbing of the fingers with difficult breathing are signs that the heart is not able to perform its function effectively. Congestive heart failure occurs when the heart becomes too weak to pump all the blood around and some of the fluid flows back into the lungs and retained in the abdomen and legs.Any history of shortness of breath should make health worker think of a heart disease.

HISTORY	EXAMINATION/ACTIONS
1.How long have you had the problem? 2.Do you have cough, chest pain, or shortness of breath? If pain where? Does deep breathing or bending make it worse? 3.Can you sleep flat or do you need several pillows at night? 4.Do you sometimes have to get up at night to breath? 5.If cough is present, do you bring up sputum? Colour? Any blood in sputum? 6.Do you eat well? 7.Have you had any night sweating or weight loss? 8.Does the pain/ discomfort start on exertion, made worse by exertion or even come at rest? Are you on any medication? If yes, what?	1.General appearance: ill, in pain, thin, pale, cysnosed 2.Neck: nodes 3.Chest: a)respiration for rate, difficulty in breathing b)crepitations, bronchi wheezes, decreased breath sounds 4.Heart: rate, rhythm, murmur 5.Abdomen: tender liver, distension 6.Legs: oedema 7.Temperature, pulse: check and record Blood pressure: measure and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Chest pain on exercises if associated with Shortness of breath, easily tired, blue lips or fingernails	Shortness of breath . Rapid or irregular heart beat . Dizziness . Fatigue. . Swellings • Coughing, • Swelling of limbs, • Fast breathing, •More comfortable sitting up, • Blueness of lips and finger nails, •Fatigue .	Severe condition suspect congestive} heart problem	It should not be considered as normal , think of :- 1. Heart 4. Lungs 5. Digestive system calm the client down, encourage rest, put patient in Cardiac position . If under Medication and encourage to take drugs If no response within few minute. Refer to hospital immediately Refer to hospital immediately in propped up position (cardiac position) If no improvement REFER
II. Easily tired, not playing with friends, squats	• Slow or fast heart rate, • Cough, • Blueness of lips and finger nails.	Severe condition Suspect congenital heart problem	Refer to hospital immediately preferably in a propped up position
III. Easily tired with any of the following: fever, or infection	Fast heart rate Normal Heart rate 4 - 8years - 90/min Warm to touch (Temp 37.5^{0}C or above)	Moderate condition Suspect Infective Heart Disease	1. If RDT is positive, treat for uncomplicated malaria 2. Cap.Amoxicillin 250mg 6hourly for 5days 3. Tab. Paracetamol 250-500mg 8hourly for 3 days 4. Tab. Salbutamol 4mg 12hourly for 3 days 5. Review in 3 days 6. If no improvement, REFER

IV.Weakness and ,fatigability	Heart murmur Hb <10g/dl	Moderate condition Suspect anaemic heart failure	1. REFER immediately

Health Education

Advise parent /Caregiver on:

1.Adequate rest
2.Proper care of person with chest/heart problem (keep warm and allow for adequate rest)
3.Adequate nutrition
4.Regular medical examination
5.Importance of taking home-based records to the clinic or hospital.

3.22 ABDOMINAL DISORDERS (INCLUDING HERNIAS)

The common causes of abdominal pain are gastro-enteritis, malaria, appendicitis, rupture of the intestine, typhoid, sickle cell crisis and obstruction, e.g. strangulated inguinal and umbilical hernias. Constipation, worm infestation may also cause abdominal pain. Abdominal pain may lead to hardness or rigidity of the abdominal wall.

An abdominal hernia is when a part of the gastro-intestinal tract protrudes through a gap or weakness in the abdominal wall. Hernia can be found on the umbilicus, groin and scrotum.

HISTORY	EXAMINATION/ACTIONS
1.Is there vomiting? How many times? 2.Is there constipation? When was the last stool? 3.Is there any abdominal pain? 4.Has he had the problem before? 5.Has any medication been given? 6.Ask the following questions: a.1 Types of pain b.2 Location of pain or site c.Duration of pain d.Accompanied symptoms e.Factor associated with pain 6 Factors relieving the pains	1.General appearance: pallor, weakness, distress 2.Respiration: difficult or rapid 3.Abdomen: a) enlarged, distended 4.Masses, enlarged spleen 5.Visible intestinal movement 6.Presence of hernia tenderness 7.Hydration status: fontanelle, eyes, mouth, tongue, skin. 8.Haemoglobin: estimate and record. Temperature: check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Abdominal pain; Not passing stool	Child ill looking, crying and in pain, abdominal rigidity.	Severe condition Suspect Abdominal Obstruction	I. REFER
II. Abdominal pain with rigidity; vomiting but no diarrhoea	Child in pain and very uncomfortable, tender abdomen	Moderate condition Suspect Worm Infestation	I. Give Albendazole according to age 2. Observe for 3 days 3. If no improvement 4. Refer
III. Abdominal pain with Constipation;	Well child; swelling of abdomen increased bowel sounds	Moderate condition Suspect Constipation	1. inncrease fluid intake 2.Liquid paraffin according to age 3. Encourage fruits and vegetable intake
IV. Abdominal pain with swelling on left side of abdomen with or without history of injury.	Ill looking, Pale, Enlarged tender mass with painful swelling on the left upper part of the abdomen	Severe condition Suspect injury to spleen	1. Do mRDT if positive manage malaria 2. If with injury Refer
V. Abdominal pain with bloody stools; Strains in passing stool	Pale looking; blood in stools, Tenesmus on passing stool	Moderate to Severe condition Suspect; Dysentery, Amoebiasis or Schistosomiasis	1. metronodazole 200mg tid daily X 5days 2. Praziquantel {20mg} per kilo 3 tid daily 3. review after 3 days 4. no improvement; Refer
VI. Abdominal pain and paleness	Thin and pale looking (Pallor of conjunctiva and fingernails). No tenderness	Mild/Moderate condition Suspect Sickle Cell disorder	1.Tab; Folic acid 20mg daily two weeks 2. Injection; Penta 3. Fluid intake 4. Review in 7 days 5. If no improvement REFER

VII. Abdominal pain with the history of passing worms. No Constipation	Appears well; Abdomen distended; no tenderness; No constipation	Mild condition; Suspect Worm Infestation	1. Dewormed the child 2. Tab; Folic acid 20mg daily two weeks 3. Review in 5 days 4. If no improvement REFER
VIII. Swelling in Scrotum	Child in Pain Scrotum tender and warm Swelling in one or both sides of Scrotum	Severe condition Suspect Epididymitis	1. Cap.Doxycyline according to age, 2. Tab Ibuprofen 50mg bdx3 days 3. Cold compress to reduce pain 4. If no improvement REFER
IX. Lump or swelling in the groin/ the scrotum	Child in distress, Presence of tender swelling	Moderate Condition Suspect: Inguinal hernia Scrotal hernia\hydrocele	1. REFER

Health Education

Advise parent /Caregiver on:

1.Adequate nutrition, prevention of constipation and adequate fibre in diet.
2.Food and personal hygiene.
3.Adequate fluid intake.
4.Discourage using force to reduce hernia.
5.Surgical intervention as appropriate.
6.Importance of taking home-based records the clinic or hospital.

3.23 ANAL DISORDERS

The Anus is the terminal end of the alimentary canal. Common problems affecting the anus include fissures, redness and prolapse. An anal fissure is painful linear sore at margin of anus. In rectal prolapse, the rectum turns inside out and protrudes out of the anus as a pink mass. It is common in young children who are poorly nourished having worms and diarrhoea frequently.

HISTORY	EXAMINATION/ACTIONS
1.How long have you had the problem? 2.Is there diarrhoea? How many times? 3.Is there constipation? When was the last stool? 4.Is there bright blood in the stool? 5.Is there abdominal pain? 6.Has the child had this problem before? 7.Has any medication been given? If yes what? 8.Is there pressure or /and bulging of the anus after stooling? 9.Is there fleshy mass hanging out of anus with leakage of mucus, stool or blood? 10.Is there pain and itching around the anus?	1.General appearance: pale weak, in distress 2.Hydration status: eyes, mouth, tongue, skin 3.Anus: prolapse, redness, fissure 4.Hemoglobin: estimate and record. 5.Temperature: check and record. Weight: check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Prolapse of rectum not reducible	Child distressed, Not dehydrated Rectal prolapse not reducible	Moderate condition Suspect prolapsed of the rectum	1. Do sitz bath 2. Observe for 2 hours, if not reduced REFER
II. Prolapse of rectum reducible with or without bleeding	Child distressed Rectal prolapse reducible	Mild condition Suspect mild prolapsed of rectum	Apply warm saline, compress Teach mother how to strap the buttocks Cap. Amoxicillin 250mg 8hourly for 5days Give Tab. Folic Acid 5mg daily for 2 weeks Tab. Ferrous sulphate 200mg (60mg elemental iron) 8hourly for 2 weeks Give LO ORS with Zinc DT Give Albendazole 200mg stat Nutritional counselling Review in 2 days If no improvement, REFER
III. Fissures or sore on anus, Redness of anus	Anus red and sore. Child uncomfortable	Mild condition suspect fissures	Clean with soap and water Dry thoroughly Zinc Oxide topically Tab. Paracetamol 250-500mg 8hourly for 5 days Dietary advice Review in 5 days If no improvement, REFER

Health Education

Advise parent /Caregiver on:

1.Personal, food and environmental hygiene.
2.Early detection and treatment.
3.Adequate nutrition.
4.Protection of genital area.
5.Discouragement of sharing of sponge, towel, underwear and clothing
6.Bowel training (going to toilet at a regular time daily).
7.Importance of taking home-based records to the clinic or hospital.

3.24 WORM INFESTATION

Worm infestation is the presence of large number of worms in the gut which causes discomfort to the patient. .Ascaris (roundworm) is the commonest worm infestation. Heavy roundworm infestation produces abdominal pains, vomiting, constipation and sometimes obstruction of the intestine. Hookworm infestation is common in rural areas where heavy infestation causes blood in stool, anaemia, malnutrition. Severe Trichuris (whipworm) infestation causes bloody diarrhoea, abdominal pain, anaemia and prolapse of the rectum. Threadworms also live in the intestine and causes' itching around the anus. Tapeworm is a flat ribbon like worm that can also be found in the intestine.

HISTORY	EXAMINATION/ACTIONS
1.When did it start? 2.Have any worms been passed? If yes, describe 3.Is there passage of watery stool? Blood in stool? 4.Is there vomiting? 5.Is there anal itching? 6.Is there abdominal pain and rectal prolapse? 7.Is there any other problem? 8.Has any medication been given? If yes, what? What are the characteristics of the worms	1.General appearance: thin, pale, 2.Eyes: pallor 3.Chest: wheezing, rhonchi 4.Abdomen: tenderness, distension, visible peristalsis 5.Anus: redness, prolapse 6.Temperature, Pulse and Respiration: check and record 7.Haemoglobin: estimate and record 8.Weight: weigh and record Stool examination: for ova of worm (if laboratory available)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Passing worms from the rectum, itchy rectum, protruding rectum, blood in stool	Rectal prolapse	Moderate condition Suspect worm infestation with rectal prolapse	Tab. Pyrantel pamoate 3 tablets stat or Albendazole 200mgstat Anusol suppository apply topically or insert into the rectum each night at bedtime Give Tab.Vitamin C 100 – 200mg 8hourly for 5days Review in 1 week If no improvement REFER
II. Passing roundworms par rectum, diarrhoea, vomiting, rectal itching, abdominal pain.	Thin, pale, restlessness There may be pallor May be underweight and have mild abdominal pain or distension ^ . Tread like worm suspect pin worm or thread worm , Rash and redness around the anus	moderate condition Suspect worm infestation	Tab. Pyrantel pamoate 3 tablets immediately (stat) LOW ORS with Zinc DT Review in 1wk If no improvement REFER Tab.Albendazole 400mg immediately (stat)
3 Passing thread like worm (think of thread worm) 4. Passing of flat white worm (think of tape worm) Expelling of segment in stool Epigatricpain, nausea, vomiting	May have face swelling. Segment of worm in stool	Moderate condition.	Tab. Pyrantel pamoate 3 tablets Stat or Paraziquantel po 5/10mg/kg single dose or niclosomide same dose.

Health Education

Advise parent /Caregiver on:

1.Personal and food hygiene
2.Adequate nutrition
3.Environmental sanitation
4.Importance of taking home-based records to the clinic or hospital.

3.25 URINARY AND GENITAL DISORDERS

Urinary and genital problems are diseases or conditions affecting the male or female urinary tract or genitalia. Children may present with abnormalities of the genitalia such as undescended testis, imperforate hymen, hydrocele, and hypospadias and may be hermaphrodite. They may also present with painful urination with or without discharge. Redness or itching of the genitalia may be due to infection.

Sexually active older children are especially susceptible to sexually transmitted infections (gonorrhea, syphilis, HIV/AIDS and other non-specific bacterial infections). Patients may complain of discharge, pain and burning sensation on passing urine and blood in urine may occur as a result of schistosomiasis infection. Other causes of urinary problems are kidney diseases and diabetes which can cause frequent micturition.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? How long have you had it? 2.Do you urinate very often? In the night or in the day? 3.Do you have pain or burning on micturition? 4.Do you drink more than usual (increased thirst)? 5.Do you have pain in the flank? 6.Is there blood in the urine? 7.Do you have night sweats? Or are you losing weight? 8.Do you have chills? 9.What area do you come from? (think of Schistospmaisis area) 10.Have you ever had this problem before? Any treatment? Do you have discharge from the vagina (or if male, from the penis)	1.General appearance: ill, thin, puffy. 2.Temperature, pulse: heck and record 3.Blood pressure: measure and record 4.Eyes: pallor. 5.Chest: crepitations, decreased breath sounds. 6.Heart: rate, rhythm, murmur. 7.Abdomen: tenderness, swelling or masses. 8.Flanks: tenderness 9.Limbs: a) legs for swelling b) Nails for pallor C) Genitals: sore, pus, or discharge 10.urine: a) colour or cloudiness b). protein sugar d) microscopy

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. No Urine for more than 12 hours	Ill, Restlessness, Anxious	Severe condition Suspect Kidney problem	REFER. immediately
II. Excessive urination with history of ants crowding round urine. Excessive thirst	Sugar in Urine	Severe condition; Suspect diabetes	1. REFER immediately.
III. Urethral or Vaginal Discharge with painful micturition	Urethral/Vaginal discharge	Moderate condition Suspect Urinary Tract Infection	Risk assessment to know if the child has been sexually abused Tab. Paracetamol 500mg 8hourly 3 days Tab. Ciprofloxacin 250mg 12hourly for 7 days Review in 5 days If no improvement, REFER
IV. Painful urination with or without frequency, cloudy urine or fever	Fever; Protein in Urine; painful Micturition	Moderate condition Suspect nephrotic syndrome	1. Encourage fluid; 2.Tab. Paracetamol 250-500mg 8 hourly for 3 days 3.Tabs Cotrimoxazole 480mg 12hourly hourly for 5 days Tab. Co-trimoxazole 480mg 6 12 hourly for 5 days 7. Review in 2 days; If no improvement, REFER

V. Pain in the flank, fever	Pain on tapping flank, muscle spasm	Severe condition Suspect Kidney Stones	Tab. Paracetamol 250-500mg 8 hourly for 3 days Amoxycillin 500mg 6 12 hourly for 5 days Encourage fluids; Review in 5 days If no improvement REFER.
VI. Redness or Swelling of Genitalia with or without painful urination, itching.	Redness and swelling of Genitalia Scratch marks Urethral Discharge with or without evidence of rape	Moderate condition Suspect Urinary tract infection Suspect fungal infection	1. Wash the area with soap and water. 2. Apply Zinc oxide and Mycostatin cream or paste to area. 3. Encourage fluids, if male retract foreskin; 4. Tab. Co-trimoxazole 480mg 12 hourly for 5 days 5. Review in 7 days; 6. If no improvement REFER.
VII. Blood in last stream of Urine with or without pain, fever or discharge. History of swimming in stream/river	Blood in Urine Terminal hematuria	Moderate condition Suspect Schistosomiasis	1. Encourage liberal fluid intake 2. give Metriphonate 200mg (2 tab) every 2 weeks for 6weeks or 3. Tab. Praziquantel 1.2g stat (2 tablets) 43. If no improvement, Refer
VIII. Painful Urination; No discharge	No Discharge	Moderate condition Suspect urinary Tract Infection	Cap. Amoxicillin 250mg for 5 days Tab. Paracetamol 250- 500mg 8hourly for 3 days Encourage fluid and fruits intake Review in 3 days If no improvement, refer.
IX. dark yellow urine, no other complaint	Dark Urine with or without yellow eyes	Mild to Moderate condition Suspect dehydration	1. If no Jaundice; encourage intake of fluids 2. If Jaundice present give, Tab. Folic Acid 5mg daily for 2 weeks 3. If no improvement, REFER.
X. abnormality of the Genitalia	Un-descended testes, imperforate hymen, Hermaphrodite, Large clitoris, Hydrocoele; Hypospadias (penis in abnormal place or several penial openings).	Mild to Moderate condition	1. Explain finding to mother and 2. re-assure her; REFER
XI. Difficulty in passing urine	Penis blocked by tight foreskin not easily retracted, bladder full	Moderate condition	Circumcise

XII. Burning sensation, itching, vaginal rashes, thick white discharge with offensive odour, painful intercourse, irritation.	White reddish in the vagina and watery, with cheese appearance	Severe condition	1 Tab. Metronidazole 20mmg 12hourly for 5days 2 Application of clindamycin cream or gel and suppositories

Health Education

Advise parent /Caregiver on:

1.Personal hygiene.
2.Adequate fluid intake.
3.Proper parental care and supervision of the child.
4.Side effects of medications.
5.Drug compliance.
6.Environmental sanitation.
7.Importance of taking home-based records to the clinic or hospital.

3.26 ARM AND LEG DISORDERS

Deformities of the arms and legs, if not corrected early could lead to permanent disabilities. The deformities or swellings may be due to bruises, fractures or a dislocation. It is important to test for normal movement of the parts whenever there is a history of injury, swelling or deformity. Swelling of a joint may be due to a sprain of the tendons surrounding the joint. Sickle cell disease may also be a cause of pain in the arms, legs and the hip bone. Tuberculosis can occur in the joints. Polio can seriously affect one or both legs or arms first causing weakness then wasting, or paralysis. Diabetes Mellitus may cause tingling sensation and numbness of the fingers and toes, foot ulcer, that fail to heal and black discoloration of toes and feet (Gangrene).

HISTORY	EXAMINATION/ACTIONS
1.When did it begin? 2.Was there an accident or injury? a.If an injury, has the child been able to use the part since the injury? 3.Is there any pain or tingling or numbness? Is it recurrent? 4.Is there a fever? 5.Is the child a Sickler? 6.Is the child passing excessive urine? Is the child having excessive thirst?	1.General appearance: ill looking, wasted or deformed 2.Limbs [a] deformity [b] swelling. [c]bleeding [d]fracture [e]tenderness [f]warmth [g]loss of function [h]wasting 3.Eyes: jaundice 4.Temperature: check and record 5.Haemoglobin: estimate and record Urine: sugar and colour

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Inability to move arm and/or leg; Pain in joints and limbs	Weakness of limbs with or without pain Paralysis with or without swelling at the joints Wasting of the limb	Severe condition Suspect Polio, Sickle Cell Disorder	1.If Polio, notify the appropriate Health authority 2.REFER
II. Swelling of one or more joints without fever or tenderness	Pain Tenderness Swelling of the joint No feve	Moderate condition Suspect Juvenile idiopatic Arthritis and Rheumatoid Arthritis	1.Asprin 150-300mg according to age or Tab. Paracetamol 250-500mg 8 hourly for 5 days 3.Tab Azithromycin 250mg - Cap. Ampiclox 500mg daily 6hourly for 7 5 days 4.Apply warm compress 5.if no improvement REFER!
III. Deformity without swelling, tenderness or pain or bowing of leg or difficulty in walking	Club foot, Bow leg, walking with difficulty	Severe condition Suspect Poliomylitis ,sickle cell disease, Rickette etc	1. Explain finding to mother/care giver 2. REFER to appropriate health facility for management.
IV.Tender lumps and nodes in groin	Painful lumps and nodes in the groin Source of infection identified	Moderate condition	Tab. Paracetamol 250-500mg 8hourly for 5 days Cap. Amoxicillin 250mg 8hourly for 5 days, Treat open wound if any. Review in 5 days If not getting better, REFER Rto appropriate Health facility

V. Swelling on and off especially of fingers, toes, back of hands and feet	Swelling of fingers, toes, back of hands and feet. Hb below 10gms	Mild to moderate condition Suspect Sickle Cell Disease	1. Tab. Folic Acid 5mg daily for 2 weeks Do mRDT if positive treat in Malaria Tab. Paracetamol 250-500mg 8 hourly for 3 days 2. If no improvement REFER
VI. chronic losses of cuteneous tissue Types a. traumatic: resulting from injuries and other factors b. Metabolic: e.g. diabetes c. Parasitic: e.g. Leishmaniasis, chancriod	Open skin, redness discharge, sensitivity,	Moderate condition	If associated with metabolic REFER- Parasites: treat parasite along with ulcer. Provide: a. Debrimentation. b. Clean with appropriate antiseptic if dirty using silver sulfadiazine, cetrimide, hydrogen, pyroxide, if multiple extentions oozing uses polyvidien. c. Give Parental Gentamycin Cap. Doxycycline if above 8yrs, 4mg/kg daily Tab. Metronidazole 30mg/kg per day in three divided dose -treat the multiple cause.
VII. Sore and ulcer	Multiple vesicular rashes with or without swelling of lymph nodes	Severe condition	1. Rrefer to lab for VDRL, syphilis, HIV, chancroid and other sexually transmitted diseases . Treat as appropriate.

Health Education

Advise parent /Caregiver on:

1. Prevention of accidents.
2. Genetic counseling.
3. Adequate diet (Calcium rich foods).
4. Proper posturing.
5. Importance of taking home-based records to the clinic or hospital.

3.27 CHANGES IN ALERTNESS AND BEHAVIOUR

Changes in alertness and behavior mean that the client does not act normally sometimes. He/she may be drowsy or give a history of fainting or there may be obvious changes such as unconsciousness or delirium which are easy to identify. All cases of change in alertness or behaviour should be taken seriously no matter how mild to prevent complications.

HISTORY	EXAMINATION/ACTIONS
1.How long has the child been like this? 2.Did anything happen to the child? Injury? 3.Did the child have a fit or convulsion, or is he clumsy? 4.Has any medication been given? 5.Has the child been ill otherwise? 6.Has he had fever, diarrhoea, vomiting or ear pain? 7.Has he been drinking more recently? 8.Has he been urinating more frequently? 9.Was there any difficulty with the child's pregnancy or delivery? 10.Was the child jaundiced at birth or shortly after? Has he been frothing at the mouth or passing urine when he faints?	1.General appearance: (especially for alertness and irritability signs of trauma). 2.Chest: a) respiratory rate, lower chest indrawing 3.Head:, signs of injury 4.Neck: stiffness 5.Ears: pus, redness or perforation of the drum 6.Limbs: arms and legs for abnormal movements 7.Temperature and pulse: check and record. 8.Haemoglobin: estimate and record. Conduct RDT.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Unconscious with fever or history of fever difficult to arouse confusion more irritable than normal abnormal movements of arms and legs	Fever of 37.5C and above Irritable. No stiffness. Patient may be delirious difficult to arouse abnormal movements of arms and legs confused	Severe condition Suspect Severe Malaria	If Rapid Diagnostic Test (RDT) is positive, give pre-referral treatment for severe malaria Obtain IV line REFER
II. Unconscious or more irritable than normal abnormal movements of arms and legs	Irritable. No stiffness. May be ; delirious difficult to arouse, seizure	Severe condition Suspect Epilepsy	1. Stabilise patient 2. Tab phenorbabitone 30mg stat 3. If no improvement REFER
III. Fainted suddenly but is arousable.	Child disoriented, with or without stiff neck	Moderate condition Suspect hypoglycemia	1. Give child sugar or glucose at once and let him rest for 30mins. 2. If no improvement, REFER. If improved, advise on proper feeding. 3. Test Urine 4. Review next day. 5. If no improvement, REFER
IV. Fainted (fit) once or twice with or without fever or frothing from mouth or family history of epilepsy	Conscious, normal temperature No fits	Severe condition Suspect epilepsy	Explain condition to mother Tab. Phenobarbitone 30mg 12hourly REFER
V. Unconscious or More irritable than normal Abnormal movement of arms and legs	Irritable, no stiffness, maybe; Delirious Difficult to arouse	Severe condition Suspect coma	1. Refer to hospital immediately

IV. a. Worries: e.g. fear of illness or fear with no clear object. b. Behavior changes. i. Nervousness and avoidance behaviour	Looks sad, lacks interest, or lacks pleasure in activities, fatigue , weight lost,	Severe condition Suspect Depresion	Identify cause and treat. Drugs: includes Tab. Paroxetine 20mg daily Cap. Fluoxetine 20mg daily (morning) Tab. Amitriptyline 25-75mg at night 2. Individual or group counselling an close monitoring
Psychotic disorder. E.g., 1. delusion, -hallucination, -agitation, -mutism, -prostration 2. Anxiety: history of sudden onset of life event. E.g., lost, stress, trauma	Looks sad, isolated, illusions. Nervous, lacks appetite, sleep-lessness, looks disturbed.	Severe condition Suspect Psychosis	1. Psycho-social support, anti-psychotic drugs. Provide home base OPD services. 2. Drugs: Tab. Risperidone Tab. Haloperidol. 3. REFER.

3.28 POISONING

Poisons are toxic substances which may enter the body through swallowing, breathing, absorption through skin or mucus membrane stings or infections. They can cause fits marked restlessness and pain. Small children are adventurous and may therefore be easily exposed to harmful substances.

Mothers should therefore be advised to keep kerosene, medicine, bleach, dye, alcohol and other harmful substances out of the reach of children. Kerosene and medicine should be labelled and must not be stored in soft drink bottles.

HISTORY	EXAMINATION/ACTIONS
1. When was the poison taken? 2. What and how much poison did the child take? 3. Has any medication been given (modern or traditional)? Has he/she vomited? Is there blood in the voimitus?	1. Try to examine poison container. 2. General appearance: consciousness, alertness, pallor, and cyanosis. 3. Eyes: size and reaction of pupils to light? 4. Chest: respiration – rate, difficulty, lower chest, indrawing, deep, sighing. 5. Abdomen: tenderness, rigidity. 6. Mouth: burns, bleeding, dribbling, colour of membrane. 7. Pulse: check rate, volume and record. Temperature: check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Ingestion of petrol or kerosine or corrosive substance e.g. alkali, caustic soda and acid.	Irritable, swelling and mouth burns	Moderate to severe condition Suspect substance poisoning	1. Do NOT induce vomiting 2. Give enema episodedm 3. Give Cap. Amoxicillin 250mg immediately (stat) 4. Give activated Charcol 5. REFER
II. Ingestion of poison with convulsion	Irritable, Restless, may or may not be convulsing No mouth burns and swelling	Severe condition Suspect Poison leading to convulsion	IM Paraldehyde 5cc stat if convulsing. Gastric lavage/washout REFER immediately NB: DO NOT INDUCE VOMITING IN A CONVULSING OR UNCONSCIOUS PATIENT
III. Ingestion of any harmful substance. E.g. Kerosene, acid or petrol	Irritable, Restless, No mouth burns or swelling May or may not be warm to touch	Moderate condition	1. Identify the poison Milk Drink or Mist Magnesium Trisilicate 5mls 8hourly for 3days 2. REFER
IV. Aspirin poison with or without fever or other signs	Temp. normal or above 37.5 0C Irritable Thirsty Restlessness	Moderate condition	1. Give Milk drink or Mist Magnesium Trisilicate 5mls 8hourly for 5days 2. Give plenty of fluids 3. Review appropriately, 4. if no improvement REFER

Health Education

Advise parent /Caregiver on:

1.Prevention of home accidents.
2.Adequate nutrition and effect on healing.
3.Avoid induced vomiting.
4.Proper storage of chemical and harmful liquid
5.Prevention of bites and stings.
6.Importance of taking home-based records to the clinic or hospital.

3.29 SCHOOL HEALTH SERVICES

The child has certain rights, which are recognized nationally and internationally, treaties and conventions. These include the right to education and access to health and other social services. Six years is the age when the Nigerian child commences primary school education in public schools. The Nigerian child has the right to quality health care which should be provided in the school health programme including routine physical examination, screening tests e.g. for hearing and vision, preventive and therapeutic dental services, management of minor ailments, immunizations, health education, provision of school meals, health promoting physical exercises and referrals.

HISTORY	EXAMINATION/ACTIONS
1.Is there any problem? 2.What type of food do you eat? (ask about main meals and snacks) 3.How many brothers and sisters do you have? 4.Whom do you live with (parents, grandparents, spouse, others, specify) 5.How many times do you eat in a day? Where do you get your food? 6.What type of work does your father do? Mother? 7.What is your source of water? 8.What type of toilet facility do you have in your house or hostel? Have you had any serious illness in the past?	1.General appearance: well nourished, ill, tired, clean, thin, congenital abnormality 2.Weight and height: weigh, measure and record 3.Head: lice, ringworm. 4.Eyes: discharge, pallor, jaundice, cataract, test of vision. 5.Nose: discharge, blockage, abnormality. 6.Ears: redness, discharge, test of hearing. 7.Mouth: odour, caries, missing teeth. 8.Gum: healthy, bleeding. 9.Throat: redness, inflamed tonsils. 10.Chest: respiratory rate, wheezes, stridor, lower chest indrawing. 11.Heart: rate, rhythm, murmur. 12.Abdomen: scars, tenderness, rigidity, masses, palpable spleen and liver. 13.Skin: dryness, sores, rashes, multiple scars or wounds (think child abuse). 14.Limbs: deformity, paralysis, oedema. 15.Pulse, blood pressure: check and record. 16.Urine: sugar, protein. 17.Haemoglobin: estimate and record. Temperature: check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I.Well, first visit	No abnormality	Healthy child	1. Encourage caregiver to maintain status quo 2. 2. Confirmed the completion of Immunization of the child Refer
II.Well visit (School Food Vendor)	No Physical Abnormality	Healthy individual	Refer vendor for the following test at least every six months Chest X-ray Sputum test for AF Blood for widal test Stool for ova and parasites If any abnormal finding, stop from providing food until treatment given and completed.

Health Education

Advise on;

1.Personal hygiene
2.food hygiene
3.Environmental sanitation
4.Safety education
5.Importance of taking home-based records to the clinic or hospital.

3.30. SEXUALITY EDUCATION

Sex education is high quality teaching and learning about a broad variety of topics related to sex and sexuality, Children, especially older ones need to have correct information about their anatomy, physiology, intimacy, sexualization reproduction and developmental changes. They should be able to handle their psycho-sexual feelings in a healthy manner and therefore require counseling.

HISTORY	EXAMINATION/ACTIONS
1.Has she started menstruation if a girl? 1.2 At what age did she experience menarche? 4Are you living with your parents? If No Who? 5Have you been sexually molested by anybody before? 2.Any other thing you may want to share with me\ 3.When did you notice changes in your physical appearance? 4.Have you had sexual intercourse before? Do you have wet dreams?	1General appearance: healthy, ill looking. 2Check temperature, pulse, respiration and blood pressure. 3Eyes: pallor, jaundice. 4Heart: Pulse rate, rhythm. 5Axilla presence of hair. 6Breast developed size. 7Pelvic public hair, vulva for perforate/imperforate hymen. 8Haemoglobin estimate and record. Weight and Height: measure and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Abdominal pain with menstruation; Foul smelling discharge per vagina; Fever	Lower abdominal tenderness, High temperature	Moderate condition Suspect infection	1. Reassure patient. 2. Counsel on personal hygiene (use of sanitary pad). 3. Prescribe Cap. Amoxicillin 250mg 8 hourly for 5 days. 4. Tab; Buscopan 10mg daily x3days 5. If RDT is positive, treat for uncomplicated malaria. Review in 5 days, 6. if no. improvement 7..REFER
II. Heavy menstrual flow with clots, prolonged menstruation	May be pale Blood clots per vagina	Moderate condition Abnormal menstrual flow	1. Reassure and counsel patient. Tab. Paracetamol 250500 mg 8hourly for 3days.
III. Lower abdominal pain associated with menstruation	Lower abdominal tenderness, no fever, No foul smelling discharge	Mild condition Suspect dysmenorrhea	1. Tab. Paracetamol 250500 mg 8hourly for 3days 2. Reassure and counsel patient. 3. Tab; Buscopan 10mg daiyx3days
IV. Mucus or blood per vagina	Normal mucus, no smell Normal menstrual flow	Mild condition	1. Counsel and reassure patient.
V. Anxiety about new relationship with the opposite sex	Normal	Mild condition	1. Reassure, counsel on physiology and developmental changes and advise on the negative consequences of early sexual relationships.
VI No sexual abuse	Healthy child	Normal	Intensify health education.

Health Education

Advise Parent /Care giver on:

1 Personal hygiene.
2 Adequate nutrition.
3 Sex education.
4 Implication of teenage pregnancy.
5 Use of LLINs.
6 Importance of taking home-based records to the clinic
7 How to have healthy relationship
8 Informed decision about sex.
9 Loving one's self.

SECTION FOUR

ADOLESCENT HEALTH (10 – 19 Years)



NATIONAL STANDING ORDERS

FOR COMMUNITY HEALTH OFFICERS

4.0 ADOLESCENT HEALTH

PREAMBLE

Adolescence is the period of physical, psychological and social changes from childhood to adulthood. It spans at the ages of 10 to 19 years.

It is subdivided into two periods: early adolescence (between 10 - 14 years) and late adolescence (between 15 - 19 years). It should also be noted that the term "young people" covers the ages between 10 – 24 years. True adolescence, therefore, is a transition that may fall anytime between years and hence whenever health status or behaviours is being considered, "youths" are quite often included among "adolescents" leading to the addressing of both categories as "young people".

It is acknowledged globally that this group form an important part of any society and in Nigeria, young people aged 10 - 24 constitute at least 30% of the population. It is also widely known that adolescence is a period of learning, excitement and overall increased prevalence of high risk behaviours.

Rapid societal changes place adolescents in a situation where they are confronted with conflicting and at times contradictory social expectations. Transition to adulthood often involves period of stress, innovation, experimentation and disorganization.

Some of the disorders with grave consequences amongst adolescents include unwanted pregnancy, rape, unsafe abortion, sexually transmitted infections including HIV/AIDS, drug and substance abuse and other behavioural changes.

To reduce these consequences, it is important to have adolescents-friendly clinic where they are provided amongst other things, information on their health needs, counselling especially during crisis situations and have access to health services. This section addresses the issues that are peculiar to adolescents. Other conditions are discussed in the child/adult section.

4.1. ADOLESCENTS CONCERNS ABOUT PUBERTY (FEMALE CONCERNS)

The period of development between the onset of puberty and adulthood (10 - 19 years), involves physical growth of an adolescent from a child into an adult. A common concern is how to distinguish between "normality" and "abnormality".

HISTORY	EXAMINATION/ACTIONS
1How are you? 2How may I help you? 3When did you notice changes in your physical appearance (presence of breast, axillary and pubic hair) 4Are you comfortable with the size of your breast, any family history of large/small breast? 5Have you had sexual intercourse before? 6Have you started menstruating? 7When was your last menstrual period? For how many days? 8Do you have abdominal pain with menstruation? 9Family history (onset of puberty, chronic illness, stress) 10What are your feeding habits like? Any other thing you may want to share with me?	1General appearance: Healthy, ill, thin, obese, anxious. 2Check Temperature, Pulse, Respiration & Blood Pressure and record, 3Eyes: Pallor, jaundice 4Axilla: Presence of hair 5Breast: Developed, Size 6Abdomen: Tenderness, palpable uterus, masses. 7Pelvic: Pubic hair, vulva for perforate/imperforate hymen, check flow if menstruating. 8Skin: Sweating, cold and clamy , oily, acne, 9Heamoglobin: estimate and record. 10Weight and height:: measure and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Early onset of physical changes of sexual maturity about age 8 years	Presence of breast, axillary and pubic hair Menstruation	Mild to Moderate condition	1. Fears and Counsel the client/parent on physiological changes of the adolescents
II. No menstruation in a child 16 years or older	Hymen may be imperforate or perforate	Moderate condition	1. Allay fear, counsel and REFER
III. No menses for more than one month after onset, abdominal pain with or without vaginal bleeding, dizziness	Abdominal tenderness, with or without vaginal bleeding, pallor, fast and weak pulse, cold and clammy hands and legs, sweating	Severe condition Suspect ectopic pregnancy	1 . Counsel and R EFER immediately
IV. No menses for more than one month after onset	Palpable uterus	Mild condition Suspect pregnancy	1. Conduct pregnancy test. Explain condition to client and parent. 2. Encourage ante natal care.
V. Prolonged and heavy menstrual flow	Heavy flow with clots May or may not be pale	Moderate to severe condition Suspect heavy menstruation	1. Explain the condition to client and parent 2. Check Hb; if below 10g/dl 3. Give Tab. Folic Acid 5mg daily for 2 4. Tab; Fasolate bd daily
VI. Scanty Menstrual flow	Small quantity of blood on pad May or may not be pale	Mild condition Suspect scanty menses	Explain condition to client and parent. Check Hb, if below 10g/dl give, Tab. Folic Acid 5mg daily for 2 weeks Ta; Fasolate bd dailyx7days
VII. No menses for more than one month. has not had sexual intercourse	uterus not palpable	Mild condition	1. Counsel and allay fears
VIII. Abdominal Pain with menstruation	May or may not have lower abdominal tenderness No mass in abdomen	Mild to Moderate condition Suspect dysmenorrhoea	1. Explain condition to client and parent 2. Encourage regular exercise 3. If below 12 yrs, give IM. Hyoscine 20mg stat then Tab Hyoscine 10mg 12hourly for 3 days or Tab.
IX. Large breasts	Anxious Excessively large breasts	Mild condition Macromastia	1. Explain condition to client (that it's normal and she will get used to a changing body).
X. Small breasts	Anxious Small Breasts No other abnormality	Mild condition	1. If pubertal signs of development are normal and has started menstruating, counsel to allay fears. If other pubertal characteristics are present;

XI Unequal breast size	Anxious No other abnormality detected	Mild condition	1. Reassure her if unequal breast size is during puberty (when the breasts are still developing). Advice if the breasts are of unequal size at the end of puberty.
XII Bedwetting	*Child is anxious & Depressed	Mild condition Nocturnal Enuresis is also called involuntary urination	1. Counsel caregiver and client on consiquences and reassure them 2. Wake them up to empty bladder. 3. Teach them early urination before bed time. 4. Reduce intake of plenty water at night

Health Education

Dietary advice/Nutritional counselling

1 Personal hygiene.
2 Solicit social & psychological support.
3 Use of LLIN.
4 Sexuality education.
5 Importance of taking home-based records to the clinic or hospital



4.2 ADOLESCENTS CONCERNS ABOUT PUBERTY (MALE CONCERNS)

The period of development between the onset of puberty and adulthood (10 - 19 years), involves physical growth of an adolescent from a child into an adult. A common concern is how to distinguish between "normality" and "abnormality".

HISTORY	EXAMINATION/ACTIONS
1How are you? 2How may I help you? 3When did you notice changes in your physical? appearance (pubic hair, axillary hair, breaking of voice) 4Any swelling/pain in the scrotum? 5Have you had sexual intercourse before? 6Any urethral discharge? 7Are you comfortable with size of your penis? 8Do you have wet dreams? 9Family history (onset of puberty, chronic illness, stress).	1.General appearance: Healthy, ill, thin, obese, anxious 2.Check Temperature, Pulse, Respiration & Blood Pressure and record 3.Eyes: Pallor, jaundice 4.Axilla: Presence of hair 5.Scrotum: presence of testes, swelling, tenderness 6.Penis: Size, deformity, pubic hair, uretheral discharge

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Early Onset of puberty at about age	Any of the following may be present: pubic hair, large penis,	Mild condition	1. Explain condition to client & parent. 2. Allay fears.
II. Late onset of pubertal changes	May or may not have family history May or may not be small for age.	Mild condition Suspect delayed	1. Explain condition to client & parent. 2. Allay fears. Follow up.
III. One or both testes missing	Anxious. One or both testes missing in the scrotum Scrotal sac retracted Scrotal sac empty	Moderate condition Suspect undescended testis	1. Explain condition to client & parent. 2. Allay fears.
IV. Painless swollen scrotum	The presence of fluid in scrotum	Moderate condition	1. Explain condition to client & parent. 2. Allay fears.
V. Scrotal swelling with pain with or without urethral discharge	Client in distress Swollen tender scrotum Presence or absence of urethral discharge	Severe condition Suspect	1. Give Tab. Paracetamol 1000mg 8 hourly for 3 days. 2. Give Caps Amoxicillin 500mg 8 hourly for 5 days. 3. Refer if no improvement
VI. Scrotal pain with or without	• Client in distress • Scrotal tenderness	Severe condition Suspect testicular torsion	1.REFERimmediately

VII. Small penis	Anxious Penis may or may not be relatively small	Mild condition	1. Reassure client and parent that size of penis does not affect sexual function.
VIII. Bed wetting	Child is anxious and depressed No other abnormality detected	Mild condition Suspect Enuresis	1. Counsel clients, parents/ guardian:Advice urinating before going to bed 2. Support by waking up to urinate

Health Education

Dietary advice/Nutritional counselling

1. Personal hygiene.
2. Solicit social & psychological support.
3 Use of LLIN.
4. Sexuality education.
5. Importance of taking home-based records to the clinic or hospital.



Figure 2- Adolescent changes to notice

4.3 CONTRACEPTIVE CHOICE FOR ADOLESCENTS

It is important to note that while abstinence should be encouraged and emphasized amongst adolescents due to the spread of HIV infection, many are already sexually active or have had sexual intercourse at least once. Helping adolescents make responsible and healthy choices in terms of the various contraceptive methods that are appropriate in preventing unwanted pregnancy is therefore crucial. Myths and misconceptions should also be dispelled and clarified.

HISTORY	EXAMINATION/ACTIONS
1.Can you tell me what you know about contraceptives? 2.Have you used any contraceptive method before (if yes, which one did you use)? 3.Which clinic did you attend? 4.Why do you want to attend this clinic? (Who referred you)? 5.Do you have regular sexual intercourse? (How many partners)? 6.Do you have any other problem?	1.General appearance: Healthy, ill, thin, obese. 2.Check Temperature, Pulse, Respiration & Blood Pressure and record. 3.Eyes: Pallor, jaundice. 4.Heart: Pulse rate, rhythm. 5.Abdomen: Tenderness, palpable uterus, masses. 6.Weight and height: measure and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Request for contraceptive advice	*No abnormalities Detected	Normal condition	1.Discuss sexuality in detail. 2.Avoid situations that can provoke sexual arousal 3.Advise abstinence as the best choice 4.Discuss various methods of contraception such as condoms (male and female), diaphragm, spermicidal, vaginal foaming tablet, pills, injectable 5.Demonstrate the proper use of CONDOM (male and female) in preventing HIV/AIDS, STIs
II. HIV positive adolescent requesting for contraceptive	• HIV positive • Looks healthy	Severe condition	1.Give adequate counseling on condom (male and female) use plus 2nd contraceptive device (i.e. dual protection). 2. use Medical Eligibility Criteria (MEC) Wheel for Contraceptive Use 3.Ensure confidentiality

Health Education

1.Counsel on choice of contraceptive (emphasizing abstinence).
2.Discuss all methods of contraception.
3.Discuss the side effect and complication of each of them.
4.Nutritional counseling.
5.Sexuality education.
6.Advise on personal hygiene.
7.Importance of taking home-based records to the clinic or hospital.

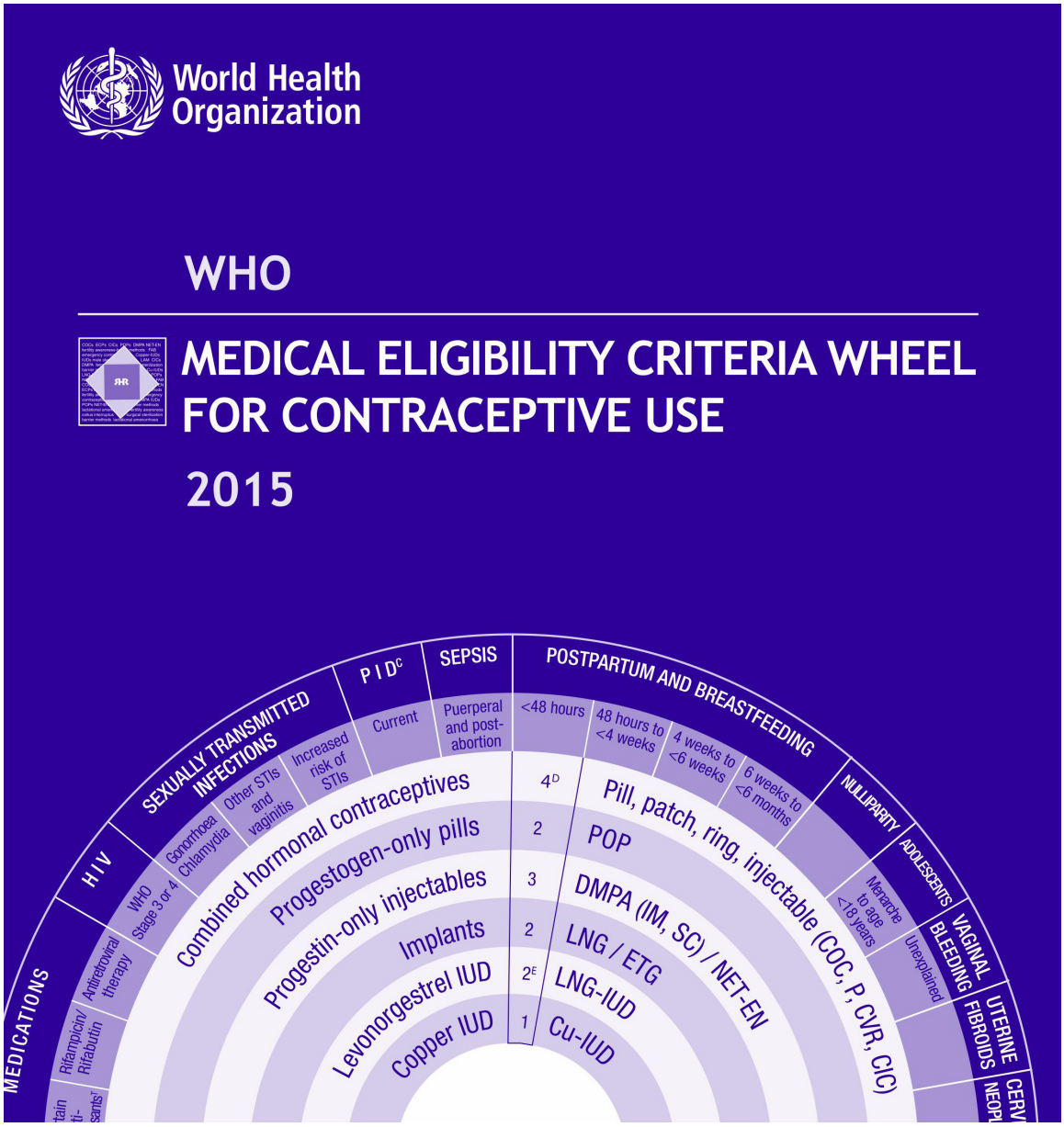


Figure 3- WHO criteria wheel for contraceptive use

4.4 SEXUAL DYSFUNCTIONS/MALFUNCTIONS

Sexual dysfunction/malfunction is difficulty experienced by an individual or a couple during any stage of a normal sexual activity, including physical pleasure, desire, preference, arousal or orgasm. There is a need to understand the human sexual response cycle to be able to help a young person with a disorder of sexual dysfunction. It may be as a result of fear, guilt, anger, anxiety, relationship difficulties or medical disorder and many of these can be solved through honest communication.

HISTORY	EXAMINATION/ACTIONS
1.What is your view about sexual intercourse? 2.Do you have a sexual partner? If yes, 3.How often do you engage in sexual practice? 4.Do you have erectile disorder? 5.Do you experience any pain during sexual intercourse? 6.Any history of sexual abuse? 7.Drug/substance history 8.How long does it take you to ejaculate? Do you have any other problem?	1.General appearance: anxious looking; angry; fearful; healthy. 2.Check the vital signs (Temperature, Pulse, Respiration & Blood Pressure). 3.Penis: turgid and erect, flaccid. Vagina: check for bleeding, bruises, tear.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
Prolonged painful erection may or may not be associated with sexual stimulation	Penis turgid and erect rigid	Severe condition. Suspect Priapism	1. Give Tab. Paracetamol 1000mg 8hourly for 3 days or Tab Ibuprofen 400mg 12hourly for 5days (After food) 2. if no improvement REFER
II. Painful intercourse may or may not have bleeding	Anxious looking	Moderate condition Suspect dyspareunia	If bleeding, REFER Counsel against pre-marital sex, dangers of STIs and HIV/AIDS Advise on the use of contraceptives Advise on sexual positioning
III. Failure to have an erection while with partner	Anxious Healthy looking	Mild to Moderate condition Suspect psychological erectile dysfunction or infection	Explain condition to client Explain that client should exercise patience when about to have sex Counsel on abstinence and risk of STIs/HIV After Review if no improvement Refer
IV. Quick ejaculations	Healthy looking May or may not look anxious	Mild condition Suspect premature ejaculation	Explain condition to client Encourage physical exercise Advise use of condom to reduce sensitivity Counsel on abstinence and HIV/AIDs Tab Diazepam 5mg stat

Health Education

1.Counsel on sexuality education.
2.Advise regular exercises.
3.Nutritional counseling.
4.Advise on personal hygiene.
5.Importance of taking home-based records to the clinic or hospital.

4.5 GENITO-URINARYTRACT DISORDER

Genito-urinary tract disorders are disorders that develop along a patient's urinary or genital tract, or both, in some cases. They are very common, especially among sexually active people and those with certain risk factors like a history of genitourinary anomalies. Genitourinary disorder can be caused by gonorrhea, syphilis, non-specific bacterial infection and schistosomiasis. Patient may complain of discharge, pain and burning sensation on micturition and blood in urine.

HISTORY	EXAMINATION/ACTIONS
1.How long have you experienced the problem? 2.Do you have pain, burning sensation when passing urine? 3.Do you pass urine normally? Is it more or less frequent? 4.What is the colour of your urine-is it reddish, cloudy, dark or yellowish? 5.Do you have urethral discharge? 6.Do you have pains on your flanks? 7.Is there swelling of your legs? Do you have shortness of breath, nausea or	1.General appearance: ill, puffy face, pale. 2.Temperature. Pulse, respiration &blood pressure, check and record. 3.Eyes: check for pallor and peri-orbital oedema 4.Chest: check breath sounds. 5.Abdomen: distension, check for flank tenderness 6.External genitalia: any abnormality, redness, swelling, discharges. 7.Limbs: check hands and feet for oedema. Urine: colour, odour, protein, cloudiness and blood.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Urinary problem with any of the following condition: Shortness of breath	Chest not clear Brown or dark urine. Swelling of legs or Abdominal distension	Severe condition Suspect acute renal failure	REFER immediately
II. Pain or burning sensation on micturition, Painful Flank,	Protein in urine Cloudy urine	Moderate to severe condition	REFER
III. Pain or burning sensation on micturation,	• urine plain colour Dysuria (pain during urination)	Mild condition Suspect STIs	1.Use syndromic Management of STI's Encourage a person to drink more fluids as this dilutes urine, making it less painful to pass. Resting and taking medications as directed can usually help relieve most symptoms.
IV. Blood in Urine (terminal) may or may not have painful micturition	Blood in Urine	Moderate condition Suspect schistosomiasis	REFER for bilharzia test If Schistosomiasis test is negative, REFER If positive give Praziquantel 60mg/kg/day in 3 divided doses or Metrifonate 4mg tab every 2 weeks rmaximum of 3 doses if necessary
IV. Cloudy or dark yellow urine without any other symptoms	Dark yellow urine No jaundice	Mild condition	1.Encourage fluid intake 2.Reviewin3days

Health Education

1.Advise on personal hygiene.
2.Advise on protection of genital area.
3.Advise on drug compliance.
4.Encourage adequate fluid intake.
5.Counsel on side effects of medications.
6.Importance of taking home-based records to the clinic or hospital.

4.6 ABORTION

Abortion is the termination of a pregnancy before the age of viability y(i.e. before the pregnancy has attained the age at which when delivered, the baby will be able to survive. In our environment, that age is 28 weeks). Abortion can occur spontaneously (unprovoked) or maybe induced (deliberately interfered with). Unsafe induced abortion is a common phenomenon among adolescents and young adults in Nigeria and has led to many cases of infertility and maternal deaths. .It is usually illegal and involves pregnancy termination by untrained person, or performed in an inappropriate place or the use of harmful instruments. Most spontaneous abortions occur in the first 12 weeks of pregnancy.

HISTORY	EXAMINATION/ACTIONS
1.When was your Last Menstrual Period (LMP)? 2.Is there bleeding? When did it start and for how long? Any injury? 3.Did you expel any blood clots/products of conception? 4.Is there abdominal pain? 5.Is there any vaginal discharge? Does it smell? 6.Do you have fever?	1.General appearance: ill, weak, pale 2.Temperature, Pulse, Respiration & Blood pressure: measure and record 3.Vagina: check for bleeding and discharge, blood clots/products of conception 4.Haemoglobin: estimate and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. History of bleeding that is subsiding without abdominal pain, Developing embryo completely expelled	Bleeding subsiding Temperature: normal No abdominal tenderness	Moderate condition. Suspect complete abortion.	1. Give Post abortion contraception 2. Give Tab. Paracetamol 1000mg 8hourly for 3 days 3. Tab. Metronidazole.400mg 8hourly for 5days. 4. Cap. Amoxicillin 500mg 8hourly for 5 days 5. Counsel to minimize emotional and physical effects of abortion Encourage abstinence.
II.Pain in the abdomen, Severe bleeding	*Abdominal tenderness *Bleeding from the Vagina	Severe condition Suspect incomplete abortion	Patient is not pale, her vital signs are normal and the uterus is not palpable in the abdomen (pregnancy was less than 12weeks in age). Counsel the patient and perform a Manual Vacuum Aspiration (MVA) Tab. Paracetamol 1000mg 8hourly for 3 days Tab. Metronidazole.400mg 8hourly for 5days. Cap. Doxycycline 100 500mg 12 hourly for 5 days
III. Severe pain in the abdomen, Nausea/vomiting with	Severe Abdominal tenderness with/without ascites	Severe condition	Set up intravenous infusion of Dextrose-saline or apply anti-shock garment on her and Refer

IV. Feverish, Foul smelling discharge from the vagina, Trauma.	Temp. 37.50C or above Abdominal tenderness with or without bleeding from the vagina. Foul smelling discharges from the vagina.	Severe condition Suspect septic abortion	1. IV Metronidazone 500mg 8hourly for 24 hoursStat 2. Give Ringers lactate 500mg 6hourly for 24 hours solution 3. Tab. Metronidazole.400mg 8hourly for 5days. Tab. Paracetamol 1000mg 8hourly for 3 days Tab. Ciproflaxacin 500mg 12hourly for 5days, Advise on personal hygiene. REFER

Health Education

1.Advise on personal hygiene
2.Sexuality education.
3.Advise on abstinence
4.Counsel on the use of contraceptive
5.Advise on adequate fluid intake.
6.Counsel on the dangers of unsafe abortion
7.Importance of taking home-based records with anytime she goes to the clinic or hospital.

4.7. NUTRITIONAL DISORDERS

Teenagers have high nutritional needs. At this stage, the bone develops to its maximum density and this (growth) depends on adequate nutrition. Adequate nutrition also provides protection against diseases. The nutritional status of an adolescent like other group is influenced by socio-economic factors.

The adolescents are very active physically and their nutritional requirements depend on such factors as physiological state, level of activities, stress, etc. Peer group influences also affect their eating habits especially girls. Therefore efforts should be made to ensure adequate nutrition for this group.

4.7.1 NUTRITION DISORDERS OF ADOLESCENT

HISTORY	EXAMINATION/ACTIONS
1Have you been eating well? 2What food are you eating? (Probe type of food for a 24 hour period)? 3What is your favourite food? 4Did have appetite? 5Are you comfortable with your weight? 6Did you have significant weight loss over several days, weeks or months? 7Have you experience vomiting after eating? 8Are you pregnant (if Female)?	1. General appearance: obese, thin, pale, anxious or aggressive looking, restless, moody, unkempt, mode of dressing 2. Orientation: person, place, time 3. Eyes: redness, pallor of the conjunctiva, dilated pupils 4. Mouth: smell of alcohol and other substances 5. Speech: slurred, incoherent in nature 6. Skin: for dryness 7. Perception: hallucination, thought disorders 8. Pockets: drugs in the pocket 9. Temperature, Pulse, Respiration & Blood pressure: check and record 10. Weight and height: measure and record (calculate Body Mass Index – BMI and or MUAC) 11. Urine test for pregnancy

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1. Excessively fat with or without increased appetite	Obese (BMI: 30 kg/m2 and above) May or may not be depressed	severe condition Suspect Obesity	Encourage exercise Nutritional counseling If no improvement, REFER
Excessively fat	Overweight (BMI: 25 to 29..9 kg/m2)	Moderate condition Suspect Over weight	Encourage exercise Nutritional counseling If no improvement, REFER
Excessively fat	Malnutrition ≥ 18.5 to < 25.0 Normal	Normal to mild	Encourage exercise Nutritional counseling If no improvement, REFER
Girl/boy losing weight Significant weight loss over several days, weeks or months Induces vomiting after eating	Thin, BMI Nutritional status < 16.0 Severe malnutrition or MUAC< 185 mm < 190 mm (SAM) May look anxious Intense fear of gaining weight. Shortness of breath. Feeling overweight or "fat," even if you're underweight. Fear of certain foods or food groups. Experiencing thoughts of self-harm or suicide. May or may not be pale	severe condition Suspect Anorexia nervosa	1. Explain condition to client and parent 2. Nutritional counseling 3. Conduct food demonstration 4. Tab. Vitamin B complex 1tab 8 hourly for 2 weeks 5. Tab. Folic acid 5mg daily for 2 weeks 6. Try to elicit the reason for inducing vomiting and counsel Approximately 7. If no improvement, REFER

Adolescent weight loss with any of the following	BMI Nutritional status ≥ 16.0 to < 17.0 Moderate malnutrition or MUAC ≥ 185 to < 220 mm Bloating and/or abdominal pain. Muscle weakness and loss of muscle mass. Dry skin, brittle nails and/or thinning hair. Poor wound healing and frequent illness. Bluish or purple coloring of the hands and feet.	Moderate condition Suspect Anorexia nervosa	Explain condition to client and parent Nutritional counseling Conduct food demonstration Tab. Vitamin B complex 1tab 8hourly for 2 weeks Tab. Folic acid 5mg daily for 2weeks Try to elicit the reason for weight loss and counsel Approximately If no improvement, REFER
Adolescent weight loss with any of the following	≥ 17.0 to < 18.5 Mild malnutrition MUAC ≥ 190 to < 230 Dizziness and/or fainting. Feeling tired. Slow heartbeat (bradycardia) or irregular heartbeat (arrhythmia). Low blood pressure (hypotension). Poor concentration and focus. Feeling cold all the time. Absent periods (amenorrhea) or irregular menstrual periods.	Mild condition Suspect Anorexia nervosa	1. Explain condition to client and parent 2. Nutritional counseling Conduct food demonstration 3. Tab. Vitamin B complex 1tab 8 hourly for 2 weeks 4. Tab. Folic acid 5mg daily for 2 weeks 5. Try to elicit the reason for weight loss and counsel 6. If no improvement, REFER

Health Education

1. Counsel on dangers of not eating balance diet.
3. Advise on personal hygiene.
4. Nutritional counseling.
5. Counsel on negative peer influence.
6. Importance of taking home-based records with anytime he/she goes to the clinic or hospital.

4.7.2 DRUG AND SUBSTANCE ABUSE

A substance could be a drug, toxin, which its abuse alter the mind, perceptions, feelings, moods and behavior of an adolescent.

The most common substance/drugs that are abuse:

• Nicotine e.g cigarettes, Caffeine e.g kolanut and coffee, miraa, rohypnol

• Cannabis e.g. Indian hemp, Heroin, cocaine, amphetamines,

• Solvents e.g. herbs, solution, petrol, gum, glue, energy drinks, Paint thinners or removers, degreasers, dry-cleaning fluids, gasoline, lighter fluids, correction fluids, permanent markers, electronics cleaners and freeze sprays, glue, spray paint, hair or deodorant sprays, fabric protector sprays, aerosol computer cleaning products, vegetable oil sprays, butane lighters, propane tanks, whipped cream aerosol containers, refrigerant gases, ether, chloroform, halothane, nitrous oxide, prescription nitrites etc.

• Hallucinogens (LSD), e.g. calamine or alum, epinephrine, Ketamine

• Barbiturates and sedatives e.g. benzodiazepines (Diazepam, Lexotan)

• Medications drugs e.g. Panadol, Sudrex (Paracetamol+caffeine), Codeine, pentazocine, tramadol, rohypnol, diazepam, codeine, cough syrup. etc.

• Aphrodisiacs or libido boosters.

Note during history taking: Denial is a common element of substance abuse and addiction, the adolescent may not easily admit to substance use because it will lead to loss of benefits from parents, guardians and families.

HISTORY	EXAMINATION/ACTIONS
1What substance do you take that may make you eat well, or sleep, or stay awake for long hours? or strong and active to perform better in skills/studies/work? 2What job are you doing? Student, physical work etc. 3What was going on in your life that makes you start using/drinking substance/drugs? 4What kind of substance/drugs are you using? 5When you use (or drink), how do you feel? 6When was your last use substance/drugs? 7What were the circumstances surrounding your last substance/drugs use? 8Do you ever feel guilty or ashamed after substance/drugs use? 9Has anyone ever expressed concern about your drinking habits? 10Have you ever stopped or tried to limit your substance/drugs use? 11Have you ever tried to quit using substance/drugs? If so, how long ago was your last attempt? 12Do you have any medical conditions that could be exacerbated by using substance/drugs? 13Have you ever experienced withdrawal symptoms when you stopped using substance/drugs? Do you have any other problem?	1. General appearance: pale, anxious or aggressive looking, restless, moody, unkempt, mode of dressing 2. Orientation: person, place, time 3. Eyes: redness, pallor of the conjunctiva, dilated pupils 4. Mouth: smell of substances etc. 5. Speech: slurred, incoherent in nature 6. Skin: check for needle pricks 7. Perception: hallucination, thought disorders 8. Pockets: search for drugs in the pocket 9. Temperature, Pulse, Respiration & Blood pressure: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1. Chronic drugs/substance intoxication	Restless Aggressive, violence Anxiety depression Bloodshot eyes (miosis). Sudden weight loss severe loss of appetite Interrupted sleep patterns: insomnia (difficulty sleeping at night) & or Hypersomnia (sleeping for too long). Disorder affecting: liver, kidney and heart disorder or cancer, Seizures, stroke, and brain damage	Severe drugs or substance abuse Chronic drugs intoxication/addiction	Counsel the patient, close friends and family as Addiction is treatable. It's crucial to seek help as soon as possible. REFER to rehabilitation centres
ii. Moderate drugs intoxication	Secretive behavior: The person withdraws from friends and family. Mental confusion Change in complexion: Repeated drug use can lead to acne, paleness and jaundice. There may also be scabs, scars, bruises and track marks on the body.	moderate drugs or substance abuse	Invite/counsel close friends and family encourage Cessation of substance use REFER to rehabilitation centres
iii. Acute drugs intoxication	Slurred speech Smell of substances Unexplained change in personality or attitude. Sudden mood swings, irritability, or angry outbursts. Poor personal hygiene: Bathing, brushing teeth and physical cleanliness may decline in anyone who is drug dependent Financial distress: (borrowing money without a reason, sell their belongings to get money for drugs). Ignoring responsibilities: avoid their daily responsibilities such as work, school or taking care of the home. Dental health disorder: dental carries Reckless behavior, isolation	Moderate condition Suspect substance abuse	Give activated charcoal If not available, give milk drink, Invite and counsel close friends and family on the dangers of substance abuse. friend or loved one should help to observe the patient under round-the-clock encourage Cessation of substance use as it help to minimize the physical harm caused by the abuse of substances. If no improvement REFER to rehabilitation centres

4.7.3 ALCOHOLISM

Alcohol use disorder:Also called alcohol dependence or alcohol addiction: is a chronic condition characterised by uncontrolled drinking and preoccupation with alcohol, resulting to inability to control alcohol drinking due to physical and emotional dependence.

HISTORY	EXAMINATION/ACTIONS
1How much alcohol do you drink? 2How often do you drink? 3What kind of alcohol do you drink? 4When was your last drink? 5What were the circumstances surrounding your last drink? 6Do you ever feel guilty or ashamed after drinking? 7Has anyone ever expressed concern about your drinking habits? 8Have you ever stopped or tried to limit your drink? 9Have you ever tried to quit using alcohol ? If so, how long ago was your last attempt? 10Do you have any medical conditions that could be exacerbated by using alcohol? 11Do you have any mental health conditions that could be exacerbated by using alcohol? Have you ever experienced withdrawal symptoms when you stopped using alcohol?	1General appearance: pale, anxious or aggressive looking, restless, moody, unkempt, mode of dressing 2.Orientation: person, place, time 3.Eyes: redness, pallor of the conjunctiva, dilated pupils 4.Mouth: smell of alcohol 5.Speech: slurred, incoherent in nature 7.Skin: check for needle pricks 8.Perception: hallucination, thought disorders 9.Pockets: alcohol in the pocket 10.Temperature, Pulse, Respiration & Blood pressure: check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Chronic Alcoholic Dependence	A patient who is dependent on alcohol, may drink: a lot of the time first thing in the morning every evening every day and with any of the following: Slurred or incoherent speech Thought disorders Looking Restless Aggressive, Violence Visual hallucinations or seizures. shaking hands (tremor)	Severe drugs or substance abuse Chronic drugs intoxication/addiction	Counsel the patient, close friends and family as Addiction is treatable. It's crucial to seek help as soon as possible. REFER to rehabilitation centres
ii. Moderate Alcoholic Dependence	Moody, Hallucinations-evidence of perception disorder seizures (fits) anxiety feeling anxious and upset feeling irritable or angry difficulty sleeping (insomnia)	Moderate to severe Alcoholic Dependence	Counseling of patient and family Detoxification (also known as 'detox') can be a key stage of treatment. Detox involves stopping drinking completely REFER to rehabilitation centres
iii. Acute Alcoholic Dependence	Looking unkempt Altered orientation Physical injury physical withdrawal symptoms, such as shaking (tremor), sweating and sickness (nausea) You may also have more severe symptoms, such as feeling sick (nausea) sweating	Mild alcoholic dependence	Counsel the patient, close friends and family as Addiction is treatable. It's crucial to seek help as soon as possible. REFER to rehabilitation centres

4.8 MENTAL HEALTH DISORDERS

Mental health disorder in adolescents are those illnesses with psychological or behavioral manifestations and/or impairment in functioning due to social, psychological, genetic, physical or biological disturbance. Mental disorders are characterised by abnormalities in a person's emotions, thoughts, cognition, sensory perceptions, beliefs and behaviour.

Mental disorder usually manifest in adolescents in many ways.

The affected may become restless, afraid, suspicious and confused. They may think that people are after them. They may imagine/or complain of seeing strange people or things.

Mental health disorder are common, A mental health disorder may start out as subtle changes to a person's feelings, thinking and behaviour. If they have ongoing and significant changes, it could be a sign that they are developing a mental health disorder. If something doesn't seem 'quite right', it's important to have a conversation about getting help.so it's important to be aware of possible signs.

They may become violent and attack other people or themselves without cause. The condition may have come on after a fever due to malaria, typhoid, or viral infection; it may also be as result of severe emotional disturbance or abnormal emotional experiences or excessive drinking of alcohol or drugs over a significant period.. If nothing is done immediately, there may be danger to life and property. Other symptoms include sleeplessness, hearing and seeing things others cannot see and suicidal tendencies.

HISTORY	EXAMINATION/ACTIONS
2What is the problem? 3Does he feel worried, depressed, guilty, worthless or feeling an exaggerated sense of 'high' 3.How long has he/she had the problem? 4.Has he/she had this or similar problem in the past? 5.Does the client use alcohol or drug such as Indian hemp? 6.How is he/she functioning at school/work? Does he/she have problem at school or workplace? 7.What is his/her relationship with his wife or any family members? 8.Is he/she in any love relationship? How steady is the relationship? 9.Is there a recently stressful event? 10.Does he/she sleep well? Or have Changes in sleep? 11.Does he/she eat well? Or have Changes in weight? 12.Any abnormal behaviour, speech or experience (e.g. excessive spending)? 13.Has he/she caused or threatened injury or thoughts or threats of suicide to others or himself / herself? Or is engaging in high-risk activities? 14.Has he/she had recent encounter with police, public?	1.General appearance: unkempt, calm, restless, threatening or physically aggressive, ill, 2.withdrawn, confused, tensed, frightened, fidgety, trembling/ shaking, sweaty, cold, worried, depressed, guilty, worthless or feeling an exaggerated sense of 'high clammy palms, acting strangely, tearful 3.Reaction to the health worker – cooperative, withdrawn, aggressive 4.Eyes: redness, staring, bulging, lid retraction, pallor 5.Mouth: smell of alcohol or other substances Speech: audible, shouting, understandable, meaningful, relevant to the questions asked or to the situation, slurred, rambling, excessive, stammering, hesitant, expressing worry, any strange statements or beliefs expressed 6.Orientation: responds to name, knows where he is, has idea of time 7.Perception: seeing or hearing things other people cannot see or hear 8.Pockets: drugs or other substances in the pocket 9.Signs of tension – suspicious, hyper-alertness 10.Thoughts: expression of hopelessness, suicidal tendencies, Don't ignore thoughts or threats of suicide 11.Weight: weigh and record Temperature, Pulse,

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
Feelings of irritability, frustration, or restlessness. Suicidal tendencies	Chronic feeling of emptiness, sadness, or inability to feel pleasure that may appear to happen for no clear reason. Thoughts of death or suicide, or suicide attempts Suicidal tendencies Persistent depressed mood irritability restlessness	Severe condition Suspect depression	1. Counsel client/family member separately and jointly 2. Group therapy 3. Involve in family activities 4. Monitor closely 1.5. Keep patient away from danger 2.6. If no improvement, REFER
1. Stressful life events such as: a) Parental disharmony b) Examination failure c) Depressed mood	Nervous Jittery Excessive worry Sad, chronic pain, headaches, and digestive issues.	Moderate condition Suspect depression	1. Counsel client/family member separately and jointly 2. Group therapy 3. Involve in family activities 4. Monitor closely 5. Keep patient away from danger 6. If necessary, intervene at home/ school 7. Home visits for follow-up 8. If no improvement, REFER
Persistent sad, anxious, or "empty" mood. Feelings of hopelessness or pessimism. Breaking of love relationship	Worry Tearful loss of interest or pleasure in hobbies and activities changes in appetite and body weight unusually slow or agitated movements decreased energy or fatigue difficulty sleeping or oversleeping excessive feelings of guilt or worthlessness difficulty concentrating or making decisions	Mild condition Suspect depression	1. Counsel client/family member separately and jointly 2. Group therapy 3. Involve in family activities 4. Monitor closely 5. Keep patient away from danger 6. If necessary, intervene at home/ school 7. Home visits for follow-up 8. If no improvement, REFER
II. Complaint of any of the following: a) Excitement, Irritability b) Sexual promiscuity, c) Excessive spending	Restless Talkative Irritable Withdrawn Weight loss	severe condition Suspect social maladjustment Chronic adjustment disorder (means symptoms last six months or longer.)	REFER to rehabilitation center
Complaint of any of the Following: Poor sleep, Talkative Changes in appetite	Loss of energy & tiredness excessive truancy, substance abuse, and/or	Moderate condition emotional or behavioral symptoms developed within three months of the start of a stressful event in your life.	REFER to rehabilitation center

Complaint of any of the Following: a persistent pattern of violating social norms" with a perpetual struggle with authority. Family causes. broken homes physical, emotional or mental disorder Hard time keeping up socially when compared to their peers. School related causes.	Talkative Moody Weight loss	Mild condition Acute adjustment disorder means your symptoms last less than six months.	1. Medications: Medications can help if depressed, anxious, or having trouble sleeping, 2. Tabs Fluoxetine 2.5mg stat 3. Counsel client/family member separately and jointly 4. If necessary, intervene at home/school 5. if no improvement REFER to rehabilitation center
III. Behavioural disorders such as: Stealing b) Truancy c) Aggression to people d) Destruction of property	Evidence of Drug use/abuse Aggressive	Severe condition Suspect behavioural disorder	Tabs Diazepam 30mg stat Counsel client/family member REFER
Tendency to use drugs, including cigarettes and alcohol, at a very early age. being aggressive to animals and other people	Poor interpersonal relation-ship Drug use/abuse sadistic behaviours including bullying and physical or sexual abuse.	moderate Behavioural disorders	Counsel client/family member separately and jointly If necessary, intervene at home/school for close monitoring REFER to Rehabilitation centre
equent refusal to obey parents or other authority figures. repeated truancy. lack of empathy for others	Easily getting annoyed or nervous. Often appearing angry. Putting blame on others. Refusing to follow rules or questioning authority.	Mild Behavioural disorders	Tabs diazepam 30mg stat Counsel client/family member separately and jointly If necessary, intervene at home/school REFER

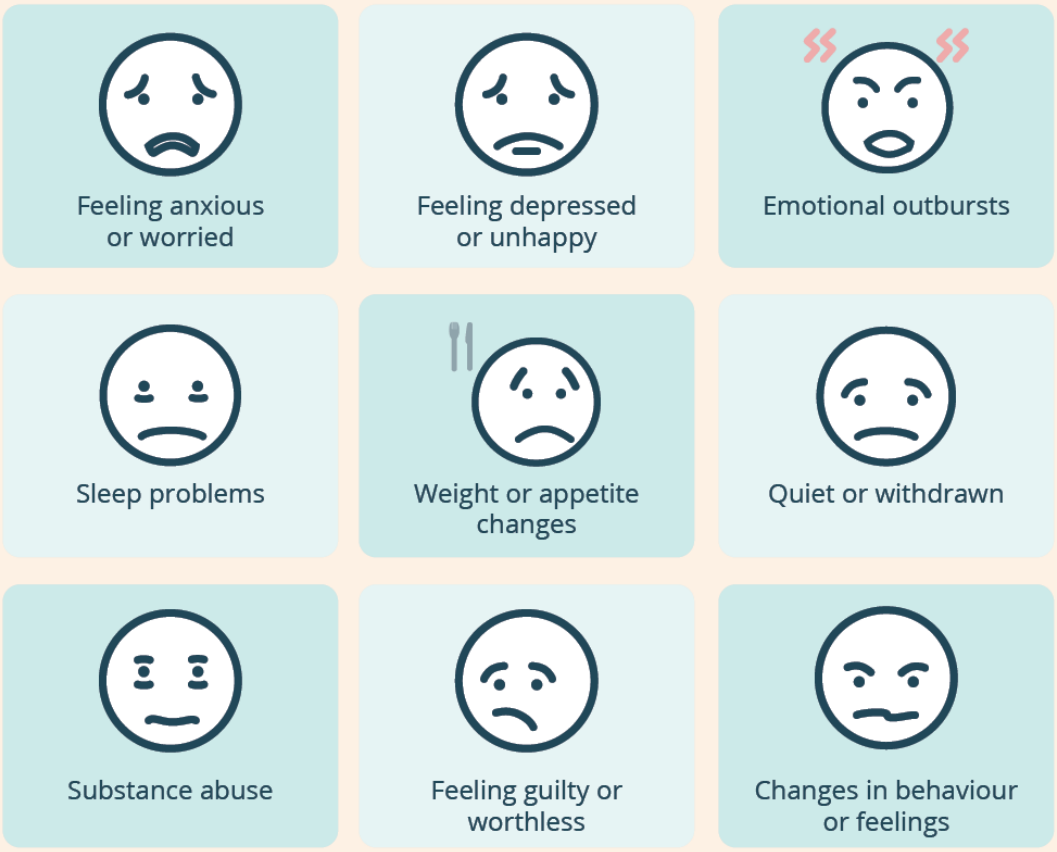
Health Education

1. Counsel on drug compliance.
2. Counsel on social behavior.
3. Counsel on dangers of drug/substance abuse.
4. Advise on personal hygiene.
5. Building a support system.
6. Engage your family, friends and peer groups to support.
7. Maintaining a healthy lifestyle: Eating healthy foods and exercising regularly.
8. Importance of taking home-based records to the clinic or hospital.

Nine signs of mental illness

healthdirect

What are the symptoms of a mental illness? If a friend or loved one doesn't seem themselves, how do you spot the difference between a bad mood and something more serious?



If you think there is an immediate risk of suicide or self harm, dial **triple zero (000)**

If you're concerned about a friend or loved one, ask them how you can help.
The first step for them should be to see a doctor or other healthcare professional.

Figure 4- Nine signs of mental illness

4.9. VIOLENCE AND ASSAULTIVE BEHAVIOURS

Violence is an outwardly directed destructive behaviour either to persons (including self) or to properties. It might be behaviour learned from the home (family) and can occur in mentally and non-mentally ill persons. It may arise from disturbance of thoughts, feelings or emotions and in adolescence, from a variety of reasons such as alcohol and other substance abuse and diseases affecting the brain.
Violence and aggression can be defined as a set of activities that may lead to harm to other persons. It can be expressed in actions or words, but the physical damage remains, and the purpose is clear.

Sexual Violence and Abuse

This is any form of sexually related activity that takes place without consent being given and may range from just touching to rape. Sexual abuse goes beyond sexual intercourse alone and includes pornographic materials, writings, verbal comments or non-verbal actions e.g. winking, licking lips, etc.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. How long has it been noticed? 3. What caused the problem? 4. Is he on any medication? If yes what type? 5. Any injury threatened or inflicted? 6. Has he/she had similar behavioural disorder in the past? 7. Any recent encounter with police or public? 8. Is there fever, stresses/or recent life disorder? 9. Is there any history of alcohol or other substance abuse? 10. Is there any exposure to pornographic materials? 11. Is there any history of fits or head injury?	1. General appearance: unkempt, restless, threatening or physically aggressive, ill, depressed, Malnourished, withdrawn 2. Level of consciousness: respond to sound, touch and light 3. Mouth: smell of alcohol or other substances 4. Speech: slurred, rambling, excessive, 5. Orientation: responds to name, knows where he is, has idea of time 6. Signs of tension – suspicious, hyper-alertness 7. Evidence of rape: e. g. blood stain, semen and physical injuries 8. Temperature, Pulse, Respiration & Blood pressure: check and record 9. urine: pregnancy test, STIs 10 blood; HIV test

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Uncooperative violent and aggressive patient at home	Irreversible harm to the health of a person; Person inflicting serious harm to himself or to others; and Person causing serious damage to property belonging to himself or to others where such behavior is believed to flow directly from the person's mental illness.	Severe condition Violence and assaultive behaviours	Emergency treatment duration is to be considered up to 72 hours from the initial evaluation by a registered Community Health Practitioner/psychiatrist. Tabs Chlopromazine 200mg dailyx3 days Close monitoring
ii. Drug abuse or substance abuse with violence and assaultive behaviours	Drug effects and withdrawals – amphetamine Substance abuse disorder (Acute intoxication/withdrawal syndrome) Alcohol/opium/ opiates/cannabis and other substances	Moderate to severe Violence and assaultive behaviours	1. Physical restrain, patient should be lying down on the bed. Both upper and lower extremities should be tied with soft clothes or bandages. 2. Constant monitoring of patients during mechanical restraint is a must. 3. If constant monitoring is not possible, the patient has to be visually observed for at least 15 minutes of restraint time and should be within eyesight. And keep watch at every 10–15 minutes for the remaining hours. 4. Counsel family members. 5. REFER immediately

III.Violence and assaultive behaviours with any of the following;;- Aggressive and Sexual disorders	Acts of physical violence. Shouting, swearing, and harsh language. Gossiping or spreading rumors about a classmate. Purposely breaking your roommate's favorite mug. Slashing your co-worker's tires.	Mild violence and assaultive behaviours	1. Counsel family members. 2. REFER immediately
iv. Uncontrolled anger a) Alcohol intoxication and substance abuse. b) Emotional outbursts e.g. impulse control disorders.	Ill looking, Aggressive Sign of recent drug use Unkempt Level of consciousness: response to sound, touch, light Mouth: smell of alcohol or other substances Tense, suspicious, hyper-alert Speech: Slurred, rambling, excessive. Orientation: Identify person, time, place. Pair influence ,violent bahaviour,assault,aggressive	Severe condition Suspect psychosis	1. Give Post-Exposure Prophylaxis (PEP) if available 2. Counsel the family/client 3. REFER for Therapist 4. Follow-up visit
II. Sexual abuse (Rape)	Anxious Withdrawn Sad Evidence of rape e.g. blood stain, semen or physical injuries	Severe condition	1. Give Post-Exposure Prophylaxis (PEP) if available 2. Record and report all cases of rape to Law enforcement agencies 2.3. Counsel the family/client 3. Emergency birth control. It is important to receive birth control and treatment for STDs within 72 hours of the assault for maximum effectiveness.Victims can get emergency contraception up to 5 days after3.4. REFER to appropriate Health facility 4. Follow-up visit
Infections, Disturbed Blood Glucose Levels, Acute Situational Reaction, Metabolic and Electrolytic Disturbances	Systemic Sepsis, Lung Diseases Survivors of Sexual Assault	Moderate to severe condition	Counsel the family/client Treat according to cause If no improvement REFER 4. Follow-up visit

Health Education

1. Advise patient not to harbour grievances or bitter revengeful thoughts.
2. Advise against alcohol and other substance abuse.
3. Advise patient not to get too angry as this may result in loss of self-control
4. Solicit religious/traditional leaders to help in counseling.

4.10.1. STIs (FEMALE)

Sexually Transmitted Infections (STIs) are infections that are spread through sexual intercourse.
Common STIs include Gonorrhoea, syphilis, herpes, chlamydia, trichomoniasis, candidiasis, genital warts, HIV/AIDS etc. STIs can be painful and uncomfortable. They can also have tragic consequences such as pelvic inflammatory diseases and infertility.
Therefore, all cases of STIs should be adequately treated. The following scenarios for the individual treatment are as follows: poor compliance – such as a patient taking a 7-day or 21-day course of doxycycline for chlamydial infections, including for lymphogranuloma venereum; reinfection – perhaps because sex took place, without a condom, with an untreated sex partner or a new partner; antimicrobial resistance – this is of particular importance in gonococcal and M. genitalium infections since antimicrobial resistance in N. gonorrhoeae and M. genitalium are being experienced with recommended treatments for these infections; and the presence of an untreated infection – such as T. vaginalis and/or M. genitalium among men with urethral discharge treated only for N. gonorrhoeae and C. trachomatis at the first visit.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. Do you have a discharge? 3. How long have you had the discharge? Is it excessive? What is the colour? Is it smelling and itching? 4. Do you soil your under wear? 5. Do you feel any burning sensation or pain on urination? 6. Do you have lower abdominal pain? If yes, for how long? 7. When was your last menstrual period? 8. Do you have any sore or wound or painful blisters in the genitalia? Any bleeding from the vagina? 9. Is there any history of sexual intercourse, pregnancy, delivery or abortion? 10. Is there fever? 11. Are you on any medication? 12. Risk assessment (see appendices)	1. General appearance: normal, uncomfortable, irritable, anxious 2. Temperature, Pulse, Respiration & Blood pressure: check and record 3. Abdomen: tenderness, masses 4. Pelvic examination: Vulva – soreness, painful blisters, redness (Speculum) Urethra – discharge Vagina – redness, discharge (colour, smell) Cervix – redness, discharge (smell, colour, purulent) Sore

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I) Vaginal discharge Itching with or without painful urination with or without lower abdominal pain	Profuse muco-purulent discharge from cervix Risk Assessment positive	Severe condition. Suspect Cervicitis	1. Give Tab. Ciprofloxacin 500mg (single oral dose) 2. Give Cap. Doxycyline 100mg 12hourly for 7 days 3. Review in 7 days 4. If no improvement, REFER
II) Vaginal discharge Profuse, thin, watery, Creamy, or thick whitish discharge with or without itching Painful Urination.	Discharge is Profuse, Thin, watery, Creamy or thick whitish Risk Assessment negative	Severe condition. Suspect the following: Candidiasis Vaginitis Trichomoniasis	Nystatin Vaginal pessaries 100,000 units x 14 days OR CLOTRIMAZOLE vaginal pessaries 100mg inserted at night for 6 days AND Tab. Metronidazole 400mg 8hourly for 7 days orally Review in 7 days If no improvement, REFER

IV) Lower abdominal pain with any of the following present: a) Late or missed period (menses) b) Recent delivery/abortion c) Vaginal bleeding d) Fever	Vaginal discharge Tenderness on moving cervix Temperature 37.50C or above	Moderate to Severe condition. Suspect Pelvic Inflammatory Disease (PID)	Give Tab. Ciprofloxacin 500mg (single oral dose) Give Cap. Doxycyline 100mg 12hourly for 14 days Tab. Metronidazole 400mg 8hourly for 14 days Review in 7 days If no improvement, REFER
V) Vaginal discharge with or without genital sore/ulcer or painful blister a) sexual exposure with or without groin swelling	Group of painful blisters (often recurrent) Risk Assessment positive	Moderate condition. Suspect Genital Herpes, HIV/AIDS	Tab. Acyclovir 400mg 8hourly for 7 days (also in pregnancy) Tab. Ibuprofen 400mg 8hourly for 3days (After food) Keep lesion dry HIV Counseling and Testing (HCT) Avoid sex during treatment 4Cs (Counsel, Compliance, Condom and Contact treatment). Review after 7 days If no improvement, REFER

Health Education

1.Advise on the 4Cs (Counsel, Compliance, Condom and Contact treatment).
2.Advise on personal hygiene.
3.Importance of taking home-based records to the clinic or hospital.

4.10.2 STIs (MALE)

Sexually Transmitted Infections (STIs) are infections that are spread through sexual intercourse. Common STIs include gonorrhoea, syphilis, herpes, genital warts, HIV/AIDS etc.
STIs can be painful and uncomfortable. They can also have tragic consequences such as infertility. Therefore, all cases of STIs should be adequately treated.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem 2.Do you have a discharge? 3.How long have you had the discharge? Is it mucoid, purulent, profuse or 4.scanty? 5.Have you ever had urethral discharge before? If yes, how long ago? 6.Do you notice the discharge after a particular sexual intercourse? 7.Have you had any sexual contact after you noticed the discharge? If yes, with who? 8.Do you feel any burning sensation or pain on urination? 9.Is there fever? 10.Are you on any medication? Risk assessment (see appendices)	1.General appearance: normal, uncomfortable, irritable, anxious 2.Temperature, Pulse, Respiration & Blood pressure: check and record 3.Throat: redness, pus 4.Skin: rashes 5.External genitalia – discharge ("milk" the urethra) 6.Uncircumcised male – retract the foreskin, be sure the discharge is from the urethra 7.Check glands for redness, ulcer. 8.Check scrotum for sores, redness, swelling, (just look, DO NOT PALPATE) 9.Check anus for purulent discharge Groin: swelling, tenderness

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I) Groin (inguinal swelling) with or without pain a) Patient not sexually active b) Under 15 years of age c) Skin rash	Inguinal swelling with or without tenderness Skin rashes	Severe condition Suspect Granuloma Inguinale	1. Give Tab. Erythromycin 500mg 6hourly for 7 days 2. Give Benzathine Penicillin G 2.4mu Intramuscularly (IM) in a single dose 3 Review in 4 days 4 If no improvement, REFER
II) Scrotal swelling with or without pain a) Urethral discharge b) Sexually active	Scrotal swelling Urethral discharge	Severe condition Suspect Urethritis	1. Give Tab. Erythromycin 500mg 6hourly for 7 days 2. Give Cap. Doxycyline 100mg tab 12hourly for 7 days 3. Review in 4 days 4. If no improvement, REFER
III) Genital sore or Ulcer a) History of sexual exposure b) Swelling in the groin c) Urethral discharge with or without painful micturition	Sore/Ulcer present Swelling in the groin Risk assessment positive	Moderate to Severe condition . Suspect Genital Ulcer Disease (GUD)	1. Give Tab. Erythromycin 500mg 6 hourly for 7 days 2. Give Benzathine Penicillin G 2.4mu Intramuscularly (IM) in a single dose 3. Review in 7 days 4. If no improvement, REFER
IV) Patient with painful micturition, no discharge	No penile discharge	Mild condition Suspect Urinary Tracts Infection (UTI)	encourage a person to drink more fluids as this dilutes urine, making it less painful to pass. Resting and taking medications as directed can usually help relieve most symptoms. Tabs. Ibuprofen 400mg 8hourly for 5days (After food) Tabs. Nitrofurantoin 100mg 8hourly for 5days 2. Review after 5 days 3. If no improvement, REFER

V) Genital sore/Ulcer History of sexual exposure With or without painful swelling in the genital	Group of painful blisters (often recurrent) present Risk Assessment Positive	Severe condition. Suspect genital Herpes, HIV/AIDS	1. Tabs Acyclovir 400mg 8hourly for 7 days (also in pregnancy) 2. Tabs Ibuprofen 400mg 8hourly for 3days 3. Keep lesion dry 4. HIV Counseling and Testing (HCT) 5. Avoid sex during treatment 6. 4Cs (Counsel, Compliance, Condom and Contact treatment). 7. Review after 7 days 8. If no improvement, REFER

Health Education

1. Advise on the 4Cs (Counsel, Compliance, Condom and Contact treatment).
2. Advise on personal hygiene.
3. Importance of taking home-based records to the clinic or hospital.

4.10.3 HIV/AIDS (MANAGEMENT OF PATIENT)

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. How long have you had the problem? 3. Are you losing weight? 4. Have you had infections? 5. Have you had blood transfusion? Scarification? 6. Do you have casual sexual partner or sexual partner of same sex? 7. Have you had contact with relatives or partners with similar problem? 8. Have you been treated for HIV/AIDs? 9. What other disorder do you have? Any cough? Diarrhoea? Repeated infections? Any fever?	1. General appearance: healthy, wasted , pale 2. Temperature, Pulse, Respiration & Blood pressure: check and record 3. Weight: weigh and record 4. Hair: thin or scanty 5. Eyes: sunken, pale conjunctiva 6. Mouth: thrush, red and white patches 7. Neck and Armpit: enlarged nodes 8. Skin: rashes, ulcers, fungal infection 9. Chest: crepitation, decreased breath sounds 10. Abdomen: palpable organ 11. Groin: enlarged nodes, ulcers 12. Genitalia: ulcers, rashes, discharges. 13. HIV screening after pretest counseling

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I) History of recurrent or persistent diarrhoea of more than 1 month duration a) Generalized itching, skin rash b) unexplained cough longer than 1 month c) Recurrent genital and mouth sores d) Prolonged fever for more than one 1 month e) Severe weight loss f) General weakness g) Ulcer that has refused to heal	Severe dehydration Temperature 37.50C or above May or may not have weight loss greater than10% of body weight Generalized dermatitis Oral candidiasis Genital ulcer Enlarged generalized lymph nodes HIV test positive	Severe condition	1. Symptomatic treatment 2. Give Paracetamol 1000mg 8hourly for 5 days 3. Give ORS solution for dehydration 4. Home-based care includes: a. Provision of counseling to patient and family members on compliance b. Provision of pastoral support/ care c. Prevention of opportunistic infections by encouraging patient to visit the H/F as soon as any complication/complaint arise 5. Management of opportunistic infections 6. REFER
II)Chronic diarrhoea Chronic cough Mouth and genital ulcer Oral thrush Skin rash	Weight loss General skin rashes Enlarged lymph mode Mouth and genital ulcers Laboratory test positive	Moderate to Severe condition	REFER
III) Claim to be HIV positive patient No other complaint	Laboratory result may or may not be positive and have weight loss.	Mild condition.	Counseling Refer for confirmation test.

Health Education

1. Advise on positive lifestyle e.g. avoidance of stress.
2. Advise on the 4Cs (Counsel, Compliance, Condom and Contact treatment).
3. Advise on personal hygiene.
4. Importance of taking home-based records with anytime he/she goes to the clinic or hospital.

4.11. INJURIES, FRACTURES AND BITES

Injuries occur in different part of the body tissues. They may be on the skin and/or deeper tissues which may be small or large and may include cutting of nerves and tendons. There could be wounds that have been for several days and have become infected or healing poorly; or Sores which are injured, infected or diseased part of the skin. Some may be Ulcers which are injuries in a bodily membrane.

A fracture is a break in the continuity of a bone. It is more serious if there is also a compound fracture, profuse bleeding resulting from wounds. There is an urgent need to stop bleeding to avoid the patient getting into shock. Early cleaning and treatment of wounds or sores is important to prevent infections including tetanus. Adolescent are prone to injuries, fractures, dislocations, wounds, bites from snakes, dogs, scorpions or humans.

Some injuries involve the head which comprises of the brain and the sense organs (for hearing, sight, smelling and tasting). Serious injury to the head may be caused by a fall or a

blow to the head. Some signs of this injury are severe headache, unconsciousness, vomiting, confusion, bleeding from the nose, mouth or ears. Delay in prompt care and/or referral of a patient with head injury may cause permanent disability or death

HISTORY	EXAMINATION/ACTIONS
1. How did the injury occur? 2. Was there any bleeding? 3. Has it bled much? (Try to carefully assess the amount) 4. Has anything been put on the wound? 5. Have you been immunized with Tetanus Toxoid? Number of doses? 6. Did anyone see it happen to describe the incident? 7. What bit him/her? A Dog, A Snake, A person? Scorpion? Or any other thing? 8. Where is the pain? How bad is it? 9. Can you move/use limb? Is there limitation of movement? 10. Which movement causes greatest pain? 11. Was there loss of consciousness? Irritability? Fits? 12. Do you have headache, nausea, or vomiting? 13. Is there any other thing you like to tell me or any additional injuries?	1. General appearance: a. Conscious, confused, irritable, unconscious b. Pale, cold or clammy c. Increased sweating. d. Increased respiration and pulse rate e. Patient agitated or restless 2. Check the vital sign: Temperature, Pulse, Respiration & Blood pressure and record 3. Head: Visible wound, swelling, bleeding a. Ear: Bleeding b. Eyes: Subconjuctival hemorrhage, pupils for size reaction to light c. Mouth: Teeth, fractures, bleeding d. Nose: Bleeding, fractures 4. Hemoglobin: Estimate and record 5. Chest: Ribs for fractures 6. Abdomen: For tenderness, rigidity 7. Urine: Monitor output WOUNDS: 8. Size and site 9. Bleeding. 10. Pus or dirt in wound. 11. Swelling, redness or warmth around wound. SPRAINS, dislocation AND FRACTURES 12. General appearance: a. Signs of shock – increased sweating, pallor, 13 restlessness, rapid pulse and rapid respiration. b. Degree of pain. c. At Limb or joint. d. Tenderness or swelling. e. Warmth or redness. f. Crepitus (crackling). g. Associated Open wound.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Wound on the limb resulting from injury	Evidence of fracture (unable to move the limb) BP below 110 / 70 mmHg or above Lethargic	Severe condition Suspect fracture	1. Immobilize the affected limb 2. Dress wound 3. Give Tetanus Toxoid 0.5ml IM stat 4. Caps Amoxicillin 500mg 8hourly for 5days
II Wound as a result of deep laceration	Bleeding from wound, Torn tendon Check hemoglobin	Moderate to Severe Condition Suspect laceration	1. Clean well with available diluted antiseptic solution 2. Control bleeding by pressure 3. Give Tetanus Toxoid 0.5ml IM stat 4. Perform stitches and dress the wound were possible 4. If hemoglobin is less than 10g/dl 5. REFER immediately
III Wound from dog bite	Evidence of bite Wound (wound from fangs seen at site of bite) With or without bleeding	Mild condition Suspect dog bite	1. Wash wound with soap or detergent/ diluted antiseptic solution 2. Advise patient to observe the dog for 10 days and report any change in the dog's behaviours or death of dog. 3. Give anti-rabies injections 1ml on days 0, 3, 7, 14 and 28 on the deltoid. Depending on the manufacturer's instruction. 4. Tab Ibuprofen 400mg 8hourly for 3 days (After food) 5. IM. Tetanus Toxoid 0.5ml stat
IV Wound from human Bite	Wound at the site of bite	Moderate condition Suspect human bite	1. Wash with diluted antiseptic solution or normal saline 2. IM. Tetanus Toxoid 0.5ml stat 3. Caps Amoxicillin 500mg 8hourly for 5days
V History of snake bite	Wound from fangs seen at the site of the bite Swelling of the affected part Bleeding gum	Severe Condition Suspect snake envenomation	1. Keep client calm, clean area thoroughly. 2. Give anti-snake venom e.g. 10mls stat Snake venom antiserum sensitivity test is done before by administering 0.1ml of the serum to the victim. (read and follow manufacturer's instructions) 3. IV Hydrocortisone 100mg stat
VI . Scorpion/Bee/Wasp sting	Site of sting swollen Itchy May be red	Moderate to severe condition Suspect scorpion/Bee/Wasp sting	1. IV Hydrocortisone 100mg stat 1.2. Administer local Anesthesia Lignocaine at the site of the sting 2.3. Tab Ibuprofen 400mg 8hourly for 3 days (After food)

VII . Ulcer with hypo-pig-mented	*Spots with or without history or sign of leprosy *Ulcers, particularly	Severe condition	1.Wash ulcers thoroughly with warm salt water 2. Clean and dress daily 3. REFER immediately
VIII. History of wound	* Greater than 4cm * No signs of shock Evidence of infection (septic wound)	Moderate condition Suspect wound	1. Clean wound with/diluted anti-septic solution 2. IM. Tetanus Toxoid 0.5ml imme-diately (stat) 3. Tab Ciprofloxacin 500mg 12hourly for 5days
III Wound from dog bite	*Wound less than 4 cm * No signs of shock *Without evidence of infection	Mild condition Suspect wound	1. Clean wound with diluted antiseptic solution 2. Stop bleeding by pressure 3. Daily dressing with Sofratulle or Vaseline gauze 4. IM. Tetanus Toxoid 0.5ml immediately (stat)
X. Injury to the head	*Bleeding from ear or nose *May or may not be conscious * Stiff neck	Severe condition Suspect fracture of the skull	1.Asses rapidly using the Glasgow coma method (GCS) method 2. REFER immediately
XI. Injury to the head with transient loss of consciousness	*With or without wound seen * Pupils reacting normally to light	Moderate to severe condition	1.Asses rapidly using the Glasgow coma (GCS) method 2. Tab Paracetamol 1000mg tds x 3days if conscious 3. REFER immediately
XII. Injury to the head without	*Pupils reacting normally to light	Moderate to severe condition	1. Monitor vital signs hourly for 4 to 6 hours1.2. Glasgow coma method(GCS) 2. If condition deteriorates, REFER
XIII. Injury to the head, No other	* Conscious * No other signs of head injury	Mild condition	1. Check vital signs and advice to return immediately if vomiting or headache or drowsiness occurs.1.2. Glasgow coma method(GCS) 2. REFER if patient returns.

Health Education

1. Advise on use of protective clothing to avoid bites.
2. Importance of compliance with treatment.
3. Give nutritional counseling.
4. Advise on environmental sanitation.
5. Advise on reducing risk of accident.
6. In cases of snake bite, if possible, take along killed snake to the clinic.
7. Advise on prevention of head injuries e.g. use of crash helmets, and seat belts.
8. Prevention of home and road accidents.
9. Importance of taking home-based records during visit to the hospital.

4.12. NECK DISORDERS/INJURIES

Neck problems can result from physical injury or infections (tuberculosis, meningitis, HIV/AIDS etc.) or it is common among adolescents, which include an injury (fall from high object, slippery, or domestic accident resulting to effect to the structures in the neck This may manifest as enlarged nodes and/or lumps around the neck which may associate with stiffness of the neck. Lumps can be located in the center or front of the necks (for example goiter).

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. Do you have difficulty swallowing? Hoarseness? 3. Do you have pain and/or stiff neck? Neck swelling? 4. Do you have pain or headache? 5. Did you injure your neck? 6. Do you have fever? 7. Do you have weakness or numbness of the arms, hands or legs? 8. Is there weight loss or night sweats?	1. General appearance: Ill-looking, distressed. 2. Check the vital sign: Temperature, Pulse, Respiration, Blood Pressure and record 3. Estimate weight and record 4. Head: Abscesses, sores or pustules, lumps a. Ears: Signs of infection. b. Mouth: Abscesses or sores 5. Neck: Swelling, nodes, stiffness, tenderness 6. Arms and Legs: Weakness or decreased sensation.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Neck swelling with or without difficulty in breathing	Palpable mass in the front of the neck	Severe condition	REFER immediately
II. Pain and stiff neck without history of fever or injury	Stiff Neck No weakness of arms and legs	Moderate condition Suspect torticollis	1. Reassure client 2. Tabs Ibuprofen 400mg 8hourly 3 days (After food)
III. Enlarged, painful nodes around the neck	Palpable, tender nodes	Severe condition Suspect infection of	1. Caps Amoxicillin 500mg 8hourly 5 days 2. Tab Metronidazole 400mg 8hourly 5 days

Health Education

1. Advise on prevention of neck injuries.
2. Advise on complications of mumps in males.
3. Advise on prevention of transmission to others.
4. Give oral hygiene instructions.
5. Importance of taking home-based records during to the clinic.

4.13 EAR, NOSE AND THROAT DISORDERS

The ear, nose and throat are linked and whatever affect one may affect the other. The ear is used for hearing and to maintain a balance, and any disease of the ear which is not urgently and adequately treated may not only lead to deafness, but also lead to inability to maintain balance.

Ear problems commonly caused by middle ear infection often accompanied by pain, dullness and redness of the ear drum with drainage of pus or fluid infections of the nose, throat or Eustachian tube may spread to the middle ear. It is important to examine the bone behind the ear if it is painful (tender to touch); it may mean that the infection has spread to the mastoid bone. Testing an adult for hearing, one is advised to sit behind the client and ask him to repeat words or numbers you whisper to him.

The nose is the organ for sense of smell, and for breathing. When the nose is affected by disease or injury, it discharges watery fluid (mucus, blood or pus.) Blocking of the nose may
lead to breathing difficulties.

The throat is the junction of the ear, nose and mouth. It contains the tonsils, pharynx, etc. Diseases affecting the throat can easily spread to the chest and brain.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? 2. How long have you had the problem? 3. Is there catarrh? 4. Is there pain? Boil? Any discharge: pus, blood or water? 5. Do you feel pain while swallowing? 6. Is there fever? Watery eyes? Headache? 7. Has it happened before? 8. Has there been any blow or injury to the head? 9. Do you have nose bleeding? Can you estimate the volume? When did it start? 10. Are you having bleeding from any other part of the body? 11. Is there pain from the teeth or gum? 12. Do you notice bad or foul breath? 13. Is there diarrhea or vomiting? 14. Do you work in a noisy environment or with noisy machines? Have you put anything into your ears, nose and throat? Did anything enter your ears, nose or throat? 16. Is there any trauma to the ear, nose or throat?	1. General appearance: Ill-looking, in distress. 2. Check the vital sign: Temperature, Pulse, Respiration and Blood Pressure and record 3. Ear: (a) Pus or blood discharge; (b) Red or bulging drum; (c) Perforated drum, foreign body (d) Cerumen (Wax); (e) Bone behind the ear (mastoid) for tenderness. 4. Eye: Redness, watery discharge 5. Nose: Mucus or catarrh 6. Neck: Stiffness or enlarged lymph nodes. 7. Throat: (a) Redness (b) Enlarged tonsils, pus. (c) Hoarseness or loss of voice. 8. Chest: Crepitations, wheezes, decreased breath sounds 9. Test for hearing and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Ear problem with vomiting	*Temperature 37.5C or above	Severe condition Suspect Mastoiditis	1. REFER to appropriate Health facility
II. Boil in external ear	*Soft, tender swelling with well-defined margins	Moderate to severe condition Suspect boil	1. Caps Amoxicillin 500mg 8hourly for 5 days 2. Tabs Ibuprofen 400mg 8hourly for 3 days (After food)
III. Pain in the ear with or without pus or water	*Temperature 37.5C or above Red, dull bulging ear drum	Severe condition Suspect Acute Supurative	1. REFER immediately

IV. Pain in the throat, Difficulty with swallowing	*Enlarged and red tonsils	Moderate condition: Suspect Tonsillitis	1. Cap Amoxicillin 500mg 8hourly for 5days 2. Tab Metronidazole 400mg 8hourly 5 days 3. Tab Ibuprofen 400mg 8hourly for 3days (After food)
V. Pus from the nose, headache, Pain	*Tenderness of the nose	Moderate to severe condition Suspect abscess	1. Caps Amoxicillin 500mg 8hourly for 5 days 2. Tabs Ibuprofen 400mg 8hourly for 3days (After food)
VII. Stuffy nose Catarrh with or without fever Sneezing with or without water	*Temperature 37.5 0 C or above *Watery eyes Catarrh	Mild condition Suspect Sinusitis	1. Tabs Chlorpheniramine maleate 4mg 12hourly 3days. 2. Tabs Paracetamol 1000mg 8hourly for 3 days 3. Tabs Vitamin C 200mg 8hourly for 7 days 4. Tabs Co-trimoxazole 960mg 12hourly for 5days 5. Advise to avoid or reduce exposure to dust or pollen 6. Review in 3 days
VIII. Pain in the throat Fever Inability to swallow Sore throat	*Throat inflamed * Tonsils not red and no pus	Mild condition Suspect pharyngitis	1. Tab. Ibuprofen 400mg 8hourly for 3days (After food) 2. Caps Amoxicillin 500mg 8hourly for 5days 3. Tabs Metronidazole 400mg 8 hourly for 5 days 4. Tabs Vitamin C 200mg 8hourly for 7 days 5. Tabs Strepsil 1bdx3days 6. Tabs Prednisolone 5mg bdx3days

Health Education

1. Advise on personal and environmental hygiene.
2. Advise on care of the ear, nose and throat.
3. Discourage traditional practices
4. 1,Advice on personal and environmental hygiene
5. Advice on the care of Ear,Nose and throat
6. Discourage traditional practices such Uvulectomy and tonsillectomy
7. Give nutritional counseling.
8. Advise on avoiding exposure to air pollution (fumes, dust and pollen)
9. Discourage home removal of foreign body.
10. Importance of taking home-based records during visit to clinic.

SECTION FIVE

ADULT HEALTH (10 – 19 Years)



NATIONAL STANDING ORDERS

FOR COMMUNITY HEALTH OFFICERS

5.0 ADULT HEALTH

This refers to an adult often recognized as somebody who is matured, well developed, a fully grown and is 18 to 59 years and above. Section 5.1 is for adults who present with no complaints. The CHO should be able to screen for health disorders in the well adults. Note that, other health conditions affecting adolescents (13 years and above) and some conditions affecting elderly are also treated in this section

5.1. ADULT FIRST VISIT

HISTORY	EXAMINATION/ACTIONS
1.How are you feeling? 2.If ill, what are your symptoms? 3.Are you taking any medication or treatment? If so, please describe? 4.Have you lost weight recently? 5.Have you had any operation(s)? If yes, what and when? 6.Are you able to cope at home? 7.Do you smoke, take snuff or alcohol? OBSTETRIC 1Have you been pregnant before? If yes, how many times? 2What was the nature of your delivery-CS, SVD or forceps delivery? 3Have you had abortion before? If yes, how many times? 4Are you pregnant now? If so, are you receiving prenatal care? 5What is your LMP? 6How many children do you have? How many are alive? IMMUNIZATION Have you ever had TD immunization? How many doses? SOCIAL 1How many wives do you have or how many wives does your husband have? 2Do they all live with you? Are you employed? What work do you do?	1. General appearance: healthy, ill-looking, thin, unkempt, or depressed 2. Temperature, Pulse, Respiration and Blood Pressure: check and record. 3. Eyes: pallor, jaundice, hyperemia, swelling, discharge. 4. Ear and Nose: examine for discharge, redness, swelling, foreign body (FB), impacted wax. 5. Mouth: conditions of the lips, mucous membranes, gums, teeth, uvula, faucal pillars and oropharynx. 6. Neck:, any stiffness, swelling or enlarged thyroid. 6. Breasts: lumps, flat or inverted nipples. 7. Chest: a) difficult or rapid breathing; b) crepitation, rhonchi, wheezing, rales or any decreased breath sounds. 8. Heart: rate, rhythm, 9. Abdomen: scars, tenderness, masses, hernias 10. Limbs: Deformities, Varicosities, Oedema. 11. Weight and height: Measure and record, determine BMI and WHR. 12. Urine: Colour, Odour, Protein, Sugar . 13. Stool: Consistency, Parasites and Ova 14. Haemoglobin: estimate and record.

Note: other health conditions affecting adolescents (13 years and above) and elderly such as fever, etc. are also treated in this section.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
First visit/Regular Check up	Well, Healthy looking. No abnormalities detected Well, Healthy looking. No abnormalities detected	Normal condition	1Encourage on: 2General medical checkup twice yearly 3Regular exercises; 4maintaining good posture; 5yearly PAP smear it female; 6regular breast examination, 7regular prostate examination in male 8Educate on 9STI HIV/AIDS prevention; 10Child spacing; 11Nutrition 12If female, within the childbearing age , advise to take 5 doses of Tetanus Diphtheria (TD) Test for urine and refer for stool test.
First Visit	Abnormalities detected	Abnormal condition	REFER to appropriate section.

Health Education

1.Healthy lifestyle.
2.Personal hygiene.
3.Regular clinic attendance.
4.Environmental sanitation.
5.Nutrition.
6.Importance of taking home-based records to the clinic or hospital.

5.2 FEVER

Fever is hotness of the body. It is body temperature of 37.2°C and above. Fever is a sign of infection or loss of body fluid. Some common causes of fever in adults are malaria, tuberculosis, respiratory tract infection, gastrointestinal tract infection and other infections. All patients with history of fever should be treated and encouraged to drink plenty of water, other fluids and fruits, and should be advised to wear light clothing. High fever (> 38.0°c) should be treated as emergencies.

HISTORY	EXAMINATION/ACTIONS
1.How long have you had this fever? 2.Is there any ear pain, cough, catarrh or headache? 3.Is there any diarrhoea, nausea, vomiting or abdominal pain? 4.Do you sleep under long lasting insecticide (mosquito) bed net? (LLIN) 5.Are there any pain(s); muscle, joints, back, arms and legs? 6.Do you have pains or burning sensation on micturition? 7.Is anyone else in the family ill? 8.Are you taking any medication or treatment? If yes, describe 9.Have you been losing weight recently? 10.Any recent travel? If yes, to where?	General appearance: ill-looking, weak, thin, unkempt Temperature, Pulse, Respiration, and blood pressure: check and record Skin: rashes, Patches Eyes: Jaundice, pallor, hyperemia, discharge, swelling. Nose: discharge, flaring, polyp, foreign body(FB), Deviated Nasal Septum(DNS). Mouth: Condition of the lips, mucous membrane, Blisters, ulcers, redness of the oropharynx swelling of the tonsils Ear: redness or dullness of drum, tenderness, discharge, foreign bodies, impacted wax Neck: stiffness, swelling or enlarged thyroid. Chest: a) crepitation, wheezes, rhonchi, rales, decreased breath sounds; b) Rapid or laboured respiration. Breasts: tenderness, swelling, flat or inverted nipples. Abdomen: tenderness, enlarged spleen or masses, scars, hernia Genitalia: discharges, sores, rashes, swelling, bruises Elicit Kernig's sign mRDT: Test and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Hotness of the body, Itching of the body, Excessive vomiting, Yellowness of the eyes	Temperature 37.2°C and above Yellowish conjunctiva, Tiredness	Severe Condition Suspect Hepatitis	1.Advise on: 2.Bed rest 3.Glucose drink and High carbohydrate diet 4.No fatty food 5.Tabs Multivitamin 1 tab 8 hourly for 2 weeks 6.Tepid sponge if temperature is high REFER immediately to secondary health facility
ii. Fever, General body Pains, Headache and Stiff neck	Temperature 37.2°C and above Tenderness of Abdomen Stiffness of neck Positive Kernig's sign	Severe Condition Suspect Meningitis	1. Inj Benzyl Penicillin 1 mega unit,IM: 6 hourly for the first 48 hours then 2.Caps. Amoxicillin 500 mg 8 hourly for 7 days 3.Tabs Phenobarbitone 60mg 12 hourly for 3days 4.Tabs Paracetamol 1000mg 8 hourly for 3days 5.Tabs Multivitamin 1 tab 8 hourly for 2weeks 6.Review in 2days 7. If no improvement, REFER to secondary health facility

iii. Fever, known leprosy patient	joint pain on pressure Reddish or Painful Skin patches Inability to close the eyes Muscle weakness	Severe condition Suspect complications of leprosy	1.Give Tab. Acetylsalicylic Acid 600mg stat, 2.If client has history of Ulcer, give Tab. Paracetamol 1000mg immediate (stat). 3.Treat according to TB / leprosy management algorithm and have necessary drugs REFER to DOT Clinic for better management.
iv. Fever longer than 2 weeks. Abdominal pain with or without diarrhea Vomiting and headache	Temperature 37.2°C and above Tiredness Tenderness in the abdomen Rapid Diagnosis Test (RDT) for malaria negative	Moderate to severe Condition Suspect Typhoid fever	1.Low Osmolarity ORS. 2.Tepid sponge if temperature is high 3.Acetylsalicylic-Acid 600mg 8 hourly for 3 days, if client has history of Ulcer, give tabs Paracetamol 1000mg 8 hourly for 3 days. 4.Caps Amoxicillin / Clavulanic acid 625mg 12hourly for 5 days. 5.Tabs Metronidazole 200mg 8 hourly for 5 days 6.Review in 3days If no improvement, REFER to secondary health facility
v. Fever, Cough, Difficulty in Breathing, Headache	Temperature 37.2°C and above Difficult or fast breathing	Moderate to severe condition Suspect Pneumonia	1.Tabs Paracetamol 1000mg 8 hourly for 3days. 2.Tabs Ciprofloxacin 500mg 12 hourly for 5 days. 3.Review 3 days If no improvement, REFER to secondary health facility
vi. Prolonged Fever for more than 1 month, with recurrent or persistent Diarrhoea, with weight loss, with or without Skin rashes	Dehydration Temperature 37.2°C and above Marked weight loss Skin rashes and/ or patches	Severe condition Suspect HIV/AIDs	1.Give Tabs Paracetamol 1g 8 hourly for 3days 2.Low Osmolarity ORS solution 3.Counsel and test for HIV 4.If positive, REFER to ART clinic If negative, REFER to Secondary health facility
vii. Fever with facial swelling, sore throat and bleeding from nose, eyes, ears, gums	Temperature 37.2°C and above Headache Joint and muscle aches Sore throat Weakness Abdominal pain Loss of appetite Vomiting or coughing up of blood Bloody diarrhoea	Severe Condition Suspect Hemorrhagic Fever	REFER immediately to secondary health facility Efforts to help those who are infected include giving either oral rehydration salts/ORS or intravenous fluids to prevent dehydration
viii. Fever: > 38°C Moderate to severe dehydration. Sub-optimal level of consciousness Cough, respiratory distress, with or without history of exposure to COVID- 19	Acutely ill- looking, to, Dyspnea as evidenced by labored breathing, Fast breathing, respiratory rate > 18 cpm etc. Moderate to severe dehydration SP02 at room temperature < 92% Fever usually high grade > 38°C Generalized body weakness	Severe Condition Suspect COVID- 19	REFER immediately to secondary facility

ix. Fever less than 1 week, no other complaints	Temperature 37.2°C or above RDT positive	Mild condition Malaria fever	Drug Age Wt.(kg) Dosage Arthemeter-Lumefantrin (AL) >14yr >35kg 4tab 12 hourly for 3days (20/120mg Tab) OR 1 at 0 hour, 8 hours later, then, 1 tab 12hourly for 3 days (80/480mg) Artesunate Amodiaquine (AA) >14yrs >40kg 4Tab (25/67.5mg Tab) once daily for 3 days Tabs Paracetamol 1000mg 8hourly for 3 days.

Health Education

1.Environmental sanitation and prevention of disease transmission.
2.Use of insecticide treated net/materials for malaria control.
3.Personal and food hygiene.
4.Nutrition.
5.Safe water and disease prevention.
6.Importance of taking home-based records to the clinic or hospital.

5.3 COUGH

Cough is a rapid expulsion of air from the lungs in order to clear the lungs and airways of fluids, mucus, food particles or other materials. Cough is a sign of irritation of the throat and wind pipe which comes suddenly. Causes of cough are: common cold, inhalation of dust, fumes and pollens, heart and lung diseases, cigarette smoking, etc. Cough sometimes helps clear the lung of accumulated mucus. It is therefore not good to give cough suppressants unnecessarily when a patient has cough.

HISTORY	EXAMINATION/ACTIONS
1.How long have you been coughing? How often? 2.Do you have difficulty in breathing? 3.Do you wake up at night with shortness of breath? 4.Do you bring up sputum? If yes, what colour is it? (Yellow, Green, White or Blood stained) 5.Is there fever or nasal discharge or Sore throat? 6.Is there any loss of appetite? 7.Have you been losing weight or sweating at night recently? 8.Does anyone else in your family have cough? 9.Have you taken any medication? If yes, what type? 10.Is your feet swollen? 11.Does the cough come while frying, sweeping, or when exposed to fumes, dust, cigarette smoke? snuff, pollen or cold	1.General appearance: ill-looking, thin, tired. 2.Vital signs: Temperature, Pulse, Respiratory rate. Blood pressure: check and record. 3.Nose: nasal discharge , flaring of alae nasi. 4.Throat: red or purulent tonsils. 5.Chest: 6.In-drawing a)Difficult or rapid breathing. b)Tenderness, crepitation, rhonchi or decreased breath sounds. 7.Heart: beats and rhythms 8.Abdomen: enlarged liver or spleen tenderness 9.Ankles: Oedema Sputum: if available, colour, quantity and appearance and AFB stain up to 3 times

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
Cough with any of the following: Swollen feet Shortness of breath Blood in sputum Severe Chest pain Easy fatigability	Pedal Oedema Cough	Severe condition Suspect heart condition	REFER immediately
Cough with wheezing, breathlessness	Cough with mild wheezing with: normal temperature (36.20C -37.2°C) respiration below 40 cycles/min no signs of heart failure	Severe condition Suspect asthma	1. Give Broncho dilator: Inj IM Triamcinolone Acetonide 10 – 40mg stat slowly 2. Give Steroids : Prednisolone 5mg 12hourly for 5 days, OR 3. IM Hydrocortisone 100mg stat, OR 4. Dexamethasone 1mg 12hourly for 5 days Review in 30-60 minutes If there is improvement, discharge on oral broncho dilators: Salbutamol 4mg 12hourly for 7 days. If no improvement, REFER to Chest Clinic

TB patient on treatment presenting with cough, blurred vision and dizziness	Jaundice, Skin rashes, Cough Severe abdominal pains Chest pain	Severe condition Suspect reaction to TB drugs	Stop all anti - TB medications REFER immediately to MDT clinic or hospital
Cough with yellowish green sputum	Temperature 37.°C or above fast breathing Chest in- drawing Yellowish-green sputum	Moderate condition Suspect Pneumonia	1.Tabs. Paracetamol 1g 8 hourly for 3days. 2.Caps. Amoxicillin 500mg 8hourly for 5 days. 3.Review 3 days If no improvement, REFER
Cough of more than 2weeks duration with or without chest pain a) Weight loss b) Shortness of breath c) Blood stained sputum	Cough Sputum may or may not have blood Weight loss Breathlessness AFB positive or negative	Moderate condition Suspect Pulmonary tuberculosis	1.If at least 2 out of 3 sputum tests are positive, treat for Pulmonary TB 2. On TB Management and have necessary drugs treat according to treatment regimen in the Annexure 3. REFER to DOT clinic for better management. If negative, refer to DOT Clinic for further investigation
Cough with numbness or tingling sensation	TB patient on anti TB medications	Mild condition Suspectside effect of anti-TB medication	1. Give Tab. Vitamin B. Complex. 1 tab 8 hourly for 3 days 2. REFER to DOT clinic for better management
Persistent cough for more than one month, Weight loss, Itching,, Generalized skin rash, Diarrhoea for more than one month, Fever	Temperature 37.5°C or above Cough Weight loss Diarrhoea Positive or negative result for AFB stain	Severe condition Suspect HIV/AIDS	1.Tabs. Paracetamol 1000mg 8 hourly for 3 days 2.Tab. Chlorpheniramine (Piriton) 4mg at night for 3days 3.Low Osmolarity ORS REFER For HCT Counsel and test for HIV If positive, REFER to ART site

Health Education

1.Avoidance of Overcrowding.
2.Personal hygiene.
3.Regular clinic attendance.
4.Adequate ventilation.
5.Nutrition.
6.How to identify and avoid trigger factors.
7.Importance of taking home-based records during visit to clinic or hospital.

5.4 DIFFICULTY IN BREATHING

Difficulty in breathing means the client has to make extra effort to breathe or a subjective sensation or perception of discomfort, effort, or impaired airflow during the process of breathing. It can be indicative of various underlying medical conditions, such as respiratory or cardiovascular disorders. Rapid breathing for in adult is breathing more than 30 times per minute. It is important to observe the client's breathing carefully as part of your general observation. Most common causes of difficult breathing are pneumonia, asthma, tuberculosis, bronchitis and heart failure Wheezing in an adult can be caused by asthma or congestive heart failure.

HISTORY	EXAMINATION/ACTIONS
1.How long have you been sick ? 2.Do you have breathing difficulty with wheezing or swollen neck? If yes, since when? 3.Did it come suddenly or slowly? At night? While walking? Lying down? Sweeping? Fumes? Dust? Cigarette smoking? Snuffing? Pollen? Cold? 4.Can you sleep flat at night? 5.Do you have fever or sweating at night recently? 6.Is there any cough? If yes, for how long? 7.Do you bring up sputum? (Yellow, Green, White or with blood stained) 8.Any weight loss recently? 9.Is there pain? Is it worse with deep breathing? 10.Is there dizziness or tingling of the fingers and toes? 11.Does your skin itch? Are you on any medication? If so, what type?	1.General appearance: ill-looking, thin, depressed, unkempt, tired. 2.Vital signs: Temperature, Pulse, Respiratory rate, Blood pressure: check and record. 3.Nose: discharge, polyp, swelling, foreign body, flaring of alae nasi. 4.Throat: reddish or purulent tonsils. 5.Chest: 6.Difficult or rapid breathing. 7.Tenderness, crepitation, rhonchi or decreased breath sounds. 8.C) Chest in drawing 9.Heart: rates and rhythms 10.Abdomen: enlarged liver or spleen, tenderness 11.Ankles: Oedema 12.Sputum: if available check colour, quantity and appearance 13.Back: straight or abnormally bent. 14.Haemoglobin/ PCV: estimate and record Stool microscopy, culture and sensitivity (if services available)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Shortness of breath, Difficulty in lying flat on the back	Shortness of breath with or without the following: Enlarged liver. Low haemoglobin of 7gm/dl or PCV of 21% Rapid pulse Goitre.	Severe condition Suspect heart condition	REFER immediately
ii. Breathing difficulty with wheezing and itchy skin rash	Wheals or hives	Moderate to severe condition Suspect Allergy	1.Tabs Chlorpheniramine maleate 4 mg 12hourly for 3 days 2. Review in 3 days 3. If no improvement, REFER
iii. Difficulty in breathing with wheezing	Difficult breathing Wheezing	Moderate condition Suspect Asthma	1. Inj. Aminophylline 250mg stat slowly over 15 minutes 2. Review in 30 minutes, if improved, Tabs Salbutamol 4mg 8hourly for 5 days. 3. Encourage fluid intake. 4. Review after 24 hours. 5. If no improvement, REFER

Health Education

1.Avoid pollutants and smoking.
2.Avoid overcrowding.
3.Advise on proper ventilation.
4.Encourage adequate fluid intake.
5.Give nutritional Counseling.
6.Importance of taking home-based records during visit to clinic or hospital.

5.5. DIARRHOEA

Diarrhoea is the passage of frequent, loose or watery stools more than 3 or 4 times within 24 hours or the passage of stool more frequent than is normal It is important to observe for mucus and blood in stool and to note the colour. Causes of diarrhoea include dysentery, typhoid fever, cholera, ingestion of poisons and contaminated foods, HIV/AIDS, anxiety, etc. Diarrhoea can lead to loss of body fluids and electrolytes. Persistent diarrhoea is one that has lasted for two or more weeks.

HISTORY	EXAMINATION/ACTIONS
1.How long have you been diarrhoea? 2.How many times do you pass stool each day? Today? 3.Have you passed any since coming to the clinic? 4.Are the stools watery? Is there blood or mucus? 5.What is your source of drinking water? 6.What type of toilet facility do you have/use? 7.Are you nauseated or have you vomited? 8.Do you have fever? Headache? Abdominal pain? Cough? 9.Are you sometimes constipated? 10.Is there anyone else in the household who is sick? 11.Have you eaten plenty of fresh fruits (e.g. mango, orange) or plenty of pepper? 12.Has the diarrhoea reoccurred or persisted for more than one month? 13.Any weight loss? Skin rashes? Oral thrush? Have you taken any medication or treatment for this?	1.General appearance: ill-looking, thin, weak, dehydrated 2.Temperature, Pulse, Respiratory rate, Blood Pressure: check and record 3.Eyes: Sunken 4.Mouth: Dry lips, tongue 5.Skin: turgor 6.Abdomen: tenderness, masses, distension. 7.Weight: Measure and record 8.mRDT(If fever present)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Recurrent or persistent diarrhoea for more than one month, with any of the following: a) weight loss, b)skin rashes, c) oral thrush, d) genital ulcers	Marked weight loss Oral thrush Lethargy Genital ulcers present.	Severe condition Suspect HIV/AIDS	1. Give Low Osmolarity ORS 2. Tabs Paracetamol 1g 8 hourly for 2 days 3. REFER for HCT
ii. Profuse diarrhoea with abdominal mass	Abdominal tenderness and palpable abdominal mass	Severe condition Suspect Abdominal tumor	REFER immediately
iii. Acute profuse diarrhea with vomiting, Weakness	Severe dehydration Sunken eyes Dry tongue Rapid pulse Low urine output	Severe condition Suspect Gastroenteritis	1. Give IV fluid (dextrose Saline) Give IV 2.Metronidazole 500mg 8 hourly for 24hours hours 3.if no improvement REFER
Persistent diarrhoea for 14 days with weight loss	Evidence of weight loss Dehydration Ova or Cysts seen or not	Moderate to Severe condition Suspect persistent diarrhoea	1. Give Low Osmolarity ORS in sips Tabs 2. Metronidazole 400mg 8 hourly for 5days 3.If no improvement REFER

iv. Diarrhoea with headache, cough and fever	Temperature 37.5°C or above Cough Mild dehydration RDT positive	Moderate condition Suspect Malaria	1. Give Low Osmolarity ORS Drug Age Wt.(kg) Dose 2. Artemisinin combination >14yr >35 kg 4 tab twice daily x3days or Day1 st 2nd 3rd 3. Artesunate Amodiaquine>13yr >40 4 Tab (AA) 4. Tabs Paracetamol 1g 8 hourly for 3 days
v. Diarrhoea with or without mucus or blood in stool	No dehydration	Mild to moderate condition Suspect Dysentery	1. Give Low osmolality ORS sips Tabs Cotrimoxazole 960mg 12 hourly for 3 days 2. Tabs Metronidazole 400mg 8 hourly for 5 days 3. If no improvement, REFER
vi. Frequent stool more than 5 times a day	Weakness Tiredness Some dehydration	Mild condition Suspect Mild Diarrhoea	Give Low Osmolality ORS for 24 hours Tabs Metronidazole 400mg 8 hourly for 5 days If no improvement, REFER

Health Education

1.Advise on personal, food, water and environmental hygiene.
2.Give nutritional counseling.
3.Advise on preparation and use of Salt-Sugar-Solution (SSS) (if Low Osmolarity ORS is not available).
4.Encourage adherence to treatment.
5.Importance of taking home-based records during visit to clinic or hospital.

5.6 VOMITING

Vomiting is the throwing up of the contents of the stomach. It may be due to illness or other conditions, for example, malaria, early pregnancy, appendicitis, eating contaminated food or water as in cholera, gastro-enteritis etc. It is a serious problem because prolonged vomiting is a major cause of body fluids and electrolyte loss.

HISTORY	EXAMINATION/ACTIONS
1.How long have you been vomiting? 2.How many times a day do you vomit? 3.What is the vomiting like? Projectile (think of obstruction), 4.Is there diarrhoea? When was the last stool? 5.Do you have any abdominal pain? Swelling? 6.Do you have any cough, fever or headache? Stiff neck? 7.When was your last menstrual period? (if female) 8.What was your last meal? 9.What is your source of drinking water? 10.Are you on any medication? If yes, what type?	1.General appearance: ill-looking, Jaundiced, dehydrated, distressed. 2.Vital signs: Temperature, Pulse, Respiration, Blood Pressure: Check and record 3.Mouth: Dry/coated tongue greenish? With Blood? 4.Eyes: Sunken, Pallor 5.Throat: redness 6.Neck: stiffness., Goiter 7.Abdomen: tenderness, rigidity, visible peristalsis. 8.Ear: redness, bulging or perforation of drum, discharge. 9.mRDT (IF fever present): Test and record 10.Pregnancy test(if necessary)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
Vomiting with any of the following: a) Abdominal swelling b) Constipation c) Stiff neck d) Severe Headache	Palpable abdominal mass Visible intestinal movement May have abdominal rigidity May have stiff neck Lethargy.	Severe condition	REFER immediately
Vomiting with moderate abdominal pain	Abdominal tenderness	Severe condition Suspect Acute Abdomen	REFER immediately
Vomiting one or two times a day with fever	Temperature 37.5°C and above No abdominal tenderness Sore throat RDT positive	Moderate condition Suspect Malaria	1. Low Osmolarity ORS 2. Tabs Paracetamol 1g 8hourly for 3 days 3. Caps Amoxicillin 500mg 8hourly for 5daysDrug Age Wt.(kg) Dose 4. Artemisinin base combination therapy (ACT) 80/480mg>14y >35kg 1 tab at 0 hour, after 8 hours, then, 12hourly for 3days. ORDay 5. Artesunate Amodiaquine>13yr >40 4 Tab(AA) 12hourly for 3days

Vomiting with frequent loose stool with or without headache	Sunken eye Dry mouth Skin pinch goes back quickly	Moderate condition Suspect Gastro enteritis	1. Low Osmolarity ORS 2. IM. Metoclopramide 10mg stat Then 3. Tab Metoclopramide 10mg 12 hourly for 2 days 4. Tab Metronidazole 400mg 8hourly for 5 days .5. Review the next day.
Vomiting with Last Menstrual Period over a month	Mild dehydration Positive Rapid Pregnancy test	Mild condition Suspect pregnancy	1. Low osmolarity ORS 2. Do Urine or Serum PT If Positive Eat dry biscuits in the morning. Do not get out of bed in a hurry 3. Take small meals (not oily food). Refer for ante-natal care.

Health Education

1.Avoid causes of vomiting.
2.Advise on personal, food and environmental hygiene.
3.Encourage adequate fluid intake and prevention of dehydration (excessive loss of body fluid).
4.Importance of taking home-based records during visit to clinic or hospital.

5.7 BODY PAIN

General body pain may be due to injuries or infectious diseases. Pain may be in any part of the body. Pain weakens the client and keeps him/her from getting normal sleep.
Strenuous work and vigorous exercise may lead to muscle strain and may cause general body pain. However, there is need to relieve pain and give comfort to the client.

HISTORY	EXAMINATION/ACTIONS
1.Where do you feel the pain? 2.How long have you had the pain? And what is it like? Any injury? 3.Does the pain worsen with movement or activities? 4.What type of work do you do? 5.Do you have fever or chills? 6.Do you have catarrh or stuffy nose? Or cough? 7.Have you been losing weight recently? 8.Does the pain radiate? Have you been bleeding from anywhere? Any black stool?	1.General appearance: ill-looking, cachexic, pale. 2.Vital signs: Temperature, Pulse, Respiration, Blood 3.Pressure: 4.check and record. 5.Eyes: pallor, jaundice. 6.Chest: a) crepitation, wheezes, decreased breath sounds; b)Rapid or laboured respiration. 7.Abdomen: tenderness, masses, rigidity, enlarged spleen. 8.Back: a) tenderness in spine on thumping; 9.deformity of the spine, masses; 10.tenderness in the flank on thumping; 11.tenderness where client says the pain is 12.tenderness on standing, moving legs or bending. 13.Limbs: swelling, pallor of fingernails, weakness, numbness or paralysis on either side. 14.Weight and height: measure and record. Urine: check for protein.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Severe back pain with generalized stiffness with difficulty in mouth opening	Stiffness and difficulty in mouth opening with or without spasm	Severe condition Suspect tetanus	1. IV Diazepam 10mg stat, 2. Give IV Phenobarbitone 30mg stat Look for injury or old wound. 3. Clean and dress the wound REFER
ii. Bone pain, fever, yellowness of eyes	Temperature 37.50C and above Yellowish conjunctiva Generalized tenderness mRDT positive	Severe condition Suspect sickle cell diseases in crises	1.Tabs Ibuprofen 400mg 8hourly for 3 days (After food) Drug Age Wt.(kg) Dose 2.Artemisinin base combination therapy (ACT) >14yr >35kg 80/480mg 1 tab at 0 hour, after 8 hours, then, 12hourly for 3days. OR 1 st 2nd 3rdArtesunate Amodiaquin (AA)>13yr 100mg/270mg dly for 3/7 3.Tab. Folic Acid 5mg daily for 5 days 4. REFER

Pain radiating down the leg, Numbness, Weakness of one or both legs	Numbness and Abdominal tenderness	Severe condition Suspect Inter vertebral disc problem	1. Tabs Diclofenac 100mg 12 hourly for 3 days (After food) or 2. Tabs Paracetamol 1000mg 8hourly for 3days 3. REFER immediately
Pain in the flanks, Frequency of Micturition, Cloudy urine	Tenderness on palpation of the flank	Moderate to severe condition Suspect Urinary Tract Infection (UTI)	1. Tabs Ibuprofen 400mg 8 hourly for 3 days (After food) 2. Caps. Amoxicillin 500mg 8hourly for 5days 3. Encourage fluids intake 4. Review in 3 days 5. if no improvement, REFER
Pain in the joint or spine	difficulty in walking	Mild condition Suspect Osteoarthritis	1. Tabs Diclofenac 100mg 12 hourly for 3 days (After food) 2. Topical application of Methylsalicylic acid cream/ ointment prn (when necessary) 3. Review in 5 days 4. Encourage moderate exercise 5. If no improvement, REFER
Back pains with menstrual period	Lower abdominal tenderness	Mild to moderate condition Suspect Dysmenorrhoea	1. Tabs Ibuprofen 400mg 8 hourly for 3 days (After food) 2. Review after menstruation. 3. If no improvement, REFER

Health Education

1. Advise on adequate rest and exercise.
2. Advise on regular clinic attendance and compliance with drugs.
3. Advice to abstain from alcohol and tobacco.
4. Give nutritional counseling.
5. Advise on proper ventilation.
6. Importance of taking home-based records during visits to clinic or hospital.

5.8 CHEST PAIN

Chest pain can appear in different forms, ranging from a sharp stabbing pain, pain that radiates to the back, shoulder or neck, to a dull ache. It could sometimes present as a crushing or burning feeling. The causes of chest pain are diverse including heart and lung issues, heavy lifting, weightlifting, trauma to the chest, swallowing a large piece of food, etc. Chest pain could sometimes be an indication of a life-threatening situation and requires urgent medical intervention.

HISTORY	EXAMINATION/ACTIONS
1. Where do you feel the pain? 2. How long have you had the pain? And what is it like? Any injury? 3. Does the pain worsen with movement or activities? 4. What type of work do you do? 5. Do you have fever or chills? 6. Do you have catarrh or stuffy nose? Or cough? 7. Have you been losing weight recently? 8. Does the pain radiate? Have you been bleeding from anywhere? Any black stool?	1. General appearance: ill-looking, wasting, unkempt, pale. 2. Vital signs: Temperature, Pulse, Respiration, Blood Pressure: check and record. 3. Eyes: pallor, jaundice. 4. Chest: a) crepitation, wheezes, decreased breath sounds. b) Rapid or laboured respiration. 5. Abdomen: tenderness, masses, rigidity, enlarged spleen. 6. Back: a) tenderness in spine on thumping. b) deformity of the spine, masses. c) tenderness in the flank on humping. d) tenderness where client says the pain is e). tenderness on standing, moving legs or bending. 7. Limbs: swelling, pallor of fingernails, weakness, numbness or paralysis on either side. 8. Weight and height: measure and record. 9. Urine: check for protein

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Chest pain with any of the following: Nausea Difficult breathing Vomiting Pain radiates to neck, arm, shoulder and back	Pulse < 50/min Or > 100/min BP > 180/110mmHg	Severe condition Suspect Angina pectoris Suspect Hypertension	1. Give Tabs Aldomet Bdx 5 days Tabs Laxis 1 daily x5day Tabs Ibrupefen 1bdx3 days Review in 3days If no improvement REFER
I I. Chest pain	BP <90/60mmHg Temperature £ 37.2°c	Severe condition	1. IV Normal saline 500ml, stop if breathing becomes worse 2. Give Oxygen if available 3. IV/IM Ceftriaxone 1g stat 4. If no improvement, REFER.
III. Chest pain becomes worse on lying down or deep breathing	Severe pain	Severe condition suspect Heart failure	REFER immediately
Persistent diarrhoea for 14 days with weight loss	In pains	Moderate to Severe condition Suspect persistent diarrhoea	1. IV Ringers lactate alter with Normal Saline 500ml 6hourlyx24 hours. 2. Low- Osmolarity ORS or 4. Review in 2 daysy 5. REFER if no improvement

V. Central chest pain, radiates to the neck, jaw and arm that is : recurrent attack (joint pain) resulting from physical activity but relieved by rest :	Central chest pain	Severe condition (suspect angina)	REFER immediately
vi. Chest pain, Joint pain with cough		Moderate condition	Refer to section on cough
vii. Epigastric pain: while hungry, eating, or lying down	Burning, aching, or cramping pain in the upper abdomen.	Moderate condition Supect pectic Ulcer Diseases(PUD)	Advise to avoid smoking, Alcohol, caffeine, Aspirin, NSAIDS(Ibuprofen naproxen, diclofenac, piroxicam) Tabs Omeprazole 20mg daily for 4 weeks Susp. Mist. Magnesium Trisilicate 20 ml 8hourly for 5 days Tabs Metronidazole 400mg tdsx5days Tabs Plasil 10mg daily x 5days Advise on intake of milk as a relieving factor If no improvement, REFER

Health Education

1.Advise on adequate rest and exercise.
2.Advise on regular clinic attendance and compliance with drugs.
3.Advice to abstain from alcohol and tobacco.
4.Give nutritional counseling.
5.Advise on proper ventilation.
6.Importance of taking home-based records during visits to clinic or hospital

5.9 HEADACHE

Headache is a common complaint among adults and may be a symptom of many diseases such as: Malaria, sinus infection, high blood pressure, ear, nose and throat problems, nervous tension, meningitis, brain abscess and typhoid fever. It can occur when there is constipation, inadequate sleep or rest.

HISTORY	EXAMINATION/ACTIONS
1.How long have you had it? How severe is it? 2.Do you have fever or chills? Convulsions? 3.Do you have nausea, vomiting or abdominal pain? (Typhoid or Malaria) 4.Is headache worse in the morning? (sinuses) 5.What time of the day is it worse? 6.Do you have constipation or diarrhoea? 7.Are you on birth control pills? 8.Have you had any head injury? If so when? Any other problem? (family, work)	1.General appearance: In distress, ill-looking, 2.Temperature, Pulse, Respiration, Blood Pressure: check and record. 3.Face: swelling, puffiness 4.Eyes: tenderness. 5.Nose: pus or mucus. Press both sides below eyes for tenderness. 6.Ears: redness or dullness of drums, mastoid pain. 7.Mouth: acetone or alcohol on breath. 8.Neck: stiffness, nodes. 9.Limbs: weakness, paralysis. 10.Urine: protein. RDT(If fever present)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
Headache with puffiness, weakness on one side of the body.	Paralysis of one side of the face Slurred speech	Severe condition Suspect Stroke	REFER immediately
Headache with any of these conditions: a) toothache, b) pus in the ear and/or pain, c) visual problems d) runny or stuffy nose e) Eye pain f) Swelling in the neck	Palpable nodes in the neck Stuffy nose Tenderness in the ear lobe May or may not have pus in the ear Diastolic BP equal to or less than 90mmHg Systolic BP equal to or less than 140mmHg	Moderate to severe condition Suspect Severe Ear Infection (Mastoiditis)	1.Tabs Cotrimoxazole 960mg 12 hourly for 5days 2.Tabs Ibuprofen 400mg 8 hourly for 3 days (After food) 3.Advise adequate rest 4.Tabs Vit C 200mg 8 hourly for 5 days 5. REFER to appropriate health facility for further treatment
Headache Disturbed sleep, Restlessness	Severe paina Diastolic BP equal to or more than 90mmHg. Systolic BP equal to or more than 140mmHg BP 140/90mmHg and above	Moderate condition Suspect Hypertension	1. Give Tabs Aldomet Bdx 5 days 2.Tabs Laxis 1 daily x5day 3.Tabs Ibruprofen 1bdx3 days 4. Review in 3days 5. REFER immediately
Pain with stuffy nose, headache, catarrh, hoarse voice	Stuffy nose	Moderate condition Suspect Sinusitis	1.Tabs Ibuprofen 400mg 8 hourly for 3 days (After food) 2.Tabs Chlorpheniramine 4mg 12hourly for 3 days 3.Tabs Vitamin C 200mg 8hourly for 5 days 4. Encourage bed rest, warm tea, honey and fluid intake d.Advise on avoidance of cold 5. Review in 3 days 6. If no improvement, REFER

Headache with Fever, aches, pain with or without pallor	Temperature 37.5°C and above, Pallor mRDT positive	Moderate condition Suspect malaria	1. Tabs Paracetamol 1000mg 8hourly for 3 days 2. Drug Age Wt.(kg) Dose 3. Artemisinin / lumerfantrine combination >80/480mg >14y >35kg 1 tab at 0 hour, after 8 hours, then, 12hourly for 3days. 4. OR Day1 st 2nd 3rd Artesunate Amodiaquin (AA)>13yr 100mg/270mg dly for 3 / 7 5. Tab. Folic acid 5mg daily for 1 week 6. Review in 3days, 7. if no improvement, REFER

Health Education

1. Advise on adequate rest and exercise.
2. Encourage regular clinic attendance and compliance with drugs.
3. Advise on abstinence from alcohol and tobacco.
4. Give nutritional counseling.
5. Advise on proper ventilation.
6. Importance of taking home-based records during visits to clinic or hospital.

5.10 PALLOR

Pallor is a major sign of anaemia which indicates low haemoglobin level below 10g/dl. The common causes of anaemia are bleeding from any part of the body, lack of iron or folic acid in food, hookworm infestation, sickle cell, and unspaced pregnancies, especially in women of child-bearing age. The place to look for pallor is the conjunctiva, the nail bed, the palm, the lips etc. Certain drugs like aspirin or phenacetin make anaemia worse, so we should not use them without good reason and never given to sickle cell and ulcer patients. Paracetamol should be used for such clients.

HISTORY	EXAMINATION/ACTIONS
1.Do you feel ill? 2.Is there any bleeding from the Nose or 3.Mouth (gums), vaginal(if female) or other parts of the body? 4.are you coughing or vomiting blood? 5.is there blood in urine or stool? Stool black or dark brown or urine black? 6.Do you have fever? 7.Do you have shortness of breath or do you wake at night short of breath? How many pillows do you need to sleep on? 8.Is there abdominal pain or have you passed any worms? 9.Do you have numbness or tingling of the feet? Sore tongue? 10.What medication are you taking? (think of Aspirin or Phenacetin -containing medicines) 11.Are you pregnant? (if female) 12.Do you have joint pains?	1.General appearance: ill-looking, thin, malnourished. 2.Vital signs: Temperature, Pulse, Respiration, Blood 3.Pressure: check and record 4.Eyes: pallor, jaundice. 5.Tongue: redness, swelling, smoothness. 6.Lips & nails: pallor. 7.Neck: nodes 8.Chest a) Breathing difficulty c)crepitation, wheezes, decreased breath sounds 9.Abdomen: spleen, liver masses. 10.Limbs: swollen hands or feet. 11.Heart: rate, rhythm and murmur. 12.Haemoglobin: estimate and record. 13.Stool: hookworm ova. 14.mRDT: Test and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1.Paleness due to acute or profuse bleeding	Bleeding due to visible injury. HB < 10g/dl	Severe condition Suspect anaemia	1.Apply pressure to arrest bleeding. 2. Start patient on IV normal saline 3. REFER immediately
2.Tiredness, Yellow eyes, Pain in the body and bone	Conjunctival pallor and lips, and mucus membrane Tenderness in the joints	Moderate condition Suspect sickle cell anaemia	Inj. I.M Pentazocine 30mg stat REFER Immediately
3. Paleness with fever	Palmar pallor	Moderate condition	REFER to section on fever
4. Paleness, tiredness with abdominal pain	Temperature 37.5^{0}c and above RDT result positive or negative ova of worms seen	Moderate condition Suspect worm infestation	REFER to section on worms

Health Education

1.Give nutritional counseling.
2.Advise on prevention and care for sickle cell disease.
3.Encourage personal, environmental and food hygiene.
4.Encourage the use of LLIN.
5.Importance of taking home-based records during visits to clinic or hospital.

5.11 WEIGHT LOSS

Weight loss is a situation where an adult weighs less than normal standard body weight for age (see BMI chart in the Annexure). The common causes of weight loss are diarrhoea, vomiting, parasitic infection, chronic diseases, tuberculosis, HIV/AIDS and poor diet. Goitre (Thyrotoxicosis), Tuberculosis, cancer, and diabetes can also cause weight loss.

HISTORY	EXAMINATION/ACTIONS
1.How long have you been losing weight? 2.How is your appetite? 3.Do you have nausea or vomiting? 4.Do you have diarrhoea or abdominal pain? Persistent or reoccurring? 5.Have you been having fever? 6.Do you have cough? How long? Is there sputum? What colour s it? 7.Do you have chest pain? 8.Are you having skin rash? Oral thrush? 9.Do you feel thirsty often? Frequent urinating?	General appearance: ill-looking, wasted. Temperature, Pulse. Respiration, Blood Pressure: check and record. Eyes: pallor. Mouth: condition of teeth and tongue Chest a) Difficulty in breathing; b) Crepitation or rhonchi, decreased breath sounds. Abdomen: tenderness, masses, rigidity. Weight and height measure and record. Haemoglobin: estimate and record. Urine: protein and sugar. Conduct HCT Sputum for AFB x 3

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
Weight loss, Persistent or recurrent diarrhoea, Generalized skin rashes, Oral thrush, Prolonged fever	Coughing Wasted Skin rash Oral thrush Temperature 37.5'C or above HCT positive or negative	Severe condition Suspect HIV/AIDS	If HCT result is positive, REFER
Weight loss, Frequent urination, Excessive drinking of water	Urinalysis – sugar in urine Waste	Severe condition Suspect Diabetes mellitus	REFER
Weight loss, Prolonged cough (more than two weeks), chest pain	Coughing Wasted Blood in sputum Sputum Test for Acid- Fast Bacilli positive or negative	Severe condition Suspect Tuberculosis	If 2 out of 3 sputum specimens are positive, commence treatment for TB. HCT, If positive, refer to the hospital If not trained, REFER to MDT Centre
Weight loss, Palpitations	Rapid pulse above 100 beats per minute Evidence of muscle wasting with or without pallor. Swelling in the neck Excessive sweating Tremor BP 140/90 mmHg or above	Moderate to severe condition thyrotoxicosis	REFER immediately

Health Education

1.Advise on personal, food and environmental hygiene.
2.Give nutritional counselling.
3.Encourage regular clinic attendance and weight measurement.
4.Advise on compliance with prescribed medications.
5.Advise on importance of taking home-based records during visits to clinic or hospital.

5.12 JAUNDICE

Jaundice is the yellow discolouration of eyes and skin due to increase in bile pigment in the body. Jaundice results from the build-up of bilirubin-a reddish pigment resulting from haeme metabolism in the body. It may or may not be as a result of liver problem. The build-up of this yellowish pigment may be due to abnormalities in its formation, transportation, metabolism and excretion. At times, the level may not be clinically reorganized until the level is so high. Jaundice occurs in liver problems, sickle cell disease, haemolysis, amoebiasis and tumor of the pancreas.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.How long have you noticed the condition? 3.Has it happened before? 4.Is there any fever or pain anywhere in your body? 5.Is there any change in colour of your body? 6.What is the colour of your urine?	1.General appearance: irritable, in distress. May be in coma. 2.Eyes: jaundice. 3.Mouth: dry, tongue coated. 4.Chest: (a) difficult or rapid respiration (b) Crepitation, wheezes, decreased breath sounds. 4. Abdomen: tenderness, rigidity, masses, spleen, liver. Skin: rash, yellow, petechial haemorrhage.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I.Yellowness of the eyes, general body weakness, may or may not have vomited	Yellow conjunctiva May or may not have swelling of the abdomen with rigidity Major may not have marked abdominal tenderness pallor, Hb less than 10gm/di	Severe condition Suspect hepatic Disorder	REFER Immediately
II.Yellowness of eyes, general body pain, dark yellow urine	Yellow conjunctiva Hb less than 10gm/di Dark yellow urine Pallor Hep B and C positive or Negative	Severe condition Suspect Hepatic disorder	REFER Immediately
iii.Fever, nausea and/or vomiting, abdominal pain. Diarrhoea with blood in stool	Yellow eyes Temperature 37.5 °C or above Abdominal pain Diarrhoea	Moderate condition Suspect Amoebic Liver Disease	REFER Immediately

Health Education

Advise Mother/Caregiver on:

1.Advise on adequate fluid intake
2.Advise on personal food and environmental hygiene
3.Give nutritional counseling
4.Advise on periodic de-worming
5.Importance to taking home-based records during visits to clinic or hospital

5.13 SKIN DISODERS

Skin disease is the presence of itchy or non-itchy, raised or non-raised lesions on the body. It may be due to poor hygiene, allergy, malnutrition or infection. Some skin diseases can easily spread to other members of the family, therefore early referral and correct treatment is important for the welfare of the whole family. Some common causes of skin diseases are: ringworm, measles, leprosy, jaundice, HIV/AIDS, diabetes, scabies and eczema. Others include tinea vesicular, vitiligo, athlete's foot, urticaria, contact dermatitis.

HISTORY	EXAMINATION/ACTIONS
1.How long has the rash been there? 2.Does anyone else in the family have a rash? Describe? 3.Does it itch? 4.Are there any other symptoms? Fever, Headache, Sore throat? 5.Any change in the skin condition: Colour? Patches? Visible veins? Swelling of the legs? Swelling with or without pus? Blisters? 6.What type of accommodation do you have? 7.How many people share the room? 8.What do you sleep on? 9.How often do you bath? 10.What medication, (including injections or home treatment) have you received lately? What cream/lotion are you using for your skin?	1.General appearance: thin, malnourished, distressed/ uncomfortable 2.Skin: 3.location, distribution and size of lesions 4.colour of rash or lesions 5.lesions or rash: flat (macules), raised, pus or fluid-filled, weeping, crusted, ulcerated, scratch marks or 6.combination of(a)to(c) 7.Ears: red or discharging 8.Eyes: redness, discolouration of eyelids (dark - due to repeated scratching) 9.Mouth: redness, discolouration of eyelids (dark-due to repeated scratching) 10.Chest: respiratory rate, wheeze 11.Vital signs: Temperature: check and record Urine: sugar, colour.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
Skin patches or lesions with light colouration with or without deformities, Fever, General malaise and or joint pains, May or may not be confirmed	Ill patient Warm and tender joints Skin Patches with loss of sensation Temperature above 37.5^{0}c	Severe condition Suspect complications of Leprosy	Treat Tuberculosis according to the treatment protocol
II. Skin patches with light colouration with or without weakness of hands and feet	Skin patches with light colouration Loss of sensation in the patch (cotton wool test), with or without weakness of foot or foot drop	Moderate - Severe condition Suspect Leprosy	Treat Tuberculosis according to the treatment protocol commence treatment according to treatment regimen in appendix 13
III. Severe itching with skin changes which disturbs sleep at night	Scratch marks Thickening discolouration of the skin Skin changes with severe itching which disturbs skin and eye with or without loss	Moderate to severe condition . Suspect herpes virus infection or HIV/AIDS	Give Ivermectin 4 tabs stat. Enrol in LGA onchocerciasis control programme for 5 years REFER immediately for further investigation
IV. Itching of affected part with blisters, fever and headache, swelling of punctured blister site	swelling of affected part Temperature > 37.2 °C	Moderate - Severe condition Suspect Onchocerciasi	1. Give Ivermectin 4 tabs stat 2. Tabs Paracetamol 1000mg 8hourly for 3 days 3. Tabs Chlorpheniramine 4mg 12 hourly for 3 days Counsel for HTS

V. Visible veins on legs	Engorged veins	Mild condition	Elastic bandage when standing.
VI. Itching of the skin followed by raised red lesions which is appears after some time	Raised lesion of skin Scratch marks	Mild condition Suspect Allergy	1. Give IM Promethazine 25mg stat 2. Chlorpheniramine 4mg 12hourly for 2days 3. Amoxycillin 500mg 6hourlyx5days
VII. Itching of the scalp or public area	Raised spots between fingers wrists waste and buttocks	Mild condition Suspect lice infestation	1. Shave the hair 2. Apply chlorhexidine gel to scalp and other areas Leave for 24 hours wash and comb
VIII. Itching rashes between the fingers, wrists and buttocks		Mild condition Suspect scabies	1. Wash with antiseptic soap and water 2. Apply Benzyl benzoate emulsion locally daily for 2days (avoid face) 3. Tabs Chlorapheniramine 4mg daily x3 days
IX. Itching with rough, raised, round rashes	Rough, raised, round rashes	Mild condition Suspect skin rashes	1 .Shave hair of affected part 2. Examine other members of the family 3. Apply miconazole ointment daily for 4-6 weeks 4. Wash add rob anti histamine
Small light patches, Not raised not itchy on the back face, etc.	small light patches , not raised	Severe condition Suspect HIV/AIDS	1. Miconazole ointment for 4-6 weeks or Whitfield ointment or 2. REFER
XI. Painful swelling with pus	Tender purulent swelling	Mild condition Suspect Abscess	1. Caps Amoxicillin 500mg 8hourly for 5 days 2. Tabs Ibuprofen 400mg 8hourly for 3 days (After food) 3. Drain and dress daily
XII. Putular rash with fever on face and trunk, loss of appetite	Pustular rashes Temperature > 37.5 °C	Mild condition Suspect Chicken pox	1. Give Tab. Chlorpheniramine 4mg 12 hourly for 3 days 2. Tabs Paracetamol 1000mg 8hourly for 3 days 3. Apply Calaminelotion on the affected area

Health Education

1. Advise on personal hygiene.
2. Discourage sharing of underwear and clothes.
3. Advise on importance of washing and ironing of clothes.
4. Give nutrition counseling.
5. Advise on the avoidance of overcrowding.
6. Advise on adequate care of beddings.
7. Importance of taking home-based records during visits to clinic or hospital.

5.14 OEDEMA

Oedema is an abnormal collection of fluid in the tissues resulting in swelling of the affected part of the body. It could be generalized or localised e.g. one foot or finger. Generalized oedema is most noticeable on the parts of the body which are lowest at that time e.g. legs when standing. Causes include congestive heart failure, severe anaemia, kidney disease, liver. disease and kwashiorkor. Oedema may also be the result of a reaction to toxin, drugs and insect sting. Ascites, which is fluid in the abdominal cavity, may accompany oedema.

HISTORY	EXAMINATION/ACTIONS
1.How long have you had this complaint? Is this the first time? 2.Have you had a good appetite? Any nausea or vomiting? 3.Have you been urinating more frequently at night? Any pain on micturition? 4.Any bleeding from nose? 5.Vomiting or coughing blood? 6.Blood in the urine or stool? Black stool? 7.Any cough? Or breathlessness at night? 8.Do you sleep with extra pillow? Have you ever been treated for jaundice?	1.General appearance:ill-looking. 2.Vital signs: Temperature, Pulse Respiration, Blood Pressure: check and record. 3.Face: puffy. 4.Eyes: pallor, jaundice. 5.Chest: crepitation, wheezes, decreased breath sounds. 6.Heart: rate, rhythm, murmur. 7.Abdomen: large, tender liver, enlarged spleen. 8.Limbs: a)legs: 9.For oedema 10.Does swelling pit on pressure(if not pitting, think of elephantiasis); b.hands for oedema. Urine: protein and sugar.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Swollen legs with large abdomen, Persistent cough, Tiredness	Swollen legs Distended abdomen Enlarged ,tender liver	I. Swollen legs with large abdomen, Persistent cough, Tiredness	REFER immediately
II. Oedema with Pallor, Cough, Breathlessness	Crepitations in the chest Pedal Oedema Paleness of the conjunctiva and mucous membrane	II. Oedema with Pallor, Cough, Breathlessness	REFER immediately
III. Swelling of the leg	Swelling of the leg Not pitting on pressure	III. Swelling of the leg	1.Test for filariasis 2.Tab Ivermectin 12mg (4tabs) once 3. Daily washing and drying of affected limb

Health Education

1.Reassure patient
2.Give nutrition counseling
3.Explain the relationship between infection, anaemia and heart failure
4.Encourage environmental sanitation
5.Importance of taking home based records during visit to the clinic or hospital

5.15 BURNS

This is injury to the skin that may be caused by boiling water, hot liquid, chemicals or fire. Serious burns can cause shock because large amount of body fluid is lost. The size of the area of body surface affected is important in deciding the severity of burns.

HISTORY	EXAMINATION/ACTIONS
How did you get burnt and with what? When and where did it happen? Has any medication been given or applied since then?	1.General appearance: restless, distressed, dehydrated. 2.Skin: (a) size and area of burns, blisters, leathery. 3.(b) signs of infection – pus, redness, warmth 4.Temperature, Blood Pressure: check and record.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Large area Burns or Scalds	Large area of burns or scald Burns 9% or more Any of the following involved: Face, hands, groin or joints Lethargy Loss of sensation Hoarse voice since burned	Severe condition Suspect severe burns	Give IM Paracetamol 600mg stat, then tab 1g 8hourly for 5 days 2. IV normal saline and maintain line (if patient not hypertensive, 3. if hypertensive use Ringers lactate solution dextrose in and if diabetic use half to full strength Ringers' lactate solution). 4. Daily dressing with Vaseline gauze under strict aseptic measures to prevent infection. 5. Apply concentrated Gentian Violet (GV) solution at the 95% wound healing state. 6. REFER immediately
II. Chemical, electrical burns	large or small area of burns or scald caused by either chemicals or electricity	Severe condition Suspect severe burns	1. Wash profusely with water. 2. IV/IM Ceftriaxone 1g stat 3. IM Paracetamol 600mg stat 4. REFER
III. Small area of burns or scalds	Burns less than 9% No blisters No lethargy Face and Hands not involved	Mild condition	1. Clean burns with cool, clean water or sterile solution 2. Do not break blister 3. Tab Paracetamol 1g 8hourly for 3 days 4. Apply clean dressing 5. Encourage fluid intake

Health Education

1. Advise on personal and environmental hygiene.
2. Advise on prevention of burns.
3. Give nutritional counseling.
4. Encourage adequate fluid intake.
5. Advise on proper storage of chemicals.
6. Importance of taking home-based records during visit to the hospital

5.16 WOUNDS, SORES, ULCERS, FRACTURES & BITES

Wounds are injuries to the different body tissues. They are breaks in the skin and deeper tissues which may be small or large and which may include cutting of nerves and tendons. There could also be wounds that have been for several days and have become infected or healing poorly. Sores are injured, infected or diseased part of the skin. Ulcer is a break or discontinuity in a bodily membrane that impedes the organ of which that membrane is apart from continuing its normal function. The client could also have bites from snakes, dogs, scorpions or humans.

A fracture is a break in the continuity of a bone. It is more serious if there is also a cut in the skin and other soft tissues (compound fracture). If there is profuse bleeding resulting from wounds, there is an urgent need to stop bleeding to avoid the patient going into shock. Early cleaning and treatment of wounds is important to prevent infection and tetanus.

HISTORY	EXAMINATION/ACTIONS
1.How did the injury occur? 2.Was there any bleeding? 3.Has it bled much? (Try to carefully assess the amount) 4.Has anything been put on the wound? 5.Have you been immunized with Tetanus Toxoid? Number of doses? 6.What bit him/her? A Dog, A Snake, A person? Scorpion? Or any other thing? 7.Where is the pain? How bad is it? 8.Can you move/use limb? Is there limitation of movement? 9.Which movement causes greatest pain?	1.General appearance: a)Pale, cold or clammy b)Increased sweating. c)Increased respiration and pulse rate d)Patient agitated or restless WOUNDS: a)Size and site b)Bleeding. c)Pus or dirt in wound. d)Swelling, redness or warmth around wound 2.Identify snake fang Mark for poisonous snakes 3.SPRAINS AND FRACTURES General appearance: Signs of shock – increased sweating, pallor, restlessness, rapid pulse and rapid respiration. Degree of pain. Limb or joint: Tenderness or swelling. Warmth or redness; Crepitus (crackling); Open wound.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Wounds on the limb resulting from injury	Evidence of fracture (unable to move the limb). BP below 110/70mmHg or above. Lethargic Wound with blood loss.	Severe condition Suspect fracture	1. Immobilize the limb. 2. Dress wound. 3. Give TT 0.5cc Stat 4. Cap Amoxicillin 500mg 8 hourly for 5 days 5.Tab. Metronidazole 400mg 8 hourly for 5 days. 6.Tab. Ibuprofen 400mg 8 hourly for 3 days (After food) or Tab. Paracetamol 1g 8 hourly for 3 days. 7.Tab ascorbic Acid (Vitamin C) 200 mg 8 hourly for 7 days. 8. REFER to higher-level health facility.

ii. History of snake bite	Evidence of bite (especially evidence of the two (2) fang marks). Wound, redness of site. Excessive thirst Swelling, pain and sweating. Tenderness at the site of bite with or without bleeding. Abnormal, spontaneous bleeding. Increased pulse rate >120 bpm. Features of hypotension and loss of consciousness in severe case of systemic poisoning from snake bite. Wound at the site of the bite x Swelling of the affected part.	Severe condition Suspect snake bite	1. Keep area of bite below level of the heart. 2.Wash bite site gently and copiously with warm water and soap. Remove foreign bodies and control bleeding if any. 3. Patient movement should be restric ted to avoid venom absorption 5. 4. IV Hydrocortissone 2mg/kg of body weight (200mg) Stat.6. IM Tetanus toxoid 0.5ml stat 5.Tab Paracetamol 1g 8 hourly for 3 days. 6. Identify type of snake (if possible), 7. If confirm snake is poisonous, initiate IV treatment with polyvalent anti snake venom 8. REFER within 24hours if there is no improvement.
Ulcer with hypopigmented spots with or without history or sign of leprosy	Ulcers, particularly on the hands and feet with or without deformity. Hypopigmented spots with or without skin patches.	Severe condition Suspect leprosy	1.Wash ulcers thoroughly with warm water. 2. Clean and dress daily. 3. REFER to TBL if with deep septic ulcers.
Wound as a result of laceration.	Bleeding from wound Torn tendon. Haemoglobin <10g/dl	Moderate to severe condition	1. Clean well with available antiseptic solution. 2. Stop bleeding by applying pressure. 3. IM Tetanus toxoid 0.5ml stat. 4. If haemoglobin is less than 10g/dl, REFER to section on low haemoglobin..
Small light patches, Not raised not itchy on the back face, etc.	small light patches , not raised	Severe condition Suspect HIV/AIDS	1. Miconazole ointment for 4-6 weeks or Whitfield ointment
Wound from dog bite	Evidence of bite. Anxious Wound, swelling and redness of site. Tenderness at the site of bite with or without bleeding	Moderate to severe condition	Wash bite site gently and copiously with warm water and soap. 2. Remove foreign bodies. 3. If skin is broken, gently press on the wound to stop bleeding. 4.Apply anti-bacterial lotion. If wound is still bleeding, gently press with clean cloth and reapply anti-bacterial lotion, then wrap in bandage. 5. Give IM Tetanus toxoid 0.5ml stat. 6. If dog is rabid or its Rabies immunization is not done or unknown: Human Rabies Immune Globulin (HRIG) on Day 0, IM 20iu/kg in anterolateral aspect of thigh. 7. RABIES VACCINE 2.5IU (3 doses) IM on Day 0, 7, 21 or 28 in deltoid region of upper arm distant from site of HRIG administration.Tab Ibuprofen 400mg 8hourly for 3 days (After food) 8. Daily dressing of wound after cleaning well.

			NB: Patients who have been previously vaccinated should only receive Rabies Vaccine. If not available, REFER patient to Centres where HRIG and vaccines are available after initial treatment. REFER patients who present first with symptoms of Rabies.
Scorpion/ Bee/ Wasp Sting	Site of sting is swollen Itchy	Moderate to severe condition	1. IV Hydrocortisone 100mg stat. 2. Tab. Ibuprofen 400mg 8 hourly for 3 days (After food) 3. Plain Lignocaine injection (qs) around the site of sting. Review within 24 hours. 4. If no improvement REFER
Wound from human bite	Wound at the site of bite.	Moderate condition	1. Wash with diluted antiseptic solution or normal saline. 2. IM Tetanus toxoid 0.5ml stat. Caps. Amoxicillin 500mg 8 hourly for 5 days. 3. Tab. Ascorbic acid (Vit. C) 200mg 8 hourly for 14 days. 4. Dress wound daily. 5. Review in 5 days. If no improvement, REFER
History of wound	Greater than 4cm. No sign of shock. Evidence of infection (septic wound)	Moderate condition	1. Clean wound with diluted antiseptic solution. 2. IM Tetanus toxoid 0.5ml stat. 3. Then, Ciprofloxacin 500mg 12 hourly for 5 days. 4. Tab. Ascorbic acid (Vit.C) 200mg 8 hourly for 5 days. 5. Dress wound daily with Eusol. 6. Review in 5 days. If no improvement, REFER.
History of wound	Wound less than 4cm. No signs of shock. Without evidence of infection	Mild condition	1. Clean wound with diluted antiseptic solution or normal saline. 2. Apply pressure to stop bleeding if any. 3. Daily dressing with Sofratulle or Vaseline gauze 4. IM Tetanus toxoid 0.5ml stat. 5. If wound is lacerated and less than 6 hours, suture, if needed 6. Caps. Amoxicillin 500mg 8 hourly for 5 days. 7. Tabs Paracetamol 1g 8 hourly for 5 days. 8. Tab. Ascorbic acid (Vit. C) 200mg 8 hourly for 14 days 9. Review in 3 days. If no improvement, REFER

Wound from circumcision	Bleeding from circumcision site	Mild condition	1. Apply pressure for 10 minutes to stop bleeding, if bleeding does not subside, identify the bleeding vessel. 2. Caps. Amoxicillin 500mg 8 hourly for 5 days. 3. Tab Paracetamol 1g 8 hourly for 5 days. 4. Tab. Ascorbic acid (Vit. C) 200mg 8 hourly for 14 days 5. IM Tetanus toxoid 0.5ml stat. 6. Review in 3 days. 7. If no improvement, REFER

Health Education

1. Advise on use of protective clothing to avoid bites.
2. Importance of compliance with treatment.
3. Give nutritional counseling.
4. Advise on environmental sanitation.
5. Advise on reducing risk of accident.
6. In cases of snake bite, if possible, take along killed snake to the clinic.
7. Importance of taking home-based records during visit to the hospital.

5.17 HEAD INJURIES

The head comprises of the brain and the sense organs for hearing, seeing, smelling and tasting. A serious head injury may be caused by a fall or a blow to the head. Some signs of serious head injury are severe headache, unconsciousness, bleeding from the nose, mouth or ears. There also may be vomiting or the patient may be confused. Any delay in referring a patient with head injury for proper medical treatment may cause permanent disability or death.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.What was the cause of the problem? 3.When did it happen? 4.Was there any loss of consciousness? Irritability? Fits? 5.Can you remember what happened? 6.Did anyone see it happen to describe the accident? 7.Do you have headache, nausea, vomiting? 8.Are there any other complaints or additional injuries?	1.General appearance: conscious, confused, irritable, unconscious. 2.Temperature, Pulse, Respiration, Blood Pressure: Check and record. 3.Head: visible wounds, swelling, bleeding. 4.Ears: bleeding 5.Eyes: sub-conjunctival haemorrhage, pupils for size, reaction to light. 6.Mouth: teeth, fractures, bleeding. 7.Nose: bleeding, fractures. 8.Chest: ribs for fractures; Abdomen: for tenderness, rigidity. Limbs: fracture or injury.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Injury to the head, Bleeding from ear or nose	May or may not be conscious Stiff neck May or may not be bleeding from ear or nose	Severe condition Suspect fracture of the skull	1. Ensure airway is clear 2. REFER immediately
II. Injury to the head with transient loss of consciousness	With or without wound seen Pupils reacting normally to light	Moderate to severe condition	1.Tab Paracetamol 1g 8hourly for 3 days 2. REFER immediately
III. Injury to the head without transient loss of consciousness	No wound seen Pupils reacting normally to light No tenderness	Moderate condition	1. Monitor vital signs hourly for 4 to 6 hours 2. If condition deteriorates, REFER
IV. Injury to the head, No other complaints	Conscious No other signs of head injury	Mild condition	1. Check vital signs and advice to return immediately if vomiting or headache or drowsiness occurs. 2. REFER if patient returns.

Health Education

1.Advise on prevention of head injuries e.g use of crash helmets, seat belts, placing children at the back seat and use child lock while driving
2.Prevention of home and road accidents
3.Explain the importance of taking home-based records during visit to the clinic.

5.18 NECK DISORDERS/INJURIES

Neck problems can be because of infections or injuries to the structures in the neck or from secondary causes. This may manifest as enlarged nodes and/or lumps around the neck. There could also be stiffness of the neck. Examples of such infections are tuberculosis, meningitis, HIV/AIDS etc. Lumps can be in the centre or front of the necks (for example goitre). There can also be an injury to the neck following a fall.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.Do you have difficulty swallowing? Hoarseness? 3.Do you have pain and/or stiff neck? Neck swelling? 4.Do you have pain or headache? 5.Did you injure your neck? 6.Do you have fever? 7.Do you have weakness or numbness of the arms, hands or legs? Is there weight loss or night sweats?	1.General appearance: ill-looking, distressed. 2.Temperature, Pulse, Respiration, Blood Pressure: Check and record 3.Head: abscesses, sores or pustules, lumps 4.Ears: signs of infection. 5.Mouth: abscesses or sores a.Neck: swelling, nodes, stiffness, tenderness 6.Arms and Legs: weakness or decreased sensation.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Neck swelling With or without difficulty in breathing	Palpable mass in the front of the neck May be in respiratory distress	Severe condition	REFER immediately
II. Injury to the head with transient loss of coavnsciousness	Stiff Neck No weakness of arms and legs No evidence of injury	Moderate condition Suspect torticollis	1.Reassure client 2.Tab Ibuprofen 400mg 8hourly for 3days (After food) 3. Methyl Salicylate Ointment
III. Enlarged, painful nodes around the neck with or without fever	Palpable, tender nodes	Suspect infection of the head and neck region	1.Caps Amoxicillin 500mg 8hourly for 5days 2.Tabs Metronidazole 400mg 8hourly for 5days 3.Tab. Ibuprofen 400mg 8hourly for 3days (After food)

Health Education

1.Advise on prevention of neck injuries.
2.Advise on complications of mumps in males.
3.Advise on prevention of transmission to others.
4.Give oral hygiene instructions.
5.Importance of taking home-based records during to the clinic.

5.19 EYE DISORDERS

Eye problems in adults can be as a result of infection, foreign body, congenital malformation not treated in childhood, deficiencies of vitamins and micro-nutrients, complication of diabetes mellitus, hypertension, thyrotoxicosis etc. Eye problems are often associated with defect of vision especially with aging. Burns from hot water and chemical can also affect the eye.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem with the eyes? 2. How long have you had the problem? Have you had it before? 3. Did the problem come suddenly or gradually? 4. Is there pain in the eye? How severe? 5. Does your eye itch? 6. Did anything get into your eye? 7. Have you (or anyone) put anything into the eye? What? 8. Has any medicine been put in the eye? What? 9. Do you have high blood pressure? 10. Do you have diabetes or any other disease? 11. Do you smoke or drink alcohol? 12. Is there any family history of blindness? 13. If so, is anyone else in the family receiving eye treatment? What type?	1.General appearance: in distress, ill-looking 2. Skin: rashes 3. Eyes: a.) Redness b) watery b.) Squint c.) Cloudiness or whiteness of pupil(s) swelling around the eye d.) Eyelid: sticky, inversion, or eversion, pus or other discharges e.) Eyelashes: irregular, missing or touching the surface of the eyeballs f.) Pupils: normal or small in size, equal, irregular in shape, reactive to light. g.) Eyeballs protruding (pushing out) h.) conduct visual acuity test of both eyes and record 4. Nose: catarrh 5.Blood pressure: measure and record 6.Urine: sugar

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1. Burns eye with chemical or hot water	*Watery, painful discharge from the eyes	Severe condition	1.Wash eye immediately with cold water 2.Tab. Paracetamol 1g 8hourly for 3days 3. REFER
Eye injury with pain	*Surface/ penetrating eyeball injury.	Severe condition	Apply pad and REFER.
Poor vision/ Blindness	*Inability to see clearly, HM, LP, NLP	Severe condition	REFER.
Cloudy central part of the eye/ poor vision	*Opaque lens/ poor vision with or without corneal scar	Severe condition	REFER
Bleeding from the eye.	*Bleeding from the eye.	Severe condition	Apply pad and REFER.
Redness of eyes, pain in the eye, worsening Vision/difficulty in seeing with one or both eyes, foreign body in the eye, swelling of eye shadowy vision	* Red eyes *Watery discharge from the eye *Pain in the eye *Squint	Moderate to severe condition	Tab. Paracetamol 1g 8hourly for 3 days. Apply pad as necessary. REFER

Discharge of pus from the eyes. Gritty sensation in the eye	*Purulent discharge *Sticky eyes *Red conjunctiva	Moderate condition (conjunctivitis)	1.Clean eyes with sterile water 2. Chloramphenicol eye ointment 6hourly for 5days
Not able to read tiny print	*Inability to read tiny prints	Moderate condition (Refractive errors)	Refer to optician
VIII.Pustule or small abscess on eyes lid (stye/hordeolum)	*Swelling on the eyelids	Mild condition Suspect stye	1.Reassure client 2.Tab. Ibuprofen 400mg 8hourly for 3days (After food)

Health Education

1.Advise on personal and environmental hygiene.
2.Encourage regular eye check-up.
3.Advise on prevention of eye injuries.
4.Give nutritional counseling.
5.Importance of taking home-based records with anytime he/she goes to the clinic or hospital.

5.20 EAR, NOSE AND THROAT DISORDERS

The ear, nose and throat are linked with each other and whatever affect one may affect the other. The ear not only helps us to hear but also helps us maintain our balance and any disease of the ear which is not urgently and adequately treated may not only lead to deafness, but also inability to maintain balance.

Ear problems are commonly caused by middle ear infection often accompanied by pain, dullness, and redness of the ear drum with drainage of pus or fluid infections of the nose, throat or Eustachian tube may spread to the middle ear. It is important to examine the bone behind the ear (mastoid bone). If it is painful (tender to touch); it may mean that the infection has spread to the mastoid bone (mastoiditis). To test an adult for hearing, sit behind the client then ask him to repeat words or numbers that you whisper to him/her

The nose is the organ for breathing and for the sense of smell. When the nose is affected by disease or injury, it discharges watery fluid (mucus, blood or pus.) Blocking of the nose may lead to breathing difficulties.

The throat is the junction of the ear, nose, and mouth. It contains the tonsils, pharynx, etc. Diseases affecting the throat can easily spread to the chest and brain

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.How long have you had the problem? 3.Is there catarrh? 4.Is there pain? Boil? Any discharge: pus, blood or water? 5.Do you feel pain while swallowing? 6..Is there fever? Watery eyes? Headache? 7.Has it happened before? 8.8.Has there been any blow or injury to the head? 9..Do you have nose bleeding? Can you estimate the volume? 10.When did it start? 11.Are you having bleeding from any other part of the body? 12..Is there pain from the teeth or gum? 13..Do you notice bad or foul breath? 14..Is there diarrhea or vomiting? 15.Do you work in a noisy environment or with noisy machines? 16..Have you put anything into your ears, nose, and throat? 17.Did anything enter your ears, nose, or throat? 16.Is there any trauma to the ear, nose, or throat	1.General appearance: In distress, ill-looking, in distress 2 .Temperature, pulse respiration, blood pressure: check and record 3.Ear: a) Pus or blood discharge b) Red or bulging drum c) Perforated drum, foreign body d) Cerumen (Wax) e) Bone behind the ear (mastoid)for tenderness 4.Eye: redness, watery discharge 5. Nose: Mucus or catarrh 6.Neck: stiffness or enlarged lymph nodes 7.Throat: a) Redness b) Enlarged tonsils, pus c)Hoarseness or loss of voice 8.Chest: crepitations, wheezes, decreased breath sounds 9.Test for hearing and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Ear problem with vomiting Tenderness behind the ear	Temperature 37.5C or above Tenderness behind the Ear Restlessness	Severe condition Suspect Mastoiditis	REFER immediately.
II. Boil in external ear	Soft, tender swelling with well-defined margins	Moderate to severe condition: suspect boil	1. Caps Amoxicillin 500mg 8hourly for 5 days 2.Tabs Ibuprofen 400mg 8hourly for 3 days (After food); ORTab Paracetamol 1g 8hourly for 3 days 3. Review in 3 days 4. If no improvement REFER
III. Pain in the ear with or without pus or water from the ear or vomiting	Temperature 37.5C or above Red, dull bulging ear drum or Perforation of ear drum Pus or blood discharge	Severe condition: Suspect Acute Suppurative Otitis Media with perforation	REFER immediately

IV. Pain in the throat, Difficulty with swallowing (dysphargia)	Enlarged and red tonsils	Moderate condition: Suspect Tonsilitis	1. Cap Amoxicillin 500mg .8hourly for 5days 2. Tab Metronidazole 400mg 8hourly for 5 days 3. Tab Ibuprofen 400mg 8hourly for 4 3 days (After food)5 Tabs Stripsil 1tdsx3 days 4. Warm saline gargle Review in 3 days. If no improvement REFER
V. Pus from the nose, Headache, Pain on either side of the nose	Temperature 37.50C or above Pus draining from the nose Tenderness of the nose	Moderate to severe condition Suspect abscess	1. Caps Amoxicillin 500mg 8hourly for 5 days 2. Tabs Ibuprofen 400mg 8hourly for 3 days (After food) 3. Tabs Vitamin C 200mg 8hourly for 7 days 4. Review after 3 days 5. If no improvement REFER
VI. Bleeding from the nose	Evidence of bleeding from the nose	Moderate condition: Epistaxis	1. Sit patient in upright position Teach patient to pinch side of the nose for 5-10 minutes 2. Re-examine, if bleeding does not stop, apply cold compress to nose for 10 minutes. 3. If no improvement, apply Adrenaline nasal pack. 4. Assess Blood Prssure 5. If no improvement, REFER.
VII. Stuffy nose Catarrh with or without fever Sneezing with or without watery eyes	Temperature 37.50C or above Watery eyes Catarrh	Mild condition Suspect Sinusitis	1. Tabs Chlorpheniramine maleate 4 mg 12hourly 3days. 2. Tabs Paracetamol 1g 8hourly 3 days 3. Tabs; Prednisolone 5mg bdx-3days 4. Tabs Vitamin C (Ascorbic acid) 200 mg 8hourly for 7 days 5. Tabs Co-trimoxazole 960mg 12 hourly for 5days 6. Advise to avoid or reduce exposure to dust or pollen. Review in 3 days.

Health Education

1. Advise on personal and environmental hygiene
2. Advise on care of the ear, nose and throat
3. Discourage traditional practices such as uvulectomy and tonsillectomy
4. Give nutritional counselling
5. Advise on avoiding exposure to air pollution (fumes, dust and pollen)
6. Discourage home removal of foreign body.
7. Importance of taking home-based records during visit to clinic

5.21 ORAL HEALTH DISORDERS

Oral health is a state of being free from chronic mouth and facial pain, oral and throat cancer, oral sores, birth defects such as cleft lip and palate, periodontal (gum) disease, tooth decay and tooth loss, and other diseases and disorders that affect the oral cavity.

The most common complaint is pain which may or may not be accompanied with swelling inside the mouth or on the face. The condition of the mouth is also a reflection of the client's general health, so it should be inspected as part of physical examination. Early recognition of oral problems with prompt attention or referral will reduce morbidity and mortality arising from otherwise simple disease condition.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.Where is the problem? 3.Is there pain from tongue, palate, mucus membrane, teeth or gum? 4.How long has it been noticed or present? 5.Are you able to eat your regular diet? 6.Does the pain increase with hot or cold drink or food? 7.Does the pain start on its own? 8.Does it keep you awake at night? 9.Is there swelling anywhere? 10.Is there any bleeding from gum or other part of the mouth? 11.Do you notice bad or foul breath? 12.What do you use for cleaning your teeth? 13.How do you clean your teeth? How many times a day? Do you notice any sore in the mouth	1.General appearance: ill-looking, facial swelling, malnourished. 2.Temperature: check and record 3.Mouth: a)Gum: sore, swelling, discharging sinus, bleeding, white patches. b)Tongue: sore, coated, redness, fissures, smoothness, dryness. c)Lips: crack; Teeth: caries, discolouration, tenderness, mobility, missing, plaque.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Severe tooth ache, disturbs patients sleep, aggravated By chewing, with or without headache	*Holes in tooth or teeth	Moderate condition Suspect irreversible pulpitis	1.Caps Amoxicillin 500 mg 8hourly for 5days 2.Tabs Metronidazole 400mg 8hourly for 5 days 3.Tabs Ibuprofen 400mg 8hourly for 3days (After food) or Tabs paracetamol 1g 8hourly for 3days 4.REFER to the dental clinic
II. Severe tooth pain, Disturbs patient's sleep, Aggravated by chewing, With or without headache, External swelling of the cheek	Hole on the offending tooth/ teeth Swelling of the jaw in relation to the offending tooth may be mobile	Severe condition Suspect Dental Abscess	1. Caps Amoxicillin 500mg 8hourly for 5days 2. Tabs Metronidazole 400mg 8hourly for 5days 3. Tabs Ibuprofen 400mg 8hourly for 3days or Tabs Paracetamol 1000mg 8 hourly for 3 days 4. Improve on oral hygiene, 5. review in 5days, 6. if no improvement REFER to the dental clinic
III. Sharp, short pain triggered by chewing, hot/cold drinks, Stops as soon as stimulus is removed.	Hole on the offending tooth/ teeth Wear of the offending tooth	Moderate condition Suspect reversible pulpitis Dentine hypersensitivity	1. Caps Amoxicillin 500mg 8hourly for 5days 2. Tabs Metronidazole 400mg 8 hourly for 5days 3. Tabs Ibuprofen 400mg 8hourly for 3days (After food) or Tabs Paracetamol 4. Improve on oral hygiene.

IV. Hole on the tooth, With or without pain	Hole on tooth/teeth	Dental caries Moderate condition	REFER to the dental clinic
V. Gum bleeding while brushing or biting into fruits e.g. apple with or without itching	Gum may be red and slightly swollen	Moderate condition Suspect Chronic Gingivitis (Gum disease)	1. Apply gentle pressure to the bleeding site 2. Caps Amoxicillin 500mg 8hourly for 5days 3. Tabs Metronidazole 400mg 8hourly for 5days 4. Tabs Ibuprofen 400mg 8hourly for 3days (After food) or Tabs Paracetamol 1g 8hourly for 3days 5. Oral hygiene instruction 6. Review in three days, if no improvement 7. REFER to dental clinic
VI. Dull, continuous pain in the gum, Food packs in the gum after eating with or without gum bleeding	Deep hole in the gum between teeth. Gum may or may not bleed and may be slightly swollen	Moderate condition Suspect Chronic Periodontitis	1. Caps Amoxicillin 500mg 8hourly for 5 days 2. Tabs Metronidazole 400mg 8 hourly for 5 days 3. Tabs Ibuprofen 400mg 8hourly for 3 days (After food) Or Tabs paracetamol 1g 8hourly for 4 3 days 5 Mouth wash bd oraldine, review after 3days 6. REFER to dental clinic
VII. Swelling bleeding of the gum with pus discharge or pocketing with or without loss of alveolar tissue.	Pus discharge, swollen gingivae, with or without halithosis and lose tooth	Moderate condition	1. Caps Amoxicillin 500mg 8hourly for 5 days 2. Tabs Metronidazole 400mg 8 hourly for 5 days 3. Tabs Ibuprofen 400mg 8hourly 3 days (After food) Or
VII. Injury to the mouth and/ or face	Broken/Missing/ Mobile tooth/teeth Lip or Tongue laceration Broken jawbone	Traumatic Injury Fracture of the teeth/ jawbone	1. Caps Amoxicillin 500mg 8hourly for 5days 2. Tabs Metronidazole 400mg 8 hourly for 5days 3. Tabs Ibuprofen 400mg 8hourly for 3days (After food) or Tabs Paracetamol 1g 8hourly for 3 days 4. REFER to the hospital
VIII. Recurrent Ulcers in the mouth, With or Without pain	Round/Oval ulcers on the lining of the mouth/tongue/ palate/gum May be associated with stress	Suspect aphthous ulcer	1. Counsel the patient that ulcers are self-limiting 2. Tabs Ibuprofen 400mg 8hourly for 3days (After food) OR Tabs Paracetamol 1g 8hourly for 3 days 3. Tabs Vitamin C 200mg 8hourly for 2 weeks 4. Use warm saline water as mouth rinse 5. REFER to the dental clinic

IX. Slow or fast growing swelling of the jaw with or without pain	Swelling may be hard or soft Located anywhere in the mouth or face	Suspect neoplasm (new abnormal growth)	REFER to the dental clinic
X. Creamy/Whitish growth on lining of the mouth/tongue/ gum/palate	Soft and friable slough on the soft tissues of the mouth - lining of the mouth/tongue/ gum/palate Not easily rubbed off	Suspect immune suppression Oral Candidiasis (Oral thrush) HIV/AIDS	1. Nystatin suspension 100,000 units/ml 6hourly after food for 7 days OR 2. Miconazole oral gel 2% place in the mouth after food 3. REFER to dentist

Health Education

1.Brush twice daily, in the morning and last thing before going to bed at night.
2.Use fluoride-containing toothpaste.
3.Change your toothbrush as soon as the bristles begin to flare/scatter.
4.Visit your dentist at least twice a year for routine cleaning (scaling and polishing).
5.Avoid use of toothpick
6.Avoid intake of sugary foods (refined sugar) in-between meals. Eat mouth cleansing fruits instead.
7.Cultivate the habit of rinsing your mouth with water after meals
8.Discourage uvulectomy
9.Emphasize the importance of bringing home-based records during visit to the clinic/hospital

5.22 CARDIOVASCULAR DISORDERS

Cardiovascular problems are increasingly becoming major health concerns in adults. These are problems that affect mainly the heart and blood vessels. Congestive heart failure occurs when the heart becomes too weak to pump all the blood around and some of the fluids flows back into the abdomen and legs. Any history of shortness of breath especially while lying down should make the health worker think of cardiovascular disorders.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? How long have you had it? 2.Do you have cough and or chest pain? 3.If chest pain, where in particular? 4.Do you use to have difficulty in breathing while lying down flat? 5.Does deep breathing and bending make it worse? 6.Can you sleep flat or do you need more than one pillow to sleep at night? 7.Do you sometimes get up at night to breath? 8.If cough is present, do you bring up sputum? Colour/any blood in the sputum? 9.Do you use tobacco and/or other substances; do you drink alcohol? 10.Do you eat well? 11.Do you have drenching night sweats? 12.Have you lost weight? 13.Does the pain/discomfort come or worsen with physical activity or when you are resting? 14.Are you on any medication? Are you on any medication? If yes,what? 15.Does the pain/discomfort come or worsen with physical activity or when you are resting? 16.Are you on any medication? If yes,what?	1.General appearance: Stable, ill-looking, in distress, wasted, pale, cyanosed, icteric, dehydrated. 2.Temperature: pulse, blood pressure, weight: check and record 3.Neck: nodes 4.Chest:(a) respiration for rate, difficulty in breathing (on sitting or lying down), wheezes (b) Crepitations, rhonchi, decreased breath sounds 5.Heart: rate, rhythm, murmur 6.Abdomen: distension, tender liver Legs: Oedema.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Severe chest pain, Pain may radiate to other parts of the body. body, Unable to sleep	Painful distress	Severe condition Suspect Angina Pectoris	REFER immediately.
II. Tightness of the chest, Discomfort, Chest pain increase on exertion and not relieved by rest, Difficulty in breathing	* Rapid heart rate less than 60 beats per min. Increases with movement Difficulty in breathing	Severe condition Suspect Myocardial Infarction	1.Give Oxygen (if available) 2. Give tabs Acetyl salicylic Acid 150mg immediately (stat) 3.REFER Immediately
III. Fast heartbeat ,feeling anxious	*Rapid heart rate 90-130 beats/min *Anxious	Severe condition Suspect cardiac Arrythmia	REFER immediately

Health Education

1.Advise on adequate rest.
2.Encourage moderate exercise.
3.Avoidance of smoking, tobacco, and alcohol use
4.Avoidance of sedentary lifestyle
5.Advise on proper management of stress, adequate sleep and rest.
6.Advise on regular clinic attendance and compliance to drugs.
7.Give nutritional counseling.
8.Emphasize the importance of home-based records during visit to the clinic/hospital.

5.23 ABNORMAL BLOOD PRESSURE

Blood pressure refers to the force with which the heart pumps blood into the body circulation. Blood pressure usually increases with age as the blood vessels become less elastic. . Women tend to have lower blood pressure than men of the same age. The normal systolic pressure for young adult is 100 - 120 mmHg and for older persons 120 - 140 mmHg. The diastolic pressure, which is the force at which blood is returned to the heart from the circulation, is normal between 70 - 89 mmHg. High blood pressure is referred to as hypertension. Hypertension is a very common cardiovascular disease affecting a lot of people. It is thought to result from the narrowing of the blood vessels. The narrowing causes the heart to beat harder in order to pump the blood. Hypotension refers to low blood pressure and can lead to shock and organ failure.

Measurement of Blood Pressure	Additional Points
Accurate diagnosis and treatment of blood pressure is hinged on accurate blood pressure measurement. This, in turn, is dependent on the use of accurate and validated BP measurement devices and the right patient technique. Proper patient preparation for blood pressure measurement includes: The patient should avoid tea/coffee, exercise, and tobacco use for at least 30 min before measurement. The patient should have rested for at least 5 minutes before measurement. Ensure s/he has emptied his/her bladder. When taking blood pressure, the patient should be seated, with the arm supported at heart level, back supported, and both feet resting on the ground. The patient should not speak while blood pressure is being measured. The ideal cuff bladder length and width are ≥80 percent and ≥40 percent of the patient's arm circumference, respectively. Ensure that the appropriate cuff size is selected for the individual who will receive the BP measurement. Arm circumference (cm) — Recommended cuff size (width x length in cm) 22-26 — 12x22 (small) 27-34 — 16x30 (medium) 35-44 — 16x36 (large) 45-52 — 45 53 (extra-large) Validated and automated/digital BP monitors are preferred for use to measure patients' blood pressure. Where this is not possible, a calibrated manual sphygmomanometer with a stethoscope can be used. **Diagnosis of hypertension** Hypertension is diagnosed if the systolic blood pressure is ≥ 140 mmHg and/or diastolic blood pressure is ≥ 90 mmHg on two different days within a week but not exceeding a month. However, if the blood pressure reading is very high (≥ 160 mmHg systolic or 100 mmHg diastolic), a second reading should be taken on the same day to establish the diagnosis at least 4-6 hours apart from the first reading. At each visit, two blood pressure measurements are taken, with at least 3-5 mins rest between both readings and the average is calculated and recorded as the BP reading for the patient.	Procedure for BP measurement Applying the cuff Explain the procedure to the patient. If the patient is wearing a thick-sleeved garment, ask the patient to uncover the arm by either removing the top garment or rolling up the sleeve. If the cuff is tied over the clothing, ensure that it is tied snugly and evenly. The patient's arm should be supported on the table and relaxed. Make sure the arm cuff is properly deflated (air is out) before placing it around the patient's upper arm. Wrap the cuff comfortably or snugly above the elbow, about 2 cm (or 2 finger breadths) from the elbow and secure the Velcro tape. Keep the cuff at the heart level during measurement. Taking a blood pressure measurement using a digital blood pressure device Turn the monitor on from the on/off button. Ask the individual to stay still, breathing normally, without talking or laughing. Press the START/STOP button to begin the measurement. The BP cuff will automatically start to inflate. This might be uncomfortable for some. Reassure the patient that the discomfort is temporary. The cuff will slowly deflate to take the measurement. When the reading is complete, the systolic (SBP) and diastolic blood pressure (DBP) readings and pulse rate will appear on the screen. Note the BP readings exactly as seen in the device in the patient record. Please do not round off. If the monitor/machine does not record the reading, re-position the cuff and try again after 1-2 minutes and repeat the above-mentioned steps. Turn the machine off and remove the cuff. Using a manual instrument when a digital, automated device is unavailable The cuff has a bulb at one end of the tube that you will squeeze to inflate the cuff. Apply the cuff on the individual's arm and place your stethoscope where you will be able to hear the sounds of blood flowing. The place is where you can feel the pulse. You can find the place by placing your index and middle fingers of one hand in the crease of a relaxed elbow. Put your stethoscope on and hold the head of the stethoscope in place. Screw the valve tight and inflate the cuff to 200 mmHg. Keep holding the stethoscope at the right spot. Slowly let the air out of the cuff while listening for the sounds of blood flowing. Keep a rate of deflation 2-3 mmHg per second (equals one line on the dial every second).

	When you start letting the air out, you should not hear any distinct sounds at first. You will then hear a thump, which will be followed by several other similar thumping sounds. The number on the dial when you heard the first thump is the patient's systolic blood pressure. The thumping sounds will eventually stop. The number on the dial when you heard the last sound is the patient's diastolic pressure. **Additional points** Although the recommendation is for multiple BP measurements and averaging them (complex computational task) to get the final BP value for patients, this may not be practical for busy primary care settings. Averaging BP readings is time-consuming and prone to errors in a busy clinic. A practical approach may be adopted below: If the first blood pressure (BP) is <140/90 mmHg, then no other blood pressure measurement is needed during that encounter. If the first BP is >140/90 mmHg, perform a second BP after 3-5 mins and use the second reading as the recorded BP for the encounter. If there is a large difference between the first and second readings (>5 mmHg), it is reasonable to do a third measurement and use the third BP as the recorded BP.

HISTORY	EXAMINATION/ACTIONS
1. What is the problem? How long have you had it? 2. Do you get headache, dizziness and swollen legs? 3. Do you use to hear of feel your heartbeat? 4. Are they related to any physical exercise? 5. Do you have difficulty in breathing? Unable to sleep at night? 6. Are you unable to sleep at night? 7. Have you ever had high blood pressure or low blood pressure? 8. Are you taking any medication? 9. Are you on any birth control pills? (if female) 10. When was your last menstrual period? (if female) 11. Do you have family history of high blood pressure? 12. Do you have any existing illness?	1. General Appearance: Stable, ill-looking, pale, flushed, anxious, cyanosed, icteric, dehydrated 2. 2. Temperature, pulse, weight, blood pressure: check and record (pay special attention to both systolic and diastolic pressure) 3. Neck: thyroid engorged veins 4. Chest:(a) respiration for rate (b) Crepitation or decreased breath sounds (c) Heart sounds, rate and murmur 5. Signs of shock: (a) pale, cold or clammy extremities (b) restless (c) Increased perspiration 6. Legs: Swellings 7. Urine: protein, sugar, colour, deposit, odour, specific gravity.

Management of Simple and Uncomplicated Hypertension	Additional points
	If the initial BP is ≥ 160/100 mmHg but less than 180/110 mmHg, start at step 2. If initial BP is ≥ 180/110 mmHg, give step 3 drugs and refer to the emergency unit of the nearest general/tertiary hospital within an hour. Tabs Telmisartan 40mg and 80mg, if available, is preferable to Tabs Losartan 50mg and 100mg. Tabs Hydrochlorothiazide 25mg, if unavailable, may be substituted with Tabs Amiloride 2.5mg/HCTZ 25 mg. Losartan and hydrochlorothiazide should not be given to pregnant women and women of child-bearing age that are not on effective contraception.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Headache	Lethargic Systolic Blood Pressure less than 90mmHg Cold extremities Feeble pulse Rapid respiration Anxious looking.	Severe condition Suspect Hypotension	1. Give IV normal Saline 1 litre stat fast 2. Give IV hydrocortisone 100mg stat 3. Elevate foot of the bed. 4. Adequate ventilation 5. Keep warm 6. REFER immediately
II. Headache, Difficulty in breathing, Unable to sleep at night. Shortness of breath on walking, Weakness, or inability to walk. Cough. Swelling of	May or may not look ill Shortness of breath Swollen feet Pitting Oedema	Severe condition Suspect Congestive Cardiac failure	1. Give IV Frusemide 80mg stat 2. REFER immediately
Medication	Starting dose	Intensification dose	Side effects
Amlodipine	5mg	10mg	Headaches (7%) and palpitations (1-3%) – usually self-limiting, observe on treatment. Lower limb/pedal edema (1-10%), especially at 10mg dosage without ARBs, e.g., losartan – Observe on treatment. Consider alternative medicine if it is excessive or persistent after using a diuretic. Muscle cramps – usually self-limiting; observe on treatment. Male sexual/erectile disorder – explore for other causes and reassure.
Losartan	50mg	100mg	Cough occasionally Hypotension with dizziness (10%) - Monitor blood pressure and consider dose reduction if systolic BP is < 100mmHg or the person is frail. Hyperkalemia (4-10%) - Monitor potassium and creatinine. Stop the drug when potassium is > 4.5mmol/L and or eGFR <30ml/min.
Telmisartan	40mg	80mg	
Hydrochlorothiazide	25mg	100mg	Frequent urination – might have to ask the patient to take the medicine in the morning rather than at night. Metabolic derangement (hyperlipidemia, hyperuricemia, etc.) – monitor while on treatment. Hypokalemia – monitor while on treatment via serum urea and creatinine.
Amiloride/Hydrochlorothiazide	2.5mg/25mg	80mg	

Laboratory monitoring		
Fasting blood sugar	Lipid profile	ECG
Urine protein.		Fundoscopy
Potassium and sodium – if on Hydrochlorothiazide/diuretic therapy		Severe condition Suspect cardiac Arrythmia
Serum urea and creatine – if on losartan/telmisartan/ARB		Fundoscopy

Health Education

1.Explain the condition and diagnosis to the patient.
2.Educate the patient that hypertension is a life-long condition that can be controlled with medications and lifestyle changes.
3.Emphasize that without treatment for hypertension, there is an increased risk of stroke, heart attack, kidney failure, vision problems, and death.
4.Ask the patient for a treatment supporter and elucidate on the importance.
5.Counsel the patient on stopping tobacco and harmful use of alcohol.
6.Encourage the patient to increase physical activity to at least 30 minutes daily and get adequate rest.
7.Counsel the patient to lose weight if overweight, using the BMI and waist circumference as a guide.Aim for BMI within 18 – 25kg/m2; or waist circumference <94cm in men, and <80cm in women.
8.Give dietary counseling (a heart-healthy diet low in salt, trans-fat, and added sugar)
I.Eat 5 servings of fruits and vegetables per day.
II.Eat nuts, legumes, whole grains, and foods rich in potassium.
III.Eat fish at least twice per week.
IV.Use healthy oils like sunflower, flax seed, soybean, peanut, and olive.
V.Limit red meat to once or twice per week.
VI.Use less salt – restrict to less than 5g daily (1 teaspoon)
VII.Limit consumption of ultra-processed, canned, and 'fast' foods.
VIII.Avoid donuts, cookies, sweets, fizzy drinks, and juice with added sugar.
9.Counsel on the importance of adherence to medications, clinic, and laboratory appointments.
10.Educate the patient on self-management practices at home.

5.24 ABDOMINAL DISODERS

Abdominal disorders affects the structures surrounding or within the abdominal cavity. They can manifest as pain, swelling or masses on the abdominal wall. The common causes of abdominal pain are gastro-enteritis, malaria, appendicitis, rupture of the intestine, typhoid, sickle cell crisis and obstruction e.g. hernia. constipation, dysentery, amoebiasis, peptic ulcer diseases (PUDs), worm infestation etc. may also constitute abdominal disorders.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? When did it begin? How severe is it? 2.Where is it? 3.Do you have or have you had nausea or vomiting? 4.Have you had any constipation or diarrhoea? If diarrhoea, how many times do you pass stool? If constipation, when was your last stool? 5.Any blood in vomiting or in stool? Is the stool black? 6.Does food make the problem better or worse? Or no difference? 7.Is there abdominal pain? Where? Is it continuous or intermittent? 8.Is there anything that relieves it or makes it worse? 9.Have you passed any worm in the stool? 10.Do you have fever or cough? Painful and/or frequent urination? 11.Are there any urinary compliant? 12.Have you taken any medication? If yes, what are they? **ADDITIONAL** **IF FEMALE** 1.Are you having any problem with menstruation? Such as dark, heavy, scanty, painful menses etc. 2.When was your last menstrual period (LMP) 3.Are you using any contraceptive? 4.Is there any abnormal vaginal discharge? Is there any history of trauma?	1. General appearance: Stable, ill-looking, pale, in distress, dehydrated, icteric, cyanosed. 2.Temperature, pulse, respiration, blood pressure, weight: check and record 3. Eyes: Pale or jaundice 4.Mouth: dry tongue, coated tonsils 5. Chest: a) difficult or rapid breathing b) Abnormal or decreased breath sounds c) Crepitations or wheezes 6.Abdomen: a) Spleen, liver, other masses b) Swelling, distension, ascites, scaphoid c)Tenderness or board-like rigidity, rebound tenderness d) Bowel sounds, absent, normal, increase, decreased 7.Flanks: tenderness over spine or lower ribs 8.Anus: prolapsed, fissures, redness, 9. Hermia: umbilical, inguinal, scrotal, etc. 10. Haemoglobin: Estimate and record 11. Stool: Sent to lab for analysis and microscopy

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I.Abdominal pain with rigidity, vomiting blood, Swelling of abdomen	Signs of shock such as low BP less than110/70 mmHg Weak pulse Rebound tenderness Rigidity of abdomen	Severe condition Suspect Acute Abdomen	1. Secure IV access IV normal saline1litre stat (determine the flow rate based on clinical assessment of the patient) 2. IV Ciprofloxacin 400mg stat 3. IV Metronidazole 500mg stat 4. IM Diclofenac 75mg stat 5. REFER immediately
II.Abdominal pain with or without vomiting,Vaginal bleeding or discharge, Last period more than one month	Abdominal tenderness with or without signs of shock Vaginal bleeding	Severe condition Suspect Ruptured Ectopic Pregnancy	1. IV normal saline 1litres stat 2. IM Diclofenac 75mg stat 3. Non Neumetic Anti shock garmet 4 .REFER immediately
III.Abdominal pain with swelling on the left side	splenomegally Warm to touch Temperature 37.50C or above Palpable mass may or may not be pale	Severe condition Suspect Rupture Spleen	1.IV normal saline 1litre stat 2. IM Diclofenac 75mg stat 3.REFER immediately

IV. Abdominal pain with jaundice with or without diarrhea, vomiting	*Slight jaundice *Upper abdominal Tenderness	Moderate to severe condition Suspect Hepatitis	1 .Bed rest, high carbohydrate diet 2. Glucose fluids orally 3. Caps Amoxicillin 500mg 8hourly for 5days 4.Tabs Multivitamin 2tab 8hourly for 1week 5.Encourage use of PPE 6.Review in 3days 7. If no improvement, REFER
V. Abdominal pain, Painful and frequent urination, May or may not have fever	Tenderness in the flank Cloudy urine May or may not be warm to touch Temp 37.50C or above	Moderate condition Suspect Pyelonephritis	1.IV Gentamicin 80mg 8hourly x48hrs continue with 2.Caps Amoxicillin 500 mg 8hourly for 5days 3.Tab. Paracetamol 1000mg 8 hourly for 3days Giive Mist Potassium citrate 15 ml in water daily x3 days 4.If no improvement in 3days, REFER
VI. Abdominal pain before or after food, Heart burn, Black stool	Tenderness in the upper abdomen	Moderate condition Suspect Bleeding	1. Susp Mist Mag Trisilicate 10mls 8hourly for 5days OR Tabs Mist Mag trisilicate 2tabs 8hourly for 5days 2. Caps Amoxicillin 500mg 8hourly for 5days 3.Tabs Metronidazole 400mg 8hourly for 5days 4. Caps Omeprazole 20mg 12hourly for10days 5. Review in 1week 6. If no improvement, REFER
VII. Non-painful swelling in the abdomen	May or may not have black stool Reducible, non tender Swelling in the groin or umbilical area	Peptic Ulcer Diseases Mild to Moderate condition Suspect Hernia	1.Explain condition to client 2.REFER
VIII. Abdominal pain with Constipation with or without blood and or mucus in stool	*Abdominal discomfort with or without distension	Mild condition Suspect Dyspepsia/ Amoebiasis	1. if abdomen is distended, REFER if not distended advise to take fruits and vegetables 2. Encourage fluid intake 3. Susp Mist Mag Trisilicate 10mls 8hourly for 5days OR Tabs Mist Mag trisilicate 2tabs 8hourly for 5 days 4. Caps Amoxicillin 500mg 8hourly for 5days OR Tabs Ciprofloxacin 500 mg 12hourly for 5 days 5.Tabs Metronidazole 400mg 8 hourly for 5days 6.Tabs Hyosine butyl bromide 10mg 8 hourly for 3 days 7. Review in 3days 8. If no improvement, REFER

Bleeding during passage of stool, Irritation around the anus	*Cut in the anus *Redness with or without visible blood	Mild condition Suspect Anal Fissure	Zinc oxide ointment locally Sitz bath with warm saline water Advise high vegetable Diet and more fluid intake Caps Amoxicillin 500mg 8hourly for 5days OR Tabs Ciprofloxacin 500mg 12hourly for 5 days Tabs Metronidazole 400mg 8hourly for 5days Tabs Vitamin C 200mg 8hourly for 7 days. 4.If no improvement in 3days, REFER. Review in 3 days, If no improvement REFER

Health Education

1.Advise on personal and environmental hygiene.
2.Give nutritional counseling (emphasize on avoiding tobacco and alcohol consumption).
3.Advise on use of portable water.
4.Encourage regular moderate exercise.
5.Advise on drug compliance.
6.Importance of home-based records during visit to clinic or hospital.

5.25 WORM INFESTATION

Worm infestation is the presence of large number of worms in the gut, which causes discomfort to the patient. Ascaris lumbricoides (roundworm) is the commonest worm infestation. Heavy roundworm infestation produces abdominal pains, vomiting, constipation and sometimes obstruction of the intestine. Hookworm infestation is common in rural areas where heavy infestation causes blood in stool, anaemia and malnutrition. Severe Trichuris trichiura (whipworm) infestation causes bloody diarrhoea, abdominal pain, anaemia and prolapse of the rectum. Thread/pinworm (Strongyloides stercoralis) also lives in the intestine and causes itching around the anus. Tapeworms is a flat ribbon-like worm that can also be found in the intestine.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.Have any worms been passed? If yes, describe the appearance? How many? 3.Do you eat well? 4.Has there been abdominal pain? Vomiting? Constipation? 5.Is abdomen tender to touch? 6.Is there any diarrhoea? Blood in the stool? 7.Is there rectal itching? 8.Is there any other illness? 9.Has any medication been used?	1.General appearance: Stable, Distressed, pale, icteric, cyanosed, dehydrated 2.Temperature, pulse, respiration, blood pressure: check and record 3.Eyes: pallor, jaundiced 4.Chest: wheezing, rhonchi, crepitations 5.Abdomen: tenderness, distended, scaphoid 6.Anus: redness, rashes, prolapsed 7.Haemoglobin: estimate and record 8.Weight and height: measure and record 9.Stool examination (analysis and microscopy): if laboratory is available

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Abdominal pain	*Abdominal tenderness with signs of obstruction *Ova or Cyst seen	Severe condition Suspect intestinal obstruction	1.IV normal Saline 1litre 2. IM Diclofenac 75mg stat 3..REFERimmediately
II. Abdominal pain	*Abdominal tenderness without signs of obstruction *Ova or Cyst seen	Moderate condition Suspect worm infestation	1.Tab Albendazole 600mg stat 2.if no improvement, REFER
III. Passing worms per rectum	*Underweight with or without mild abdominal pain or distention *Presence of ova or Cyst from laboratory result	Mild to moderate condition Suspect worm infestation	1.Tab Albendazole 600mg stat OR 2.Tab. Pyrantel pamoate 4-6 tablets stat 3. If tapeworm give niclosamide 200mg stat

Health Education

1.Advise on personal, food and environmental sanitation (proper disposal of excreta)
2.Advise on periodic deworming
3.Give nutritional counseling (wash fruit and vegetable before eating).
4.Advise on proper care of feet (use of foot wears)
5.Importance of taking home-based records during visit to the clinic or hospital.

5.26 GENITO-URINARY DISORDERS

Genito-Urinary problems are common and can be due to metabolic/degenerative changes, infections, abnormal growths, etc. Examples of metabolic/degenerative changes include diabetes, kidney stones, vaginal/uterine prolapse etc. Examples of abnormal growths include enlarged prostate, cervical cancer etc. Examples of common infections include gonorrhea, syphilis, non-specific bacterial infection, schistosomiasis etc...
Patient may complain of discharge, pain and burning sensation on passing urine (micturition), blood in urine., frequent micturition, urinary retention etc.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? How long have you had it? 2.How old are you? 3.Do you urinate often? How often: night and day? 4.Do you have swelling in the scrotal sac with or without pain? 5.Did you have any trauma? Or Mumps? 6.Do you have abnormal pain or burning sensation on micturition (dysuria)? 7.Do you have weight loss? 8.Do you drink more than usual? (Explain) 9.Do you have pain in the flank? 10.What is the colour of your urine? Is there blood in the urine? 11.Do you have Ulcer in the mouth? Oral thrush? Bleeding gum? Skin rash? Night sweats? Weight loss? 12.Do you have fever? Nausea? Vomiting? 13.Do you have history of wading in stream during childhood? (schistosomiasis) 14.Are you on any birth control method? (If female) 15.Have you ever had this problem before? Any treatment? 16.When was your last menstrual period? (if female) 17.Do you have discharge or itching from the vagina (female) or penis (male)? 18.Are you having any problem with menstruation? Such as dark, heavy, scanty, painful menses etc.(if female) 19.Do you experience unusual pains during intercourse (if female)	1.General appearance: Stable, distressed, pale, icteric, cyanosed, dehydrated ill-looking, wasting, puffy 2.Temperature, pulse, respiration, blood pressure, weight: check and record 3.Eyes:pallor, jaundice, discharges 4.Chest: crepitations, decreased breath sounds, wheezing, rhonchi 5.Abdomen: tenderness, swelling or masses 6.Flanks: tenderness 7 .Limbs:(a) legs for swelling (b) palms, nail bed, for pallor 8. Genitals: sores, cracks, odour, discharges (pus, blood, fluid). If pus, note the colour 9. Urine: (a)colour or cloudiness, odour, deposit, specific gravity, reaction, volume/amount. (b)protein (c)Sugar (d) Sent to lab for analysis and microscopy.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Swelling in one or both scrotal sacs with pain	*Swollen Scrotum *Tender	Severe condition Suspect strangulated Scrotal Hernia/Hydrocele	1.Tab Ibruprofen 400mg 8hourly for 3days (After food) 2.REFER immediately
II. Blood in urine with history of trauma to the external genitalia	*Blood urine (initial)	Severe condition	1.REFER immediately
III. Swollen scrotum not painful	*Swollen scrotum, may or may not be Reducible	Moderate to severe condition	1.Reassure client 2.REFER immediately
IV. Swollen painful scrotum with or without trauma	*Swollen tender scrotum	Severe condition Suspect Testicular Torsion	1.Apply cold compress 2.Untwist the testicular cord (left or right Until comfortable) 3. IM Diclofenac 75mg stat 4 .REFER in no improvement

V. Swollen painful scrotum with or without discharge	*Swollen tender scrotum *with or without discharge	Moderate to severe condition Suspect epididymo-orchitis	1. IM Diclofenac 75mg stat Tabs Paracetamol 1g 8hourly for 3 days 2. Tabs Metronidazole 400mg 8hourly for 7days T3. abs Vitamin C 200mg 8hourly for 7days
VI. Urinary problem with any of the following: a) Shortness of breath b) Nausea and/or vomiting	Chest may or may not be clear *Brown or dark urine *Swollen legs or Abdomen	Severe condition Suspect Acute Renal Failure	REFER immediately
VIII. Blood in urine, painful micturition, losing weight weak	*Protein in urine *Cloudy or clear urine	Moderate to severe condition Suspect urinary tract infection	1.Tabs Ciprofloxacin 500mg 12 hourly for 7days 2.Tabs Metronidazole 400mg 8 hourly for 7days 3.Tabs Vitamin C 200mg 8hourly for 7days 4.Paracetamol 1g 8hourly for 3 days 5 .Encourage fluid intake s 6.If no improvement in 3days, 7. REFER
VIII. Blood in urine, painful micturition, losing weight weak	*Blood in urine *Pallor	Severe condition Suspect bladder tumou	1. 1. Conduct urine test for schistosomiasis 2. If positive Give Tabs praziquantel 240 mg of 4 tabs stat 2 .If negative, REFER
IX Blood in urine	*Blood in urine (terminal)	Moderate condition Suspect schistosomiasis	1.If positive, give praziquantel 240mg of 4 tabs stat 2 Tabs Vitamin C 200mg 8hourly for 7 days 3.Review in 2 weeks 4.If no improvement, REFER to the hospital
X. Dark yellow urine without any other symptom	*Dark yellow urine *No jaundice	Mild condition Suspect dehydration	1.Encourage high fluid intake 2.Review in 3 days 3.If no improvement, REFER to the hospital

Health Education

1.Advise on personal hygiene
2.Advise on the protection of genital area
3.Advise on drug compliance/adherence
4.Encourage adequate fluid intake
5.Advise on side effects of medications
6.Advise on the treatment of partner
7.Importance of taking home-based records during visit to the clinic or hospital

5.27 UPPER AND LOWER LIMBS DISORDERS

The limbs may be affected by arthritis of various kinds, trauma and localized or systemic infections which may manifest as dislocation, fracture, swelling pain and deformity in adults. Infection in the bone and sickle cell crisis can also cause pain in the limbs.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? When did it start? 2.Has it happened before? 3.Was there any accident or injury? 4.If there is injury, have you been able to move the part since? 5.Is there any pain? Tingling? Numbness? Swelling? 6.Do you have joint pains or are they stiff? 7.Do you have back pain? Do you have fever??	1.General appearance: Stable, ill-looking wasted, deformed, distressed, pale, icteric, cyanosed, dehydrated 2.Temperature, pulse, respiration, blood pressure weight: check and record 3.Eyes:jaundice 4.Spine: localized tenderness 5.Limbs: a) deformity b) tenderness and swelling c)bleeding d) fracture e) not able to move part f) wasting g) varicosity h) sign of paralysis i) sign of foot drop j) Trans illumination test k) Fluctuation test 6. Haemoglobin: estimate and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Painful swelling on the upper and/or lower limbs as a result of an accident (home or road traffic)	Painful swelling of the upper and/or lower limbs with deformity	Severe condition Suspect fracture	1.Splint to affected limb/s 2.Tabs Paracetamol 1000mg stat 3.IM Diclofenac 75mg stat 4. Refer 3.REFERimmediately
II. Painful swelling on the upper and/or lower limbs as a result of an accident (home or road traffic)	Painful swelling of the upper and/or lower limbs without deformity	Moderate condition Suspect soft tissue injury	1.Splint to affected limb/s 2.Tabs Diclofenac 50mg 12hourly for 5 days (After food) 3.Tabs Chymoral 2 tabs 12hourly for 5 days 3.Review in 3days 4.If no improvement, REFER
III. Pain in the bone or Muscle with localized redness, pus accumulation and tenderness.	Local Redness Swelling and tenderness of bones. If one of these is found positive:Trans illumination test or Fluctuation test. Positive paracentesis test (MCS recommended).	Mild condition Suspect infection (suspect pyoderma)	1.Tabs Ciprofloxacin 500mg 12hourly for 5days 2.Tabs Metronidazole 400mg 8hourly for 5days 3.Tabs Ibuprofen 400mg 8hourly for 3days (After food) 4. Incision and drainage (I & D) 5. Daily dressing 6.Review in 5days 7. If no improvement, REFER
IV. Pains and swellings in the joint with difficulty in moving the affected part.	Tenderness in the joint Stiffness in the joint	Mild condition Suspect Arthritis/sprain.	Splint to affected limb/s Tabs Ibuprofen 400mg 8hourly for 3days (After food) Cold compress/poultice alternate with warm compress. Review in 3 days If no improvement, REFER

Health Education

1.Advise on prevention of accidents
2.Counsel on adequate diet (calcium rich foods)
3.Advise on proper posturing
4.Importance of taking home-based records during visit to the clinic

5.28 MENTAL HEALTH DISORDERS

Mental health conditions showing with psychological or behavioral manifestations and/or impairment in functioning due to social, psychological, geriatric genetic, physical, or biological disturbance. Mental disorders are characterised by abnormalities in a person's emotions, thoughts, cognition, sensory perceptions, beliefs, and behaviour. Mental disorders usually manifest in many ways.

The affected patients may become restless, afraid, suspicious, and confused. They may think that people are after them. They may imagine/or complain of seeing strange people or things. They may become violent and attack other people or themselves without cause. The condition may have come on after a fever due to malaria, typhoid, or viral infection. It may also be as result of severe emotional disturbance or abnormal emotional experiences or excessive drinking of alcohol or drugs over a significant period. If nothing is done immediately, there may be danger to life and property. Other symptoms include sleeplessness, hearing and seeing things others cannot see and suicidal tendencies.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.How long has the client had the problem? 3.Has the client had this or similar problem in the past? 4.Did he recently have febrile illness? If yes describe 5.Any head injury? Accident? Severe stress? E.g. loss of job, divorce, arm robbery attack, loss of relation or loved one etc. 6.Does the client use alcohol or drug such as Indian hemp? 7.Any headache? Jerking of limbs? Clinching of teeth? Biting of tongue? 8.How does the client function at work? With relations at home? 9.Has the client had prolonged fasting and praying or ritual dancing? 10.Is there change in the way the client speaks? 11.Does any other person in the family have the same problem? What makes you think that the client's behaviour is abnormal?	1.General appearance: unkempt, calm, restless, threatening or physically aggressive, ill, withdrawn, confused, tensed, frightened, fidgety, trembling/shaking, sweaty, cold clammy palms, acting strangely, tearful, hallucinating. 2.Temperature, Pulse, Respiration, Blood pressure: check and record. 3.Reaction: to the health worker – cooperative, withdrawn, aggressive 4.Eyes: redness, staring, bulging, lid retraction, pallor 5.Mouth: smell of alcohol or other substances 6.Speech: audible, shouting, understandable, meaningful, relevant to the questions asked or to the situation, slurred, rambling, excessive, stammering, hesitant, expressing worry, any strange statements or beliefs expressed 7.Orientation: responds to name, knows where he is, has idea of time 8.Pockets: drugs or other substances in the pocket 8.Signs of tension – suspicious, hyper-alertness 9.Thoughts: expression of hopelessness, suicidal tendencies 10.Weight: weigh and record 11. Urine: (a)colour or cloudiness, odour, deposit, specific gravity, reaction, volume/amount. (b)protein (c)Sugar (d) Sent to lab for analysis and microscopy

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. low energy, fatigue, sleep or appetite problems, persistent, sad or anxious mood, irritability, low interest or pleasure in activities that used to be interesting	Weak, unkempt, withdrawn, sad, irritable, smell of alcohol/ tobacco or other substances.	Severe condition Suspect depression	counsel on sleeping habit and drug usage and participation in community activities group psychotherapy
II. Incoherent or irrelevant speech, Delusions (force, firmly held beliefs), Hallucination (hearing voices or seeing things that are not real), Neglecting usual Responsibilities	Unkempt, self-neglect, aggressive, hallucinating	Severe condition Suspect psychosis	1. IM Chlorpromazine 100mg stat 2. REFER immediately

iii. History of frothing in the mouth after signs or premonitions, loss of consciousness with any of the followings: Jerking of limbs Clenching of teeth biting of tongue Incontinence (urine or stool)	May or may not be Jerking Frothing Biting of tongue Unconscious or drowsy	Moderate condition Suspect epilepsy	1. Prevent Client from injury to self 2. Keep airway clear 3. Check if client is on medication 4. If not on any medication , give IV Diazepam 10-20mg stat (if CONVULSING) 5. Tabs Phenobarbitone 30 mg stat 6. REFER
IV. Violent behaviour	Injection marks Drugs/substance in pockets Agitated Violent behavior May or may not be drunk Perceive smell of alcohol and/or other substances	Moderate condition Suspect alcohol or drug abuse	1. Restrain and observe until effects have worn off 2. IV Diazepam 10-20mg stat 3. Seek police and Community/religious Leaders assistance if necessary 4. Review in 2 weeks 5. Counsel client and family members.
v. Confusion, Restlessness, Dizziness or Convulsing, Known or unknown Diabetic	Restless or confused May be unconscious or convulsing Ketone smell (sweet breath) Check urine, blood	Severe condition Suspect hyperglycemia	1. Give IV normal saline 1 litre stat 2. REFER immediately
VI. Confused, Restlessness, Drowsy or unconscious, Known or unknown Diabetic	May be unconscious Sweating profusely Anxiety Restlessness No sugar in urine Check blood sugar	Severe condition Suspect hypoglycemia	1. Give IV 50% Dextrose stat 2. Give sugar in water until gets to the clinic 3. Re-assure client 4. REFER

Health Education

1.Counsel client and family members.
2.Give nutrition counseling.
3.Discourage abuse and use of alcohol, drugs and other substances.
4.Advise on Psychotherapy.
5.Encourage participation in community and religious activities.
6.Encourage to visit friends and relatives.
7.Importance of taking home-based records during visit to the clinic.

5.29. POISONING

A poison is a toxic substance which may enter the body through swallowing, breathing, absorption through skin or mucus membrane or infections. It can cause fits, marked restlessness, abdominal pain, diarrhoea, and vomiting. Patient may feel dizzy, weak, lose consciousness and it may result in death. Therefore, it is advisable to keep, kerosene, medicine, bleach, dye, alcohol insecticides, chemicals, and other harmful substances properly. The type of poison consumed should be communicated to the health worker.

The principles guiding treatment are the following: Ability to identify ingested poison, use of activated charcoal and providing support to vital functions (A B C D).

Airways - Relieve obstruction;

Breathing - Check the person's breathing, maintain an open airway and normal body temperature;

Circulation - treat hypotension;

Dextrose - avoid severe hypoglycemia by giving glucose or neutralizing the effect of poison by ingestion of milk or magnesium tablets.

HISTORY	EXAMINATION/ACTIONS
1.What did you ingest? 2.When was the poison taken? 3.Have you taken any medication? If yes, what? Did you vomit? If yes, any blood in the vomitus?	1.Examine poison container (if available). 2.General appearance: conscious, drowsy, sleepy, confused or unconscious, Stable, ill-looking, wasted, deformed, distressed, pale, icteric, cyanosed, dehydrated. 3.Temperature, Pulse, Respiration, Blood Pressure: check and record. 4.Eyes: size and reaction of pupils to light. 5.Chest: lower chest indrawing, fast breathing. 6.Abdomen: tenderness, rigidity, distended Mouth: burns, bleeding, dribbling, colour of membrane.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. History of poisoning with any sign	Temp less than 37.50C or above Irritable No other sign	Moderate condition	1. Induce vomiting by giving salty water or use spatula Milk drink, if not available, 2. Susp. Magnesium trisilicate 10ml 8 hourly for 2days 3. Review 2nd day 4. If no improvement, Refer
II. Ingestion of Kerosene or petrol or corrosive substance	irritable consciou	Moderate condition	1. Do not induce vomiting 2.Milk drink, if not available 3. Susp. Magnesium trusilicate10ml 8hourly for 2days Procaine penicillin 600,000iu daily for 7days Review 2nd day If no improvement, Refer
III. Iron poisoning	*Irritable, restless	Severe condition	1.Refer immediately

Health Education

1.Advise on proper storage and labeling of chemicals and harmful substances.
2.Advise against use of bleaching creams.
3.Advise to keep chemicals and harmful substances out of reach of children.
4.Advise on personal and environmental sanitation.
5.Advise on proper use of drugs.
6.Importance of bringing home-based records during visit to clinic.

5.30 MANAGEMENT OF HIV/AIDS PATIENTS

HIV/AIDS is of public health concern. It is responsible for increased number of deaths. It affects not only the patient but relations and friends. In recent times, with the introduction of Anti Retro Viral drugs (ARVs) the condition can be successfully managed while lives of the infected can be prolonged. The health worker therefore, should be able to provide care and support for people living with HIV/AIDS and those affected by it.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.How long have you had the problem? 3.Any weight loss? General weakness? 4.Have you had infections? 5.Have you had blood transfusion? Scarification e.g tribal mark? Circumcision? Tattoo? 6.Are you married? 7.Do you have a sex partner? How many sex partners do you have? 8.Have you had contact with relatives or partner with similar problems? 9.Have you been treated for HIV/AIDS before? 10.Doyou have genital and/or mouth sores that have refused to go? 11.Do you have persistent cough? Diarrhea? Repeated infections? Fever? For how long? 12.Did it persist or reoccur after treatment? 13.Doyou have itchy skin or skin rash? 14.What other problems do you have? Have you taken any drug? If yes. what type of drug?	1.General appearance: Stable, ill-looking, wasted, deformed, distressed, pale, icteric, cyanosed, dehydrate, healthy, thin 2.Weight: weigh and record. 3.Temperature, pulse, respiration, blood pressure: check and record 4.Hair: thin, scanty. 5.Eyes: sunken, pale conjunctiva. 6.Mouth: thrush, red and white patches. 7.Neck and Armpit: enlarged nodes 8.Skin: rash, ulcers, fungal infection, kaposis around the mouth. 9.Chest: crepitations, decreased or increased breath sound, respiration rate. 10.Abdomen: palpable organs, tenderness, distended 11.Groin: enlarged nodes, ulcers. 12.Genitalia: ulcers, rashes, discharge. Blood: RDT for HIV confirmation

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. History of recurrent or persistent diarrhoea of more than 1 month duration Generalized itchy skin rashes unexplained cough longer than 1 month Recurrent genital and mouth sores Prolonged fever for more than one 1 month Severe weight loss General weakness Ulcer that has refused to heal	Severe dehydration Temp. 37.5°C or above May or may not have weight loss greater than10% of body weight Generalised dermatitis Oral candidiasis Genital ulcer Generalized enlarged lymph nodes *HIV test reactive	Severe condition Suspect HIV/AIDS	1. Observe universal precautions with patient's body fluids. See Appendix. 2. EnsureCONFIDENTIALITY 3. Confirm contact address 4. Follow up patient SYMPTOMATIC TREATMENT 5. For fever, diarrhoea, cough, sore mouth and throat, skin problems, pains, anxiety and depression etc. Low -ORS 6.Tab. Chlorpheniramine 4mg at night for 3 days 7.Apply Nystatin oral Gel to mouth REFER for HIV Counseling and testing
II. Chronic Diarrhoea, Chronic cough, Mouth and genital ulcer, Oral thrush, Skin rash	Marked weight loss Generalised skin rashes Enlarged lymph nodes Mouth and genital ulcers	Severe condition	Give Symptomatic Treatment REFER to the hospital
III. Reported to be HIV positive, No other complaint	Laboratory result may or may not be positive weight loss may not be present	Mild condition	1. Give symptomatic treatment REFER to the hospital.

Health Education

1.Counselling on HIV/AIDS prevention.
2.Counselling on nutrition.
3.Counselling on adherence to treatment.
4.Contact tracing and treatment.
5.Compliance with drugs.
6. Condom use.
7.Counseling on the risk of re-infection.
8.Importance of taking home-based records during visit to the clinic or hospital.

5.31 SEXUALLY TRANSMITTED INFECTIONS (STIs) (FEMALE) REPRODUCTIVE TRACT INFECTIONS (RTIs) FEMALE

Reproductive Tract Infections are caused by organisms normally present in the reproductive tract or introduced from the outside during sexual contact, medical procedures or as a result of an imbalance in vaginal bacterial flora. The common organisms responsible are Neisseria gonorrhoeae, Chlamydia trachomatis and Trichomonas vaginalis. Others are Human Immunodeficiency Virus (HIV), cytomegalovirus, streptococcus and Hepatitis B. The rationale for syndromic management is that it is effective, prevents complications or mother – to – child transmission, encourages treatment of partner, promotes safe sex habits and use of cheap, locally available, and effective drugs. The common symptoms are; vaginal discharges, itching, ulcer, rashes, lower abdominal pains among other as presented in the table below.

SYNDROME	STI/RTI	ORGANISMS	TYPE
i. Genital Discharge	Bacterial vaginosis Yeast infection Gonorrhea Chlamydia Trichomoniasis	Gardnerella vaginalis Anaerobe Candida albicans Neisseria gonorrhoea Chlamydia trachomatis Trichomoniasis	Bacteria Fungal Bacterial Protozoan Bacterial
ii. Genital ulcer disease (GUD)	Syphilis Chancroid Herpes rash Granuloma inguinale (donovanosis) Lymphogranuloma venereum (LGV) Moluscum contagiosum	Treponema palladium Hemophilus ducreyi 2 Herpes simplex virus (HSV-2) Inguinale granuolomatis Chlamydia trachomatis M. contagiosum	Bacterial Bacterial Bacterial Viral Bacterial Bacterial
Lower abdominal pain (LAP)	Chlamydia Gonorrhoea Scabies(crab) Body lice Genital warts	Chlamydia trachomatis Neisseria gonorrhoea Sarcoptes scabies Phthirus pubis Human papilloma virus	Bacterial Bacterial Tick Tick Viral
Adopted from WHO (2005)			

HISTORY	EXAMINATION/ACTIONS
1.How long have you had the discharge? 2.Is the discharge purulent, mucoid, abundant or scanty? 3.Have you ever had a urethral discharge before? How long ago? 4.Did you notice the discharge after a particular sexual contact? 5.Have you had any sexual contacts after you noticed the discharge? 6.Do you feel any burning or pain on urination? 7.What treatment have you taken? 8.Are you having any problem with menstruation? Such as dark, heavy, scanty, painful menses etc. 9.Do you experience unusual pains during intercourse	1.General appearance: Stable, distressed, pale, icteric, cyanosed, dehydrated ill-looking, wasting, puffy 2.Temperature, pulse, respiration, blood pressure, weight: check and record 3.Eyes: pallor, jaundice, discharges 4.Chest: crepitations, decreased breath sounds, wheezing, rhonchi 5.Abdomen: tenderness, swelling or masses 6.Flanks: tenderness 7.Limbs:(a) legs for swelling (b)palms, nail bed, for pallor 8.Genitals: sores, cracks, odour, discharges (pus, blood, fluid). If pus, note the colour, (milk the urethral) 9.Anus: Check for purulent discharge, ulcer. 10.Throat: Redness, pus 11.Urine:(a) colour or cloudiness, odour, deposit, specific gravity, reaction, volume/amount. (b)protein (c)Sugar (d) Sent to lab for analysis and microscopy.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I.Vaginal discharge	Discharge is profuse Muco-purulent discharge from Cervix Risk Assessment positive (unmarried, under 21 and Sexually active, more than 1 partner in the last 12 months, new partner in past 3 months)	Severe condition Suspect cervicitis Vaginitis, urethritis	1.Tabs Ciprofloxacin 500mg 12hourly for 7 days 2. Caps Doxycycline 100mg 12hourly for 7days 3.Pessaries vaginal nystatin 100,000 units inserted every night for 14days 4.Tabs Metronidazole 2gm orally stat then 400mg 8hourly for 7 days 5.Tabs Vitamin C 200mg 8hourly for 7 days 6. Bring male partner for treatment 7. Review after 7days 8.If no improvement, REFER to the hospital 9. Refer to appendix on syndromic management 10. HCT, if positive REFER for ART and follow-up
II.Vaginal discharge Profuse, thin, watery or Creamy, thick whitish with or without, itching Painful Urination	Discharge is Profuse Thin, watery or Creamy or thick whitish with or without itching *Risk assessment negative	Severe condition Suspect the following: Candidiasis Vaginals Trichomoniasis	1.Pessaries vaginal nystatin 100,000 units inserted every night for 14days 2.Tabs Metronidazole 2gm orally stat then 400mg 8hourly for 14 days 3.Tabs Ciprofloxacin 500mg 12hourly for 14 days 4.Caps Doxycyline 100mg 12hourly for 14days 5.Tabs Vitamin C 200mg 8hourly for 14 days 6.Bring male partner for treatment 7. Review in 14 days, if no improvement REFER to the hospital 8.Refer to appendix on syndromic management 9. HCT, if positive REFER for ART and follow-up
III. Severe lower abdominal pain with Vaginal Discharge	Pain with Vaginal Discharge Pain on moving Cervix	Severe condition Suspect cervicitis	1.Tabs Paracetamol 1000mg 8hourly for 3 days 2.Tab Ciprofloxacin 500mg 12hourly for 7 days 3.REFER immediately to the hospital 4.Ifreferral is difficult, stat 5.Tabs Doxycycline 100mg 12hourly for 7days 6.Tabs Metronidazole 400mg 8hourly for 7days 7.Bring male partner for treatment 8. Refer to appendix on syndromic management HCT, if positive REFER for ART and follow-up

IV. Lower abdominal pain with any of the following: History of late or missed period (menses) Recent delivery/abortion Vaginal bleeding Fever	Vaginal discharge Pain on moving cervix Temp. 37.5^{0}C or above	Moderate to severe condition Suspect Pelvic Inflammatory Disease (PID)	1. Tab Ciprofloxacin 500mg orally stat 2. Tab Doxycycline 100mg 12hourly for 7 days 3. Tab Metronidazole 400mg 8hourly for 14 days 4. 4Cs (counselling, contact treatment, compliance with drug , condom use) 5. Review in 3 days 6. If patient improves, continue treatment until completion. 7. Bring male partner for treatment. HCT, if positive REFER for ART and follow-up 8. If no improvement, REFER to the hospital

Health Education

1. 4Cs (Counseling Contact treatment
   Compliance with drug, Condom use)
2. Avoid sexual intercourse during the period of the treatment
3. Advise of completion and adherence to drugs
4. Importance of taking home-based records to the clinic or hospital

5.32 SEXUALLY TRANSMITTED INFECTIONS [STIs (MALE) REPRODUCTIVE TRACT INFECTIONS [RTIs (MALE)]

Sexually Transmitted Infections (STIs) are infections that are spread through sexual intercourse. Common STIs include gonorrhea, syphilis, herpes, genital warts, HIV/AIDS etc. STIs can be painful and uncomfortable. The common syndromes are discharge, genital ulcer and lower abdominal pain. They can also have tragic consequences such as infertility. Therefore, all cases of STIs should be adequately treated.

HISTORY	EXAMINATION/ACTIONS
1.How long have you had the discharge? 2.Is the discharge purulent, mucoid, abundant, or scanty? 3.Have you ever had a urethral discharge before? How long ago? 4.Did you notice the discharge after a particular sexual contact? 5.Have you had any sexual contacts after you noticed the discharge? 6.Do you feel any burning or pain on urination? 7.What treatment have you taken?	1. General appearance: Stable, distressed, pale, icteric, cyanosed, dehydrated ill-looking, wasting, puffy 2.Temperature, pulse, respiration, blood pressure, weight: check and record 3.Eyes:pallor, jaundice, discharges 4.Chest:crepitations,decreased breath sounds, wheezing, rhonchi 5. Abdomen: tenderness, swelling or masses 6.Flanks:tenderness 7.Limbs: (a)legs for swelling (b)palms, nail bed, for pallor 8.Genitals:sores, cracks, odour, discharges (pus, blood, fluid). If pus, note the colour, (milk the urethral) 9. Uncircumcised male: retract the foreskin, be sure discharge is from urethra. 10. Check glans for redness, ulcer 11. Anus: Check for purulent discharge, ulcer. 12 Throat: Redness, pus 13.Urine:(a)colour or cloudiness, odour, deposit, specific gravity, reaction, volume/amount. (b)protein (c)Sugar (d) Sent to lab for analysis and microscopy.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
Groin (inguinal) swelling, Patient not sexually active No ulcer No rashes	Inguinal swelling	Severe condition	1. REFER immediately to the hospital
II. Swelling in the groin, Genital sore or Ulcer, History of sexual exposure	Sore/Ulcer present Swelling in the groin Risk assessment positive	Moderate to severe condition Suspect genital ulcer disease (GUD)	1. IV Benzathine Penicillin G.2.4 mega unit stat 2.Tab Erythromycin 500mg 6hourly for 7days 3.Tabs Metronidazole 2gm orally stat then 400mg 8hourly for 14days 4.Caps Doxycycline 100mg 12hourly for 14 days 5.Tabs Vitamin C 200mg 8hourly for 14 days 6.Bring female partner for treatment 7. 4Cs-counseling,compliance,contact treatment And condom use. 8. Review in 7 days 9.HCT, if positive refer for ARTS Treatment 10. if no improvement, REFER the hospital

III. Patient with Scrotal swelling and painful Urethral discharge, Patient sexually active	Examine the genitals (just look (DO NOT PALPATE)	Severe condition	1 Tabs Metronidazole 2gm orally stat then 400mg 8hourly for 14 days 2. Tabs Ciprofloxacin 500mg 12hourly for 14 days 3.Caps Doxycycline 100mg 12hourly for14days 4.Tabs Vitamin C 200mg 8hourly for 14 days 5.Bring female partner for treatment 6. 4Cs-Counselling, compliance, contact treatment And condom use 4.Advise to return 7days; if no improvement refer 5. REFER immediately to the hospital
IV. Urethral discharge and or painful urination, Genital Ulcer, Sexual exposure	Penile discharge Ulcer present Risk Assessment positive	Moderate condition Suspect genital ulcer disease (GUD)	1.Tabs Ciprofloxacin 500mg 12hourly for7days 2.Caps Doxycycline 100mg 12hourly for 7days 3. Tabs Vitamin C 200mg 8hourly for 7 days 4.Tabs Ibuprofen 400mg 8hourly for 5 days (After food) OR 5. IV Benzathine Penicillin G 2.4 Mega units 6 hourly for 3 days 6.Tabs Erythromycin 500mg 6hourly for 7 days 7.Tabs Metronidazole 400mg 8hourly for 7 days 8. Bring female partner for treatment 9. 4Cs-counseling compliance contact treatment And condom use 10. Review in 7 days 11. If no improvement, REFER to the hospital
V. Patient with painful urination no discharge	No penile discharge	Moderate condition	REFER for Urine Microscopy Treat based on result.
VI. Genital sore/Ulcer, Group of painful blisters (often recurrent), History of sexual exposure, With or without painful swelling in the genital area	Group of painful blisters Risk Assessment Positive	Severe condition Suspect genital herpes HIV/AIDs	1. Tabs Ibuprofen 400mg 8hourly for 5 days (After food) keep lesion dry 2. Tabs Vitamin C 200mg 8hourly for 7 days 3. 4Cs-Counseling Compliance Contact treatment and condom use 4.HCT 5.Review after 7days 6.If there is no improvement, REFER to the hospital

Health Education

1.Counselling
2.Contact treatment.
3.Compliance with drug.
4.Condom use.
5.Avoid sexual intercourse during treatment and relapse.
Importance of taking home-based records with during visit to the clinic or hospital.

SECTION SIX
MATERNAL HEALTH



NATIONAL STANDING ORDERS
FOR COMMUNITY HEALTH OFFICERS

6.0 MATERNAL HEALTH

6.1 PRENATAL (ANTENATAL) CARE

The prenatal period, also known as pregnancy, is the time from conception to childbirth. This period typically lasts around 280 days or 40 weeks. During pregnancy, a woman undergoes numerous physiological changes that ensure the survival of the fetus and prepare the body for labor and childbirth. These changes affect every system in the body, leading to altered metabolic rates, biochemical processes, and hormone levels. Normal pregnancy signs include various discomforts. However, persistent or severe symptoms Prenatal or antenatal care (ANC) Prenatal care encompasses health guidance and professional advice, monitoring of maternal health, necessary medical interventions, and maintaining the health and well-being of both mother and child. ANC is essential for detecting and managing medical and obstetric complications, addressing mental health issues such as anxiety, depression, or substance abuse, and screening for domestic violence.

The 2016 WHO ANC model aims to provide pregnant women with respectful, individualised, person-centred care at every contact and to ensure that each contact delivers effective, integrated clinical practices (tests and interventions), provides relevant and timely information, and offers psychological and emotional support by practitioners with good clinical and interpersonal skills working in a well- functioning health system. The WHO recommends a minimum of eight contacts: five contacts in the third trimester, two in the second trimester, and one in the first trimester. Quality of care should be emphasised, and great focus should also be maintained on birth planning and emergency preparedness.

HISTORY	EXAMINATION/ACTIONS
At all visits; 1.How are you feeling? 2.If ill, what are your symptoms? 3.Do you have problems sleeping? Do you sleep well? 4.Do you get easily irritated, anxious or depressed? 5.How is your appetite? 6.Have you lost weight? 7.Any bleeding? 8.What arrangement have you made for delivery? 9.Any other problem (s)? **Social History** 1.Any history of Alcohol Use? 2.Any history of Substance Use/Abuse? 3.Any history of Domestic violence? 4.Any history of Mental Problem? **At First Contact Only** **Medical History** 1.Do you have any particular medical condition, e.g hypertension, diabetes, sickle cell, epilepsy, asthma/allergies, kidney disease etc. 2.Have you had any Surgery (C/S, Hernioraphy, Appendectomy etc)? 3.Have you ever had T.D immunization? How many does? (ask for card) **Obstetric History** 1.How many pregnancies have you had? (livebirth stillbirth and miscarriage) 2.How many of your children are alive? 3.How many died? Age at the time of death and cause of death? 4.How many miscarriages have you had? 5.Any special maternal complications in previous pregnancies? E.g pre-eclampsia/eclampsia, bleeding, assisted delivery etc. 6.LMP: Obtain the date of LMP and calculate the EDD by adding 7 days to the first day of the LMP and subtract 3 months from the month of LMP i.e. LMP + 7 days -3months = EDD	1.General appearance: well or ill looking, facial expression 2.Skin: pallor, rashes, sores, oedema. 3.Check skin for signs of domestic violence (like scars and broken teeth etc) 4.Eyes: redness, jaundice, pallor, discharges 5.Nose: discharge, polyps 6.Mouth: condition of mucous membrane, tongue, gum & teeth 7.Chest: for respiratory rate & sign of respiratory distress 8.Breast: tenderness or lumps, nature of nipples whether normal, flat or inverted, scar 9.Height: (first visit only) Check and record 10.Weight: weigh & record (initial weight will serve as baseline for comparison with subsequent readings) 11.Blood pressure: measure and record (initial reading will serve as baseline for comparison with subsequent findings) 12.Temperature: Check and Record 13.Pulse Rate: Count and record **Obstetric Examination** Abdomen: Inspect for scar, shape and size Fundal height: measure and record Palpation: foetal parts, lie, presentation, position, masses Auscultation: foetal heart sound **Pelvic examination:** (after the 35th weeks) Inspection: check external genitalia Speculum should be correctly inserted for inspection of cervix for colour, smoothness, any lacerations, discharge, or bleeding. Pelvic measurement: diagonal conjugate, height of symphysis, prominent spines **Laboratory Investigation** **First Visit Only** HCT: for positive client to be enrol for ART VDRL/RPR (Rapid Plasma Reagin): for syphilis Blood grouping and rhesus determination Any appropriate investigation as indicated by patients' condition e.g HBs Ag, Widal test, M RDTetc HBs Ag for Hepatitis

	At All Visit: Perform the following for patient Urinalysis Haemoglobin estimation or PCV count Ultrasound Scan Refer clients for Uss at least once before 24 weeks

ANC CONTACTS

First Trimester

- Contact 1: up to 12 Weeks

Second Trimester

- Contact 2: 20 Weeks
- Contact 3: 26 Weeks

Third Trimester

- Contact 4: 30 Weeks
- Contact 5: 34 Weeks
- Contact 6: 36 Weeks
- Contact 7: 38 Weeks
- Contact 8: 40 Weeks

WELL VISITS

- 1ST Contact (up to 12 Weeks) or as soon as pregnancy is confirmed
- ·Establish a friendly atmosphere.
- ·Obtain comprehensive history.
- ·Perform head-to-toe examination.
- ·Request relevant laboratory investigations.
- ·Carryout Ultrasound Scanning.
- ·Provide Routine Drugs: Tabs Folic Acid 5mg daily, Ferrous Sulphate 200mg 8 hourly, Multivitamin daily and Vitamin B Complex 8 hourly for 4 Weeks.
- ·Provide the first dose of T.D. immunization.
- Provide LLITN (if available).
- ·Educate and counsel the client on any topic relevant to her condition.
- ·Initiate discussion on birth and emergency planning.
- ·Give an appointment for the next visit.

2ND and 3RD CONTACT (20 - 26 WEEKS)

- ·Maintain a friendly atmosphere
- ·Take history and perform physical examination.
- ·Abdominal examination and fetal heart sound.
- ·Take action on the results of laboratory investigations.
- ·Check for anaemia, measure B.P., and check urine for sugar and glucose.

- ·Provide a second dose of T.D. immunization.
- ·Provide first dose of S.P. for IPT of malaria in pregnancy at 13 Weeks and repeat after 4 Weeks.
- ·Provide Routine Drugs (Folic acid and ferrous sulphate).
- ·Continue the discussion on birth and emergency planning.
- ·Give the next appointment.

4TH -7TH CONTACT (30 - 38 WEEKS)

- ·Maintain a friendly atmosphere.
- ·Take history and perform physical examination.
- ·Re-check Hb and treat for anaemia if present.
- ·Check urine for sugar and glucose.
- ·Measure B.P.
- ·Provide routine drugs (Folic acid and ferrous sulphate).
- ·Provide a second dose of S.P. for IPT of malaria
- ·Check the uterine size and palpate the fetal position to exclude multiple pregnancies and abnormal presentation.
- ·Discuss lactation and contraception.

8TH CONTACT (40 WEEKS)

- ·Maintain a friendly atmosphere.
- ·Take history and perform physical examination.
- ·Check the uterine size and palpate fetal lie, presentation, engagement, and Fetal Heart Sound.
- ·Conduct Pelvic examination
- Check Hb to exclude anaemia,
- ·Measure B.P.
- ·Provide Routine Drugs (Folic acid and ferrous sulphate).
- ·Determine decision on Postpartum Contraception.
- ·Educate and counsel the client on signs/symptoms of labour.
- ·Review the birth plan with the client.
- ·Review the Birth Plan with the client.
- ·If not delivered by the end of 4 weeks, advise the client to report back and take a decision on delivery.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Difficulty breathing, persistent cough, yellow eyes, pallor, abdominal pain	Might be in pain Rapid breathing above 30/min Yellowness of eyes Pallor, HB < 7g/dl, HBsAg+	Severe condition Suspect Severe Anaemia, Hepatitis B, Severe Chest Infection	1. Reassure Patient 2. Refer to hospital immediately
II. Bleeding from the vagina before 20 weeks of gestation with or without lower abdominal pain	II. Bleeding from the vagina before 20 weeks of gestation with or without lower abdominal pain	Severe to moderate condition (suspect inevitable abortion if Os is dilated and threatened abortion when closed)	1.Assess vaginal bleeding for passage of clots or products of conception. If inevitable abortion without shock and pregnancy is below 12 weeks, 2. perform MVA and give Post abortion care (If trained) If pregnancy is over 12 weeks and/ or patient is in shock, set intravenous infusion of Normal Saline and refer 3.If bleeding is not much and cervical Os is closed advise on Complete Bed Rest 4,Tab Folic acid 5mg daily for 4 weeks 5.Tab Ferrous sulphate 200mg 8 hourly for 4 weeks 6.Review in 3 days
III. Bleeding from the Vagina after 20 weeks of gestation (exclude provocation by sex, foreign body or trauma)with or without abdominal pain	Vaginal Bleeding Low BP, high pulse rate, cold clammy skin	Severe condition Antepartum Haemorrhage (think of abruptio placenta or placenta previa	Do Not Do Vaginal Examination (V/E) Set IV line Normal Saline Refer to hospital immediately
IV. Unconscious, convulsion/ fits (severe muscle spasm), No previous history of epilepsy	Gestational age >20 Weeks BP > 140/90mmHg Noisy breathing Protein in urine 1+ or above With or without Unconsciousness Facial contraction and mouth foaming	Severe condition Suspect ante partum Eclampsia	1. Position pt. in left lateral position, clear and maintain airways 2. Give Oxygen if available Set up IV infusion of 500ml Dextrose-Water with cannula/needle Give 50% MgSo4 loading dose (IV 4gm in 12mls of distilled water over a period of 10-15minute, then IM 5gm + 1ml of 1% Xylocaine into each buttocks) 3.Tabs Nifedipine 10 mg under tongue (sublingual) or Hydralazine 5-10 mg IV slowly over 10-15 minutes 4. Refer to Hospital If BP is_> 160/110mmHg give Tabs Nifedipine 20mg under tongue (sublingual) stat. 5. If woman is in advanced stage of labour deliver the baby, if not 6. Refer immediately

V. Headache, visual disturbance, Epigastric pain, vomiting/ Nausea	Gestational age>20weeks Hyperreflexia BP > 160/110mmHg Protein 3 + in urine	Severe condition Suspect severe pre-eclampsia	1. Set up IV infusion of 500ml Dextrose-Water 2. Give 50% MgSo4 loading dose (IV 4gm in 12mls of distilled water over a period of 10-15minute, then IM 5gm + 1ml of 1% Xylocaine into each buttocks) 3. Tabs Nifedipine 20mg sublingual stat (if conscious) or Hydralazine 5-10 mg IV slowly over 10-15 minutes 4. if unconscious Refer to hospital immediately
VI. Pregnant woman with any two of the following present: Weight increase of 1 kg or more in a week Urine with protein of 1+ or more Pitting oedema	Raised BP, protein of 1+ or more in urine Pitting oedema	Severe condition Suspect toxaemia in pregnancy	1. Set up IV infusion of 500ml Dextrose-WaterGive 50%MgSo4 loading dose (IV 4gm in 12mls of distilled water over a period of 10-15minute, then IM 5gm + 1ml of 1% Xylocaine into each buttocks) 2. If BP exceeds 160/100, give Tabs Nifedipine 20mg under tongue (sublingual) stat (if conscious) or Hydralazine 5-10 mg IV slowly over 10-15 minutes if unconscious 3. Reduce Salt intake 4. Refer to hospital immediately
VII. Pregnant Woman with any of the following present: substance use/abuse (alcohol, cough syrup, etc) Scars suspicious of domestic violence Prenatal anxiety/depression	Disorientation Incoherent speech Physical Scars/bruises Looks anxious or scared Looks distracted with no interest to her surroundings	Moderate/Severe condition Suspect Common Perinatal Mental Disorders (CPMD)	1. Council on implications of substance use/abuse on the fetus 2. Review in 3days 3. Refer for Appropriate Care
VIII. Headache, Puffiness of the face or, generalized body oedema Pregnant Woman with any of the following present: Headache or Puffiness of face Polydipsia Polyuria Polyphagia (2nd or 3rd trimester)	Gestational age > 20 weeks BP > 140/90mmHg Protein 2 + in urine BP < 140/90mmHg Protein – No trace RBS - > 11.0mmol/l Or FBS > 7.0mmol/l BMI > 25kg/m2	Moderate to Severe Condition Suspect Gestational Diabetes and mild preeclampsia	1. Advice on the danger signs of imminent eclampsia 2. Tabs Nifedipine 20mg stat 3. Refer to hospital Advise on lifestyle modification; – Eat healthy diets (low carbohydrate with high fibre) – Adequate exercise etc – Reassure and refer
X. Constant Fatigue, Family History of Diabetes Prior history of Macrosomic baby (> 4kg at birth) Known diabetic Unconsciousness	BP 140/90mmHg OR less Protein – Present or not present RBS - > 11.0mmol/l Or FBS > 7.0mmol/l BMI > 25kg/m2	Severe to Moderate Condition Suspect Gestational Diabetes	Advise on lifestyle modification; – Eat healthy diets (low carbohydrate with high fibre) – Adequate exercise etc – Reassure

XI Headache pedal oedema	Protein 1+ in urine, with or without pitting oedema BP not more than 140/90mmHg	Severe condition Suspect mild pre-eclampsia	1. Advice on adequate bed rest 2. Tabs Nifedipine 20mg daily for 5 days 3. Restrict salt intake 4. Review in 3 days 5. If improving, continue on bed rest If no improvement, Refer to hospital
XII. Headache, Pedal oedema, epigastric pain Known hypertensive	No Protein in urine BP=/> 140/90mmHg	Mild-Moderate Condition Suspect Essential Hypertension	1. Advice on adequate bed rest 2. Tabs Nifedipine 20mg daily for 5 days 3. Restrict salt intake 4. Review in 3 days 5. If improving, continue on bed rest 6. If no improvement, Refer to hospital
XIII Weakness, Dizziness, Breathlessness at rest, palpitation	Pallor- paleness of the conjunctiva, tongue or palms of the hands etc. Haemoglobin below 7g/dl or PCV of less than 30%	Severe condition Suspect severe anaemia in pregnancy	1. Reassure Patient 2. Advice on hospital delivery 3.4. Refer immediately to hospital
XIV. Discharge of watery fluid from the vagina, Not in labour	Wet pad or cloths No sign of labour, Abnormal colour and odour of liquor FHS = 160b/m and above	Severe to moderate condition pre mature rupture of membrane (PROM)	1. Apply clean sanitary pad 2. IM Gentamicin 80mg (immediately) stat 3. Caps Amoxicillin 500mg (immediately) stat 5 Encourage bed rest 6 Review after 5 days
XV. Cough, Fever, sneezing, difficulty in breathing	Cough chest not clear Temp 37.5^{0}C or above BP 140/90mmHg and above Hb between > 7g/dl mRDT = +ve	Moderate condition Malaria in Pregnancy Hypertension in Pregnancy	1. Caps Amoxicillin 500mg 8hourly for 5 days 2. Tabs Paracetamol 1gm 8hourly for 3 days 3. Tabs Artemether/Lumefantrine AL 80/480mg 1 tab at 0 hour, 8 hours later, then, 12 hourly for 3days if pregnancy is within 2nd and 3rd trimester 4. Tabs Nifedipine 20mg daily for 5 days 5. Continue routine haematinic 6. . Encourage more fluid intake. 7. Nutritional counselling 8. Review in 5 days 9. If no improvement Refer
XVI. Pain or burning sensation on passing urine, increased frequency of micturition after 12 weeks gestation, lower abdominal pain	Urine test protein 1+ or more With or without high temperature	Mild condition Suspect t Uninary transmitted infections UTI	1. Encourage more fluids intake 2. Tabs Erythromycin 500mg 6hourly for 7 days 3. Tabs Metronidazole 400mg 8houly for 7days 4. Tabs Vitamin C 200mg 8hourly for 7 days 5. Tabs PCM 1g 8hourly for 3 days Review in 3 days, 6. if no improvement Refer

XVII Vaginal discharge, painful sores/rashes. With or without Lower abdominal pain	Excessive vaginal discharge Abnormal colour or cloudy Abnormal odour	Mild condition Suspect STIs	1. Treat according to National will the guideline be provided? in the appendix perhaps 2. Avoid drugs that are contraindicated to Pregnancy 3. Education on the need to treat partner and cowives 4. Review in 3 days. 5. If no improvement Refer
XVIII. Pain in the abdomen or back	No abdominal tenderness With or without uterine contraction Intermittent abdominal pain after 28 weeks gestation Pain often associated with blood stained mucus discharge (show) Watery vaginal discharge or a sudden gush of water Cervical effacement i.e., the progressive shortening and thinning of the cervix during labour; and cervical dilatation i.e., the increase in diameter of the cervical opening measured in centimetres.	Moderate condition Suspect labour	1. Admit in labour ward 2. Monitor vital signs PCV or HB estimation Urinalysis (sugar, protein & ketones) Allay fears of woman Open Labour Care Guide if/when cervix is 5cm or more dilated (see Appendix)
XIX. Nausea, Vomiting in the morning may or may not have seen period (menses)	Anxious No other findings PT = +ve	Mild to moderate condition Suspect morning sickness	1. Advise on small frequent meals especially dry foods (energy foods) 2. Give Low Osmolality ORS Advise to get up slowly from bed in the morning 3. Review in 3 days 4 If vomiting persists Give IV Glucose (Hyperemesis graidarum) Refer 4. If there is history of fainting, review Hb, BP, Urine for sugar Ask for black stool or other bleeding 5. Nutritional counseling 6. Review weekly
XX. Heartburn	No other findings	Mild condition (suspect reflux)	1. Advice to avoid spicy and too oily food 2. Ask to take cold milk during attack 3. Tabs Mist Magnesium Trisliciate 2 8hourly for 5 days or when necessary (PRN) if condition is severe 4. Tabs Metoclopramide10mg daily x3days 5. Advise to stay chest-up after meals 6. Avoid fizzy drinks 7. Eat few hours before bed
XXI. Inability to defecate for some days, abdominal discomfort	No other findings	Mild condition Suspect Constipation	1. Advice to take plenty of water, fruit, leafy vegetables, and fiber rich foods

XII. Visible blood vessel and leg cramps	Visible veins on the lower extremities	Mild condition Suspect Varicosity	1. Keep leg elevated while sitting or lying down 2. Advise on calcium rich foods (sea foods, soyabeans) if in 3rd trimester
XIII. Pregnant Woman with sleeping difficult	No other findings	Mild condition	1. Advice on sleeping less during the day. (take a nap only) 2. Avoid going to bed hungry and do not eat late. 3. Encourage to take enough fluids to aid digestion 4. Ensure the sleeping place is comfortable. 5. Avoid tight clothes, allow fresh air, take a shower, and relax before going to sleep.
XXIV. Pregnant (well), woman for registration or routine visit	Pregnant well No other finding Urine normal	Normal condition	Refer to the page on well visit (ANC) depending on the number of visit.

Health Education

1. Ensure adequate rest (at least 8 hours at night and 2 hours during the daytime).
2. Provide nutritional counseling.
3. Emphasize personal, food, and environmental hygiene.
4. Offer mental health counseling.
5. Encourage drug compliance and discourage the use of unprescribed and herbal drugs.
6. Provide advice on maintaining good posture.
7. Advise against lying flat on the back in late pregnancy, as it reduces blood supply to the placenta.
8. Stress the importance of hospital delivery and being prepared for emergencies.
9. Educate on the danger signs in pregnancy, such as severe headache, vaginal bleeding, severe abdominal pain, convulsion, etc.
10. Highlight the importance of bringing home-based records to the clinic or hospital at all times.
11. Provide guidance on newborn care and promote exclusive breastfeeding.
12. Offer postpartum family planning services.

6.2 MATERNAL HEALTH (LABOUR AND DELIVERY)

Normal labour is characterized by the onset of strong, painful, and regular uterine contractions and retraction, resulting in the delivery of a live baby without injury to the mother or baby. Labour progresses in two phases: latent and active. The latent phase is marked by cervical effacement (0-4cm dilated), while in the active phase, cervical dilatation occurs much faster. The active phase of labour occurs when the cervix is dilated between 5 and 10cm.

Labour is divided into four stages:

First Stage: This stage begins with the onset of strong, painful, and regular uterine contractions, leading to the dilation of the cervix and the appearance of the show. It should not last for more than 12 hours.

Second Stage: It starts from full dilatation of the cervix to the expulsion of the baby and should not last for more than 1 hour.

Third Stage: This stage begins after the delivery of the baby and continues until the complete expulsion of the placenta and membranes, as well as the control of bleeding, which should not last for more than 30 minutes.

Fourth Stage: It encompasses the first hour after delivery, during which both the mother and baby are allowed to rest and are closely observed. This period is critical, as complications can occur.

HISTORY	EXAMINATION/ACTIONS
1.How many months pregnant are you? 2.How are you feeling? Do you have any concerns? 3.Do you have problems sleeping? Do you sleep well? 4.Do you get easily irritated, anxious or depressed? 5.Are you having strong contractions? When did it start? How often? Any now? 6.Is there any show (thick blood stained mucus)? 7.Have the membranes ruptured? When? Are you draining now? 8.Do you feel the baby's movement? 9.Have you had any abdominal operations before? (unbooked) 10.Did you attend the ANC Clinic? What tests were done, what information/instructions and treatments were given to you?	General appearance: in pains, distressed, anxious Temperature: check and record Pulse: check and record Respiration: check and record Blood pressure: measure and records Eyes: pallor, jaundice Mouth: pallor, dry tongue Heart: rate, rhythm, murmur Abdomen: (a) inspect- shape, scars, distention, masses, whether it moves with respiration. (b) palpate- presenting part, foetal parts, position, foetal lie (c) Contraction- frequency, duration, rhythm (at least 2 in 10 minutes each lasting for 40 — 45 secs) Foetal Heart sound: heard, rate and rhythm Limbs: oedema Vulva: protruding or ruptured membranes, liquor draining, bleeding Vagina: (a) Condition of vagina (b) Cord in vagina (c) Cervical dilation (d_ presenting part, foetal parts, position, cord prolapse Urine: protein,sugar, acetone, katone, etc Haemoglobin: estimate and record Use the Labour Care Guide to monitor the progress of labour once the woman is in the active phase. (sample of Labour Care Guide).

MANAGEMENT OF NORMAL LABOUR

First Stage of Labour:

- Explain the condition to the client.
- Explain all procedures and seek permission for examination.
- Ensure and respect the privacy of the woman.
- Discuss findings with the client and keep her informed about the progress of labour.
- Encourage active movement alternating with rest between contractions and frequent emptying of the bladder.
- Encourage the woman to bathe or wash herself and her genitals at the onset of labour.
- Ensure infection prevention.
- Avoid vaginal examination during contractions.
- Do not shave the perineal area.
- Do not administer an enema during labour.
- Start a Labour Care Guide when labour is in the active phase.
- Allow the client to choose a more comfortable position.

Second Stage of Labour:

- Continue to monitor maternal and fetal vital signs.
- Swab the vulva with a mild antiseptic lotion.
- Discourage bearing down until the head is at the perineum (teach the woman to breathe through the mouth to prevent pushing before the head has distended to the perineum).
- Ensure controlled delivery of the head by taking the following precautions:
- Keep a hand on the head as it advances with contractions.
- Cover the anus with a perineal pad using your other hand.
- Observe the perineum for signs of tear.
- Ask the mother to breathe deeply and steadily with her mouth open and not to push during the delivery of the head.
- Feel gently around the neck for the presence of the cord; if the cord is loosely felt, deliver the baby through the loop, but when it is tight, clamp and cut it, then unwind it from the neck.
- Allow for natural restitution.
- Deliver the anterior shoulder by applying gentle pressure downward.
- Lift the baby upward toward the abdomen to deliver the posterior shoulder and the rest of the body.
- Place the baby on the mother's abdomen.
- Note the time of delivery.
- Tie/Clamp The Cord in 2 Places and Cut the Center After About 2-3 Minutes.
- Provide immediate newborn care (Refer to Newborn Section).
- Avoid the following:
- Rectal examination.
- Lithotomy position with or without stirrups.
- Fundal pressure during labour.

- Verbal abuse.

Third Stage of Labour (Use Active Management of Third Stage of Labour - AMSTL):

- Palpate the abdomen to rule out the presence of another baby.
- Administer a uterotonic IM oxytocin 20mg on the thigh within one minute after birth or IM or IV Carbetocin 100ug stat OR Tabs Misoprostol 600ug (3 tabs) under the tongue (sublingual).

Perform Controlled Cord Traction (CCT) as follows:

- Clamp the cord close to the perineum.
- Use the other hand to apply counter traction on the uterus, just above the pubic bone.
- Use a forward and downward motion to deliver the placenta.
- Maintain steady but gentle pressure.
- Cup the placenta in your hands and deliver it gently once it is visible at the perineum to ensure that the membranes are intact.
- Examine the placenta and membranes to ensure completeness.
- Clean the mother and check the perineum for laceration or bruising.

Fourth Stage of Labour:

- Massage the uterus every 15 minutes for the next 2 hours to ensure that it is well contracted.
- Continue to check vital signs.
- Ensure breastfeeding is initiated to promote bonding.
- Monitor the amount of bleeding.
- Observe for any danger signs in both mother and newborn.
- Teach the mother about perineal care.

Best Practices:

- Close monitoring during the first 6 hours.
- Monitor BP, Pulse, Vaginal bleeding, and uterine hardness; every 15 minutes for 2 hours, every 30 minutes for 1 hour, and every hour for the last 3 hours.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. Labour pains	Mal-presentation Transverse lie Shoulder presentation Brow or face presentation Breech presentation	Abnormal Condition	1. Explain the condition to client and relation 2. Set up intravenous infusion of Dextrose-Saline and Refer to hospital immediately
II. Labour pains for more than 12 hours	Non descent of presenting parts Vomiting Rapid pulse Dehydration	Abnormal condition Prolonged labour Maternal exhaustion	1. Examine for cervical dilatation 2. Set IV line Dextrose saline 3. IM Paracetamol 600mg stat 4. Elevate the woman's leg 5. Refer to hospital immediately 6. Assign a skilled CHEW to accompany her if possible
IV. Labour pains with cord seen outside the vagina	Cord still pulsating Transverse lie	Abnormal condition Cord prolapse	1. Set up intravenous infusion of Dextrose-Saline 2. Give Oxygen to mother (if available) 3. Replace Cord up in the vagina and pack introitus with moist gauze 4. Refer Perform External cephalic version pack cord inside and initiate delivery 5. If no improvement REFER
IV. Labour pains with cord seen outside the vagina	Cord may or may not be pulsating Cephalic presentation Cervix fully dilated	Abnormal condition Cord prolapse	1. If cord is pulsating : Do generous Episiotomy Push cord back up beyond the head or apply two clamps on cord and cut in-between Deliver the baby Repair episiotomy 2. If cord is not pulsating on knott use vaccum to deliver the baby or REFER :
V. Labour pains, pregnancy not up to term	Gestational age less than 37 weeks Labour in active phase	Abnormal condition Preterm labour	1. Explain the condition to the patientt and relation 2. IV Hydrocortisone 100mg stat 3. Set up intravenous infusion of Dextrose-Saline 4. Conduct deliver
Labour pains, no foetal movement felt for some time	Cord may or may not pro-lapsed Smelly or unclear liquor FHR <120 beat or >160 beat per minutes	Abnormal condition Foetal distress	1. Explain condition to patient and relation 2. Administer Oxygen to mother 3. Set up intravenous infusion of Dextrose-Saline 4.ReferTransport patient in left lateral position Conduct delivery Only when it becomes imminent and no time to get to the hospital

VII. Labour pain, with fits or convulsion	conscious BP> 140/90mmHg Protein in urine	Abnormal condition Intrapartum eclampsia	1. Set up IV infusion, 500ml DextroseWater with cannula/ needle 2. Give 50% MgSo4 loading dose (IV 4gm in 12mls of distilled water over a period of 10-15minute, then IM 5gm + 1ml of 1% Xylocaine into each buttocks) 3. Tabs Nifedipine 20 mg under tongue (sublingual) stat (if conscious) or 4. Hydralazine 5-10 mg IV slowly over 10-15 minutes 5. if unconscious REFER TO HOSPITAL
VIII. Bleeding after expulsion of placenta	Bleeding in excess of 500mls or any sign of shock (rapid pulse, low BP, dizziness, Or unconsciousness) Uterus not firm (above umbilicus)	Abnormal condition Primary Postpartum Haemorrhage (PPH)	1. Shout for help Uterine massage 2. Encourage woman to empty her bladder or catheterize if necessary 3.IM oxytocin 10iu stat or Tabs Misoprostol 800 mcg sublingual (under tongue) or rectal Set IV 3. Normal saline with 20iu oxytocin at 60 drops per minutes If bleeding continues apply NASG (if available) 7. Resuscitate if in shock 8. Refer immediately to hospital
IX. Bleeding after 24 hours Fever	Slight or heavy bleeding Foul smell from the vagina	Abnormal Condition Suspect Puerperal Sepsis	1. IV Ceftriaxone 1gm Stat, then 2. Tabs Cefixime 400mg 12hourly for 5 days 3. Tabs Metronidazole 400mg 8 hourly for 5 days 4. Tabs Vit C 200mg 8 hourly for 7 days 5. Tabs PCM 1gm 8 hourly for 3 days if Ceftriaxonewas not given during delivery 6. Tabs 6 7. Misoprostol 600mcg under tongue (sublingual) stat 7. Advice on Personal Hygiene 8.If no improvement Refer
X. Perineal tear with or without bleeding	Cervical laceration present Degree of perineal or vaginal laceration (1st, 2nd,or 3rd degree tear)	Abnormal condition	1. Clean the area, apply pressure with a perineal pad 2. Repair cervical tear 3Repair 1st and 2nd degree perineal tear4If tear is extending to the anus (3rd degree), clean and apply perineal pad and the woman to put her legs together (Do Not Attempt to Repair) 5.Refer to VVF hospital

XI. Retained placenta or placenta fragment	Placenta products are visible through the vaginal orifice May or may not be bleeding	Abnormal condition	1. If retained placenta and not bleeding, attempt delivery of the placenta manually. 2. If bleeding, manage as in PPH above 3. Encourage client to pass urine or catheterize 4. Leave uterus alone for 10 minutes Attempt Control contraction traction (CCT), 5. if not successful, add 20 IU Oxytocin to infusion and Sunction the woman

Health Education

1. Adequate rest.
2. Vulva hygiene.
3. Nutritional counseling.
4. Exclusive breastfeeding.
5. Routine immunization.
6. Importance of postnatal visits.
7. Importance of taking home-based records during visits to the clinic or hospital.
8. Care of the newborn.
9. Counseling on postpartum contraceptives.
10. Appropriate mental health counseling.

6.3 POSTNATAL DISORDERS

The postnatal period spans 6-8 weeks after delivery. During this time, lactation becomes well established, and the uterus and other reproductive organs begin to return to their pre-pregnancy state. It is advisable to encourage at least three postnatal visits for check-ups to assess the well-being of both the mother and baby. These visits also provide a crucial opportunity to screen postnatal mothers for signs of mental health conditions such as puerperal psychosis, involvement in substance use/abuse, and signs of domestic violence, enabling appropriate management and referral when necessary.

HISTORY	EXAMINATION/ACTIONS
1.How are you feeling? Is there any problem? 2.How many days or weeks since the delivery? 3.Is the baby being exclusively breast fed? Any feeding problems? 4.Do you observe any changes in his/her skin or umbilicus? 5.How is your appetite? 6.Is there any pain or discomfort? 7.Do you sleep well? Is there a sleeping challenge? 8.Do you get easily irritated, anxious or depressed? 9.How long do you want to rest before having your next baby? 10.What is the nature of lochia?	General appearance: tired or pale check skin for signs of domestic violence. Temperature: Check and record Blood pressure: measure and record Breast: -tenderness, redness, lumps, engorgement Nipples: soreness, cracks, Abdomen: size and firmness of uterus, Legs - any local inflammation Pelvis: (internal examination not done before six weeks postpartum Condition of perineum Cervix Os Uterus involution Discharge, Bleeding Vulva: Nature of lochia, condition of repaired episiotomy/ tear (if any) Weight: weigh and record Haemoglobin: estimate and record Urine: protein, sugar and record Examine the child: Refer to appropriate section)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
1. Severe headache convulsion/fits at home or on the way to clinic delivered few days ago	BP 140/90mmHg or above Protein in urine	Severe condition suspect postpartum eclampsia)	1. Resuscitate if patient is unconscious 2. Set up intravenous infusion of 500ml of Dextrose- Water 50% MgSo4 loading dose (IV 4gm in 12mls of distilled water over a period of 10-15minute, then IM 5gm + 1ml of 1% Xylocaine into each buttocks) 3. Tabs Nifedipine 20mg stat under tongue (sublingual) 4. Refer immediately
ii. Continuous vaginal bleeding	Bleeding not profuse Frank bleeding Primary PPH within 24 hours or Secondary PPH after 24 hours	Moderate condition	1. Check for cervical tear or erosion 2. Apply clean pad to perineum 3. IM oxytocin 20iu stat or tabs misoprostol 600mcg stat sublingual (under tongue) 4. Set intravenous infusion of Dextrose-Saline 5. Refer to hospital immediately if bleeding is profuse

iii. Intermittent vaginal bleeding, fever, vaginal discharge with or without smell	Discharge with or without smell Temp. 37.5C or above BP 90/50 or above Spotting	Moderate - severe condition	1. IV Ceftriazone 1gm Stat 2. Caps Amoxicillin 500mg 8hourly for 5 days 3. Tabs Metronidazole 400mg 8 hourly for 5 days 4. Tabs Ferrous sulphate 200mg 8 hourly for 5 Weeks 5. Tabs Paracetamol 1gm 8hourly for 3 days 6. Review daily 7. If no improvement after 3 days Refer to hospital
iv. Vaginal discharge, not smelly. no fever	Discharge present Not smelly Temp, less than 37.5C	Mild condition suspect Reproductive tract infection (RTI)	1. Tabs Erythromycin 500mg 6hourly for 7 days 2. Tabs Metronidazole 400mg 8hourly for 14 days 3. Tabs Vit C 200mg 8hourly for 7 days 4. Use 4cs 5. Review in 3 days 6. If improvement continue treatment Bring male partner for treatment 7. If no improvement, Refer to hospital
v. Excessive tiredness or fatigue with one or more of the following: Dyspnea on exertion Palpitation Facial or pedal oedema Pallor	P>140/90mmHgor <90/50mmHg	Severe to moderate condition Suspect heart conditions or Severe Anaemia	1. Reassure Patient 2. Refer to hospital
vi. Breast Engorgement	Breast full and painful Temperature > 37,5oc	Mild condition	1. Teach proper positioning of the baby on breast and ensure good attachment 2. Encourage exclusive breast-feeding 3. Advice on the use of firm Brassier 4. Warm compress 5 Tabs Paracetamol 1000mg 8hourly for 3 days
vii. Sore or cracked nipples	Pain on nipples while breast-feeding	Mild condition	1. Observe the mother breastfeed 2. Give appropriate health education 3. Tabs Paracetamol 1000mg 8hourly for 3 days If sores are extensive, Tabs Erythromycin 500mg 6hourly for 5 days Tabs Vit C 200mg 8hourly for 7 days Teach how to express milk and feed with cup and spoon Review in 3 days If no improvement Refer

viii.Breast abscess/ inflammation	Tenderness Pus formation	Mild condition	Warm compress Tab Paracetamol 1000g 8hourly for 3 days Tabs Erythromycin 500mg 6hourly for 5 days Tabs Vit C 200mg 8hourly for 7 days If fully ripe do an I & D Dress daily Encourage more fluid If no improvement Refer
ix. Vaginal wall comes down with coughing	Visible vaginal protrusion	Mild condition Suspect Vaginal prolapse	Explain cause to woman Teach postnatal pelvic floor exercise, and advice against lifting heavy objects check in 1 month if no improvement, Refer.
x. Uterus still large or tender after 8th day and up to 2l days	Tenderness Uterine size still above the umbilicus	Moderate condition Suspect Sepsis and sub- involution	1. Advice restricted activity 2. Caps Amoxicillin 500mg 8hourly for 7 days 3. Tabs Paracetamol 1000g 8hourly for 3 days 4. Tabs Metronidazole 400mg 8hourly for 7 days 5. Tabs Vit C 200mg 8hourly for 7 days 6. Review in 3 days, if improved, continue treatment 7. If no improvement Refer.
xi. Headache, Fever with or without nausea and vomiting, Joint pains, Cough	BP < 140/90mmHg Temp > 37.5^{0}c Hb>7g/dl Lochia not smelly or profuse mRDT =+ve	Mild condition Malaria	1. Tabs Paracetamol 1000g 8hourly for 3 days 2. Tabs Artemetherlumefantrine 80/480mg1 tab at 0 hour, 8 hours later, then 12 hourly for 3 days Or Tabs Artesunate+ Amodiaquine 100 mg AS/270 mg AQ base daily for 3 days 3. Encourage to take plenty of fluid 4. Review in 5 days
XII. Sleeping difficulty	No other findings	Mild condition	Advice on good nap and being active during the day Avoid going to bed hungry and do not eat late. Find time to rest between baby care to avoid stress. Ensure the baby is well fed, dry napkin, no heat and comfortable for sound sleep, to allow the mother to sleep well. Take enough fluids to aid digestion and breastmilk formation Ensure the sleeping place is comfortable. Avoid tight clothes, allow fresh air, take a shower and relax before going to sleep.

XIII. Sleeping difficulty with any of the present: serious substance use/abuse (alcohol, cough syrup, etc) Scars of confirmed domestic violence Puerperal Psychosis	Suspect Common Perinatal Mental Disorders (CPMD)	Moderate/Severe condition	Advice on avoidance of substance use/abuse Refer for Appropriate Care Review in 3days
XIV Fever after child birth	Temperature 38oc or more mRDT = -ve	Moderate condition	1. Lower temperature with fan or tepid sponging or antipyretic 2. Adequate hydration by mouth or IV 3 .Encourage bed rest 4.Monitor vital signs and urine output 5.If blood transfusion is needed refer 6.Administer therapeutic antibiotics Cap. Amoxycillin 500mg 6hourly x5 days

Health Education

1. Prompt return for postnatal visit.
2. Adequate nutrition and hydration.
3. Immunization for both mother and baby.
4. Emphasize the importance of exclusive breastfeeding for six months, followed by the introduction of appropriate complementary foods at 6 months.
5. Personal hygiene practices (such as washing breasts with water only before feeding the baby) and pelvic exercises.
6. Educate on proper positioning and ensuring good attachment to the breast during breastfeeding.
7. Stress the importance of using a firm brassiere for breast support.
8. Awareness of danger signs in both mother and newborn.
9. Provide counseling on and referral for cervical cancer screening.
10. Offer counseling and contraceptive services.
11. Highlight the importance of maintaining home-based records during visits to the clinic or hospital.
12. Provide appropriate mental health counseling support.

6.4 FAMILY PLANNING

Family Planning is the process of assisting clients in achieving their desired number of children and determining the timing of pregnancies, or avoiding conception altogether. While oral contraceptive pills are a common method, some women may encounter difficulties in consistently taking them, leading to potential adverse effects. These issues can range from mild to severe, requiring the health worker to provide follow-up care and take appropriate action. Additionally, family planning consultations present an opportunity to screen mothers for signs of mental health conditions such as anxiety, depression, substance use/abuse, and domestic violence. Identifying these issues allows for their proper management and referral to relevant services.

HISTORY	EXAMINATION/ACTIONS
1.Have you used any contraceptive method before? 2.What method were you using? Which Clinic did you attend? Why do you want to attend this particular clinic? 3.Who referred you to this Clinic? 4.How many children have you had? How many are alive? How many dead? When did you have your last baby? Did you have a normal vaginal delivery or were you assisted (caesarean section, forceps delivery or vacuum extraction). 5.Are you breastfeeding? 6.Are you menstruating now? When did you start? If not, when was the last menstrual period? 7.Do you normally have; 8.Heavy periods with cramping pain or clots? 9.Scanty period? 10.How often do you bleed? How long do you bleed? 11.Have you had any operations? When? Where? 12.Do you have any pains in the lower part of your abdomen with fever and discharge? Do you have severe headaches? 13.Are you on any special drugs or receiving treatment anywhere else? 14.Do you have any medical condition such as diabetic, sickle cell, disease, cancer, kidney issues etc? 15.Do you sleep well? Is there a sleeping challenge? 16.Do you get easily irritated, anxious or depressed? Social History 1.Any history of Alcohol Use? 2.Any history of Substance Use/Abuse? 3.Any history of Domestic violence? 4.Any history of Mental Problem?	1.General appearance: anxious thin, obese, ill, 2.Skin: scars (signs of domestic violence) 3.Weight: weigh and record 4.Blood pressure: measure and record 5.Hair: hair loss especially in temporal region 6.Face: swelling 7.Eyes: pallor or jaundice 8.Neck: goitre, lymph nodes 9.Chest: shape, respiratory difficulty or rapid creps or wheeze 10.Heart: rate, rhythm, murmur 11.Breast: lump, dimpling of the skin, abnormal appearance of the nipples, retraction of the nipples. 12.Abdomen: scars, tenderness, masses 13.Legs: varicose veins, oedema. 14.Thigh: excoriation, ringworm lesions, warts. 15.Pelvis: 16.Bimanual palpation: position and size of uterus, tenderness 17.Speculum: condition of the cervix, cervical polyps, lesions, vaginal infection, menstrual bleeding or signs of pregnancy. 18.Haemoglobin estimate and record 19.Urine; protein, sugar

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
First visit for Child Spacing	Well (may or may not be pregnant)	Normal condition	If Pregnant, advise to book for ANC and come for child spacing after delivery If not Pregnant, counsel on various family planning methods using the counselling tools Use MEC wheel for any client with a known medical condition to determine the appropriate method
ii. Intrauterine Device (IUD) thread not felt, suspected pregnancy, severe lower abdominal pain	Intrauterine Device (IUD) thread not seen Abdominal tenderness No bleeding PT = +ve	Severe condition Suspect pregnancy	1.Tabs Paracetamol 1000mg immediately (stat) 2. Conduct a speculum examination, if tip of IUD stem is visible, use forceps to slowly draw it out 3. If not visible, Refer immediately to the hospital

iii. Intrauterine Device (IUD) in place with sign and symptoms of pregnancy. With or without bleeding	(IUD) in place Pregnancy Bleeding	Moderate to Severe condition	1.Tabs Multivitamin 8 hourly for 2 weeks 2.Tabs Ferrous sulphate 200mg 8 hourly for 2weeks 3.Tabs Folic acid 5mg daily for 2 weeks 4.Refer immediately to the hospital
iv. Heavy menstrual bleeding for 10days or more, Intra (IUD in place with or without cramping	Bleeding with IUD in place	Moderate condition	1.Tabs Multivitamin 8 hourly for 2 weeks 2.Tabs Ferrous sulphate 200mg 8 hourly for 2weeks 3.Tabs Folic acid 5mg daily for 2 weeks 4.Explain findings to client 5.Administer Tab Tranexamic acid 1500 mg 8 hourly for 3 days Then 1000mg once daily for 2 days beginning when heavy flow stat 6.Tabs Ibuprofen 400mg 12 hourly for 5days (After meal) 7.If no improvement Refer immediately to the hospital
v. IUD thread not felt	IUD thread seen	Mild condition	Explain finding to client Ask client to re-examine herself and feel the identified string or replace the IUD with another without trimming its string
vi. Back pain during or between menstrual period with IUD in place	Pain IUD in place	Mild condition	1.Tabs Paracetamol 1000mg 8 hourly for 3days 2. Review during the next menstrual period
vii. Requested for IUD insertion Menstruating with any of the following: Foul smelling discharge Vaginal itching Heavy bleeding last 7 days or more with severe cramping History of D & C, or caesarian section in the last 6 months	Signs of infection in the uterus, Pallor Heavy bleeding with or without fever (>37.5°C)	Moderate Condition Suspect RTIs	Do not insert IUD Advise on use of alternative method based on Medical Eligibility Criteria (MEC) Caps Doxycycline 100mg 12 hourly for 14days Tabs Metronidazole 400mg 8 hourly for 14days Tabs Ciprofloxacin 500mg immediately (stat) Review in 7days If there is no improvement, Refer to the hospital
viii. Bleeding or spotting with any of the following after sexual intercourse, lower abdominal pain, fever, IUD in place	warm to touch Temp 37.50C or above Bleeding or spotting mRDT = +ve or –ve	Severe condition Suspect PID Malaria	If mRDT =+ve, treat for malaria as appropriate Caps Doxycycline 100mg 12 hourly for 14days Tabs Metronidazole 400mg 8 hourly for 14days Tabs Ciprofloxacin 500mg immediately (stat) Review in 7days If there is no improvement, Refer to the hospital

Lower abdominal pains during or after insertion IUD Increase amount of bleeding	Fainting during or after insertion, pains, rapid pulse or respiration, fatigue	Severe Condition Suspect Perforation	Stop insertion If IUD is already inserted, remove Place client in a horizontal position and observe for an hour Monitor vital signs Check for intra-abdominal bleeding If no sign of bleeding, observe for 1hour and counsel on abstinence from sex for two weeks Help her choose another method If vital signs are getting down, refer to a higher facility.
Irregular, heavy or prolonged bleeding continuous after several months of normal bleeding of long after IUD was inserted	Vaginal examination shows no signs of malignancy Abnormal discharge from the vagina PT = +ve or -ve	Severe Condition Suspect PID	If PT is +ve send for USS to rule out pregnancy If PT is –ve do not remove IUD. Treat for PID as in action VIII above
Pain, redness, swelling or pus at the site of sub-dermal implant insertion	Tenderness Redness Visible pus Tip of the implant may or may not be visible	Moderate to Severe Condition	If not infected and the tip of the implant is visible, remove and replace a new implant on the other hand on the client's request If there are signs of infection, do not remove the implant Clean the infected area with antiseptic Make an incision to drain the pus (if abscess has formed) Caps Amoxicillin 500mg 8hourly for 7days Tabs Metronidazole 400mg 8hourly for 7 days Tabs Vitamin C 200mg 8hourly for 7 days Tabs Ibuprofen 400mg 12hourly for 5 days (After food) If no improvement Refer
Irregular or heavy vaginal bleeding after initiation of implant	No underlying medical condition Implant in place	Mild to Moderate Condition	Reassure the client that bleeding changes are common in women using implant, but they gradually subside with time If the client finds the bleeding unacceptable, offer one cycle of low dose combined oral contraceptive (based on MEC)
Severe lower abdominal pain on implant Abnormal vagina bleeding or no monthly bleeding	Signs of pregnancy Abdominal tenderness Fainting	Severe Condition Suspect Ectopic Pregnancy or Ovarian Cyst	Leave implant in place Reassure client Refer for pelvic USS If ovarian cyst or ectopic pregnancy, Refer to hospital

Headache, with or without aura (blurred vision , temporary loss of vision or seeing flashy light) on implant	BP > or < 149/90mmHg PT = -ve No underlying medical condition	Mild to Moderate condition	If headache without any other symptoms Tabs PCM 1g 8 hourly for 3 days Or Tabs Ibuprofen 400mg 12 hourly for 5 days (After food) If headache with other symptoms after implant insertion Reassure client Remove the implant Advise other non-hormonal method Review after 3 days If no improvement Refer
Menstruation with any of the following: Visible blood vessel History of jaundice or blood clots Breast abnormalities Breastfeeding less than 6 months	Varicose veins seen on legs Jaundice Blood clots Lump in breast inverted nipples cracked nipples BP 140/90 or above	Severe condition	Advise on elevation of feet Reduction of period of standing Proper position and attachment of baby to breast If on hormonal contraceptives. Stop use. Use a non-hormonal method as a backup. Refer immediately to the hospital
On oral Contraceptive with mild, moderate or sudden severe headache	No sign of pregnancy Normal BP	Moderate condition	If headache is sudden and severe, take client off oral contraceptive immediately Use back up method Tabs Paracetamol 1000mg stat Review in 2 days If no improvement, Refer to the hospital
On oral contraceptive with any of the following; Scanty menstrual bleeding Brownish discharge or spotting No signs of pregnancy	No sign of pregnancy Blood spot seen	Moderate condition	Counsel client Reassure Review during next menstrual period if no improvement, take off oral contraceptive Counsel on barrier methods (condom and diaphragm
On oral Contraceptive with heavy bleeding 7 -10 days	Bleeding BP normal	Moderate to severe condition	Tabs Multivitamin 8 hourly for 2weeks Tabs Ferrous sulphate 200mg 8 hourly for 2weeks Tabs Folic acid 5mg daily for 2weeks Explain findings to client Administer Tranexamic acid 1500mg 8 hourly for 3 days Then 1000mg once daily for 2 days beginning when heavy flow stat Tabs Ibuprofen 400mg 12hourly after meal for 5days If no improvement Refer immediately to the hospital

xix. On oral Contraceptive with heavy bleeding 7 -10 day	Bleeding Systolic BP lower than 120mmHg and diastolic lower than 70mmHg	Severe condition Suspect hypo tension	Set up intravenous infusion of Normal Saline and Refer immediately to the hospital
xx. On oral contraceptives, Persistent irregular bleeding bleeds after intercourse	No abnormality	Mild to Moderate Condition Suspect Dyspareunia	Change pills If heavy bleeding persists Tabs Mefanamic Acid 100mg 12 hourly for 5 days If bleeding persists or become heavy review contraceptive method If there is no improvement, Refer to appropriate health facility.the hospital
xxi. On oral contraceptive, with or without complaint	BP 140/90mmHg or above	Mild-Moderate to severe Condition	Stop pills Change to barrier method If BP is not more than 140/90mmHg, Review in 2 weeks If BP is still raised or higher, Refer to the hospital
xxii. On oral contraceptive and bleeds after intercourse	No abnormalities	Mild condition	Reassure client Advise on proper position during intercourse Refer to the hospital review in 2 weeks
xxiii. Delayed Pregnancy with Regular menstrual period stopped depo provera injection 15 months ago and anxious to get pregnant	No obvious abnormalities or infection	Mild condition	Explain lo client that condition could be the side effect of Depo Provera Instruct her to have intercourse during fertile period Advice to remain lying down 1-2 hours after intercourse Invite partner for counselling and investigation Advise to keep a record of menstrual period Review in 3 months
xxvi. Irregular menstrual bleeding, spotting or scanty bleeding injectable contraceptive	No sign of infection PT = -ve	Mild Condition	Re-assure Explain condition to client If still worried discuss about other methods
xxv. Weight gain. On Injectable contraceptives	PT = -ve No underlying medical conditions	Mild Condition	1.Re-assure 2.Explain condition to client 3.Advice on lifestyle modification (eat healthy and exercise)
xxvi. Headaches, hot flashes, mood changes and vaginal irritation	No underlying medical condition No signs of STIs	Mild to Moderate Condition	Reassure Tabs Ibuprofen 400mg 12hourly for 3 days (After food) Review in 3 days If no improvement discuss on other methods

xxvii. Menstrual irregularity	No evidence of other abnormalities No infection	Mild condition	Advise to keep record of menstrual period for 3 months Review in 3 months If irregularity continues Refer to the hospital
xxviii. On oral contraceptives, Nausea and or vomiting	Pregnancy test Signs of pregnancy	Mild condition	Review contraceptive method If test is positive , counsel and provide Antenatal care
xxix. On oral contraceptives. Nausea and or vomiting	No signs of pregnancy Pregnancy test is negative	Mild condition	If pregnancy test negative, change to lower dose pills (if available) If nausea persists, Avoid oily food and Encourage dry food
xxx. On oral contraceptives, Excessive weight gain after starting pills, Increased appetite, Weight gain cyclical	Increased weight	Mild condition	Encourage diet control Advise regular physical exercise Change to lower dose pill If weight gain persists, change to other method
xxxi. On oral contraceptives. Depression or mood changes after starting pills, No associated social problems	Depressed Unhappy	Moderate condition	Stop pills Change to barrier non hormonal method If depression persists, refer to the hospital.
xxxii. On oral contraceptives, Breast tenderness or fullness after starting pills, No signs of pregnancy	Tender full breast, no lump Pregnancy test negative	Mild condition	Advise client to wear firm, supportive brassiere Give Tabs Paracetamol 1000mg tds for 3 days If discomfort persists, change to lower dose combined contraceptive, progestin only pills or barrier method
xxxiii. On oral contraceptives, Acne (Pimples) following use of pills	No other abnormality	Mild Condition	Change to lower dose oestrogen pills If acne persists or become more severe, stop pill change to barrier method and refer to the hospital
xxxiv. Sleeping difficulty	No other findings	Mild condition	Reassure the patient If condition persist refer to Postnatal Disorders
xxxv. Sleeping difficulty with any of the following Noticeable anxiety serious substance use/abuse (alcohol, cough syrup, etc) Scars of confirmed domestic violence	Patient looks anxious, worried, scars on the body	Moderate/Severe condition suspect Anxiety/Psychosis/ Depression	Refer to appropriate health facility

Danger Signs

1Severe Abdominal Pain
2Severe Chest Pain
3Severe Headache
4Eye Problems (Blurred Vision, loss of vision, flashes of light)
5Severe Leg Problems (swelling, pain in calf or thigh).
6Restlessness and visible signs of anxiety
7Severe back pain
8Heavy/prolonged bleeding

Health Education

1Healthy timing and spacing of pregnancy (HTSP).
2Personal, food and environmental hygiene.
3Nutritional counselling.
4Compliance to drugs.
5Adhere to treatment.
6Appropriate Mental Health Counselling.
7Importance of taking home-based records to the clinic or hospital.

6.5 INFERTILITY

Infertility is defined as the inability of a couple to conceive after one year of regular unprotected intercourse. Male infertility may arise from various factors, including deficiencies in sperm production, blockage of the sperm delivery system, testicular injury, hormonal disorders, or poor testicular descent. Similarly, women may encounter challenges in conceiving, which could stem from factors such as infections, lack of awareness regarding fertile periods, or irregular menstrual cycles. Mental health conditions like anxiety, depression, substance use/abuse, and domestic violence can also impact a couple's ability to conceive.

However, community health practitioners are equipped to provide appropriate advice and take necessary actions to address infertility issues. Moreover, infertility consultations present an opportunity to screen mothers for signs of mental health conditions, substance use/abuse, and domestic violence, facilitating their proper management and referral when needed.

HISTORY	EXAMINATION/ACTIONS
Menstrual History: When was her last menstrual period (LMP) Gynaecological History: Any signs of infection, vaginal discharge, lower abdominal pain, pain or bleeding on intercourse? How was the infection treated? Any operation D & C. Caesarian section. Abortion? Where and when was it treated? Any problems after it such as bleeding, discharge, fever, pain. Obstetric History: Has she ever been pregnant? With this man? Any problems after the pregnancy - bleeding, discharge fever, lower abdominal pain? What happened to the pregnancy? How long ago was it? Contraception History: What method of birth control has she used? When did she stop? Any problems with the method - bleeding, discharge, fever, lower abdominal pain, no menstrual period? Sexual History: How often do they have intercourse? Does her husband have other wives? Does her husband have children? Does her husband live with her? How long have they been trying? Any sexual problem – pain on intercourse? Sexual history is important in her husband. Medical History: How old is she? How does she feel now? Has she come to the clinic or seen the doctor in the last 6 months? For Husband; Has the husband had mumps before? Any history of injury to the genitals? Any undescended testicles? Social History: Any history of alcohol use? Any history of substance use/abuse? Any history of domestic violence? Any history of mental problem?	General appearance: tired or pale check skin for signs of domestic violence. General Inspection: evidence of sexual maturity (breast development, womanly shape, female hair distribution), obstacles to sexual intercourse (obesity, physical abnormalities) or general poor physical health or small vaginal opening. Abdomen: scars, masses. Speculum examination: position of the cervix, signs of infection, erosions, lacerations. Bimanual examination: The presence of a normal uterus, the size of the uterus, masses, tenderness of the uterus or adnexa. Blood pressure: measure and record. Temperature: - check and record Hemoglobin: estimate and record. Urine: protein, sugar.

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Inability to impregnate a woman	Undescended or very small testis	Severe condition	Refer to the appropriate health facilities
ii. Unable to get pregnant, May or may not have commenced menstruation	Sexual immaturity Too young	Mild condition	If she is ovulating reassure. Explain that fertility might come with age (if she is under 16 years).. Refer to hospital if the women is 18 years or more and has never menstruated

iii. Unable to get pregnant, May or may not be menstruating, above 18 years	Young woman No abnormality	Mild condition Suspect infertility	If she has commenced menstruation, instruct her to have intercourse during her fertile period Advise on sexual position Have her remain lying down for 1-2 hours after intercourse Invite husband for counselling and investigation Review in 3 months, if no menses Refer to hospital
iv. Unable to get pregnant	Obese Anxious Emaciated	Mild condition Suspect infertility	Nutrition counselling Have her to keep dates of menstrual period Advise on fertile period Review in 3 months If no improvement, Refer to the hospital
v. Inability to impregnate a woman	Adult male No abnormality detected	Mild condition	Counsel client on best time for intercourse. Frequency of intercourse. Advice to take wife to the clinic. Review in 3 months. If no improvement, Refer to hospital.
vi. Couple unable to conceive for over a year of regular sex with the following; Painful urination Vaginal/Urethral discharge	Vaginal/Urethral discharge	Moderate to Severe Condition Suspect Infertility secondary to STIs	Reassure client Treat according to Syndromic Management of STIs If no improvement. Refer

Danger Signs

1Severe abdominal pain.
2Severe chest pain.
3Severe headache.
4Restlessness and visible signs of anxiety.
5Eye problems (Blurred Vision, loss of vision, flashes of light.
6Severe leg problems (Swelling, pain in calf or thigh).

Health Education

1. Maintain a record of menstruation.
2. Advocate against early marriage.
3. Provide nutritional counseling.
4. Stress the importance of maintaining home-based records during visits to the clinic or hospital.
5. Educate on the fertile period and the importance of intercourse within that timeframe.
6. Advise against the use of herbal medication.
7. Offer appropriate mental health counseling.

6.6 PRE-MENOPAUSE

Pre-menopause is a natural process marking the transition to the menopausal stage, characterized by the cessation of a woman's menstrual periods due to advancing age, signifying the conclusion of her reproductive years. This phase entails fluctuations in the menstrual cycle, accompanied by various physical and emotional symptoms, all attributed to hormonal changes within the body. Pre-menopause typically spans between 2 to 10 years before the onset of menopause.

HISTORY	EXAMINATION/ACTIONS
Personal History: Age? When did symptoms start? Menstrual History: When was your last menstrual period (LMP), how often do you menstruate Gynaecological History: Any signs of infection, vaginal discharge, lower abdominal pain, pain or bleeding on intercourse Obstetric History: Have you ever been pregnant? Any problems after the pregnancy - bleeding, discharge, fever, lower abdominal pain Contraception History: What method of birth control have you used? For how long? When did you stop? Any problems with the method - bleeding, discharge, fever, lower abdominal pain, no menstrual period? Sexual History: How often do you have intercourse? Any sexual problem – pain on intercourse, decrease libido Medical History: How regular is your period How do you feel now? Do you have any known medical condition e.g. Diabetes, Hypertension, Liver or Kidney Problem? Social History: Any history of alcohol use? Any history of substance use/abuse? Any history of domestic violence?	General appearance: anxious thin, obese, ill, Skin: scars (signs of domestic violence) Weight: weigh and record Blood pressure: measure and record Hair: hair loss especially in temporal region Face: swelling Eyes: pallor or jaundice Neck: goitre, lymph nodes Chest: shape, respiratory difficulty or rapid creps or wheeze Heart: rate, rhythm, murmur Breast: lump, dimpling of the skin, abnormal appearance of the nipples, retraction of the nipples. Abdomen: scars, tenderness, masses Legs: varicose veins, oedema. Thigh: excoriation, ringworm lesions, warts. Pelvis: Bimanual palpation: position and size of uterus, tenderness Speculum: condition of the cervix, cervical polyps, lesions, vaginal infection, menstrual bleeding or signs of pregnancy. Haemoglobin: estimate and record Urine; protein, sugar

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I. A woman over 40 years with Sleep problems with two or more of the following; - Poor sleep - Hot flashes - Weight gain - Headache - Hair loss - Body itching - Anxiety - Mood changes	No underlying medical condition No signs of Infection No skin lesions	Mild to Moderate Condition Suspect Premenopausal Syndrome	1. Reassure 2. Nutritional counselling 3. Encourage simple exercise
II. A woman over 40 years with any of the following; - Low libido - Painful intercourse - Not pregnant or breast feeding	No signs of Infection PT negative Pelvic examination –Normal	Mild to Moderate Condition Suspect Premenopausal Syndrome	1. Reassure 2. Nutritional counselling 3. Encourage simple exercise 4. Advice on use lubricants like K-Y Jelly for Painful Intercourse

III. A woman over 40 years who develop Irregular menstrual periods with heavy bleeding	No signs of Infection PT negative Pelvic examination -Normal	Mild to Moderate Condition Suspect Premenopausal Syndrome	1. Reassure 2.Nutritional counselling 3.Tabs Ibuprofen 400mg 12hourly for 3 days (After food) 4. Review in 3 days, if no improvement refer
IV. Decrease libido, Erectile dysfunction Fatigue Sleep distunbance Decrease cognitive functions	Weigt gain Osteophorosis Mood change Irritability	Mild to moderate Suspect Andropause	Explain the condition to the patient Reassure the patient Life style modification Advice on healthy diet and exercise, Stress management If no improvement REFER

Health Education

1. Ensure adequate rest.
2. Identify and avoid triggers of hot flashes.
3. Provide nutritional counseling.
4. Increase fluid intake.
5. Avoid the use of substances in the vagina and maintain vulval hygiene.
6. Offer appropriate mental health counseling.

SECTION SEVEN

ELDERLY HEALTH (60 YEARS ABOVE)

7.0 GERIATRIC HEALTH

The World Health Organization (WHO) defines the elderly as people whose age is 60 years and above. It's important to note that there are a significant and growing number of elderly persons in the population. This means that the PHC worker can expect an increasing number of elderly patients. This group of patients (clients) are vulnerable and at risk of having chronic diseases such as hypertension, diabetes, osteomyelitis, cataracts, nyctalopia, hearing impairment, dementia, nutrition and social problems. They spent most of their lives in nation-building, either in the public or in the private sector, when they were much younger. They deserve the best of care from the country they served.

There are differences in the signs and symptoms of diseases presented in the elderly and young people. Therefore, diagnosis in the elderly may require more precise history/laboratory tests. Due to some changes associated with the ageing process, such as difficulty in hearing, speech problems, memory loss, etc., it would be necessary for a family member, who plays a crucial role in the care of the elderly, to be involved in their care. An understanding of the needs of the elderly is essential for PHC workers, as is the recognition of the invaluable support that family members can provide in this process.

7.1. MUSCULO - SKELETAL SYSTEM DISORDERS

As adults age, they experience a gradual loss of minerals and living materials from their bones, resulting in weaker and more brittle bones. This physiological process increases the likelihood of fractures and broken bones among the elderly. Additionally, osteoarthritis, a condition associated with aging, can cause pain and difficulty in normal joint movement.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.When did it begin? 3.Has it happened before? 4.Was there any accident or injury? 5.If injury happened, have you been able to move part since? 6.Is there any pain, swelling and numbness? 7.Do you have joint pain or stiffness? 8.Do you have back pain? 9.Do you have fever? 10.Are you on any medication?	1.General appearance: look out for distress 2.Temperature: check and record 3.Limbs: check for deformity, swelling on the affected part, localized severe tenderness/pain, inability to move part

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Severe pain on limb on falling even on a flat surface	General appearance: ill- looking and/or in pain Temperature: 37.5°c or above Limbs: deformity or swelling on the affected part or localized tenderness, inability to move part	Severe condition Suspect a fracture or osteoporosis	Paracetamol 1.0g immediately (stat) or Non- Stereoidal Anti-inflammatory e.g Ibuprofen 400mg immediately (After food). Apply splint and firm bandage on affected limb and REFER
ii. Bone pains: Stiffness and cramps of extremities Pains in the joints aggravated by use and relieved by rest	General appearance Ill-looking and in pain Joints: swelling and tenderness movement of the joint	Moderate to severe condition Suspect osteoarthritis	Tabs Paracetamol 1g 8hourly for 3 days Methyl salicylate ointment, apply twice daily topically. Tabs Vitamin C (Ascorbic Acid) 200mg 8hourly for 5 days Advise on weight reduction (if necessary) Keep joints warm; Moderate exercise e.g. walking and use of walking sticks to support REFER, If no improvement after 2weeks

iii.General body pains No fever No headache	General appearance not in any pain Limbs: No swelling, no localized tenderness	Mild condition Suspect fatigue	Tabs Paracetamol 1g 8hourly for 3 days Tabs Multivitamins 1 daily for 2 weeks Adequate rest Small frequent meals, fruits and vegetables

Health Education

1.Explain process of ageing to client.
2.Advise on gentle exercise for moderate condition.
3.Advise on intake of large quantity of water, vegetables and fruits as well as intake of frequent small meals.
4.Advise client to avoid risk of falling.
5.Encourage outdoor relaxation
6.Encourage compliance with prescribed drugs (use of right dose at right time)
7.Encourage support using walking sticks
8.Encourage adequate rest.
9.Advise on periodic medical check-up.

7.2. CARDIOVASCULAR SYSTEM DISORDERS

Cardiovascular problems are conditions affecting the heart and blood vessels. When these vital components function optimally, blood flows smoothly throughout the body. However, disruptions in the circulatory system can occur due to either poor heart function or blockages in blood vessels, leading to inadequate blood flow through arteries and veins and potentially resulting in serious health complications.

While some individuals are born with cardiovascular issues, most develop over years. Consequently, there is a higher prevalence of cardiovascular diseases among the elderly population. Common conditions such as arteriosclerosis, atherosclerosis, heart attack, heart failure, and stroke are prevalent in older adults. High blood pressure is a major contributor to these diseases and is more prevalent in the elderly compared to younger individuals. Other contributing factors include high blood sugar or diabetes, high cholesterol, and dyslipidemia.

Fortunately, early detection and appropriate management of these risk factors can often prevent death from cardiovascular diseases. Therefore, it is crucial to identify and treat these risk factors promptly and refer individuals to appropriate healthcare providers for further management.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.How long have you had it? 3.Do you have cough, chest pain, or shortness of breath? 4.If cough is present, do you bring out sputum? Colour? 5.And you sleep flat or do you need several pillows to sleep? 6.Do you sometimes have to get up at night struggling to breathe? 7.Have you ever lost consciousness? 8.Any sudden fall? 9.Do you have difficulty in moving any part of the body? 10.Do have any history of high blood pressure or diabetes? 11.Are you on any medication?	General Appearance: look out for respiratory distress Temperature: check and record Cardiovascular System (CVS). a. Pulse: check for rate, rhythm, volume b. Blood pressure: check and record c. Heart sound: check for heart sounds, murmur Chest: check for respiratory rate, signs of dyspnea Abdomen: check for distension, tenderness, enlarged liver Central Nervous System (CNS): check for level of consciousness and paralysis Lower limbs: check for pedal swelling/oedema Urine – Urinalysis – protein, glucose, etc Blood – Haemoglobin determination Eye – Visual acuity Ears – Test for hearing Weight – weigh and record (BMI)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. First Visit (Elderly) at the age of 60 years	General appearance: Ill- looking or uncomfortable CVS: Pulse: fast, irregular Blood Pressure: high (above 140/90 mmHg) CNS: Partial or total loss of consciousness, partial or complete paralysis (one sided); Loss of speech (if paralysis is on the right side); Memory may or may not be affected Urine – Urinalysis – protein, glucose, etc Blood – Haemoglobin determination Eye – Visual acuity Ears – Test for hearing Weight – weigh and record (BMI)	No abnormality detected Abnormality detected	Counsel on subsequent visit or when there is any problem. Treat as appropriate below.

a) History of loss of consciousness b) History of sudden fall c) Sudden inability to move any side of the body d) Loss of speech e) Slight deviation of the mouth or fac	General appearance: Ill- looking or uncomfortable CVS: Pulse: fast, irregular Blood Pressure: high (above 140/90 mmHg) CNS: Partial or total loss of consciousness, partial or complete paralysis (one sided-hemiplegia); Loss of speech (if paralysis is on the right side); Memory may or may not be affected	Severe condition Suspect stroke	Give first Aid management for unconsciousness by keeping always free, keep warm and make patient comfortable. Treat any sustained injury. Measure blood pressure intermittently. Then REFER appropriately within one hour to the emergency department of the closest secondary or tertiary health care facility.
a) Headaches oedema b) Breath lessness at rest or on mild exertion e.g. after walking c) Awareness of tachydardia. d) Inability to lie flat on bed, Coughing Easy tiredness	General appearance: Ill- looking or uncomfortable CVS: Pulse: fast, irregular; BP: 140/90mmHg and above Chest: difficult or fast breathing, CNS: changes in level of consciousness Lower limbs: pitting oedema	Severe condition Suspect hypertension with heart failure	IM. frusemide, (laxis) 40mg stat REFER immediately within one hour to the emergency department of the closest secondary or tertiary hospital.
a) Difficulty in falling asleep b) Waking up after very few hours of sleep c) Associated Social problems d) History of high Blood Pressure	General appearance: pale anxious and not happy CVS: BP 140/90mmHg	Mild condition	Drugs that can be used in management of cardiovascular system problem in primary health settings; 1. Antihypertensives: Used to control high blood pressure. Examples include ACE inhibitors (e.g., lisinopril), beta-blockers (e.g. metoprolol), and diuretics (e.g. hydrochlorothiazide). 2. Antiplatelet Agents: Help prevent blood clots. Aspirin and clopidogrel are examples. 3. Statins: Used to lower cholesterol levels and reduce the risk of cardiovascular events. Atorvastatin and simvastatin are commonly prescribed. 4. Beta-blockers: Besides hypertension, they may be used for conditions like angina and heart failure. Examples include metoprolol and carvedilol. 5. Angiotensin II Receptor Blockers (ARBs): Similar to ACE inhibitors, they help manage blood pressure and may be prescribed for heart failure. Losartan is an example. 6. Calcium Channel Blockers: Used for hypertension and certain heart conditions. Amlodipine and diltiazem are examples. 7. Antiarrhythmics: Prescribed for irregular heart rhythms. Amiodarone and flecainide are examples. Counseling on sleeping habits, drug usage, food intake and participation in community work Occasional Tranquilizer e.g. Tabs Diazepam 5mg at night for 5 days Tabs Ferrous gluconate 8hourly for 2weeks.

			Tabs Multivite 12hourly for 2weeks Review after 3 days, if no improvement, refer. OR Counseling on sleeping habits, drug usage, food intake and participation in community work. Occasional Tranquilizer e.g. Tabs Diazepam 5mg at night for 5 days. Tabs Ferrous gluconate 8hourly for 2weeks. Tabs Multivite tab 12hourly for 2weeks. Review after 5days, if no improvement; REFER to the emergency department of the closest secondary or tertiary health facility.
a) Oedema of both feet b) Breathless ness at rest or on mild exertion e.g. after walking , c) Awareness of rapid beating of the heart. d) Inability to lie flat on bed e) Coughing f) Easy tiredness	looking or uncomfortable CVS: Pulse: fast, irregular; BP: 140/90 mmHg and above Chest: difficult or fast breathing, CNS: changes in level of consciousness Lower limbs: pitting oedema	Suspect hypertension with heart failure	Encourage very low salt diet Discourage alcohol Intake REFER immediately within one hour to the emergency department of the closest secondary or tertiary hospital.

Health Education

1.Counsel client on the need to have adequate rest and go to bed early.
2.Advise to avoid mental and emotional stress.
3.Encourage moderate exercise e.g-walking.
4.Advise on avoiding consumption of alcohol and stopping the use of tobacco products.
5.Advise on low salt/low fat diet (plant fats are better) and to reduce dietary sugar intake.
6.Encourage Intake of fluids, and vegetables
7.Encourage intake of complex carbohydrate whole gain e.g sorghum, wheat, guinea corn.
8.Encourage keeping personal records of BP and blood sugar.
9.Avoid self-medication.
10.Advise adherence to medication, clinic, and laboratory appointments.

7.3 GENITO- URINARY DISORDERS

Genito-urinary problems are prevalent among the elderly and can stem from various factors such as metabolic changes, age-related alterations, infections, and abnormal growths. Examples include diabetes, vaginal or uterine prolapse, enlarged prostate, cancers, and schistosomiasis.

Patients may present with symptoms such as discharge, pain, and a burning sensation during urination (dysuria), blood in the urine, frequent urination, and urinary retention, among others

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.How long have you had it? 3.Do you urinate very often? How often? Night and day? 4.Do you have difficulty in passing urine or pain/burning sensation when passing urine)? 5.Do you have feeling of incomplete passage of urine? 6.Are you able to control urine? 7.Do you drink more than usual? 8.Do you have pain in the flank? 9.Is there blood in your urine? 10.Do you have fever? 11.Do you have a history of wadding in streams in childhood? (Schistosomiasis) 12.Have you had this problem before? Have you had any form of treatment? If yes, which? 13.Do you have any vaginal discharge or itching (if male, any discharge from the penis?) 14.Any history of weight loss 15.Are you on any mediation?	1.General appearance: look out for discomfort, 2.Temperature :check and record 3.Pulse: check and record 4.Respiration: check and record 5.Blood pressure: measure and record 6.Eyes: check for pallor, jaundice etc 7.Abdomen: check for tenderness, swelling or masses 8.Flanks: check for tenderness 9.Limbs: check for any abnormality 10.Legs: for oedema 11.Palms and nails: for pallor 12.Genitals: check for sores, pus or discharge 13.Urine: check for- 14.Colour or cloudiness 15.Protein 16.Sugar

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Sudden inability to pass urine	General appearance: in distress, ill-looking CVS: - Pulse: fast or normal - Blood pressure: above 140/90 mmHg Abdomen: swelling/ distension, lower abdominal tenderness Flanks: tenderness	Severe condition Suspect acute renal failure	REFER immediately.
ii. Severe lower abdominal pain Painful urination with or without urethral discharge. There may or may not be Fever.	General appearance: in painful distress Temp.: 37.5°C and Above Abdomen: Lower abdominal tenderness Genitals: normal or urethral discharge Urine: clear or cloudy (presence of protein)	Severe condition Suspect urinary tract infection (UTI)	Tabs Ciprofloxacin 500mg 12hourly for 10 days Tabs Paracetamol 1g 8hrly for 5 days If no improvement, REFER OR Tabs Nitrofurantoin 50-100 mg 6hourly with food for 5-7 days. Tabs Trimethoprim/sulfamethoxazole (TMP-SMX 80/400 mg 12hourly for 3 days to 14 days. If no improvement, refer.

iii. Blood in the Urine	General appearance: normal or ill-looking and pale Abdomen: hard lower abdominal mass schistosomiasis which may or may not be tender discomfort Temp.: 37.5°C and above Urine: blood in urine or terminal haematuria.	Severe condition Suspect bladder cancer or schistosomiasis	For bladder cancer, REFER For schistosomiasis, Give Tabs Praziquantel 1.2g stat then, 0.6g daily for 5days Tabs Paracetamol 1000mg 8hourly for 5days If no improvement after 3 days REFER
iv. Urine incontinence	General appearance: smell of urine with wet pants and/or clothing	Moderate to severe condition Suspect urinary incontinence due to ageing or UTI	Tabs Ciprofloxacin 500mg 12hourly for 10 days Counsel patient and relatives and explain that the changes are due to ageing Encourage to avoid constipation Personal hygiene Reduce quantity of fluid intake after 6pm. Encourage to wear adult diaper if necessary If severe, REFER immediately.
v. Difficulty in passing urine in the male; has to wait a while before urine begins to come out, may or may not have pain and .blood in urine	General appearance: normal or anxious Abdomen: lower abdominal tenderness Urine: presence or absence of blood	Moderate to severe condition Suspect enlarged prostate	Tabs Paracetamol 1g 8hourly for 5 days Tabs Co-trimoxazole 960mg 12 hourly for 5days Encourage fluid intake REFER and follow up
vi. Passing a lot of urine. Getting up more than 4 times at night to pass large quantities of urine but without pain. Feels very thirsty and drinks a lot of water/fluid, loss of weight	General appearance: ill- looking. weak or normal Urine: presence of sugar.	Severe condition Suspect Diabetes mellitus	Explain the condition fully to the patient and how he is responsible for about 75% of his own treatment Give dietary advice and menu guideline Give diet sheet and map out food menu. Do a Fasting or Random Blood Sugar REFER - and arrange follow up

Health Education

1.Advise on compliance with prescribed drugs and diet.
2.Encourage regular medical checkups.
3.Encourage client to take care of feet.
4.Encourage client to avoid and report any form of injury that results from loss of sensation.
5.Encourage personal hygiene.
6.Emphasize on the importance of taking along home-based records anytime he/she goes to the clinic or hospital.

7.4 GASTROINTESTINAL DISORDERS

Gastrointestinal issues affecting the elderly may arise from various causes including infections, ulceration, and malignant changes. These conditions can manifest as constipation, cholera, diarrhea, gastritis, peptic ulcer, or cancer of the colon, stomach, or esophagus.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.For how long? 3.Do have nausea or vomiting? If vomiting, how many times? 4.Do you have constipation? When was your last stool: 5.Do you have diarrhea? If yes, how many times do you pass stool in a day? 6.Any blood in your vomit or stool? What colour is your stool? 7.Does food make the problem better or worse or no difference? 8.Is there abdominal pain? Where is it located? Is it continuous or intermittent? 9.Have you passed any worms in stool? 10.Do you have fever? Have you taken any medications? What are they?	1.General appearance: look out for discomfort 2.Eyes: check for palour and jaundice 3.Temperature: check and recorded 4.Blood pressure: measure and record 5.Abdomen: check for: 6.Swelling 7.Tenderness 8.Hernia: check for umbilical, inguinal, scrotal hernia 9.Anus: check for prolapse, fissure, redness 10.Hemoglobin: estimate and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Inability to pass stool, Abdominal fullness, discomfort, and flatulence	General appearance: restlessness Abdomen: distended, hard, generalized tenderness	Moderate condition Suspect constipation	Encourage plenty fluids intake Eat fruits and vegetables Give liquid paraffin 10ml as necessary for one week De-worm at least once yearly (Albendazole 400mg stat) If no improvement, REFER
ii. Epigastric pain worsened by eating, Epigastric pain worsened by hunger/ disturbs sleep and relieved by eating, with or without blood In stool or black coloured stools	General appearance: distressed Eyes: pale Abdomen; epigastric tenderness Stool: blood stained or black coloured	Moderate to severe condition Suspect Peptic Ulcer Disease (PUD)	Encourage eating small meals regularly (every three hours) Drink milk for temporary relief Give Tabs Antacid 2tab 8hourly for one week Give Folic Acid 5mg daily for two weeks If no improvement after 5 days, REFER
iii. High temperature not responding to anti-malaria drugs	Loss of appetite general weakness loss of weight severe headache	Severe condition	REFER

Health Education

1.Discourage use of laxative.
2.Encourage moderate exercice to facilitate bowel movement.
3.Encourage consumption of fluids, fruits and vegetables.
4.Encourage food hygiene and good personal hygiene (e.g regular and washing).

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
i. Difficulty in falling asleep or waking up after very few hours of sleep, weakness, fatigue, Sadness, anxiety and irritability, Low interest in activities that used to be interesting, Difficulty in carrying out usual activities. Poor appetite, Use of medications	General appearance: unkempt/ untidy, ill- looking, anxious and unhappy, weak, irritable, smell of alcohol or tobacco Temperature: normal or high Eye: pale Blood pressure: 140/90mmhg or above	Mild, moderate to severe condition Suspect depression	Counsel on sleeping habits, drug usage, food intake and participation in community work. REFER
ii. Complaints by relatives of decline or problem with memory (severe forgetfulness), Loss of orientation (nonawareness of time, place and person), Loss of emotional control (easily upset, irritable or tearful), Loss of interest in the environment, Difficulties in carrying out usual work), Wandering about	General appearance: unkempt, appearing uninterested or irritable CNS: cannot remember important events and dates, not oriented in time, place and person,	Severe Condition (Suspect dementia)	Counsel patient and relatives REFER
iii. Increase consumption of alcohol and use of drugs such as pain killers, sleeping tablets, coffee, tobacco leaves, cigarettes and snuff, Poor sleep, tiredness, headaches, Poor appetite, nausea, vomiting, indigestion, Difficulty in carrying out .usual work	General appearance: unkempt/ untidy, weak, agitated, smell of alcohol or tobacco, unsteady gait, may have some drugs in the pocket of their shirt and/ or trousers, purse or bag, skin infection, injuries etc CNS: Nervous/fidgeting, slurred speech	Moderate to severe condition Suspect alcohol or drug use disorders	Counsel patient and his family/ friends Encourage participation in social activities within the family and the community. REFER Follow up on return from referral
iv. Not happy Loneliness Isolated Dejected	Not cheerful Weeps during the interview Looks dirty with sores on the body Poor oral hygiene with long dirty nails, dirty clothes May or may not be blind or deaf	Moderate condition Suspect depression	Counsel family of patient Encourage him to participate in religious and community activities. Encourage family involvement in the therapy REFER

Health Education

1.Counsel family members to give both physical and emotional support to the elderly.

2.Encourage good nutrition.

3.Encourage the elderly to participate in religious and community activities and encourage visiting friends and associates.

4.Encourage family members to show love, affection and provide for the general needs of the elderly.

5.Discourage use of alcohol, tobacco and un-prescribed drugs.

7.5 SPECIFIC PROBLEMS OF THE ELDERLY

Elderly individuals commonly experience age-related issues such as hearing and vision difficulties, along with complaints like coughing and high blood pressure. Additionally, conditions such as heart attack, osteoarthritis, arthritis, and peptic ulcers can arise due to heart or respiratory problems. Irregular meals are also a concern for many elderly individuals, as they may lack the strength to prepare their own meals, leading to malnutrition in some cases. Moreover, diabetes is prevalent among the elderly population.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.How long have you had the problem? 3.Is there pain in the eye/ear? 4.Did you (or anyone) put anything into the eye/ear? If yes, what? 5.Do you have any chronic disease such as high blood pressure or diabetes? 6.Is there any family history of blindness, deafness, high blood pressure or diabetes? 7.Do you observe ants around your urine? 8.Do you eat food regularly? Any change in appetite?	1.General appearance :look out for distress, 2.Temperature: check and record 3.Blood pressure: measure and record 4.Eyes: check for: 5.Pallor 6.White patch in both pupils (cataract) 7.Visual acuity using Snellen's chart 8.Ear: check for wax 9.Chest: check for difficulty in breathing

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
II. Difficultly in reading and seeing near objects	•General appearance: normal •Eyes: cannot see near objects	Moderate condition Suspect presbyopia (difficulty in seeing near objects)	1.Counsel on process of ageing (suspect 2.Give Tabs Multivite 8hourly for 2weeks 3.Give Tabs Yeast 8hourly for 2weeks 4. Encourage adequate in-take of foods like green and yellow/leafy vegetables, fruits, fish, and dairy products. 5.Advise to read with good light and in larger prints. 6.REFER
II. Difficultly in seeing far objects	•General appearance: Normal •Eye: Normal or white patch in the black of the eye; decreased vision with Snellen'schart:	Moderate condition(suspect cataract)	1.REFER
III. Bumping into objects	•General appearance: Normal •Eye: Normal	Severe condition (suspect glaucoma)	1. REFER IMMEDIATELY
IV. Difficulty in hearing	•General appearance: normal •Ear: hearing with some difficulty, wax in the ear.	Moderate to severe condition Suspect partial or total ear blockage))	1.Apply a drop of warm olive oil to ears and clean with cotton bud for about 3-4days to soften wax. 2.REFER

V. Meals are irregular and not on time Loss of appetite, Body weakness ,and tiredness, Weight loss	• General appearance: ill both looking • Mouth: Angular Stomatitis (sores at the sides of the mouth), • Eye: pallor -Blood test shows anaemia	Moderate to severe condition (Suspect malnutrition)	Counsel on adequate nutrition for both patient and care providers Tabs Multivite 8hourly for 2 weeks Tabs Vitamin B Complex 8hourly for 2 weeks Tabs Vitamin C 200mg 8hourly for 2 weeks Follow up visit
VI.Fever, headache, Rigor, Loss of sweating high (Temp:above 37.5^{0}c) Eye: Pallor	General appearance: ill-looking	Moderate - Severe condition (Suspect malaria)	Conduct malarial Rapid Diagnostic Test (mRDT) Give ACT if positive mRDT, according to the malarial treatment guideline Paracetamol 1000mg (2 tabs) for 3 days If no improvement, REFER.
VII. Temperature not responding to antimalarials	General appearance :- ill lookingand weak Temp: 37.5oc	Moderate condition (Suspect typhoid fever)	1. Tab Ciprofloxacin 500mg 12hourly for 5days 2. Tab Paracetamol 1g 8hourly for 3days 3. Tab Multivite 8hourly for 5days 4. Review after 3 days 5. If no improvement, REFER
VIII. Chronic cough (dry and productive), difficulty in breathing, history of tobacco use (smoke or smokeless forms) for many years.	General appearance: ill-looking, in respiratory distress with bouts of cough Chest: difficulty in breathing , increased respiratory rate	Moderate or Severe condition (suspect chronic bronchitis)	1 Cap Ampiclox 500mg6hourly x5 days 2 1 Tab Prednisolone 5mg daily x5 days 3. REFER
IX a.Passing stool infrequently b.Abdominal discomfort lack of appetite	Nil of note; healthy looking No complaints	Mild condition	Counselling – nutrition education Consumption of adequate diet containing fruits and vegetables Take plenty of water Reduce intake of bread and other pastries Moderate exercise Syr Liquid paraffin 10mls daily for 1 week De-worm at least once a year (Tabs Pyrantel 4-6 tabs stat) Discourage laxatives

Health Education

1.Encourage use of LLINs.
2.Encourage consumption of fluid, fruits and vegetables.
3.Discourage use of tobacco and avoid exposure to second hand tobacco smoke.
4.Discourage exposure to smoke from household cooking stove, lamp, fire wood etc.
5.Advise on the importance of taking home-based records anytime patient visits the clinic or hospital.

7.6 MENTAL DISORDERS

The risk of developing mental health disorders rises as individuals' age. The World Health Organization (WHO) estimates that mental health issues affect over 20% of the elderly population. Despite this significant prevalence, mental health problems often go unrecognized among older adults. The most common mental health disorders in this age group are dementia and depression, with others including anxiety disorders, substance use problems, and self-harm. Additionally, elderly individuals are vulnerable to neglect and maltreatment by family members, which can exacerbate or contribute to the development of mental health challenges.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.For how long has this problem been there? Has it happened before? If yes, when? 3.Is there persistent tiredness 4.Is there persistent sadness or anxiety? 5.Is there loss of interest or pleasure in activities that used to be interesting and enjoyable? 6.Is there loss of appetite, nausea? 7.Is there headache and/or fever? 8.Is there difficulty in carrying out usual work or daily task? 9.Are there any problems with sleeping or waking up? 10.Is there frequent complaint of multiple symptoms with no obvious physical cause? Is there decline or problems with remembering things (memory loss) and orientation (loss of awareness of time, place and person)? Is there loss of emotional control (easily upset, irritable or crying)? Is there any family history of similar problems?	1.General appearance: look out for state of personal hygiene, , irritability, smell of alcohol or tobacco, signs of injury 2.Temperature: check and record 3. Blood pressure: measure and record 4. Eyes: check for pallor 5. CNS: check for – ability to remember (memory) awareness of time, place and person (orientation)

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
I- Difficulty in falling asleep or waking up after very few hours of sleep, fatigue, Sadness, anxiety and irritability, Low interest in activities that used to be interesting. Difficulty in carrying out usual activities. Poor appetite, Use of medications	• General appearance: unkempt/untidy, ill -looking, anxious and unhappy, weak, irritable, smell of alcohol or tobacco •Temperature: normal or high • Eye: pale • Blood pressure: 140/90mmhg or above	Mild, moderate condition to severe Suspect depression	1.Counselon sleeping habits, drug usage, food intake and participation in community work 2. REFER
II. Complaints by relatives of decline or problem with memory (severe forgetfulness), Loss of orientation (awareness of time, place and person), Loss of emotional control (easily upset, irritable or tearful). Loss of interest in the environment.	• General appearance: unkempt appearing uninterested or irritable • CNS: cannot remember important events and dates , not oriented in time, place and person,	Severe condition Suspect dementia)	1. Counsel patients and relatives. 2. REFER

III. Increase consumption of alcohol and use of drugs such as pain killers, sleeping tablets, coffee, tobacco leaves, cigarettes and snuff, Poor sleep, tiredness, headaches, Poor appetite, nausea, vomiting, indigestion, Difficulty in carrying out usual work	General appearance: unkempt/ untidy, weak, agitated, smell of alcohol or tobacco, unsteady gait, may have some drugs in the pocket of their shirt and/ or trousers, purse or bag, skin infection, injuries etc. CNS: Nervous/fidgeting, slurred speech	Moderate to severe condition Suspect alcohol or drug use disorders	Counsel patient and his family/ friends Encourage participation in social activities within the family and the community. REFER Follow up on return from referral
IV. a)Not happy b) Loneliness c) Isolated) Dejected	• Not cheerful • Weeps during the interview • Look dirty with sores on the body • Poor Oral Hygiene with dirty nails, dirty clothes.	Moderate condition	1.Counsel family of patient 2.Encourage him to participate in religious and community activities 3.Encourage family involvement in the therapy 4.REFER

Health Education

1..Counsel family members to give both physical and emotional support to the elderly.

2..Encourage good nutrition.

3.Encourage the elderly to participate in religious and community activities and encourage visiting friends and associates.

4.Encourage family members to show love, affection and provide for the general needs of the elderly.

5.Discourage use of alcohol, tobacco and unprescribed drugs.

7.7 ORAL DISORDERS

Oral health encompasses freedom from chronic mouth and facial pain, oral and throat cancer, oral sores, birth defects like cleft lip and palate, periodontal (gum) disease, tooth decay, tooth loss, and other disorders affecting the oral cavity.

Among the elderly, oral health issues are common physiological changes, often evidenced by high levels of tooth loss, dental caries, and periodontal diseases. Early recognition of oral problems, coupled with prompt attention or referral, can significantly reduce morbidity and mortality arising from otherwise manageable conditions.

HISTORY	EXAMINATION/ACTIONS
1.What is the problem? 2.Where is the problem? 3.Is there pain in the teeth or gum? 4.When was it noticed? 5.Are you able to eat your regular diet? 6.Does the pain increase with hot or cold drink or food? 7.Does the pain keep you awake at night? 8.Is there bleeding from gum or any other part of the mouth? 9.What do you use for cleaning your mouth? 10.How do you clean your teeth? How many times a day?	General appearance: ill-looking, malnourished, facial swelling, Mouth: Gum: look out for sores, swellings, discharging sinuses, bleeding and white patches. Tongue: look out for sore, coating, redness, dryness, c. Lips: look out for cracks Teeth: Look out for caries, discoloration, shaky or missing teeth, plaque. Temperature: Check and record

FINDING ON HISTORY	FINDINGS ON EXAMINATION	CLINICAL JUDGEMENT	ACTION
Severe pain in the mouth; with or without facial swelling a) Pain disturbs patient's sleep and aggravated by chewing b) May or may not be associated with headache	General appearance: ill- looking, and in pain, malnourished, swollen face Mouth: bad oral hygiene: foul smell gum: sores, swelling, discharge, blood stain/bleeding teeth: holes, discoloration, shaky or missing teeth, plaque	Severe condition Suspect dental abscess	Tabs Paracetamol 1g 8hourly for 3 days or Ibuprofen 400mg 8hourly for 3 days (After food) Tabs Vitamin B complex 8hourly for 2 weeks. Tabs Vitamin C 100mg 8hourly for 2 weeks Use warm saline water as mouth rinse REFER
Continuous dull pain in the mouth, Bleeding in the mouth	General appearance: normal Mouth: Teeth: shaking tooth or tender/painful denture Gum: sores and/or bleeding	Moderate condition Suspect chronic periodontitis	Tabs Paracetamol 1g 8hourly for 3 days Tabs Vitamin B complex 8hourly for 2 weeks Tabs Vitamin C 100mg 8hourly for 2 weeks Use warm saline water as mouth rinse REFER if no improvement
Injury to the mouth, tongue and/or face, In-ability to chew properly	General appearance: traumatized face/mouth Mouth: Jaw: broken Lip: laceration or bruises Teeth: broken/missing/ shaky tooth/teeth Tongue: laceration	Moderate to severe condition Suspect fracture of the teeth/jaw bone	Tabs Ibuprofen 400mg 8hourly for 5 days (after meals). Tabs Vitamin C 100mg 8hourly for 2 weeks Use warm saline water as mouth rinse REFER

Health Education

1.Explain process of ageing.
2.Brush twice daily (morning and last thing before going to bed at night).
3.Encourage regular intake of food rich in calcium.
4.Avoid intake of sugary foods (refined sugar) in-between meals.
5.Encourage to rinse mouth with water after meals.
6.Encourage good personal and oral hygiene.
7.Report any tooth/teeth pains.
8.Advise on regular dental check-up at least twice a year.
9.Discourage use of toothpick.

SECTION EIGHT
HEALTH FACILITY MANAGEMENT



NATIONAL STANDING ORDERS
FOR COMMUNITY HEALTH OFFICERS

8.0 HEALTH FACILITY MANAGEMENT

INTRODUCTION / PREAMBLE

The National Health Policy (1988) formally embraced Primary Health Care (PHC) as the cornerstone and entry point to health for all Nigerians, aligning with the national philosophy of social justice and equity. It is a vital component of the country's health system, with a central role and primary focus, contributing significantly to the overall social and economic development of the community. Primary Health Care is not just a strategy for achieving universal coverage, but also a philosophy that involves members of the political wards in which the Primary Health Centre is located, in management and decision-making for the health facility in their location.

Over the years, the Primary Health Care system has not only received increasing government support through policy reviews, improved funding, and implementation frameworks, but it has also demonstrated its potential for success. These interventions, such as the Primary Health Care Under One Roof (PHCUOR) and the Basic Health Care Provision Fund (BHCPF), have been instrumental in revitalizing and improving PHC service delivery, painting a promising picture for the future of healthcare in Nigeria.

Given the current global health crisis and the need to optimize PHC service delivery in a post-COVID era, it is crucial to prioritize the training and development of Officers-in-Charge (the OICs) of Primary Health Centre facilities. These individuals, who are at the forefront of our platform for achieving the Sustainable Development Goals (SDGs) and Universal Health Coverage (UHC), must be equipped with the right skills and tools for effective management. This is not just a recommendation, but a necessity for the future of healthcare in Nigeria.

It is widely acknowledged that the majority of Primary Health Centres across the country are led by health personnel in the cadre of Community Health Extension Workers (CHEWs), Community Health Officers (CHOs), and Community Nurse/Midwife Officers. Their capacity in managerial skills is a critical factor for organizational success and improving it will have a positive impact on service delivery at the PHC facilities. This section is dedicated to enhancing the managerial capacities of these Community Health Practitioners, empowering them to effectively manage these healthcare facilities and deliver quality care to the populace. The key management activities, derived from various operational guidelines, manuals, and job aids developed by the National Primary Health Care Development Agency (NPHCDA), are designed to provide basic tools for governance, financial management, infrastructural maintenance, quality improvement, reporting, monitoring, supervision, and mentoring tasks at the PHC level.

8.1 HEALTHCARE FACILITY MANAGEMENT

8.1.1 Concepts, Definitions, Principles and Functions of Management.

Healthcare Facility Management:

There are many definitions for different purposes. Perhaps the shortest is that management is "getting things done."

The principle underlying this definition is "commitment to achievement," i.e., commitment to purposeful action, not action for its own sake. To stress this notion of purpose, the definition may be rewritten as follows: "Management is saying what one wants to be done and then getting it done." In other words, management ensures that objectives are specified (i.e., specifically what is to be achieved) and then that they are achieved. Health facility management coordinates facilities, resources, and people to provide the best possible patient care. The goal of managing a healthcare facility is to enable the highest standard of care, whether the facility

is a large hospital with 1,000 beds or a small health post. The goal is to enable caregiving at the highest level by ensuring the facility runs smoothly.

The Facility Management Committee (FMC) is a crucial component of all Healthcare Facilities. It is responsible for ensuring the smooth operation of the facility and comprises the following key members;

- Officer In-charge (OIC) of health facility.
- Heads of PHC Departments.
- A representative of WDC/VDC, as necessary.

During their regular monthly meetings, the Ward Development Committee (WDC) and Village Development Committee (VDC) should, depending on the situation, receive updates and feedback from the Facility Management Committee (FMC).

Two committees provide support for the WHS:

The Ward Development Committee (WDC) – operates at the ward level, and

The Village Development Committee (VDC) – operates in urban and rural community settings.

Principles of Management

Management by Objectives:

A management principle implies that one should decide and say what is to be accomplished by setting an objective (a goal, a purpose, an end, a target). Objectives should state what is to be done, how much is to be done, where it is to be done, when it is to be completed and the standard by which it will be possible to tell whether or the extent to which it has been achieved. Setting objectives helps those who are to produce the result to organize, distribute and coordinate their efforts. It helps determine when the intended result is being achieved or has been achieved. In other words, a clear statement of objectives is essential for effectiveness.

Effectiveness is not just a measure of how well an objective has been achieved; it is a constant goal that management strives to improve. For instance, if a health team in PHC, Cana sets the objective of immunizing 100 children against diphtheria by providing Penta vaccinations during the following week, and succeeds in immunizing 95, the work has been managed effectively. However, if an epidemic of diphtheria were to occur a few weeks later, how would you judge the team's performance? The urgency and importance of effectiveness is further highlighted when considering a team objective like holding daily clinics from 9.00 – 5.00pm, and providing 24/7 services. If, by the end of 2022, the children under 5 of Cana have not been fully immunized, and the objective was for 85% to be fully immunized, it might seem very likely that those in charge did not manage their programme effectively. If the health centre at Cana remains closed for two days during the week ending 27 July, can it be said that the objective of holding daily clinics has been achieved effectively? This constant evaluation and striving for effectiveness is what drives continuous improvement in management.

The management principle that underlies the comparison of objectives with their achievement in order to judge effectiveness is known as 'learning from experience'. In the context of management, 'learning from experience' refers to the process of reflecting on past actions and outcomes, and using these insights to inform future decision-making and improve performance. It derives from and justifies - proves the value of - the first management principle, namely, management by objectives.

Learning from Experience

How is this principle applied? When there is a gap between objectives and results (or achievement), management analyzes why only the observed results were achieved and why they fell short of the set objectives. Some causes can be easily remedied, and action is taken accordingly. Others cannot be removed

in the short term and are called constraints.

Management learns from this process and uses what it has learned to achieve its objectives. This process is sometimes called 'feedback' (of information from experience to decision for action).

A second reason for specifying objectives is that it enables management and health workers to decide how to achieve them. In other words, stating the objective, or end, helps select the means of achieving it. Sometimes, however, it needs to be clarified which ends and which means.

When pipe-borne water has been installed, when a health worker has been trained as a skilled birth attendant, and when a guide for primary healthcare management has been produced, are these ends or means?
When people have accepted the use of latrines when children have been immunized and when health workers evaluate their effectiveness, can we talk of ends or means?

The answer is, of course, both. Every end may be seen as a means to a higher end, almost indefinitely: trained staff is a means of delivering better services; better services, in turn, are a means of improving a community's health; better health is a means of ensuring a better quality of life.

The following definition of management mentions both ends and means:

Management is: "Getting things done through people."
This does not mean that someone orders and others do what they are ordered to do, although this is often what happens. It means that people are the most important means, or resource, for getting things done.

Getting things done through people means that people must work, performing certain activities and tasks to reach certain ends or objectives. When we are concerned with how means are used to reach ends, we are concerned with efficiency.

Efficiency is about reaching ends by only the necessary means or the least wasteful means. It measures the relation between the results obtained and the effort expended (by the health team, for example). This concept has many implications for resource use.

Almost any work involves more than one person. When two or more people are involved in work or activity, management must apply two complementary principles: division of labour and convergence of work.

Division of Labour:
When work is divided, or distributed, among group members, and the work is directed and coordinated, the group becomes a team. In a team, and generally, when there is specialization and division of labour, with each category of staff exercising its skills towards achieving the objectives, management entails assigning a balanced proportion of each kind of staff to the work to be done.
Efficiency implies that all necessary means (but no more than necessary) are used to achieve objectives and that the least expensive is chosen if there is a choice of equally effective means. The team approach is how management attempts to bring about balance among the different members of the team and the work they do. For instance,

- For a surgeon to operate, someone must give the anaesthetic.
- Someone must lay out the instruments.

- Someone must ensure that the instruments and the theatre clothes have been sterilized.
- Electricians must ensure the power supply, and other staff must be responsible for other tasks.

The general principle of management illustrated by this example is division of labour: work must be shared by, or divided among, several categories of technically skilled people. As pointed out earlier, one of the advantages of defining objectives and considering resources is that this helps. In deciding how to reach the objectives, i.e., the activities needed to achieve them and the methods and organization that will enable people to work harmoniously towards the objectives while using resources efficiently. The study of work activities and the organization of work relations form a significant area of management. Its governing principle is that of convergence of work.

Convergence of work

Convergence of work means that the activities of the various people who do the work come together to achieve objectives. The activities should be designed, assigned and directed to support each other in moving towards a common goal. It also implies that working relations - how the team members interact with one another- should contribute to the success of each activity and, thus, to general effectiveness. In general, health activities are studied, described, and performed under three main headings, namely:

- Service activities
- Development activities
- Support activities

A service activity (e.g. immunization) usually requires some preceding development activity (e.g. training of immunizers) and some continuous support activity (e.g. provision of supplies). These three kinds of activity need to be managed to bring about convergence of work, balance of resources and harmonious work relations, and ultimately, the intended results. Specific work activities must be brought into logical relations with one another. For instance, trained staff is needed to provide services. Specific objectives are set for training staff. To achieve these objectives, various preliminary development activities are carried out in relation to one another, e.g., designing a curriculum, preparing learning materials, selecting trainees, and recruiting trainers.

Activities must also have a time (or temporal) relation or sequence. For instance, trainers can be recruited first and asked to prepare the curriculum and the learning materials, then select trainees and begin teaching. However, other time sequences are possible. Also, training can begin either before staff recruitment (before beginning to provide the services) or after staff recruitment, when training is given 'on-the-job', i.e. at the same time as services are provided.

Frequency is an aspect of the time relationship. In the work of a health team, some activities (such as certain development activities) occur only once or rarely; some (such as services) occur more often, and some (such as support) occur continually.

There is also a place (or spatial) relation between activities. For instance, Services are given in families' homes (e.g. treatment of illness); on-the-job training may take place elsewhere in the district; formal training is given in a classroom at the health centre or a training college; the curriculum may be designed in the capital city.

Management is: "the efficient use of resources"

It is well recognized in health work, as in other kinds of activity, that the different types of resources used for achieving objectives must be carefully balanced. Use of the team approach to bring about balance among the

different people (human resources) concerned has already been mentioned. However, other resources must also be managed to achieve objectives; examples are materials (equipment and supplies) and money (without which people and materials often would not be available). Beyond these '***three Ms' - manpower, materials, and money*** – there are time and information, which the health team must also have. These other types of resource must also be kept in balance. For instance:

If there is insufficient equipment (e.g. syringes) or a shortage of supplies (e.g. petrol), a health team may not be able to reach its immunization target.

When money runs short, salaries cannot be paid on time and staff morale drops, and this may result in targets not being reached.

A prime concern of management is therefore to ensure adequate funds and stocks of materials (commodities, diagnostic supplies and drugs) to permit the staff to provide the planned services. This is one meaning of "efficient use of resources" in the above definition of management.

If all resources were plentiful and free, the concept of efficiency might have no practical importance. However, some - often many - resources are scarce and costly. For instance:

Wazobia State has an ample supply of trained people to staff the health services. But the local authorities are poor and their budget does not allow them to fill vacancies without cutting down on supplies, such as petrol for transport or essential drugs (both of which are imported and therefore costly).

This is a typically difficult position for management, which illustrates one more concept - economy - and one more principle - substitution of resources.

In everyday life some commodities become scarce, either because production or supply falls or because demand for them rises. As a result, they become more expensive. When a commodity becomes scarce or more expensive, ordinary people buy less of it; in other words, they economize. The concept of economy - or careful use of resources - is applied by almost everyone in daily life, with more or less success. The related principle, substitution of resources, is also widely applied.

Often, when the resources normally used to provide services become scarce or too expensive, different resources or a different balance of resources may be used to produce the intended results. For instance:
When purchasing food for a hospital kitchen, cheap produce in season can be substituted for other similar but more expensive food if it is equally nourishing.

When petrol continues to go up in price, bicycles may be substituted, at least partly, for motorized transport. The savings on petrol, vehicle maintenance and drivers may be better spent on more field staff (perhaps saving time previously lost on travel), thus achieving the same coverage of the population as before. This leads to the fifth principle of management, substitution of resources.

Substitution of resources:
Labour substitution, a pivotal resource management strategy in healthcare, is the utilisation of trained auxiliary personnel or volunteers for tasks previously handled by professionals. This practice, similar to the use of generic drugs in health management, is a potent tool for cost reduction without sacrificing quality, particularly in countries where generic drugs are dependable alternatives to brand-name preparations.

One more definition may be considered briefly, which brings together many of the points discussed so far:

Management is:"getting people to work harmoniously together and to make efficient use of resources to achieve objectives."

This definition underscores the vital role of teamwork in healthcare management. The working relations between individuals, often referred to as functional and structural, are a primary focus for managers. Functional relations are directly derived from the technical aspects of the work, including where, when, and in what sequence it is performed. An example is the working relationship between the members of a surgical team. Structural relations, on the other hand, are related to administrative rules and standards, and particularly to the authority and responsibility assigned to individuals - for instance, who hires and fires staff, how much a medical officer may spend without consulting higher authorities, or to whom the public health nurse must submit her reports and accounts.

The two key concepts that underlie functional and structural relations within a health team are responsibility and authority, which can hardly be considered separately in day-to-day teamwork management. For instance: A member of the staff of a health centre who is entrusted with the responsibility for the immunisation coverage of the community must also be given the authority to requisition vaccines from the central cold storage depot, to order transport from the vehicle pool for the immunisation team; to have the refrigerator repaired if necessary; to replace leaky syringes and used needles with new materials from the stores. This empowerment ensures that the immunisation programme will be carried out effectively. This stress on the importance of responsibility and authority should empower healthcare managers in their decision-making roles.

An officer responsible for "directing and supervising" staff but with no say in their leave, transfer, or promotion would be unable to fulfil his or her responsibilities.

The management principle derived directly from the need to associate responsibility and authority may be stated as functions determine structure.

Functions determine structure:
When work is clearly defined, i.e., the function and duties of individual team members are clearly defined and known to all, and the working relations (the structure) follow.
However, what is the exact nature of authority? One more definition of management will help clarify this point:

Management is:"to make decisions"
This definition is more general than those given earlier. It stresses the most essential element of management, namely decision-making. A decision is a choice between two or more courses of action. It can be a resolve to act - "this shall be done" - or, sometimes, not to act - "this shall go on as before". Some decisions deal with quantity- "more of this, less of that, none of the other"; some deal with quality- "higher coverage, better services". In management terms, a decision is an answer to a question about possible courses of action, an answer that can be stated simply as yes, no, more, or none. "I don't know" in this context is indecision.

The authority of a health team member can be defined as the decisions that that member is authorised to make. A common management issue is the lack of clear responsibility for urgently needed decisions or a situation where someone is responsible but lacks the necessary authority to act. For instance:

The ambulance has broken down in the bush. The driver, upon securing police watch over the vehicle, reports to his officer-in-charge, who is "responsible for the maintenance and upkeep of transport". However, the breakdown truck belongs to the State Primary Health Care Development Board, and the charge for picking up the ambulance will likely be more than, say, N25,000. The officer-in-charge has the authority to spend up to N10,000 on repairs, and the Local Health Officer up to N15,000 only and LGHA has already spent its budget allotment for transport. The questions are: Who is the decision-maker, and where is he or she?

A primary concern in working relations is enabling the most suitable person to make decisions where and when necessary. Any health team member may be called upon to decide at one time or another, especially when the team leader is absent; otherwise, the team would be paralysed. A health worker who strictly follows the rules may do nothing and is then blamed for doing nothing, say in an emergency. What is needed, then, is the application of the principle of delegation.

Delegation:

Delegation occurs when someone with authority 'lends' the authority to another person, conditionally or not, to enable that person to take responsibility when needed. This management principle has been stated with a humorous twist: "Never do yourself what another can do for you as well as you would." This advice is for the busy manager, but it could also apply to anybody's family life and the performance of the tasks of any member of a health team. Information and decision rules are needed to decide between different courses of action. For instance:

A management decision for Health Team X is whether to transfer the responsibility for managing labour and delivery from one staff category (A, which could be nurses) to another (B, which could be midwives). The relevant decision rule might be: "If managing deliveries takes more than 20% of A's work time, and if B, after training, can conduct ten consecutive deliveries without fault, then A may authorize B to take over this task". Information is needed to show that the deliveries take over 20% of A's time, that B has been trained, and that B has passed the test.

Another crucial element of decision-making is ensuring that the decision, once taken, is effectively communicated to all concerned. This is not just about sharing information; it's about making everyone feel valued and integral to the process. Many people would even say that all concerned should know beforehand that a decision is about to be made. Often, knowledge about the need for a decision, and taking part in making it, are the best assurances that the decision will subsequently be acted upon. One thing is essential, however, and that is communication between those who make decisions, those who implement them, and the people affected by the decisions. At least, the decision itself should be communicated, but normally the supporting information, the decision rules, and how the decision is to be implemented, should also be included.

Management by exception:

Management by exception means two things:

First: be selective. Do not become overloaded with routine and unnecessary information. Keep your mind available for critical information, on which you will be required to act. For instance: If you were in charge of fifty beds, you would not want to know every day every patient's blood pressure, temperature, pulse rate, respiration rate, bowel movements, volume of fluids passed, etc. Just one of these elements of information might be all you would need to watch in each case to manage most patients effectively and efficiently. This is the principle of management by exception, which is particularly relevant in a healthcare setting where

managers need to focus on critical patient information rather than being overwhelmed by routine data.

Second: Make big decisions first. To be overloaded with petty decisions (such as scheduling staff breaks) may result in more important ones (such as implementing a new patient care protocol) being neglected, or what has been called "postponing decisions until they become unnecessary". In short, management by exception means selectivity in information and priority in decision.

Shortest decision-path:
The last management principle discussed in this session, the principle of the shortest decision path, deals with the issue of who should make which decision. (and often when and where, as well). Answering this type of question will help clarify the delegation of authority referred to earlier. Applying this principle means that decisions are made as close as possible in time and place to the object of the decision and those affected by it. This saves time and work (e.g., in transmitting information) and also ensures that decisions can take full account of the circumstances that make the decisions necessary and in which they are put into effect.

The main functions of management:

A function may be defined as a group of activities with a common purpose.

Management is not just a set of tasks; it's a series of functions that are crucial to the success of a health team. A health team, in particular, has three main management functions: planning, implementation, and evaluation. These functions are not just about getting things done, they are about inspiring and motivating the team to achieve their goals.

Thus, the principle of management by objectives requires the specification of what and how much is to be done, and where and when it is to be done. Each of these four questions needs one or more planning decisions. For instance, a planning decision could be deciding the number of staff needed for a particular shift. The sum of the planning decisions constitutes the planning function of management in a health team.

The principle of delegation is concerned with authority and responsibility, i.e., with functional relations between people working together to achieve some purpose. The types of decisions involved are concerned with the organization of working relations to ensure effective and efficient work, i.e., implementation. Organizational decisions are part of the implementation function of management.

Applying the principle of learning from experience requires the analysis of gaps between desired results and actual results or achievement and the use of the information obtained from the analysis in decision-making. This is, in other words, a measurement and a judgement of performance, or the evaluation function of management, a function that contributes significantly to the success of a health team.

8.1.2. LEADING AND MANAGING HEALTHCARE FACILITY

I. Distinction between Leaders and Managers:

Leaders: Have a vision of what can be achieved, then communicate this to others and evolve strategies for realising the vision. They motivate people and can negotiate for resources and other support to achieve their goals.

Managers: Ensure the available resources are well organised and applied to produce the best results. In resource-constrained and complex environments, a manager must also be a leader to achieve optimum results.

Attributes of a good leader should include:

- A sense of mission.
- Be charismatic.
- Ability to influence people to work together for a common cause.
- Be decisive.
- Ability to use creative problem-solving to promote better care and a positive working environment.

b)Attributes of a manager should include:

- Clarity of purpose and tasks.
- Good organisational skills.
- Ability to communicate tasks and expected results effectively.
- Ability to negotiate various administrative and regulatory processes
- Good delegation skills

I. Conditions for Good Management:

Certain conditions are essential for creating good management, including:

- Managers and team members need to be selected on merit
- Managers need to earn the respect of their staff, patients, and supervisors
- Managers need to have the knowledge, skills and understanding of the role, tasks and purpose of the services they deliver.
- Basic support systems function well; clear staff administration rules and regulations; well-planned and timely delivered supplies, equipment, and drugs; clear and transparent financial processes; and well-planned and monitored activities.

II. Leadership and Management Roles of the Officer-In-Charge (OIC):

The Officer-In-Charge of the PHC Health Centre is responsible for providing leadership and managing all activities in the PHC. The OIC performs four significant functions, namely;

- Planning
- Organising
- Leading
- Controlling

Planning Functions of the OIC

Planning in healthcare management is a systematic process that involves identifying future goals, outlining appropriate actions, and determining the resources needed to achieve them. It is a decision-making process that considers past and current circumstances. Planning is the foundation of all management functions and precedes all others.

The OIC may be engaged in the following types of planning:

a) Strategic planning: a dynamic and deliberate process that allows the PHC to invent the future, usually in about 3-5 years. This deals more with outcomes and impacts

b) Operational planning is establishing plans that relate to running a PHC on a short-term basis. It deals more with services, outputs and setting targets. This is the most critical part of the PHC's running and is supposed to be led by OIC in conjunction with the WDC and FHC.

The significant steps in planning are as follows:

Step 1. Problem identification: Define the present situation of the PHC (situational analysis) by providing answers to the following questions

1. Where are we now?

a. Geographic coverage

b. Target population

2. What is the magnitude of the problem?

3. How visible is it?

4. Is it new or old?

5. Are there effective interventions

Step 2. Stakeholder's analysis and analysing strengths and weaknesses (SWOT) of the PHC: This is applied by examining the following:

1. Strengths: What are the strengths of the PHC? What do we do well?

2. Weaknesses: What are the weaknesses of the PHC? What do we need to improve?

3. Opportunities: What are the external trends that offer opportunities?

4. Threats: What are the local, State and National threats that pose risks?

Step 3. Establish goals and objectives: Always make objectives SMART:

Specific: What is the specific change desired? Who is the target group? What is the location?

Measurable: What is the measure (number or percentage) to monitor progress/performance?

Appropriate: to goals and strategies

Realistic: Given the limited amount of resources

Time-Bound: What is the date by which the change will occur? This will help with resource allocation and activity planning.

Step 4. Develop action plan (road map): The action plan describes what to do, who is to do it, how to do it, when, and the outputs/deliverables. A sample action plan is in Annex VI.

Step 5. Develop budgets: After developing activities to be executed, cost them to determine the resources needed to implement the plan. Any plan that is not costly is not practical. The Account Officer and other unit heads will be beneficial in determining unit costs and cost elements of particular activities. At this point, it is also important to identify where the required resources will likely come from.

Step 6. Implement plans: The OIC should ensure that the plans are implemented. The OIC is responsible that resources are available to implement the plan. He should also liaise with relevant stakeholders to ensure it happens.

Step 7. Monitor plans: While the heads of units are responsible for monitoring the plans of their various units, the OIC is also responsible for monitoring the overall plan. The OICs are encouraged to extract monthly plans from the annual plans and display them in conspicuous places in their offices. The use of pictograms and charts is encouraged to make this happen. Equally, all the Heads of the Unit and OIC are advised to have a weekly and daily to-do list. The weekly management meetings and any other meetings also allow for monitoring of the plans.

Organising Functions of the OIC

This function refers to how the OIC distribute resources, delegates tasks, structures departments, sets staffing levels, etc. This function encompasses everything from assigning right fit tasks to the appropriate team members to deciding how those team members relate to each other in an organisational structure.

The OIC should ensure that all staff members are aware of the organisational structure of the PHC and the reporting relationships. Despite being in different units, there is a need to work harmoniously to achieve the objectives of the health facility, which is to provide quality healthcare services that meet the needs of the people they serve. Orientation is important for any new staff in the facility. Where the staff available are short in supply of requirements and task-shifting and task sharing are to be practiced in line with the relevant policies, there should be constant meeting with the staff to explain why this is necessary and clarify roles fully. The checklist should also be provided to staff for any new task assigned to provide appropriate guidance on proceeding with such tasks until the staff masters the processes involved.

Organising the different units within the health facility to ensure effective workflow processes and patient flow to reduce the stress patients go through is critical. For instance, the facility records section, which is the first place of contact with the client, should be conspicuous and easily accessible. The waiting section should also be spacious to accommodate clients (based on service demand projections) almost always. There should also be a designated place where patients could stay when there is an overflow, especially on ante-natal days.

Leading Functions of the OIC

Leadership is a process by which the OIC influences others (other medical staff, patients, community and stakeholders) to accomplish the required activities and outcomes. Leading is not about what one can achieve on their own; it is about inspiring people to be the best they can be at what they do.

In leading, the OIC must be able to;

Effectively plan and manage work and resources, lead people, and guide future leaders or people who can take over from him.

To effectively plan and manage work and resources, the OIC must;

- Plan work to be done

- Maximise resources (both human and material)
- Take decisions with the information available on issues like procurement, maintenance, and
- Staffing and staff deployment
- Be flexible enough to change direction if need be

To lead the other staff members, the OIC should:

- Provide clear guidance
- Build a team
- Acknowledge the teammates
- Provide feedback, and above all
- Set good examples.

To guide the future leaders, the OIC should:

- Hold people accountable, including him/herself
- Delegate to individuals and teams
- Coach individuals and teams
- Encourage development and publicly reward high performance in staff members.

Controlling Functions of the OIC

In the performance of its controlling functions, the OIC should;

1. Continuously check progress against plans.
2. Set up individual and organisational performance standards and see that they are achieved per the plan.
3. Detect irregularities either in data submitted or service delivery procedures, amongst others.
4. Identify opportunities for resource mobilisation and partnerships working in consultation with the WDC and other stakeholders.
5. Minimise costs and identify and eliminate wastages, particularly in the area of administrative expenses and consumables and drugs.
6. Develop coping strategies to deal with uncertainty regarding human, financial, and material resources as they will never be in the necessary proportions.

IV. Management Tasks

1. Priority setting

One of the most significant problems of management and leadership is the art of setting priorities for daily work and projects because everything seems to be essential and requires attention and time. However, the truth is that a lot of the work we do every day does not really need to be done, or at least not as important, that needs to be done right away. Prioritisation of the work is critical. Simply defined, prioritisation is the action or process of deciding the relative importance or urgency of a thing or things.

While prioritisation seems simple, i.e., knowing what needs to be done, it can be complex. It is essential to take the following steps:

1. Make a list of what needs to be done in the facility for the day, week, month or year.

2. Highlight the ones that need to be completed urgently. These include things that, if not done, will have negative consequences on the facility and quality of care provided
3. Identify which is most important, keeping in mind that the primary goal is to ensure an efficient facility that offers patients quality care.
4. If activities have the same weight, start with the one that takes the most effort to complete or is most beneficial to the patients.
5. Be aware that there could be uncertainties and changes, so know that priorities might sometimes change, so be flexible but focused.

2. Setting and sharing objectives

People work well together when they agree with one another and share in setting and achieving the objectives of the whole team. To fulfil the objectives of an organisation, those who work for the organisation should know its objectives. People who have yet to share in setting objectives or been told what these objectives are may waste much effort on activities that do not bring achieving objectives any nearer. People who oppose the objectives may obstruct the organisation's work. It is challenging, if not impossible, for real progress to be made in health care without the active cooperation of everyone concerned. People do not give their best when ordered to do things. One of the best ways, perhaps the only way to ensure that people agree on objectives and that they are motivated to achieve them is to make sure that they take part in setting them.

3. Motivating team members

Motivation is an inner impulse that induces a person to act in a certain way. It is a series of internal drives within a person at different levels:

Level 1: To obtain the necessities of life: food, shelter, clothing, rest and safety.

Level 2: To satisfy social needs such as companionship, love, and a position of respect.

Level 3: To ensure some degree of personal satisfaction and to pursue ideals. People need to feel reasonably satisfied with themselves, what they make of their lives, and their talents and abilities.

A team leader should understand what encourages people to apply their ability and energy to work and what makes people dissatisfied. These two groups of factors may be called motivators and dissatisfiers, respectively.

The six primary motivators in work are:

- **Achievement:** Most people like to do things well and succeed. Their satisfaction in success and in getting things done well comes primarily from achieving what they expect to be able to achieve and what they aim to achieve. Thus, achievement is significant to the individual. Good team leaders help their team members achieve by giving them clear instructions, suitable training, and the necessary facilities and supplies.

- **Recognition:** Very few people are satisfied with simply knowing in their minds that they have been successful. Most people like others to know about their success. Individuals can be sure that others know of their success only when they respond to their achievements, i.e., acknowledge their success. It should be easy to praise and publicise good work, but many leaders find it very hard to do so. Perhaps they fear their achievements will be belittled if they recognise good work from those they lead. Lack of recognition can be very discouraging; leaders should be willing to praise others when praise is due.

- **The work itself:** People like to do valuable and worthwhile work, be helpful to other people and help themselves achieve their ideals. The staff of an organisation likes to do work that they can see as contributing to the organisation's objectives. The appeal of curative medical or caregiving work is self-evident. However, health workers may find it hard to believe that such dull jobs as accurate record-keeping and inventory

control are worth doing. A good leader will try to keep dull, repetitive tasks to a minimum and take every opportunity to assure the team members of the value of their work.

- **Responsibility:** To have responsibility is to accept the consequences of a decision or an action, good or bad. Most people welcome responsibility; some fear it. Most people like to make decisions about their lives and accept responsibility. A leader can quickly destroy or, at least, reduce staff members' sense of responsibility by making decisions for them. This should be avoided, especially when managing isolated clinics where staff work primarily without direct supervision. A good leader takes the opportunity in such a clinic to increase the staff's sense of responsibility for work rather than diminish it with petty criticisms.

- **Advancement:** It is a form of recognition. Recognition without reward is not very convincing. People prefer recognition that comes in a tangible form, such as an increase in salary or more responsibility, with freedom to use their initiative, which leads to job satisfaction. People's satisfaction often depends as much on what they expect to get as on what they get. In acknowledging good work, the team leader or supervisor does not make promises of advancement that are impossible to keep. Instead, he or she should encourage people to increase their knowledge and skills to become ready for promotion.

- **Self-improvement:** People like to become mature and develop as people. Many make great sacrifices to improve themselves and their families. Many people want opportunities to discover, develop, master, and use their abilities to the fullest extent in their work and private lives. Since health workers deal with people, there are endless opportunities for self-improvement. Team leaders can help their health workers take such opportunities by providing learning material, challenging them with questions at work, helping them set up research projects (which can be very simple), and giving them high-quality supervision. Given such opportunities to develop, health workers will likely bring energy and enthusiasm to their work.

4. Delegating authority and responsibility

The leader of a team is responsible for the work the team undertakes. The leader must ensure that the team has the necessary equipment and resources and that team members are adequately trained and can do the work. The leader is responsible for the team's failures and successes. At the same time, the health workers in the team are responsible for the particular tasks of their jobs.

One way of using authority is to delegate it. To delegate means to give another person some of one's authority or, in other words, to give another person the power to make decisions.

Delegation has certain advantages:

- Delegating some decision-making saves time for other duties.
- When work is spread over a large area, as in rural health work, the health workers on the spot must be able to make decisions according to circumstances.
- Delegation of responsibility saves long delays when awaiting decisions from a central office or other distant authority.
- Health workers who can make decisions enjoy their work more and become more knowledgeable and skilful.

It also has disadvantages:

- If wrong decisions are made, the work may not be done or done less well.

•A leader who does not delegate properly may pass all the work on to the team members, leaving little more to do.

•A leader may delegate decisions to people with insufficient experience.

Deciding what to delegate and to whom

In deciding what to delegate and to whom, a leader can use a table like the following, listing current tasks in the left-hand column.

Table 1 How to decide to delegate

Current Tasks	Skill Used	Health Workers who can do this for me now	Health Workers who could be trained to do this for me/
Separate weekly meetings with Village health Workers in 3 villages.	Supervisory Educational Consultative	Medical Assistant and Auxiliary Sanitary Workers in turn	

Rules for delegating authority and responsibility

1. Be clear about precisely what is delegated.
2. Select the person who can do the work.
3. Explain to others the work that has been delegated and to whom.
4. Only interfere if asked to, and be prepared for some mistakes.
5. Give support as needed and follow up on the work progress.

5. Coordinating activities

Coordinating activities or groups of activities involves bringing them into proper relation with each other to ensure that everything that needs to be done is done and that no two people are trying to do the same job. To make coordination effective, seven well-recognised principles of organisation must be applied:

1. Objective: The objective of each group of tasks must contribute to the organisation's objectives as a whole.
2. Definition: Each group of tasks must be clearly defined so that everyone knows precisely the tasks.
3. Command: Each group of tasks must have one person in charge, and all concerned must know who this person is.
4. Responsibility: The person in charge of a team is responsible for the performance of its members.
5. Authority: Each person responsible for a group of tasks must have authority equal to the responsibility.
6. Span of control: No person in charge of a group of tasks should be expected to control more than six to ten others.
7. Balance: The person in charge of several groups must ensure that they are balanced. For instance, case-finding must not be so extensive that more disease cases are found than can be treated.

A coordinating checklist

A health worker responsible for an action (any action) will find it helpful to apply the following checklist:

What is to be done? Where will the action take place? When will the action take place? What equipment is needed? How will the action be arranged? Which members of the health team will take part? Who outside the health team will take part? Who will do what? Who will lead?	Coordinating the activities Coordinating the people
Is all necessary information available? Has the information been communicated?	Communication

6. Communication

Successful teamwork hinges on strong relationships among team members, particularly between the team leader and others. It is a common scenario that team dynamics can be intricate. Challenges often stem from or are exacerbated by ineffective communication within the team and between the leader and team members. Similarly, complex relationships can lead to or worsen poor communication. Therefore, a team leader must prioritise the quality of team relations and communication to uphold positive relations.

To encourage communication, the team leader should always observe certain principles:

All team members should be free to express and explain their views and should be encouraged to do so. Their unique perspectives and insights are invaluable to the team's success, and their contributions should be acknowledged and appreciated.

The team leader is also responsible for ensuring that all messages, whether spoken or written, are articulated clearly and in a manner that all can comprehend. Your guidance and support can significantly boost the team's communication, making them feel supported and guided.

Communication is a two-way process - sending and receiving. When the message that is sent is not received, communication has not taken place. Therefore, as a team leader, it is crucial always to use some means of checking that the intended effect has taken place. This can be done through feedback, asking questions to ensure understanding, or using visual aids to reinforce the message. By checking that the intended effect of the communication has taken place, one can ensure effective communication and understanding, making the team feel heard and considered.

Conflict or disagreement is a natural part of human relationships. However, it should be managed to yield constructive outcomes. The team leader plays a pivotal role in facilitating open and respectful discussions, promoting compromise, and ensuring that conflicts are resolved in a just and timely manner. This approach fosters a sense of security and harmony within the team.

How to manage space to assist communication

Always try to arrange rooms, offices, classrooms, other group-education spaces, tables, and seats to communicate effectively. For instance, if the purpose of the meeting is to encourage open discussion, a circular seating arrangement can facilitate equal participation. On the other hand, a theatre-style seating arrangement can be more suitable if the meeting involves a presentation. The key is to adapt the seating arrangements to the purpose of the meeting, not the other way around. This can help create a more conducive environment

for effective communication and engagement.

7. Conducting meetings

Meetings are a necessary part of health work, especially when the work is with rural communities. Meetings are of many kinds and can have many different purposes. Large public meetings, often held in the open air, may be held to encourage people to express their views on a new project or inform them of and explain something new. Small meetings may be held with community leaders to identify health problems and needs. Health education may be taught in meetings with special groups, such as patients or mothers. The health team also has regular meetings. Sometimes, committee meetings are held to decide on a new project, educational meetings, or discussion groups to learn new skills and approaches.

Preparing for a meeting

When preparing for a meeting, it is useful to check the following items:

- Purpose of meeting
- Main subject-matter
- Type of meeting
- Size of meeting
- Place, time, and duration of the meeting
- Who is convening and organising the meeting
- Announcements or information about the meeting.

Purpose: The purpose of the meeting should be clear. For a formal committee, the agenda should state the purpose. However, it is worth writing a summary of the purpose, stating what it is hoped to achieve. Understanding the purpose of the meeting is crucial as it helps you set the correct agenda, invite the relevant participants, and prepare the necessary materials. Some meetings are called to communicate information, others to exchange views and ideas, and others to make decisions about plans or activities.

Subject Matter: If a meeting is to be useful, each person present must have as much information as possible about the subject to be discussed. Also, the facts, principles or ideas needed as a basis for the discussion must be provided, sometimes before the meeting by means perhaps of a book or a working paper, or at the meeting by a knowledgeable person or by use of a tape recording or a film. If the subject for discussion is new to several group members, someone with exceptional knowledge of the subject should be asked to give a brief introduction.

Type of Meeting: Meetings may be small or large, formal or informal, and open to all or only to members.

Size of the Meeting: The Size of a meeting largely determines how a subject is discussed, whether everybody takes part, and whether it is easy to reach decisions or take votes. Two or three people may be enough when the group is meeting to share ideas or for a spontaneous discussion, searching for new ideas. A group of five to seven people is usually large enough to provide a reasonable variety of experiences and personalities while enabling everyone to participate. Generally, a group should be at most ten to twelve people for group discussion or training. Even with this number, a small subgroup may form and dominate the discussion, while the rest of the group may become mere observers. Large meetings are more helpful in communicating information and exchanging views than discussing or making decisions.

Place and Time: The choice of meeting place should avoid the need for most people to come a long way; a hall or open square at the centre of a village is often the best place. The time chosen for a meeting is significant. The time must suit the public rather than the organiser for a public meeting. People who work all day prefer evening or weekend meetings.

The Convener and Organizer: The convener is the person who calls people together for a meeting and maybe the health worker, a community leader, or the chairman of the group that is holding the meeting. The convener plays a crucial role in setting the agenda, ensuring that all necessary participants are invited and the meeting's objectives are met. The organiser, who may be the same person as the convener or a helper, makes the practical arrangements for the meeting, such as hiring or borrowing the meeting place, informing the members, inviting the speakers, and providing information in advance about the meeting. Their roles are essential in ensuring the meeting is well-organised and productive.

Announcements or Information about the Meeting: All those concerned with a meeting should know about it in plenty of time. Notices about a meeting may be written invitations but are usually posted on walls and doors in public places such as shops or post offices.

Conducting a committee meeting

How a committee meeting is conducted depends on whether it is large, small, formal, or informal. The chairperson should be someone who can encourage good communication. In conducting any meeting, certain factors must be kept in mind:

- The agenda should be followed.
- All participants should be allowed to speak.
- Decisions should be made based on consensus whenever possible.

These practices ensure that the meeting is productive and that all participants feel heard and valued.

Communication - The success of any meeting depends on the quality of communication. Suppose the purpose is to pass on information or to explain something. In that case, speakers should find out whether it has been successful, for example, by allowing questions and discussion, which will show whether the subject has been understood. If the purpose is to seek the views of those present at the meeting, the chairperson or secretary should summarise the views expressed or put them in different words to obtain agreement about what has been said. Quarrels and shouting at meetings are not helpful and are often the result of poor communication. When people understand each other clearly, there are fewer disagreements.

The role of the chairman -The chairman keeps the meeting to its primary purpose, gives everyone a fair chance to take part, controls the timing, and keeps order. There are three simple rules for group meetings:

1. There must be no rudeness or personal remarks; no member must ever make another appear foolish.
2. The chairperson must have the absolute right to control the discussion, rule out irrelevant remarks, and stop the proceedings if necessary.
3. The chairperson should be able to keep the discussion going when necessary, e.g. by raising questions or new topics. At a small meeting, everyone present should be stimulated to take part. Those who are too talkative should be discouraged from talking too much, and those who are hesitant about joining in should be encouraged.

Controlling the time -Controlling the time that discussions and questions take is essential to conducting a meeting. Members' questions and discussions must be kept within a definite time limit to give everyone a chance to speak. If the subject permits or the discussion is likely to become lengthy, a definite time limit should be set for a small meeting, with target times set at intervals. Using time in this way helps people bring their ideas into focus. If a decision cannot be reached within the time limit set, it is advisable to postpone the attempt, thus giving time for further thought or preparation.

Role of the Secretary -The secretary records the decisions. Like the chairperson, the secretary may be appointed by those who set up the committee or elected by the committee members. The secretary's main task is to make a record of the main events of the meeting, and in particular to record:

- ·The names of members present.
- ·The names of any visitors.
- ·The names of absent members and of those who apologised for their absence.
- ·The exact wording of any decisions taken.
- ·Summaries of discussions leading to these decisions.
- ·The date of the next meeting.

In recording discussions, it is customary not to record the names of members who made particular points.

8. Dealing with problems and conflicts

Helping staff solve personal problems - A good supervisor tries to understand that health staff may have financial problems (for instance, because a landlord has increased the rent) or worries because a child is sick, a relative has died, or a child has failed an examination. Personal worries are many and varied and may interfere with work. A good supervisor listens sympathetically to personal problems and, whenever possible, tries to help solve them. A person whose problems are considered with sympathy can work better. One who feels that a supervisor does not care and is unsympathetic may feel resentful and angry, which may affect the quality of his or her work.

Dealing with disputes

Disputes often occur when people work together in groups and teams. The team leader or supervisor is expected to prevent or settle them.

Preventing disputes

The most typical cause of disputes in a work team is confusion, caused by people having different ideas about what should be done and how it should be done. Such disputes can be minimised or prevented by:

- ·Frequent meetings of health team members
- ·Allowing people to express views openly and letting the whole group decide what should be done.
- ·Sharing agreed objectives.
- ·Having clear and detailed job descriptions.
- ·Having clear instructions and procedures to follow.
- ·Distributing tasks fairly.
- ·Creating work schedules that distribute work fairly.
- A common cause of argument is jealousy and favouritism. A supervisor must behave towards each team member with complete fairness and justice and never publicly criticise staff.

Settling disputes

A serious argument can affect the morale and performance of the whole health team. Arguments must be stopped quickly, and the people concerned must be reconciled. What should the supervisor do in such cases?

First, all the people involved in the argument should be separately interviewed so that all the relevant facts are known. Second, every effort should be made to discover the actual cause of the argument. There is much truth in the saying, "It takes two to quarrel"; while more blame may attach to one side than to the other, there are likely to be faults on both sides. Third, the people involved should be asked - again separately - for their views on how to resolve the argument and whether they are willing to be reconciled. If a solution is acceptable to all concerned, productive and friendly working relations can be re-established. If agreement cannot be reached, the best solution is to persuade those involved to 'agree to differ' - to acknowledge the differences in their views but to stop arguing about them.

Summary;

- The supervisor tries to understand the personal problems of health workers
- The supervisor tries to prevent quarrelling and lousy feelings among health team members.
- The supervisor tries to reconcile people who quarrel, to bring them together.
- Clear instructions, job descriptions, and sympathetic supervision help prevent disputes.

V. Healthcare Facility Management Tool

1) Quarterly Business Plan

This is a quarterly document that describes all planned activities, needs, and requirements and estimates costs across all key priority areas for the quarter in view. It is prepared by the facility and WDC with support from the LGHA. When finalized, it is signed by the OIC and WDC chairman and submitted to the LGHA for approval.

VI. Community Engagement in Governance of PHCs

Community Engagement (CE) is the process of working collaboratively with and through groups of people affiliated by geographic proximity, particular interest, or similar situations to address issues affecting the well-being of those people. It is a powerful vehicle for bringing about environmental and behavioural changes that will improve the community's and its members' health. It often involves partnerships and coalitions that help mobilize resources and influence systems, change relationships among partners, and serve as catalysts for changing policies, programs, and practices. Therefore, community engagement in the governance of PHCs seeks to work collaboratively with all the established groups in the catchment area, e.g. WDCs/VDCs. So, community engagement is neither synonymous nor interchangeable with community mobilization, community participation or community health programmes (CHPs).

The Core Principles of Community Engagement are as follows;

Careful Planning and Preparation: Through adequate and inclusive planning, ensure that the design, organization, and convening of the process serve both a clearly defined purpose and the needs of the participants.

Inclusion and Demographic Diversity: Equitably incorporate diverse people, voices, ideas and information to lay the groundwork for quality outcomes and democratic legitimacy.

Collaboration and Shared Purpose: Support and encourage participants, government and community institutions, and others to work together to advance the common good.

Openness and Learning: Help all involved listen to each other, explore new ideas unconstrained by predetermined outcomes, learn and apply information to generate new options, and rigorously evaluate community engagement activities for effectiveness.

Transparency and \Trust: Be transparent and open about the process, and provide a public record of the organizers, sponsors, outcomes, and range of views and ideas expressed.

Impact and Action: Ensure each participatory effort has real potential to make a difference and that participants are aware of that potential.

Sustained Engagement and Participatory Culture: Promote a culture of participation with programs and institutions that support ongoing quality community engagement.

Activities that align with the community engagement approaches include:

1. Provision of services, counselling and behavioural change communication through home visits by Community Health Influencers, Promoters and Services (CHIPS) agents and other mobilizers
2. Promoting active male involvement through various interpersonal communication strategies – community sensitization workshops and discussions.
3. Town announcements by town announcers
4. Mosque and church announcements.
5. Compound meetings with heads of households, women, and other key stakeholders.
6. Edutainment activities – Rallies and road shows, among others.
7. Community dialogue is a process that encourages communities to plan and implement their solutions to health issues.

Multiple channels should be used to complement community efforts (social and traditional media, production, and distribution of IEG materials).

VII. Ward Health System:

The Ward Health System has embedded community engagement principles within it. It aims to improve and ensure sustainable health services with full and active participation and involvement of people at the grassroots level. The system has a governance mechanism through which communities are constantly engaged and involved in governance; hence, it has the Ward Development Committee (WDC) at the ward level, the Village Development Committee (VDC) at the village level, and the Facility Management Committee (FMC) for every primary health care facility.

Ward Development Committee

The ward has the lowest level of healthcare and governance, and to be effective, the ward development committee is supposed to be constituted to assist the PHCs in demand creation and health education and provide an enabling environment for the PHC. The WDC serves as the apex body for the functions of the PHC and other development activities carried out in the ward. It has a direct link with the LGHA through a representative of the WDC serving on the LGHA Advisory Committee.

Roles and Responsibilities of the Ward Development Committee

1. Identify the health and development needs of the ward and plan for them.
2. Develop and forward the Community Action Plan (for the ward) to the LG Health

Authority.

3. Identify local human and material resources to meet these needs.
4. Ensure implementation of the Community Action Plan
5. Ensure ownership and sustainability of programs and projects.
6. Mobilize and stimulate active involvement of prominent and other local people in project planning, implementation, monitoring, and evaluation.
7. Take an active role in supervising and monitoring the Drug Revolving Fund (DRF) and the proper use of free/donated health commodities in the ward.
8. Funds should be raised for health and development programs at facility, village, and ward levels when necessary.
9. Ensure accountability by providing regular feedback to the ward on how funds raised are utilized.
10. Ensure that women, children and the poorest can access the health services they need.
11. Liaise with local government and other voluntary agencies to find solutions to health and development challenges in the ward.
12. Supervise the activities of the VDCs/FMCs/CHEWs/JCHEWs/CHIPS Agents/CEFPs
13. Oversee the functioning of all health facilities in the ward.
14. Provide necessary support to CHIPS Agents and CEFPs.
15. A dedicated bank account for the selected Primary Health Centre in the ward with a reliable commercial bank for the BHCPF funds. The signatories shall be the WDC Chairman and OIC Primary Health Centre
16. Maintain a separate WDC bank account for WDC fundraising activities. The signatories shall be the Chairman, Secretary, and Treasurer of the WDC
17. Monitor essential medicines and consumables, essential equipment and infrastructure at monthly intervals.
18. Ensure the proper functioning of health facilities in the ward using a maintenance plan.
19. Monitor the timely submission of monthly service data at community and facility levels.

Village Development Committee

The ward should be divided into a minimum of 2 and a maximum of 6 villages or neighbourhood areas (for urban LGAs) based on either traditional authority or accessibility.

Roles and Responsibilities of the VDC

The committee shall:

- Identify health and health-related needs in the village.
- Plan for the health and welfare of the community.
- Identify available resources (human and material) within the community and allocate as appropriate to PHC programmes.
- Supervise the implementation of the PHC work plan.
- Monitor and evaluate the progress and impacts of the implementation of health activities.
- Mobilize and stimulate active community involvement in the implementation of developed health plans.
- Determine exemptions for drug payment and deferment, but provide funds for the exemptions/deferments.
- VDC collaborates with health workers to determine the prices of essential drugs.
- Supervise all account books (Monies at hand should be deposited in a bank within 24 hours or 72 hours

at weekends)

- Supervise and monitor the quantity of drugs in the supply.
- Select appropriate persons within the community to be trained as Village Health Workers (VHWs/TBA) for PHC, AIDS/STD and other programmes.
- Supervise the activities of the Village Health Workers and Traditional Birth Attendants, including reviewing the monthly work record.
- Remunerate in cash or kind, the Village Health Worker for his/her work in the community.
- Agree with the Village Health Worker on the number of hours he/she should work daily.
- Establish a village health post, where there is none already.
- Ensure that VHW/TBA Kits are stocked to top-up level for drugs.
- Liaise with other officials living in the village to provide health care and other development activities.
- Provide necessary support to VHW for the provision of health care services.
- Forward local community health plan to ward level.
- Provide feedback to their community on their activities regularly (at least quarterly)

Facility Management Committee

A Facility Management Committee (FMC) represents another layer for promoting community accountability and improving community confidence in and use of health services. An FMC should be established for every Primary Health Care facility.

Roles and Responsibilities

1. Work with other health facility staff to promote improvements in health services and client satisfaction.
2. Ensure maintenance of the health facility infrastructure, vehicle and any other equipment.
3. Designate the Officer-in-Charge of the facility to follow up on the Planned Preventive Maintenance requirements for the facility.
4. Identify resources from the government and other stakeholders to improve services within the facility.
5. Ensure CHIPS agents/ Community Engagement Persons submit/transmit data of their work and validation is done by the supervising CHEW.
6. Monitor the activities of the CHIPS personnel (CHIPS agents and CEFPs) in the ward.
7. Ensure the effective running of the DRF.
8. Provide monthly reports of the committee's activities to the Ward Development Committee or Village Development Committee as applicable.

8.1.3 MANAGING HUMAN RESOURCES FOR HEALTH (HRH) IN THE PRIMARY HEALTH CENTRES (PHCs)

Human resources are the essential ingredient for all care delivery. The PHCs need adequately trained and motivated staff to provide quality services. Managing human resources is a complex task that requires national-level policy and planning for long-term sustainable impact. The following are some steps that the staff of a primary health centre can take to help:

1. Ensure an adequate number of staff.
2. Make task-shifting and task sharing effective.
3. Make sure staff has appropriate training.
4. Ensure supportive supervision and mentorship.
5. Improve staff motivation and retention.
6. Establish a safe workplace.

S/No.	Type of HRH Requirement	Community	Health Outreach	PHCs Level I	PHCs Level II	PHCs Level III
1.	Medical Officer II (if available)				1	1
2.	Community Health Officer					1
3.	Nurse/Midwife				1	Minimum 2
4.	CHEW			1	2	Minimum 2
5.	Pharmacy Technician				1	2
6.	JCHEW		2	2	2	Minimum 4
7.	Environmental Health Officer					1
8.	Health Record Technician				1	2
9.	Medical Lab Technician				1	2
Support Staff						
1.	Account Clerk				1	1
2.	Driver					1
3.	Health Attendant/Assistant			2	2	2
4.	Security Personnel			2	2	3
5.	Cleaner		1	1	2	3
6.	CHIPS	10				
7.	CEFPs	2				

Human Resources for Health Requirement for PHCs

The HRH requirements for PHC would vary depending on the type of PHC but the HRH requirements according to the National Standards and Regulatory Framework for PHCs in Nigeria are shown in the Table below:

Where the OIC has less than these minimum number of staff in the facility, continuous advocacy by the WDC is recommended for government (LGHA/SPHCB) to recruit staff to make up for the number. However, facilities that are benefiting from the BHCPF are allowed to hire additional staff after they must have hired 2 midwives. Generally, recruitment and hiring are carried out at state and LGA levels, not the health centre.

However, there are steps the officer-in-charge (OIC) can take to improve the chances of receiving the number of staff needed at the health centre. These include:

1. Contact your LGHA office to learn the positions (number and cadre) assigned to your health centre

- How does the number and cadre of positions actually assigned compare with recommended staffing (see below)?
- Are any assigned positions unfilled at your health centre? If so, how many?
- Which positions are vacant, and for how long have they been vacant?

2. Locate job descriptions for each position assigned to your health centre

Job descriptions will help you determine the qualifications and positions for staff you wish to recruit to your health centre. These job descriptions are usually standardized across health facilities, and should be on fi le at your health centre or at your district health office. Once staff is hired, job descriptions can be used to help assess employee performance.

3. Learn how the local hiring process works

Hiring is usually carried out by the State and the LGHA, but you can increase your odds of obtaining the staff you need by: being informed about the hiring procedures that apply to your health centre, advocating for your team, and pursuing alternative hiring procedures when needed particularly as it relates to engaging the volunteers. For example, you may need to send an official request to the State PHCDA or LGHA create a position or fill the vacancy.

4. Communicate regularly with the local recruiting authorities

Keep in contact with people in the personnel office at your LGHA so that you know of upcoming changes that could affect staffing of positions at your health centre. Keep these people informed of your staffing needs, using the information you gathered, and citing the vacancies that exist and the recommended staffing tables above to justify your need for additional staff.

5. Be persistent!

If budget ceilings or other limitations prevent you from hiring staff:

- Ask the WDC to hire qualified temporary staff from their internally generated income/contributions.
- Try alternative procedures. Contact local NGOs and donors to ask if they can hire and pay the salaries for new staff at your health centre.
- Recruit volunteers. Make sure you develop good relations with local communities and community groups so you can recruit volunteers in times of high workload (see below).

Roles and Responsibilities of Officer-in-Charge in the PHCs:

The roles and responsibilities officer-in-Charge are as stated below:

a. Provide services beyond the level of expertise of other staff according to guidelines (Standing Orders).
b. Encourage the practice of rational drug use.
c. Ensure management of clients according to Standing Orders or national guidelines.
d. Establish and implement a two-way referral for clients, including appropriate. links between facility staff and the CHIPS Agents.
e. Ensure adherence to standards of care for provision of high quality of care.
f. Responsible for day-to-day management of the facility.
g. Assign responsibilities to other health workers.
h. Supervise functions assigned to other health workers.
i. Maintain discipline in the facility.
j. Oversee the upkeep of the facility and its surroundings.
k. Develop ward action plans and draw up schedule of activities with the WDC.
l. Supervise data collection, collation, analysis and transmission especially the NHMIS.
m. Organize regular staff meeting and feedback on on-going activities.
n. Organize continuous education for the health team and the community.
o. Conduct community diagnosis exercises.
p. Plan for health interventions with other health workers and the community.
q. Ensure community participation in health and health-related programmes.
r. Ensuring adequate record keeping, ordering and accountability of health equipment, supplies and medicines.
s. Ensure that Facility Management Committee (FMC), VDC and WDC meet regularly.
t. Ensure conduct of verbal autopsy for maternal and perinatal deaths (CMPDSR).

Role and Responsibilities of Other Staff of the PHC

The other staff at the PHC includes the Community Health Officers, Nurses/Midwives, CHEW Health Education Officer and Administrative Assistant. Their roles and responsibilities are discussed in the following sections.

Medical Doctor (if available)

A medical doctor where available will be the head of the facility and the health team in the ward.

- Plan and provide oversight for implementation of activities at the facility and community levels in the ward.
- Attend to all referrals from the ward.
- Provide technical oversight for service provision and data management.
- Ensure that essential medicines, vaccines, consumables, and essential equipment are always available in the facility.
- Conduct supervisory visits to other health facilities in the ward.
- Provide on the job training and mentoring for the health workers.
- Establish referral linkages with other facilities.
- Hold regular meetings with staff on the smooth running of the facility.

Community Health Officer (CHO)

- Spend 20% of working time in the community.
- Ensure provision of integrated PHC services in the health facility
- Ensuring quality and continuity of care.
- Initiate, direct, and work with community and other staff to find solution to identified health problems.
- Carry out with other senior health professionals (where available) day to day health administration services in the target population.
- Organize and participate in health education for patients and their families.
- Carry out community mobilization for health action including coordinating the community-based efforts of CHEWs.
- Participate in house mapping, numbering, placement of home-based card and facility master register.
- Initiate community diagnosis and continuous health needs assessment.
- that supportive supervision is given to health workers at community and facility levels using standard guidelines and protocols.
- Carrying and supervising administrative activities and procedures (client record, consumption statistics, registers etc.)
- Participate in the monitoring and evaluation of health activities.
- Identify training needs, organize, direct and conduct training and continuing education programs including on the job training for members of health team including community-based workers.

Nurse/Midwife

- Provide maternal and child health services including antenatal care and conducting normal deliveries.
- Provide Basic Emergency Obstetric and Newborn Care BEmONC
- Assess all pregnancy/labour cases with a view to referring complicated cases.
- Provide family planning services.
- Provide adolescent reproductive health.
- Screening and management of NCDs.
- nutrition services.
- Provide immunization services.
- Diagnose and treat malaria, tuberculosis, and HIV/AIDS.
- Provide basic oral, eye and mental health services.
- Manage minor/common ailments using the National Standard Treatment Guidelines.
- Provide maternal and child health services including antenatal care and deliveries.
- Manage NTDs.
- Provide care for the elderly.
- Ensure quality of care

- Participate in outreaches and conduct follow up home visits.
- Perform administrative functions where applicable.

Community Health Extension Worker (CHEW)

- Spend 50% of working time in the community.
- Manage clients according to Standing Orders
- Ensure all pregnant women are identified and have access to antenatal care service.
- Maintain high quality of care.
- Establish and implement a two-way referral for client.
- Perform administrative functions including attending WDC meetings.
- Visit communities regularly to assess community-based services.
- Ensure community participation in health and health-related activities.
- Provide community-based health care services.
- Mentor and provide integrated supportive supervision to CHIPS agents.
- Develop work schedule and monthly work plan with the approval of the supervising officer.
- Carry out community mobilization for health action.
- Supervise and participate in community house numbering and placement of home base records.
- Train, support and supervise community-based health workers and JCHEWs.
- Work with other health workers and community in identification of health problems, prioritization and planning solution to the problems.
- Prepare or assist the supervising officer to develop budget for implementation of the plan.
- Manage and monitor utilization of resources in health facility (drugs, equipment, and other commodities)
- Organize and run integrated PHC services in the health facilities (where CHOs are not available) based on their skills and competences.
- Keep adequate records, analyse and transmit data and compile monthly reports.

Junior Community Health Extension Worker (JCHEW)

- Spend 80% of their working time in the community.
- Work with other health workers and the community to carry out community diagnosis and health needs assessment in the community and plan solutions to the identified health needs.
- Mentor and provide integrated supportive supervision to CHIPS Agents
- Identify learning needs of community-based health workers and participate in their training.
- Collect and collate health data for monitoring and evaluating PHC activities and the community.
- Ensure community participation in health and health-related activities.
- Carry out basic clinical procedures excluding patient management in health facilities and assist in maintenance of facility records.
- Conduct home visits and provide basic preventive and curative care and essential community health services with guidance from the Standing Orders
- Promote two-way referral

Pharmacist/Pharmacy Technician

- A Pharmacist or Pharmacy technician where available will oversee both the pharmacy and store.
- Receive written prescription or refill requests and verify that information is complete and accurate.
- Maintain proper storage and security conditions for medicines.
- Ensure rational dispensing of medicines.
- Provide medication counselling when necessary or requested by clients.
- Receive and store incoming supplies, verify quantities against invoices and inform head of facility of stock needs and shortages.
- Monitor medicines use and requisition appropriately to ensure regular availability.
- Maintain adequate record including bin card, requisition forms etc.
- Keep proper records of purchase using ledgers.
- Support in the implementation of Drug Revolving Fund (DRF) policy

Medical Laboratory Scientist/Technician

- A Medical Laboratory Scientist or Technician where available will oversee the laboratory.
- Perform technical procedures according to protocols.
- Maintain quality results by adhering to standards and controls and utilizing approved testing procedures.
- Ensure a safe and secure environment for clients and co-workers by following established standards and procedures and complying with legal regulations.
- Maintain patient confidence by keeping laboratory information confidential.
- Conduct Planned Preventive Maintenance of laboratory equipment.

Health Information Officer

- Health Information Officer where available will oversee the medical records unit.
- Keep records of patients
- Fill completely and accurately all forms, cards and registers used in client management.
- Ensure that information contained in the facility record is complete, accurate and only available to authorized personnel.
- Enter data into DHIS2 platform.
- Collate, validate and transmit community- based data.
- Conduct basic analysis of health facility and community-based data to inform local action.

Health Education Officer

Coordinates the provision of health information to clients through different medium in different forum and facilitates health promotion and health-seeking behaviors of staff.

Duties performed:

- Plans and directs statewide program of public health education and promotes establishment of local health services:
- Directs workers engaged in preparation and distribution of health information materials,

such as brochures, films, weight charts, and first-aid kits.

- Promotes establishment or expansion of local health services and provides technical assistance to individuals and groups conducting [health conferences, workshops, and training courses.
- Answers health information requests received by department or reviews correspondence prepared by others.
- Coordinates special health education campaigns during epidemics, rabies outbreaks, instances of food poisoning, and similar emergencies.
- May direct health education activities in public schools.

Community Health Influencers, Promoters and Services (CHIPS) Agents

- Conduct house mapping, numbering, household listing and completion of household register.
- Conduct home visits.
- Line list pregnant women and infants
- Counsel and health educate household members.
- Identify and refer pregnant women to PHC facilities.
- Identify and refer children to PHC facilities for immunization and track defaulters.
- Identify and refer children with severe acute malnutrition.
- Provide basic treatment of sick children without danger signs.
- Provide first aid to household members.
- Counsel clients on family planning, Water, Sanitation and Hygiene (WASH)
- Distribute basic commodities (LLIN, misoprostol, chlorhexidine, Mama kits)
- Conduct community surveillance and report disease outbreaks in the communities.
- Document and transmit data of the work done.
- Document and report maternal and child mortalities as well as participate in the conduct (CMPDSR)
- Encourage birth registration.

Community Engagement Focal Person (CEFP)

- Conduct community mobilization.
- Promote male participation.
- Engage with men to address barriers to service uptake including immunization.
- Support CHIPS Agent in house mapping, numbering, household listing and filing household registers.
- Verify, validate, and transmit data collected by the CHIPS agents.
- Facilitate monthly immunization data reconciliation between village heads, health facility and community leader.
- Track immunization defaulters
- Participate in review meetings

Ward Focal Person

- Provide oversight for service delivery in the ward.
- Participate in development of work plans and distribution of resources for outreaches and campaigns.
- Identify and prioritize settlements into low, medium and high risk.
- Participate in Integrated Supportive Supervision activities.
- Participate in community engagement meetings.
- Collate and analyse ward level data.
- Participate in WDC meetings.
- Provide feedback to the Local Government health authority.
- Participates in CMPDSR activities.

Administrative Assistant

- Duties perform:
- Drafting of memos, official letters, circulars, minutes of meetings etc.
- Organizing and planning the activities of the department.
- Conduct inductions for new staff.
- Issuance of staff identity card.
- Filling and documentation of all unit's information and materials

Developing Duty Rosters

A duty roster is a roster that schedules staff based on tasks and duties. This type of roster ensures that you have particular duties covered. It is unlike regular rosters that scheduling is based on availability and hours. Please see annex I for sample Duty Roster template. Duty rosters are important because they help schedule staff based on your business's demand instead of your team's availability.

Principles of developing duty rosters

The principles are as follows:

a) Delivery of service to patients is always the first consideration when developing a roster.
b) Rosters must ensure that there is sufficient and appropriately skilled staff rostered to work in order to provide appropriate patient care and to meet anticipated service demands.
c) Rosters must conform to relevant regulatory frameworks, including antidiscrimination, work health and safety legislation among others
d) Rosters should ensure staff is treated fairly, while still providing appropriate flexibility to facilitate meeting unit staffing needs.
e) Rosters must make appropriate provision for adequate staff supervision, training and clinical handover
f) Developing rosters is based on co-operation between the managers (OIC in this case) and staff, in order to promote fairness and to deliver appropriate care to patients.

Steps in developing Duty Rosters

Step I: Start by listing all the tasks and services provided by the health facility. This will vary depending on whether it is type I, II or III PHC.

Step II: Track when these tasks occur. Think about the tasks listed you created and when the tasks/services are most likely to be performed/demanded. It could be during morning, afternoon and evening of a day or what days of the week or month.

Step III: Schedule based on skill. After determining the tasks and the required period of coverage (hours/days), you must schedule staff for each duty based on their skill and qualifications.
A suggested template for developing duty roster is found in Annex .

Task Shifting and Task Sharing

Task shifting is "the rational redistribution of tasks among health workforce teams. Specific tasks are moved, where appropriate, from highly qualified health workers to health workers with shorter training and fewer qualifications in order to make more efficient use of the available human resources for health."

Task sharing is allowing a wider range of cadres to offer certain services, when this can be done safely and effectively as a means of rapidly expanding access and improving health care. The goal of task shifting or sharing is simply "to get the right workers with the right skills in the right places doing the right things"4 without necessarily abandoning other methods of increasing the number of qualified health workers It is the policy of training specific health cadre on tasks that are ordinarily meant for other health cadres (Particularly those above them) for them to perform such tasks when those that are designated for such tasks are not available. For instance, Voluntary Health Workers can be trained on counseling to enable them counsel pregnant women in the absence of a substantive counselor.

Please find in Annex II a table indicating tasks that may be performed by different cadre of staff.
Below are steps on how to make "task-shifting" effective:

1. Establish a clinical "team-based approach" through regular clinical team meetings and good communications between staff. Conduct a weekly meeting of all staff at which you can openly discuss patient cases and issues that arise, and work together to solve problems. Encourage regular dialogue between staff about how to improve tasks to increase service efficiency and quality.
2. Establish regular performance measurements to assure adherence to clinical and other standards (see Quality Improvement chapter) Implement strategies to motivate your staff and to prevent 'burnout'. When staff is required to take on new tasks in an already heavy workload, they can suffer increased anxiety, stress and burnout. Work together as a team to determine how you can keep each other motivated. Task shifting can be a real asset to your health centre, but it takes teamwork, supervision and constant communication!

Using Job Descriptions
The distribution of tasks among the members of a health team is one of a manager's most important functions. When work is distributed unfairly it causes dissatisfaction and sometimes quarrelling. Health tasks are very varied. A nurse, for example, may have to take temperatures, deliver babies, conduct clinics and write reports. This variation in tasks makes organization in health work a challenge to the team leader. Job descriptions are one means of helping to distribute tasks among the health team. The purpose of job descriptions is to define exactly for the holders of different posts, their fellow workers and their supervisors:

- What the holders of the posts are expected to do.

- What standards they are expected to reach.
- To whom they are responsible.
- Whose work they supervise.

Uses of Detailed Job Descriptions

Job descriptions are a valuable tool for the organization of work. A job description:

a) States clearly what each health worker must do and is expected to achieve.

b) Help prevent arguments between people about who should do what. They help also in the distribution of the equipment needed to do the work.

c) Help prevent gaps and overlaps. It often happens that certain tasks are not done because nobody accepts responsibility for them. These gaps can be prevented by a clear job description for each team member. Without specific job descriptions, two people may think each is responsible for the same job. Such confusion should be avoided.

d) Can show the need for training, for instance if it includes a duty for which the health worker has not been trained nor needs further training.

e) Are useful as a basis for evaluating team members' performance. However, they should be interpreted flexibly, as guidelines, rather than too strictly or literally. Thus, team members should be ready to help with one another's work when necessary.

Contents of a Job Description

A job description should be written under specific headings to make sure that it gives all the necessary information.

Job title: This is the standard title for the person doing the work or job, e.g. Nurse/midwife, Grade 2.

Date: The date is included because a job description is not final. People and roles change, and job descriptions should be reviewed and, if necessary, revised at least once a year.

Job summary: This is a brief summary of the main responsibilities of the job.

Duties: This is the central and most important part of the job description. Each duty should be an identifiable entity, are cognizable part of the jobholder's work. Each duty should correspond to one or more programme objectives, which should be listed. The health worker can then see how his or her duties contribute to the improvement of the health of the community.

Relations: These are simple statements concerning:

- The title of the person to whom the jobholder is accountable. For a particular task one person can be responsible to only one superior. However, one person can hold more than one job, and for different jobs may be responsible to different superiors.
- The titles of people supervised by the jobholder. For example, a nurse/midwife may supervise several community nurses in different villages. She is responsible for seeing that they perform their work properly.

Qualifications: A section on qualifications describes the basic training and level of experience required for the job.

Training and Development: Every job description should be accompanied by a programme for the further training and development of the person holding the job. This may be, for example, a regular programme of reading or of in-service training, and opportunities to attend professional meetings. Like every other element of the job description, this should be discussed in full and worked out with the jobholder.

Review and appraisal: This is a statement describing the process for review and appraisal of the performance of the jobholder. Normally the supervisor is expected to carry out such a review periodically. The review and appraisal statement should state clearly who has this responsibility. The appraisal may take the form of an annual confidential report written by the supervisor, a simple statement that work is satisfactory, or a recommendation for a change in the duties, or the promotion, of the job holder.

How to Outline Duties

The duties of a job are usually determined by:

Tradition and training. These define how similar jobs have been done previously or are done elsewhere. For example, midwives are expected to perform certain duties according to the way they are trained and the way other midwives practise.

The interpretation of the person who holds the job. Experience often leads people to change their way of work or to adapt to new circumstances. For example, a midwife in a village must work in a different way from a midwife in a large hospital.

Because rural health work and community work are developing and changing, supervisors and health workers should keep job descriptions under review, so that they remain useful. One way to do this is to get health workers themselves to define their duties.

A supervisor will find it very useful to ask health workers to list their duties and then to discuss these duties with them; it may be possible to rearrange or reallocate certain duties so that they are performed more efficiently.

The requirements of programmes and services. The objectives and strategies of the services and programmes in which the health workers are employed are the determining factors in the definition of duties and tasks.

Ensuring staff have appropriate training

As the in-charge you play an important role in planning and tracking the training your staff receive. You should ensure that health centre staff has the right training at the right time to provide the quality basic services outlined in this manual. You also should ensure that training opportunities are provided fairly and do not interfere with service delivery. By helping your staff gain access to training opportunities in an equitable way, you also help promote their career development and improve motivation and morale. The following steps are recommended:

Steps needed to manage training at health centre:

1. Identify training needs: Review your training logs to identify the required and optional training courses that each staff member currently needs.
2. Identify available training: Contact the LGHA office, State hospitals and donors in your area to identify available training that matches your needs. Does the training take place on-the-job or offsite? Is any follow-up support provided? Is training accredited? Determine how training will be paid for, including fees, travel, and meals.

3. Determine how you will maintain service delivery while your staff is away at training.
4. When training is completed, keep a record of it in a training log. This log can be completed by a clerk, but as supervisor you should ensure that this recordkeeping happens on a regular basis.
5. Training debriefing: Ask staff who received the training to share key lessons learned and training 'likes and dislikes' with other health centre team members. This can take place during a team lunch or meeting.
6. Do not be afraid to ask. If the LGA or donors do not provide the training your staff needs, tell them what you need. They may be able to help - lend you staff from the LGA to fill in for your staff while they are on training.

Improving staff motivation and retention

Employee satisfaction is directly linked to employee motivation, performance and quality of care. If you do not pay attention to employee satisfaction, your staff will be disgruntled and perform poorly. Unhappy employees can lead to unhappy and dissatisfied patients. The OIC needs to consider the following:

1. Provide opportunities for staff to grow and develop (participating in training, mentorship, new tasks, and chances to be promoted)
2. Treat staff fairly (fairness in how praise or criticism is delivered, and how training, salaries, promotions, and other opportunities are provided)
3. Ensure staff has clear job descriptions and appropriate tools to complete the tasks.
4. Provide regular feedback to staff on his/her performance.
5. Share with staff how their work contributes to the team and to the health of patients.

Staff Handing Over Notes

Staff movement (redeployment, retirement, transfers etc) are realities of the workplace but which may result in disruption of workflow if not properly handled. The officer-in-charge should ensure that proper handing over notes is prepared by the departing or out-going staff for the in-coming. The comprehensiveness of such notes and the diligence in which it is prepared will greatly assist the new staff in settling down to his/her duties.

Follow the following steps:

Step 1. Discuss with the handover with the employee
Step 2. Draw up an early outline of your handover document (use a standard template see annex 3)
Step 3. Staff tie up loose ends (clear all outstanding issues)
Step 4. Communicate with your colleagues (for any areas to highlight)
Step 5. Complete a formal handover document with sample content below:
Step 6. Allow as much time as possible for a full handover.
Step 7. Hand over key documents personally
Step 8. Make the effort to identify and help resolve any occurring issues.
Step 10. Have a final handover meeting.

The content of the hand over document will include:

1.A list and timetable of actions.
2.A briefing of ongoing issues.
3.A calendar of forthcoming events and deadlines.
4.Passwords and login information.
5.A list of useful contacts.
A guide to navigating files and folders on the computer system.

8.2 FINANCIAL MANAGEMENT IN PHC

8.2.1 Overview (Reasons, Principles, Pillars, Tools)

For PHCs financial management entails planning, organizing, controlling, and monitoring the financial resources of the health Centre to improve service delivery and accountability. The OIC is ultimately responsible for managing the finances of the PHC, however, every staff member and the WDC are also accountable and should help in ensuring the effective controls are in place.

At the heart of financial management is the concept of financial control. This describes a situation where the financial resources of the health facility are being correctly and effectively used. Financial management will only happen if strong and relevant financial policies and procedures are put in place. Poor financial control in the health facility means that:

- Assets will be put at risk of theft, fraud, or abuse.
- Funds may not be spent in accordance with the NGO's objectives or donors' wishes and
- The competence of managers may even be called into question.

Ideally, an Accounts Officer or Clerk is responsible for the accounting function of healthcare facilities, which includes recording of all financial transactions related to assets, liabilities, income, and expenses of the PHC and maintaining the books of accounts in a complete and accurate manner. At a minimum, education up to ordinary diploma (ND) level is required to take on this position. An accounting clerk job usually requires a few years of experience in accounting duties, and a firm understanding of numbers and ledgers and how they relate to invoices and receipts. However, if the PHC does not have an Accounts Officer or Clerk, they may want to request help from the LGHA accounts clerk, and it that is not possible, the OIC must take responsibility.

The financial management process is continuous process of: Plan, Do and Review.

Decentralised Facility Financing (DFF)
As part of efforts to address the limited funding of PHCs, the Basic Health Provision Fund (BHCPF) was established in the 2014 National Health Act. A percentage of the resource under BHCPF was earmarked to improve PHC service delivery. NPHCDA adopted the Decentralised Facility Financing (DFF) an innovative and tested approach for improving quality of primary health delivery in Nigeria. DFF entails direct transfer of funds to facilities and giving PHC facilities some level of autonomy in the management and use of resources for improved service delivery.

Three key principles guide the implementation of DFF:
Facility autonomy: The principle of facility autonomy ensures a health facility has the status to receive, manage and account for funds from any legal source or funds flow. Increasing facility autonomy requires strengthening

governance including realigning institutional roles and relationships across all levels of government. Facility autonomy can help to establish or invigorate the governance structure of facilities through the separation of financial management responsibilities such as authorization and payments. Increased community participation in planning can create new civic space to enhance transparency, accountability and engagement.

Purchase health services using output-based payment. The output-based payment principle enables health purchasers to "buy the right thing" and better match payment to prioritized services. Output-based payment permits health facilities to determine and procure the best mix of inputs to produce and deliver service outputs to clients and communities.

Sound facility financial management. The autonomy to directly receive funds paid for outputs and the flexibility to manage procurement of inputs to produce and deliver desired services must be underpinned by sound health facility financial management and accountability. Establishment of good financial management systems and processes at provider level enables facilities to diligently account for the inputs procured, report on the use of funds and provide an audit trail to facilitate financial accountability. Health facilities can function as businesses or management entities when they take advantage of these robust financial management systems and processes.

The autonomy principle delegates to health facilities the right to procure inputs to produce and deliver outputs, the output-based payment (financing) principle determines what facilities can procure, and the good financial management principle sets the guidelines for how facilities procure with associated internal controls. The Immunization Plus & Malaria Progress by Accelerating Coverage and Transforming Services (IMPACT) project was conceptualized, with funding from a World Bank IDA credit facility, to support the Government of Nigeria's commitment to improving its human capital with the goal of reducing under-five mortality by 40 percent over the next decade. More importantly, both interventions have within their mandate a decentralized facility financing system for PHC operations through direct funds transfer to each health facility.

Performance and Financial Management Officers (PFMOs)

Therefore, to help optimize PHC performance and improve financial accountability and transparency, the Performance and Financial Management Officers (PFMOs) are engaged to serve as performance improvement TAs and fiduciary agents responsible for overseeing, co-managing, monitoring and evaluating PHC performance against all financial inflows to primary health care centers nationwide, prioritizing IMPACT and BHCPF facilities which are currently receiving DFF. The Performance and Financial Management Officers engaged and domiciled within the PHC Department, LGA Headquarters, will provide TA to PHCs and LGAs on the use, documentation and reporting of PHC funds. It is expected that they will support PHC facilities and LGA in BHCPF and IMPACT States; working with State PIUs and States Projects Financial Monitoring Unit (SPFMU) in conducting sound financial management processes in all the PHC facilities that are beneficiaries of BHCPF, IMPACT funds and other resources, in every LGA.

The PFMOs will promote a collaborative, client-oriented approach consistent with the public financial rules and regulations and shall focus on the following:

- Evidence-based performance management of PHC programmes
- Support the LGA Finance Officers for the management of financial documentation of PHC facilities (i.e., retirement documentations)
- Financial knowledge management and capacity building of PHC financial focal persons and LGAs finance officers

- Providing financial advice and guidance to PHC officers-in-charge within their LGA of assignment
- Financial management, financial reporting and controls

The PFMO directly impacts the effective management and financial integrity of primary health care centers through the effective stewarding of government and partner resources including the World Bank IMPACT funds, maintaining appropriate controls and integrity of financial management in accordance with relevant rules and regulations, enabling prudent management financial resources and building an informed and shared management engagement on financial issues in support of the PHC revitalization strategy. Also, the PFMOs will be responsible for supporting programmatic performance improvement across the selected facilities. Specific programmatic and performance indicators will be included in the TORs and KPIs of the PFMOs.

FMOs Scope of Work

Under the guidance and supervision of the SPHCB (IMPACT PIU and BHCPF Core team) and NPHCDA, the PFMO's duties include: Technical support to PHC facility finance officers and ensure:

- Oversight of the PHC facility finance officers, establishing strategic team objectives, aligning with BHCPF key performance indicators, measurement standards, and expected results, offering guidance to enable responsible, effective and efficient performance.

- Effective planning and allocation of resources in line with the facility business plan and ensure the cost effective and appropriate use of resources in accordance with financial standards of ethics, integrity and accountability framework.

- Continuous assessment of the performance and effectiveness of financial services and recommending changes and/or corrective actions in the finance operations to enhance and maintain effectiveness in changing environment.

- Fostering a positive work environment, respectful of both men and women, and ensure that the highest standards of conduct are observed.

Financial Knowledge Management and Capacity Building

- Build the capacity of the primary health care facilities' Finance Officers/Officers-in-Charge on financial management in line with the health facility Financial Management guidelines.
- Lead, organize and conduct of trainings or workshops to build capacity of stakeholders on financial accounting, including the identification of expert resources for the trainings/ workshops, synthesis and dissemination of lessons learned and best practices in Finance, and contribute to knowledge networks and communities of practice.
- Coach and mentor, the health facilities' Finance Officers and Officers in Charge on proper financial management and supervise related financial activities.
- Put in place a need-based capacity-building plan for financial management for staff of the health facilities including support with development of quarterly business plans and use of financial management tools deployed at the facility.
- Provide hands-on support to the health facility management and finance officers to function as required.

Providing financial advice and guidance to PHC officers-in-charge within their LGA of assignment

- Financial advisory services to PHC facilities, and the LGA department of health to enable informed and shared management engagement on financial issues.
- Capacity building of LGA finance team.

Financial management, financial reporting and controls

- Track all funds (investment funds, DFF, capitation etc.) disbursed to the health facilities and all administrative procedures in line with BHCPF and World Bank requirements as well as routinely (monthly and quarterly) report on their utilization.
- Review and ensure that expenditures at the health facilities are in line with the approved plans.
- Ensure that Goods, Works, and Services have been procured in accordance with the relevant financial guidelines.
- Monitor and escalate any case(s) of misuse of funds through established channels to the SPHCB (IMPACT PIU and BHCPF Core team) and NPHCDA.
- Ensure the financial management processes of the health facilities adheres to the protocol and regulations and that the financial procedures approved for the health facilities were followed and used.
- Support the safety of the health facilities' information by ensuring electronic information is properly backed-up and transmitted to State and National as at when due.
- Support the establishment of Assets Registers and its effective utilization.
- Provide monthly activity reports. Based on agreed template and share with the state IMPACT and BHCPF core team leads while copying the National team.
- Undertake other duties as assigned from time to time by the State.

Terms of Engagement

This engagement at this stage is for a period of one (1) year.

Reasons for Good Financial Management

- To be accountable, funders with good financial reporting systems, it is easier to show fund providers that funds are being used for the purpose intended.
- To be accountable to the communities we work with We have a moral obligation to show that funds raised in the community's name are being used correctly.
- To be able to produce financial statements for funders, regulatory, supervisory bodies and other stakeholders As part of the transparency and accountability process, PHCs may be required to be accountable for the money they raise and spend.
- To minimise fraud, theft and abuse of resources Good financial management includes internal controls. When these are in place, they help to stop fraud and protect the staff as well as the assets.
- To plan for the future and become more financially secure We have to plan to make sure we have enough money to carry out our objectives now and in the future. Budgets help us plan for activities and manage cash. Financial information helps us to identify potential financial risks and the need for savings (reserves). We need financial information about where we are now and where we want to be in the future, to help identify our long-term financing needs.
- To enable health facilities to make better decisions on the use of funds Complete, up-to-date and timely reports enable managers to plan their activities according to the budget available and take decisions to fulfil

objectives. Good cash flow management enables activities to be planned, items purchased when needed.

- To achieve the objectives of the organisation Good financial management will give the management team the information they need to ensure they are fulfilling their objectives and following the budget.
- To enjoy confidence and trust from their stakeholders Health facilities that keep good accounts, create great budgets and produce accurate and timely financial reports, receive confidence and trust from their stakeholders.
- To strengthen fundraising efforts Health facilities that present good budgets and financial reports with funding proposals are more likely to receive a favourable response.
- To get value for our money Financial information allows us to compare and assess spending plans to make sure we make efficient and effective use of financial resources.

Principles of Financial Management

Use these guiding principles as standards to assess your current practice and goals to aim for.

Consistency: Consistent use of financial policies and procedures are important for efficient operations. This assists the financial reporting process and promotes transparency (one of the best ways to hide irregularities is to change the way figures are reported).

Accountability: This is the moral or legal duty, placed on health facilities to explain how funds and equipment in their custody have been utilized for the benefit of all stakeholders.

Allowability: Basic financial management cost principles define any expenditure incurred as either allowable or unallowable. Allowable costs are expenditures incurred that can be covered under the terms of the funds received grant. Unallowable costs are expenditures incurred that cannot be covered or reimbursed under the funding agreement.

Transparency: PHCs must be open about their work, providing information about activities and plans to all stakeholders. This includes preparing accurate, complete and timely financial reports. If an organization is not transparent, it may give the impression they have 'something to hide'.

Viability: To be financially viable, a PHC's spending must be kept in balance with money coming in, both at the operational and the strategic levels. Viability is a measure of the PHC's financial continuity and security, the Business Plan will help in ensuring financial viability of PHCs.

Integrity: On a personal level, individuals must operate with honesty and propriety. Staff and WDC members must lead by example in following policy and declare personal interests that might conflict with their official duties. The integrity of financial records and reports is dependent on the accuracy and completeness of financial records.

Stewardship: Financial stewardship involves taking good care of the financial resources we are entrusted with, to make sure they are used for the intended purposes. The WDC has overall responsibility for this, they should work together with staff in assessing financial risks and setting up appropriate systems and controls.

Accounting Standards: The system for keeping financial records and documentation must meet generally accepted accounting procedures. Any person from anywhere around the world should be able to understand your financial accounting systems.

Pillars of financial management

The following are four basic building blocks which must be in place to achieve good practice in financial management:

Accounting records: Every organization must keep an accurate record of financial transactions that take place to show how funds have been used. Accounting records also provide valuable information about how the organization is being managed and whether it is achieving its objectives.

Financial planning: Link to the PHCs business and operational plans, the budget is the cornerstone of any financial management system and plays an important role in monitoring the use of funds.

Financial monitoring: Providing the organization has set a budget and has kept and reconciled its accounting records in a clear and timely manner, it is then possible to produce financial reports for all stakeholders. Internal budget monitoring reports help managers to monitor the progress of projects and annual financial statements provide accountability to external stakeholders.

Internal control: A system of controls, checks and balances – collectively referred to as internal controls – are put in place to safeguard an organization's assets and manage internal risk. Their purpose is to deter opportunistic theft or fraud and to detect errors and omissions in the accounting records. An effective internal control system also protects staff involved in financial tasks.

Tools of Financial Management

There are many tools, not necessarily financial, which managers can use to help achieve good practice in financial management and control. We can identify these tools under each of the four functions of financial management (as included in the definition above):

Planning is basic to the management process and involves looking ahead to prepare as well as possible for the future. In the course of putting a plan together managers will consider several possible alternatives and make a number of choices or decisions. Planning must always precede the doing. Tools for planning include business plan, activity plan, budgets, work plans, cash flow forecast etc.

Organizing The resources of the organization – staff and volunteers, vehicles, property, money – have to be coordinated to ensure implementation of the overall plan. It needs to be clear what activities and responsibilities are to be undertaken, when and by whom. The tools used include organisation charts, flow diagrams, job descriptions, Chart of Accounts, Finance Manual, budget etc.

Controlling A system of controls, checks and balances are essential to ensure proper application of procedures and resources during programme implementation. Tools used include Budgets, delegated authority, procurement procedure, reconciliation, internal and external audit, fixed assets register, vehicle policy, insurance etc.

Monitoring This involves producing regular and timely information for managers and stakeholders for monitoring purposes. Monitoring involves comparing actual performance with plans to evaluate the effectiveness of plans, identify weaknesses early on and take corrective action if required. Tools used include evaluation reports, budget monitoring reports, cash flow reports, financial statements, project reports, donor reports, audit reports, evaluation report etc.

8.2.2 Financial Record Keeping at PHC

The Chart of Accounts

There are many kinds of financial transaction taking place in a health facility, to make sense of all of this financial activity, it helps to 'sort' the different types of transaction into a series of predetermined descriptive categories or 'accounts. These accounts are listed in the Chart of Accounts and are typically arranged in a logical order as follows:

- Income (money received by the facility such statutory allocation, see below)
- Expenditure (things we spend money on such as repairs, fuel, utilities commodities
- Assets (things we own such as vehicle, equipment, stock of drugs etc.).
- Liabilities (things we owe such as utility bills payable, vendor for supplies not yet paid etc.)

Income of PHCs

It is advised that each funding source should have a separate bank account and cashbook, the income sources include the following:

- Gateway (monthly) operational funds: Received from NPHCDA through the SPHCDAs to PHCs. This should have a dedicated bank account.
- NHIA reimbursement
- Revenue from charges for services not covered under the BMPHS
- Sale of drugs under the Drug Revolving Fund (DRF)
- Statutory allocation from state budgets received through the SPHCB and LGHA
- Others (including funds received from community members, development partners and non-state actors).

The conditions for receipt of the Gateway funds and the NHIA reimbursement are that the facility must have been selected by the State under the principle of one PHC per political ward and is accredited under the BHCPF or empaneled by the State Health Insurance Agency that services can be accessed at the facility.

Expenditures

- Provision and maintenance of facilities, equipment & transportation.
- Cost of fuel and diesel for power generators and vehicles
- Cost of human resources (training and hiring of ad-hoc staff and allowances)
- Procurement of essential drugs and commodities.

Recording Income and Expenditure in the Cashbook
Which accounting records to keep

For small PHCs with very few financial transactions, a simple bookkeeping system is all that is needed, refer to figure 8 above. As an organization grows and takes on a number of projects and different sources of funding, its reporting requirements, and therefore its financial systems, will become more sophisticated. Accounting records fall into two main categories: – Supporting Documents Every organisation should keep files of the following original documents to support every transaction taking place:

- Receipt or voucher for money received.
- Payment voucher for money paid out.

- Invoices – certified and stamped as paid.
- Bank paying-in vouchers stamped and dated when money is taken to the bank.
- Bank statements.
- Journal vouchers – for one-off adjustments and non-cash transactions.

With these documents on file, it will always be possible to construct a set of accounts. Other useful supporting documents include:

- Local Purchase Orders (LPOs)
- Goods Received Notes (GRNs).

Books of Account

Each facility should keep record of all financial transactions. The bookkeeping tools described below should be available and used in PHCs:

- Bank (or Cash) Book for each bank account: To record all receipts and payments of all cash, cheques and bank transfers of all financial transactions of a health facility. It is important to ensure that all financial transactions pass through the bank accounts for accountability completeness of the accounting records.

- Cheque Register: This is used to record all cheques issued and received by the facility.

- Fixed Assets Register: This is where all assets bought or donated to the facility are recorded. The register spells out the type of asset, cost (or market value where cost is not known especially if donated), quantity, condition, and location of the asset. It is important that assets bear special identifying marks (tags) to distinguish them.

- Bank Reconciliation Statements: These show reconciled figures between the cash book and bank statements. It should be prepared monthly (at the latest) for every bank account operated by the facility.

- Purchase Day Book: This is used to record credit transactions. It contrasts with the cashbook which records cash/cheque/transfers. It is written from invoices received from suppliers and vendors.

Recording in the Cashbook

All money that comes into the facility in the form of cash, cheques or bank transfers, and expenses incurred should be recorded in the Cash Book. The PHC can purchase a pre-printed cashbook from a stationary store, or a hard cover notebook and draw out the cashbook as shown below. Take the following steps:

- On receiving notification from bank of receipt of funds (e.g., NPHCDA Gateway, NHIS Pathway reimbursement etc.), obtain the statement.
- Record the receipt (using same description as on bank statement) on the left-hand side (Debit) of the cash book.
- Record payments on the right-hand side (credit) referencing the PV number used for the transactions. It is important that details in the cashbook should correspond with that of the payment voucher.
- Sum up the entries in left-hand side (incomes) and insert the amount in row where you have Total. Balance carried down (c/d) is the difference between the total income and the total payments.
- Sum up all the entries on the right-hand side included the balance c/d and insert in the cell next to Total. The Balance b/d is the same as the Balance c/d and is the figure that begins the next month's entries.

8.2.3 Estimating Costs for Business Plan, Monthly Expenditure Reporting and Quarterly Financial Reports

Financial Reporting

PHCs are expected to develop plans and report on the implementation of such plans. The Quarterly Business Plan, Monthly Expenditure Report and the Quarterly Financial Reporting are essential reports expected at the PHC level.

Preparing Monthly Expenditure Reports

The report is prepared by health facility account clerk, it is signed by the OIC and the WDC Chairman and submitted to the LGHA for approval and implementation support.

- Complete General Information Table: this captures the basic data such as name of health facility, Ward, LGA, State and Zone. It also captures the month of the year under review and the date the plan was developed.
- Complete Plan of Activities Table: this captures the relevant information regarding any activity planned for the month. The required fields are explained thus:
- Needs/requirements: these include all items and services required for the planned activity to be carried out as part of strategy implementation process. Please refer to the Expenditure Categories below for guidance.
- Description of Activity: this involves a brief narrative of the planned activity. Please refer to the Expenditure Categories below for guidance and consistency.
- Cost: this indicates the amount required for executing the activity. Please refer to the section on Budgeting below for guidance on how to estimate costs.
- Beneficiary: indicates the vendor/individual to which the amount indicated will be paid to render the service/ make procurements to execute the activity. e. Beneficiary's Contact Details: this includes the house/business address and phone number(s) of the vendor/individual to whom money is paid to render the service/make procurements.

Preparing Quarterly Financial Reporting

Please note that for ease of reporting, timeliness, completeness and accuracy, the total figures to be reported should be obtained from the individual expense ledgers using the sum of the monthly totals for the months in the quarter. For instance, the amount for stationeries for Quarter ending March 31 will be made up of total stationeries expense figures for January, February and March). The following steps should be followed:

- Complete General Information Table: this captures the basic data such as name of health facility, Ward, LGA, State and Zone. It also captures the months and the year of last quarter and current quarter under review and the date the plan was developed.
- Complete Revenue Table: this captures the projected and actual revenues for the last quarter and the projected revenue for the current quarter under review for all the sources for the facility such as:
    1. NHIS Pathway (total reimbursement expected from NHIS)
    2. NPHCDA Pathway (total operational advance expected from NPHCDA
    3. Voluntary donations (total funds expected from voluntary donations)
    4. Revenue from other services (revenues from charges on patients for services not covered under the BMPHS etc.
- Complete Expenses Table: this captures the projected and actual expenses for the last quarter and the projected expenses for the quarter under review for the facility such as:

- Human resource development (total cost incurred on training or hiring of staff)

1. Procurement of essential drugs and commodities (total incurred on purchase of drugs and other essential commodities and consumables)
2. Maintenance and transport (total cost incurred on repairs of infrastructure and equipment, transport costs etc.

- Complete Summary Table: this shows a summary of the total revenues and expenses of the health facility. It captures total projected and actual revenues and expenses for the last quarter and total projected for the current quarter under review. It also shows the cash and bank balance at the end of the last quarter brought forward into the current quarter.

Preparing Budgets/Cost Estimates

A budget describes an amount of money that an organisation plans to raise and spend for a set purpose over a given period of time. It is a financial plan that relates directly to the activities of the organisation, it is important to involve those who will be responsible for implementing in writing the budget. If this is not done, the budget could be less accurate and staff less likely to appreciate the need to spend within budget or to reach fund-raising targets.

Things to be aware of when estimating costs

A good practice in the budgeting process should imbibe the following:

- Clarity: Since many different people will need to use the budget for different purposes, they should be able to understand it (and adapt it, when necessary) without any additional explanation beyond what is written down. So, keep notes on budgeting assumptions and calculations.

- Get organised: There are several stages involved in constructing a budget before it can be submitted for approval to the WDC, so it is a good idea to prepare a budgeting timetable and commence the process early. This could be up to three months before the start of the financial year, depending on the size of the PHC and what approach has been adopted.

- Budget Headings: One way of achieving consistency is to design a Budget Preparation Sheet for your facility, which will act as a memory-jogger and prompt staff to include all relevant costs. It will list all of the main types of income and expenditure that a PHC might have in a typical year.

- Estimating Costs: It is important to be able to justify calculations when estimating costs. Do not be tempted to simply take last year's budget and add a percentage amount on top for inflation. While last year's budget could be very helpful as a starting point, it could also be very misleading and contain historical inaccuracies. From this detailed working sheet, it is then a simple matter to produce a summarised budget for each line item. It is also easy to then update if quantities or unit costs change. See below for detailed guidance on how to use a budget worksheet.

- Contingencies: Contingencies are items included in a budget for unforeseen expenses. It is best to include these for the budget lines that might need a 'cushion' (e.g. fuel) rather than adding a 'bottom line' percentage. Every item in your budget must be justifiable – adding a percentage on the bottom is difficult to justify – and difficult to monitor.

- Forgotten costs: There is a tendency to under-estimate the true costs of running a health facility due to omission of some costs such as: i. Vehicle running costs. ii. Equipment maintenance and repairs (e.g. for air-conditioners, photocopiers and computers) iii. Governance costs (e.g. WDC meetings) iv. Book-keeping and Audit fees etc.

- Check the figures: Double check and ask colleagues to check your draft cost estimates with a critical eye. Ask them to check the figures for accuracy, that you have included all of the resources needed for the facility and if the total cost looks reasonable.

Purchasing of Goods and Services

Procurement of goods and services should provide good value for money spent through a transparent process. The procedures outlined in below describes the procurements steps:

- Specify goods or services to be purchased, check budget: The standard, quantity and price of goods or services required, as described in the activity plans, is specified clearly. The amount currently available in the budget for the item to be purchased should be checked at the specification stage in case the price has changed since the budget was prepared.

- Prepare Purchase Requisition An internal request is prepared (usually on a standard form for that purpose) to formally request the purchase of the goods or services specified. The request will include a description of the purchase and state why it is required.

- Authorize Purchase Requisition The purchase requisition will usually be checked and authorized by the budget holder or other nominated person to verify that there is a genuine reason for the purchase. The available budget will usually be checked again at this stage.

- Obtain Quotations from reputable independent suppliers are requested (in accordance with internal procedures and donor rules) to make sure the organization gets best value for money and to minimize the risk of collusion.

- Select Supplier Quotations are reviewed and a supplier is selected based on price, quality, delivery times and 'after sales' terms to ensure value for money. For larger purchases, it is usual to have a Procurement Committee (comprising of a small group of staff and WDC members) who take responsibility for selecting the supplier.

- Issue (Authorized) Purchase Order (PO) The authorized Purchase Order is sent to the selected supplier with a copy kept on file with the supplier's quotation. This is a legally binding contract.

- Receive Goods from Supplier When supplies are delivered and received, a Goods Received Note (GRN) is usually signed to confirm receipt and a copy filed for later reference.

- Receive and Check Supplier Invoice The invoice should be checked and matched up with the GRN, PO and quotation, usually by the finance team.

- Prepare and Authorize Payment Authority The Payment Authority is attached to the invoice and all the supporting documents. It includes budget and accounting codes and must be checked and authorized by the budget holder or other nominated person.

- Pay Supplier Invoice Payment should be made to the supplier within the specified payment terms, usually 30 days.
- Enter Payment into Cashbook The final stage is to record the payment in the books of account.

8.2.4 Sundry Finance Functions: Banking Functions

Opening Bank Accounts

It is advisable that the selected bank is easily accessible from the location of the PHC, to this end, account must be opened with a Bank located or having a branch in the local government area where the PHC is located. Where this is not possible, then the bank must be in the nearest local government area. The account to be opened must be in the name of the PHC. Follow the following procedures:

- Obtain authorization letter from the SPHCDB authorizing the opening of the bank account and stating the signatories to the account.

- Obtain accounts opening form from the Bank.

- Fill the account opening form: Name of Account (health facility name), address, type of account (corporate), specify telephone number to receive bank alerts (this should be a person separate from the Accounts Clerk), request for cheque book, online banking access.

- Fill the bank Mandate cards for the account (The account operating mandate to the bank should consist of a combination of signatories of categories A and B: the "A" signatories shall be from the health facility; these will be the OIC (main) and another staff from the same facility (alternate). The "B" signatories shall be from the WDC, these will be the WDC Chair (main) and the WDC Treasurer (alternate).

- Obtain passport photographs and copies of valid means of identification (National Identity Card, Driver's License, International Passport or Voter's Card) of each of the signatories in panel A and B.

- Submit account opening form, completed mandate cards, copies of means of identification of signatories, copies of PHC letter headed paper (copy of which future instructions from the PHC to the bank will be made) to the bank for processing.

Changing Signatories to a Bank Account

This could happen in a situation where one or more authorized signatories of the PHC bank account are no longer available or unable to perform their duties. Follow the following steps:

- Obtain letter from the SPHCDB authorizing change of signatories, stating the name of the signatory that needs to be changed and the new one(s) as replacement(s).

- Obtain the passport photograph and copies of valid means of identification of the new signatory.

- Fill a new mandate card and capture the signature of the new signatory on the card.

- Submit SPHCDB letter, completed new mandate card and other documents (copies of valid means of identification) to the bank.

Closing a Bank Account

- Obtain a letter from the SPHCD/A authorizing closure of the PHC account.
- Authorized signatories sign a letter requesting the bank to close the account.
- Present documents in steps 1 and 2 to the bank to effect closure of the account.

Requesting Bank Statements

- Prepare letter using the PHC letterhead addressed to the Bank Manager requesting the Bank Statement and stating the period for which the statement is required.
- Get authorized signatories to sign the request letter.
- Present the signed request letter to the Bank.

Making Bank Transfer Instructions

- Prepare a letter addressed to the bank using the PHC letterhead instructing the bank to transfer funds based on the schedule included of beneficiary(ies) showing names, account numbers, beneficiary bank and amount to be transferred to each.
- Get the authorized signatories to sign the letter according to the bank mandate. – Present letter to the bank and obtain signed and Stamped (acknowledged) copy.
- Confirm debit alert with transfer instruction.

Making Payments

Depending on volume of transactions and to reduce frequency of traffic to the bank, the Accounts Clerk may schedule two days of the week when payments are made. He/she will compile payments scheduled to be made so the approvals and instructions to the bank are made once or twice in the week.

Step 1. Compile vendor documents (approved invoice, purchase order or instruction to provide services of supplies ensuring supplies are in line with the purchase order/instructions in terms of quantity and quality), or service provided certified to be satisfactory by the user or consumer of the service)

Step 2. Raise payment voucher (PV) showing the name of the beneficiary (vendor), amount (in words and figures), purpose of payment, date of payment voucher and the PV number.

Step 3. Attach documents in step 1 to the PV raised in step 2 ensuring adequacy and appropriateness of the supporting documents and present to the OIC for approval.

Step 4. Record the name of beneficiary, amount date and cheque number and PV number in the Cheque register and raise cheque in the name of the beneficiary (same name on the invoice) where payment is to be made with cheque. If payment is to be made by bank transfer, prepare instruction to the bank on PHC

letterhead stating the beneficiary's details.

Step 5. Present the cheque or bank transfer instruction to the authorized signatories for signing.

Step 6. Get vendor to acknowledge receipt of the cheque on the copy of the cheque (stating "original cheque collected by me") or on the cheque register showing name of person collecting the cheque, date of collection and signature.

Preparing the Bank Reconciliation Statement

The PHCs financial transactions are recorded by both the bank where the transactions take place and the PHC itself. The bank records these transactions in the bank statement) while the PHC records in the Cashbook, which means that the same transactions are recorded in in different books by different entities (the Bank and the PHC). It is necessary to prepare a reconciliation of both records to ensure that all transactions are captured in both books for completeness and accuracy and to prevent financial fraud. Bank reconciliation statements are usually prepared monthly, but they can be prepared weekly or daily depending on the volume of transactions. Maintain a file for bank reconciliations and bank statements. Follow the steps below in preparing the bank reconciliation statement.

- Obtain the bank statement for the period for which the bank reconciliation is to be prepared.
- Tick off the items on the credit side of bank statement (that is incomes) against the items on the left-hand side of the cashbook. Note items outstanding on both records as follows:
- Items outstanding on the credit side of bank statement but not in the Cashbook will be: i. Direct deposit or transfers into the bank account that the PHC is yet to recognize in the cash book. Update the cashbook and the Ledger Account of the depositor.
- Items outstanding on the left-hand side of the cashbook but not on the bank statement will be: i. Cheques received but not yet deposited in the bank.
- Tick off items on the debit side of the bank statement (that is expenses) against items on the right-hand side of the cashbook, note any outstanding items.
- Items outstanding on the debit side of bank statement but not on cashbook will be: i. Bank charges (maintenance fee, sms alert charge, cheque book charge etc) for which the PHC will only be aware when it receives the bank statement. Enter these in the cashbook before closing the book for the month and update the Expense ledger account.
- Items outstanding on the right-hand side of the cashbook but not on the bank statement will be: i. Payments made such as cheques issued to vendors by the PHC which are yet to be presented for payment.

8.3 HEALTH INFORMATION SYSTEM (HIS) FOR PHCs

8.3.1 Background to Health Information System (HIS)

Health Information System (HIS) is an integral and major thrust of any health system. The goal of the National Health Management Information System (NHMIS) is to establish an effective Health Management Information System at all levels of government in the Federation. The NHMIS provides information on health service delivery and the health system that supports it, in a timely and reliable manner. It is equipped to improve data capture, storage, analysis and report generation for health data in Nigeria.

Nigeria uses DHIS2 as the instance for reporting health data at all levels of government and facilities in the country. There are currently several forms available on the NHMIS DHIS platform. These paper-based monthly reports are collated and populated by the M&E focal person of the health facility for the previous month and is submitted to the LGA office by the 7th day of the current month which is the deadline for submission. The submitted information is then entered into the DHIS2 system by the M&E focal person at the LGA health office by the 14th day of the current month. To enhance data accuracy, validation rules have been set up in DHIS2. Tracer data elements have been made compulsory before data can be submitted to enhance data completeness.

All Health facilities report through these HMIS tools, and no other person, project, or partner is allowed to update, add, or edit the HMIS tools or introduce parallel data collection and reporting tools without approval. At the health facility levels, the following steps are involved in HMIS data collection:

1. Health data are captured from the clients at the health facility (outpatient or inpatient) and recorded on the service forms.
2. Data recorded on the service forms are immediately entered into the appropriate health facility registers.
   Steps 1 and 2 are done by the health workers who are providing the health services. The facility in-charge should ensure that the right HMIS tools are available and used by the health workers collecting health data in Steps 1 and 2.
3. The facility in-charge or designated health facility staff should obtain the different registers and respective tally sheets to tally the recorded health data and generate daily summaries. Make sure that no field in the register is left blank. Include total number of clients served for each service delivery area for every given month to the relevant service report for the month.
4. For any unclear information, the staff completing the tool should consult the health worker before the end of the day to verify the information recorded in the registers. Check whether the client is an old or new. If it is a new client, assign a client number and issue folder and proceed with registration. If it is an old client, locate the folder number, retrieve the folder, and proceed with registration.
5. Use appropriate standard registers to register the client. Record details of the client using indelible ink/pen with sufficient detail; ensure details are readable. Record in legible handwriting. Record all fields in the register.

Steps Involved in Reporting and Transmitting HMIS Data

The steps are detailed below:

1. Using the daily tally sheet, compile the weekly and monthly health summary reports.
2. The facility in-charge or the ward in-charge (in case of hospital) should cross-check the weekly or monthly HMIS reports for errors before submission to the next level.
3. Entries are to be made and regularly updated in the health facility DHIS2 databases at the LGA level.
4. Transmission of reporting forms to the next levels are done within the following timeframes:
    - Hospital wards/units to M&E units: second day of each month.
    - Health facility to LGA: seventh day of each month.
    - LGA to national: 14th day of each month
    - National to programs and partners: 25th day of the ensuing month

Steps Involved in Validating HMIS Data Before transmitting HMIS data to the next level

Each preceding level should cross-check the weekly or monthly health summary forms for errors. Specifically, the data validation should follow these steps:

1. Form a data validation team at each level. At the health facility, the facility in-charge should be the chairperson of this team.
2. Establish a culture of routine data validation, recounts, and verification.
3. Meet monthly to validate data before transmission.
4. Cross-check total figures on the reporting forms.
5. Check for accuracy and completeness of reports.
6. Cross-check data consistency across reports.
7. Look for unusually low or high values for events and diseases.
8. Compare with previous months and the same period the year before.
9. Do necessary corrections before transmission.
10. Sign off on the reports after they are validated by the facility in-charge.

8.4 Health Management Information Systems (HMIS)

Health Management Information Systems (HMIS) are one of the six building blocks essential for health system strengthening. HMIS is a data collection system specifically designed to support planning, management, and decision making in health facilities and organizations.

MEASURE Evaluation worked in Ethiopia to implement a HMIS in the Southern Nations, Nationalities, and Peoples' Region (SNNPR) and Oromia. The following training materials were used to support implementation of the HMIS including data quality assurance and information use procedures, the electronic HMIS system for managing data processing, reporting, analysis and use; and the and the community health information system (CHIS), which organizes information on individuals and families—information related to vaccines, family planning, maternal and child health, HIV treatment and support, and other services—and makes it available all in one place, such as a family folder.

The National Health Management Information System is intended to provide information on Health Service delivery and the health system that supports it in a timely and reliable manner. The efficiency and effectiveness of health services delivery is assessed by availability of quality, and timely information.

A. Routine Data collection summary forms

1. NHMS 000: Community based summary forms

This is used by the community Village Health worker to summarize activities at monthly intervals sent to the health facility through the Community Health Extension worker (CHEW)
Revised NHMIS forms 2

2. NHMIS summary forms 001A & 001B

- Health facility community outreach forms
- Used by the health facilities to summarize community outreach activities at monthly intervals
- Aggregate transferred to health facility form 001
- Revised NHMIS forms 3

3. NHMIS summary form 001

- Health facility based forms
- Used by health facility to summarize activities in the health facility at monthly intervals from a daily register
- Sent to the LGA
- Revised NHMIS forms 4

4. NHMIS summary form 002

- Used by the LGAs to summarize data from all health facilities in their area at quarterly intervals.
- Sent to the SMOH, HMIS unit.

5. NHMIS summary form 003

- Used by the state to summarize data from all LGAs.
- Sent to the Federal NHMIS every 6 months.

HIM FORMS

- ANC register
- Immunization Tally Sheet
- General Attendance Register
- OPD Register
- Monthly Summary Form (MSH)
- IPC Register
- Immunization Register
- Severe Malaria Register
- NHMIS for Secondary and Tertiary Health Facility
- Health Facility Monthly Data Triangulation Form

NOTE: This entire Section is adapted from publications of the National Primary Health Care Development Agency (NPHCDA) (i.e. a) National Primary Heath Care Development Agency (2023). Facility and Financial Management Toolkit For Primary Healthcare Centres Participants' Manual, and b) National Primary Heath Care Development Agency (2023). Operation Manual for Facility and Financial Management of Primary Healthcare Facilities).

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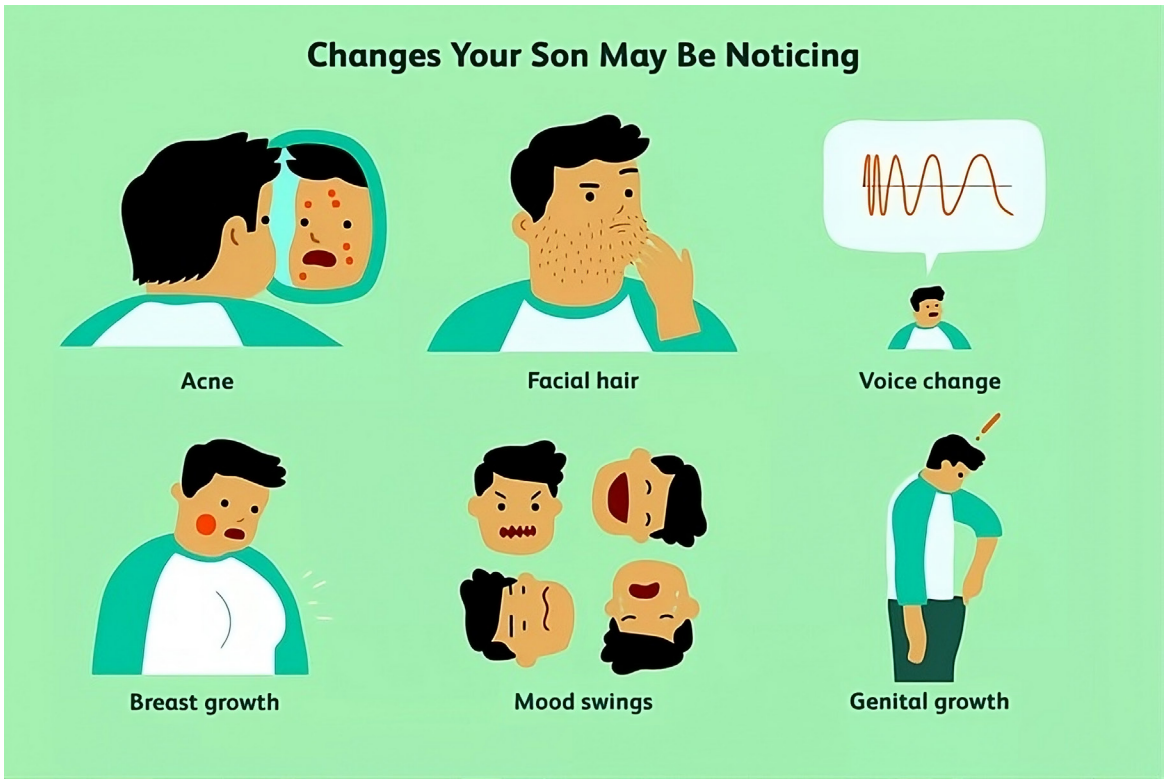
APPENDICES

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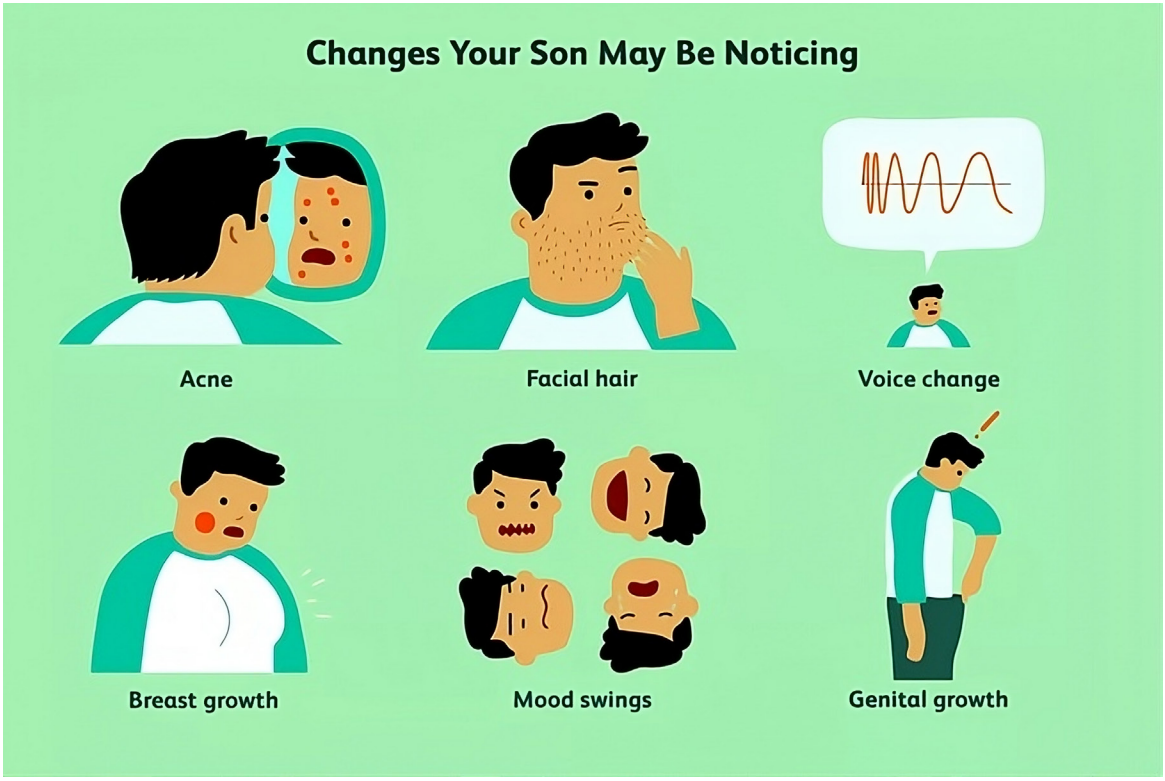
Appendix A: Integrated Mamagement Chart



Appendix B: Essential Care for Every Baby (ECEB) Chart



Appendix C: Good Attachment to the Breast



Appendix D: Essential Care for Every Baby in the First 90 Minutes of Life

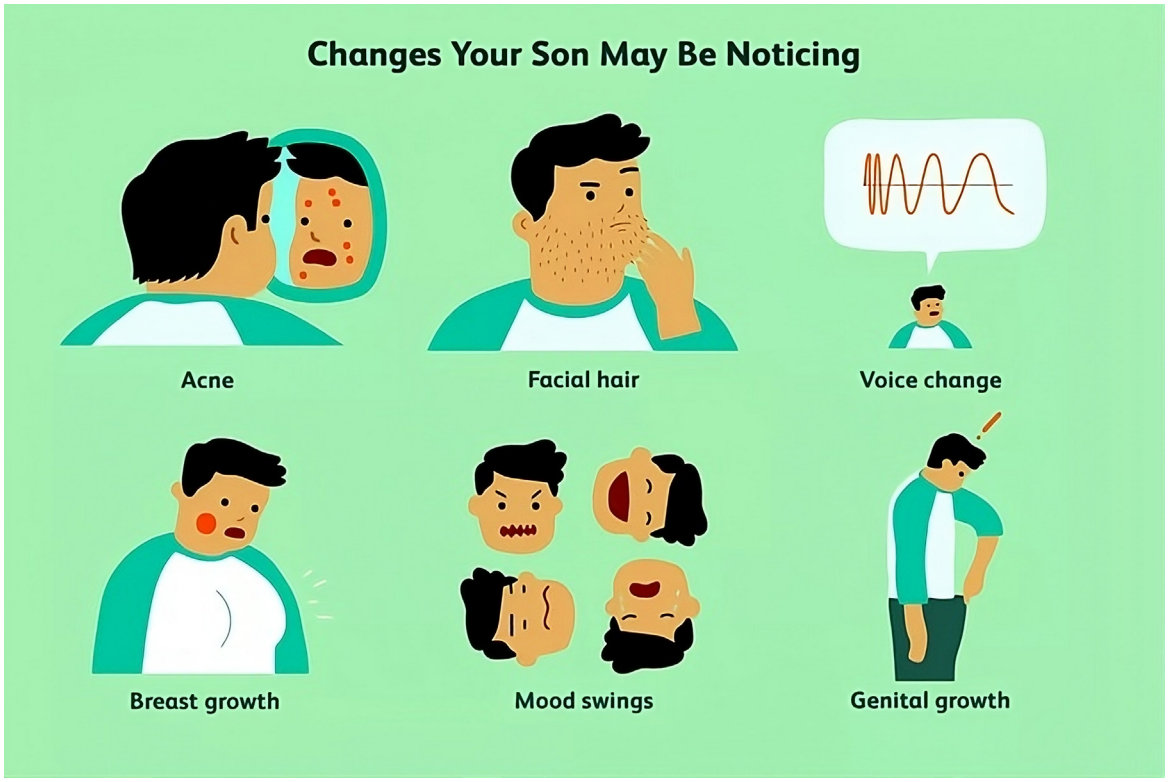


FIGURE 10. NEONATAL CONJUNCTIVITIS

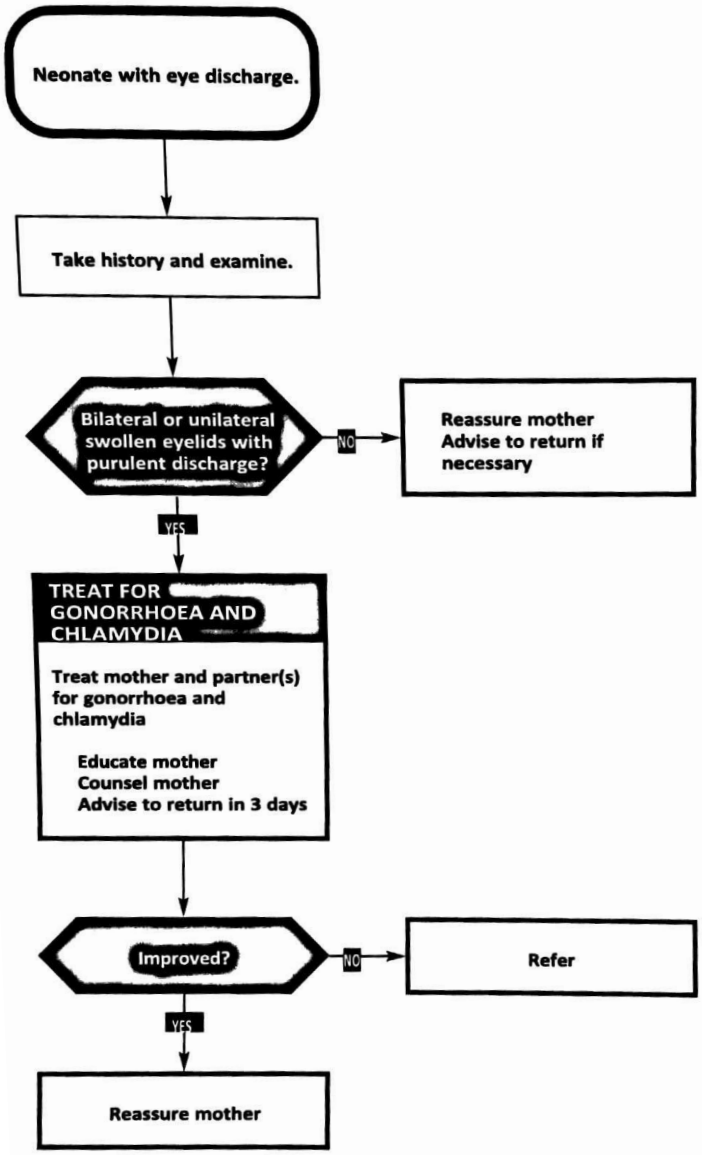


FIGURE 2. PERSISTENT/RECURRENT URETHRAL DISCHARGE IN MEN

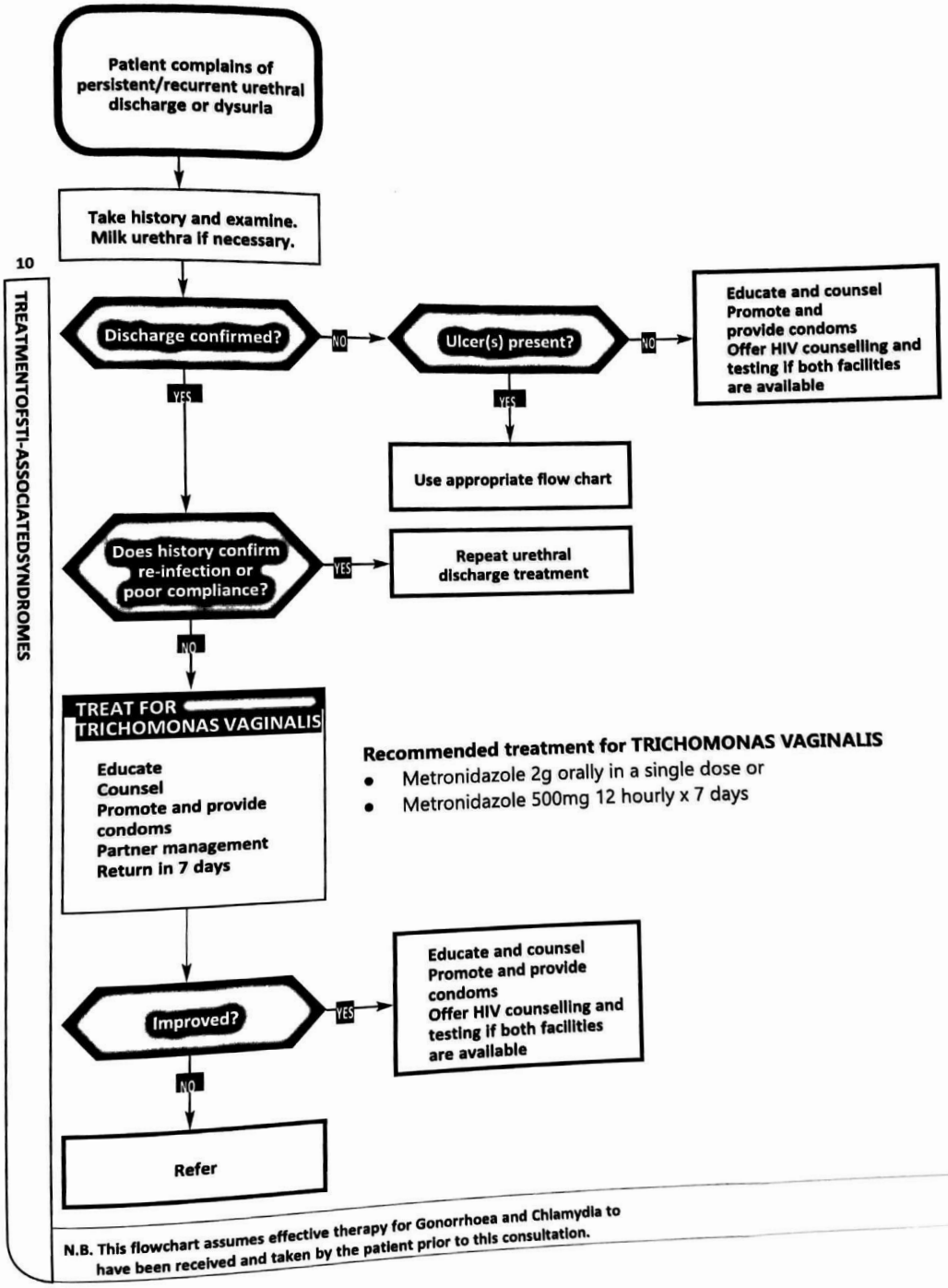
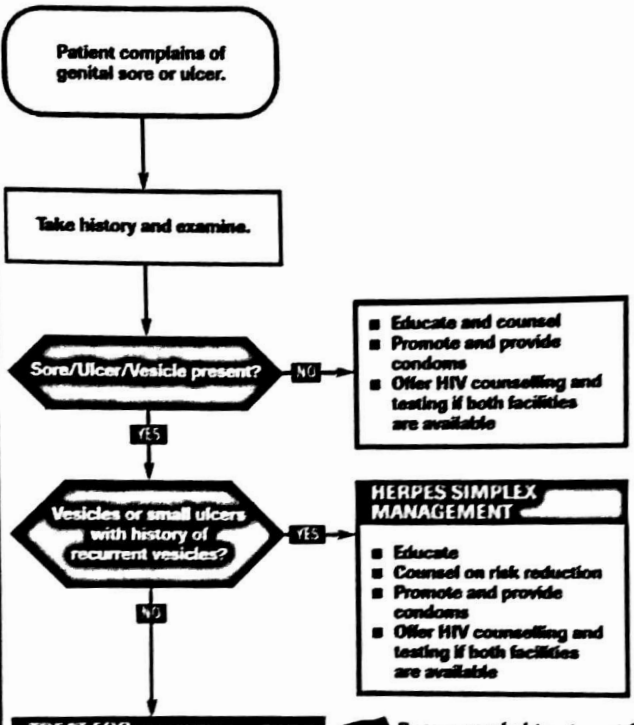


FIGURE 3. GENITAL ULCERS



Recommended treatment for Syphilis

Primary, secondary or early syphilis (> 1 year)

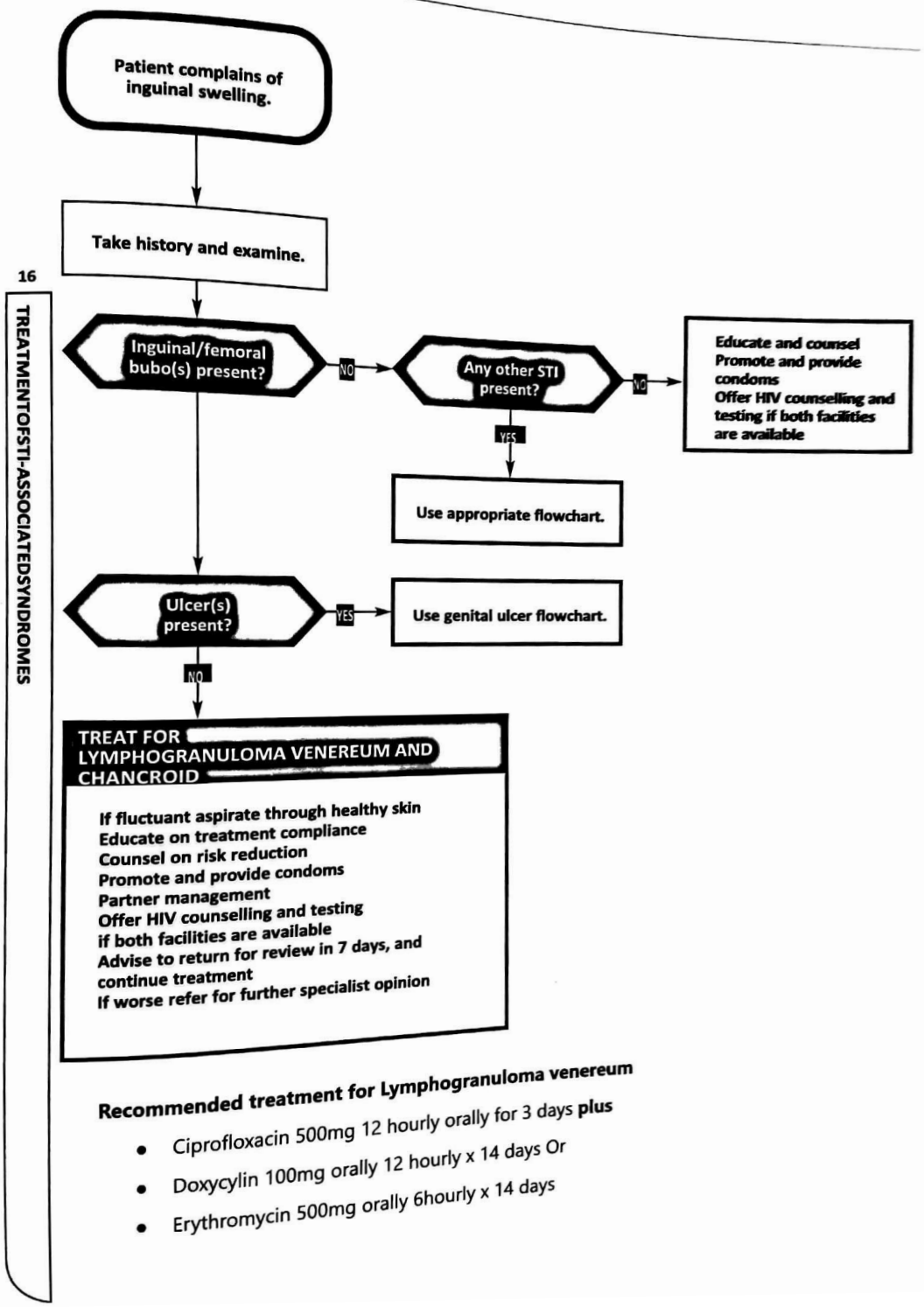
- Benzathine penicillin 2.4 million units IM X 1 dose
- For penicillin allergic;
  Doxycylin 100mg orally 12 hourly x 14 days Or
  Erythromycin 500mg orally 6hourly x 14 days Or
  Tetracycline 500mg 6 hourly x 14 days plus
- Acyclovir 400mg 8 hourly for 10 days (or 200mg 5 x daily for 10 days)

Late syphilis (> 1 year) or unknown

- Benzathine penicillin 2.4 million units IM weekly x 3 doses **Or**
- Procaine penicillin 1.2 million units IM daily x 20 days **Or**
- Doxycylin, Tetracycline or Erythromycin orally x 28 days

Needs adaptation to local epidemiological situation.

FIGURE 4. INGUINAL BUBO

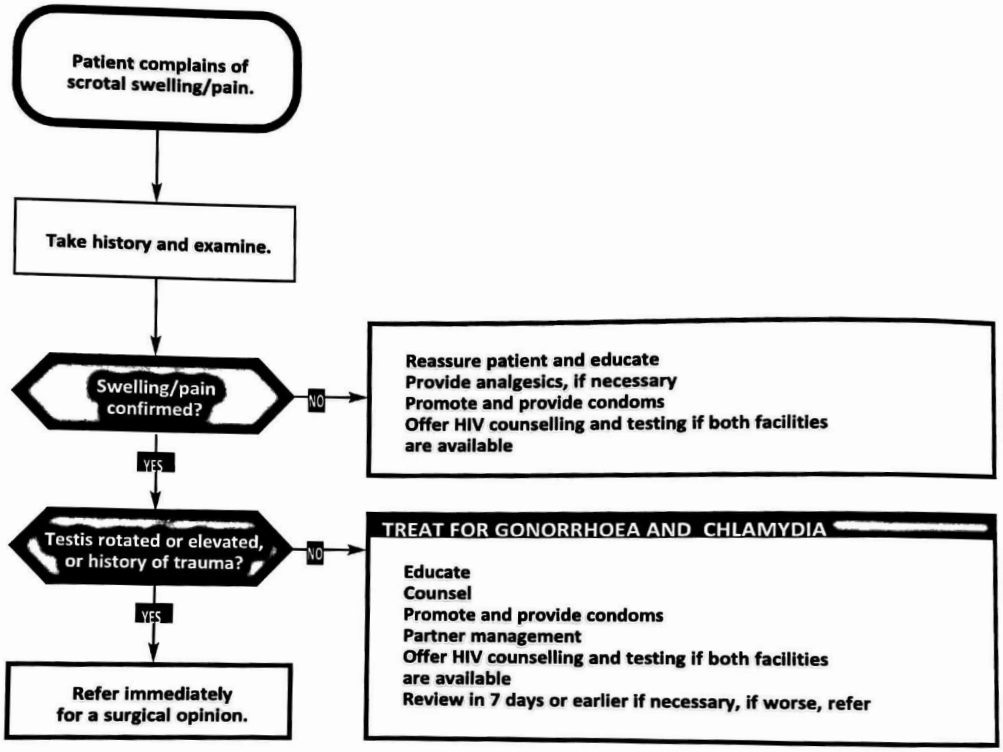


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Recommended treatment for Lymphogranuloma venereum

- Ciprofloxacin 500mg 12 hourly orally for 3 days **plus**
- Doxycylin 100mg orally 12 hourly x 14 days Or
- Erythromycin 500mg orally 6hourly x 14 days

FIGURE 5. SCROTAL SWELLING



Recommended treatment for Gonorrhea
Treatment options:

- Cefixime 400mg orally x 1 dose
- Ceftriaxone 250mg IM x 1 dose
- Spectinomycin 2 gram IM 1 dose

Always treat Chlamydia as well

Recommended treatment for Chlamydia

- Doxycylin 100mg orally 12 hourly x 7 days
  Or (alternatives)
- Erythromycin 500mg 6hourly x 7 days (safe for pregnant women)
- Tetracycline 500mg 6 hourly x 7 days

Always treat Gonorrhea as well

FIGURE 6. VAGINAL DISCHARGE

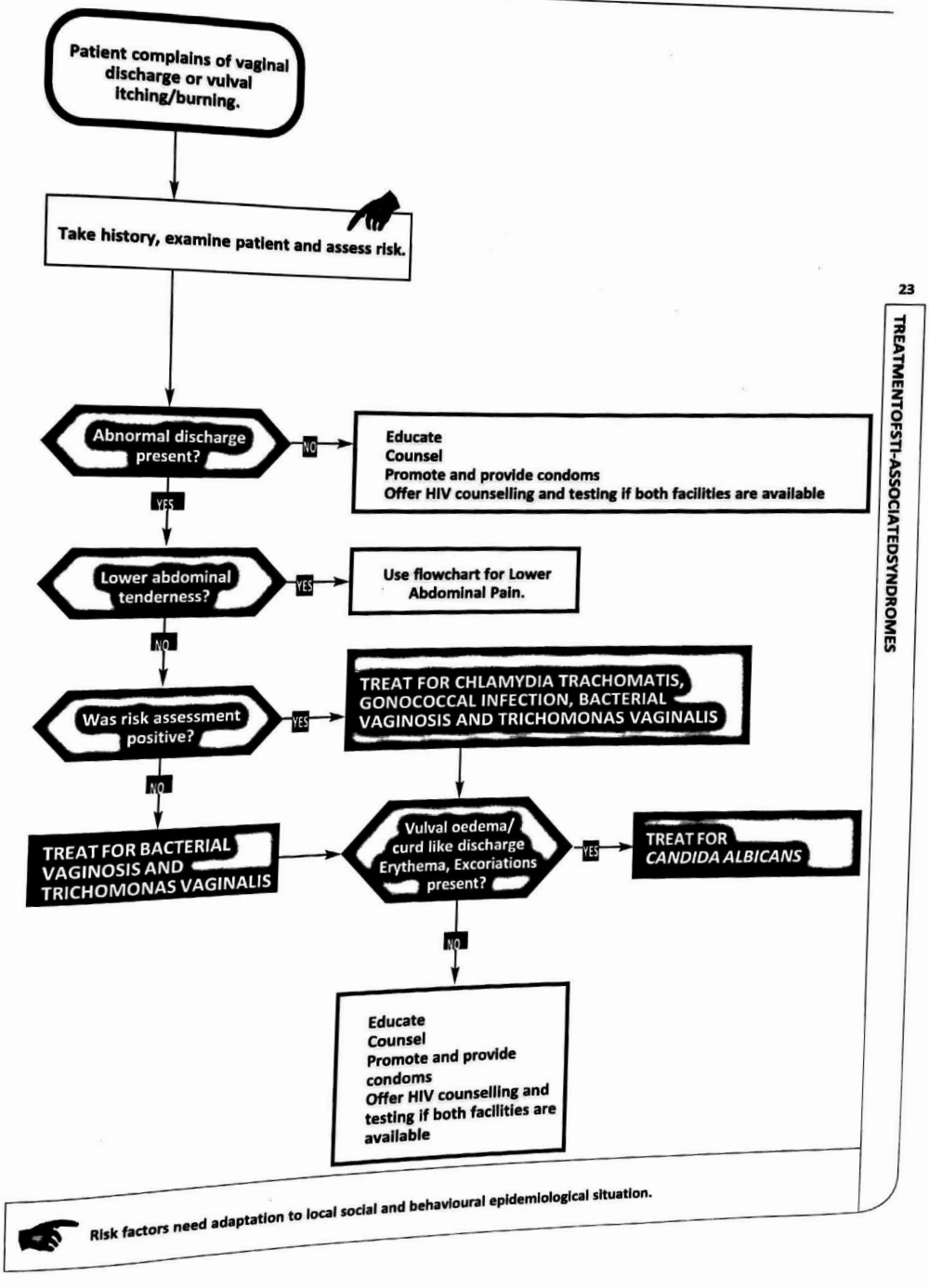


FIGURE 6. Vaginal discharge

Recommended treatment for Gonorrhea
Treatment options:

- Cefixime 400mg orally x 1 dose
- Ceftriaxone 250mg IM x 1 dose
- Spectinomycin 2 gram IM 1 dose

Always treat Chlamydia as well

Recommended treatment for Chlamydia

- Doxycylin 100mg orally 12 hourly x 7 days
  Or (alternatives)
- Erythromycin 500mg 6hourly x 7 days (safe for pregnant women)
- Tetracycline 500mg 6 hourly x 7 days

Always treat Gonorrhea as well

Recommended treatment for Bacterial vaginosis or Trichomonas

- Metronidazole 2g orally in a single dose or
- Metronidazole 500mg 12 hourly x 7 days

Recommended treatment for Candida infection

- Clotrimazole or miconazole topical x 3- 7 days
- Fluconazole 150mg oral x 1 dose
- Nystatin 100.000IU intravaginal daily x 14 days

FIGURE 7. VAGINAL DISCHARGE (SPECULUM AND BIMANUAL)

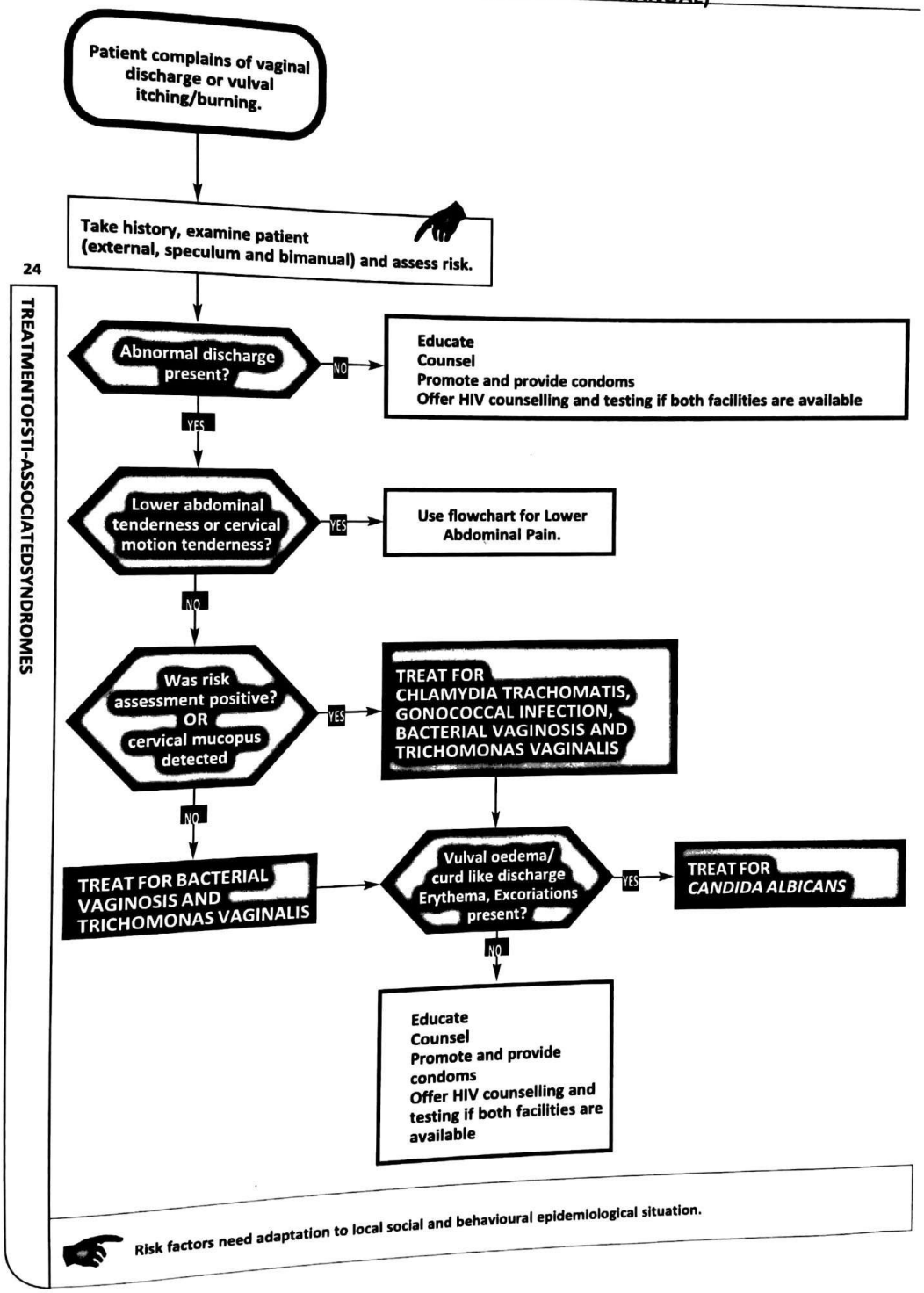
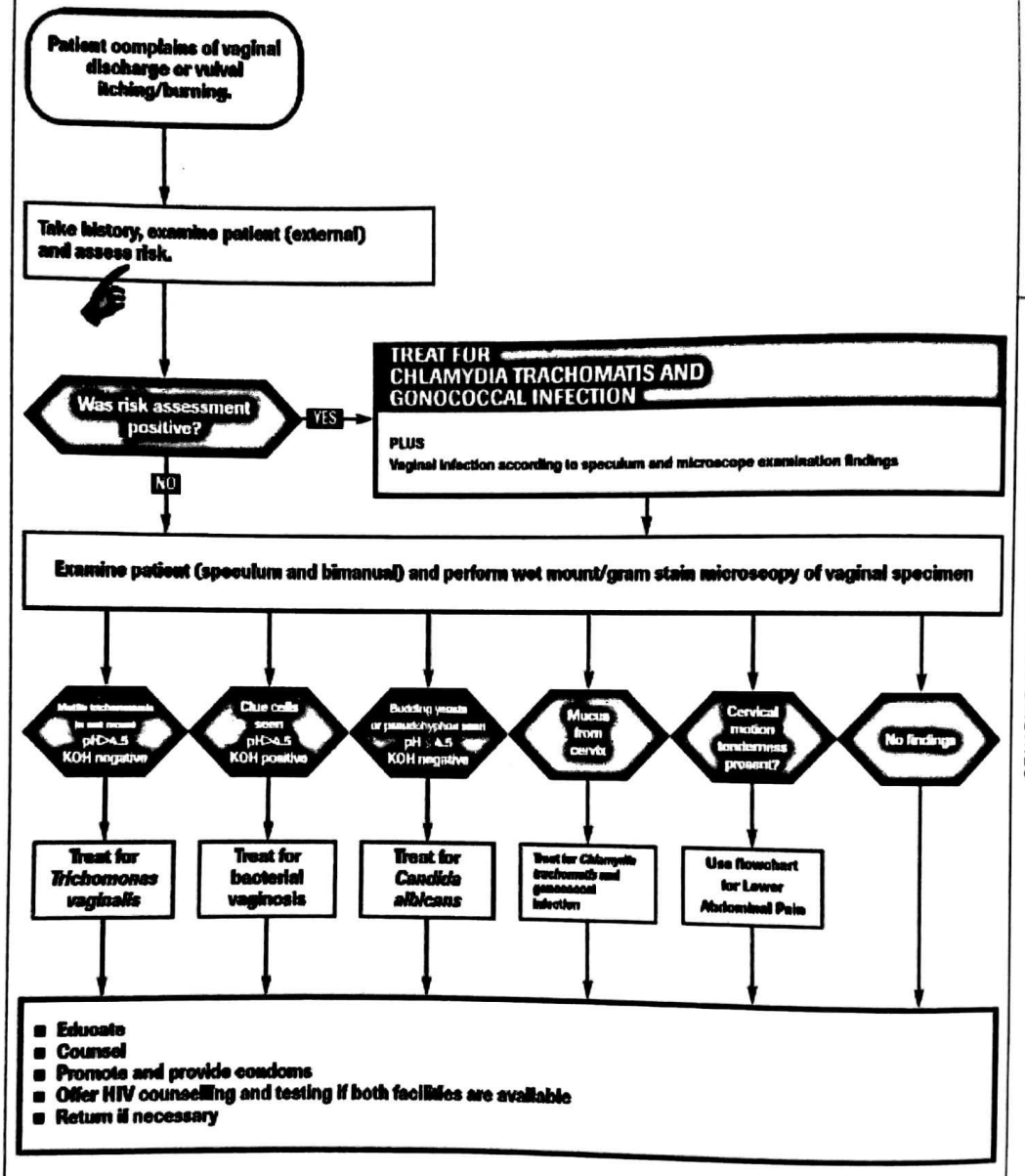


FIGURE 8. VAGINAL DISCHARGE (SPECULUM AND MICROSCOPE)

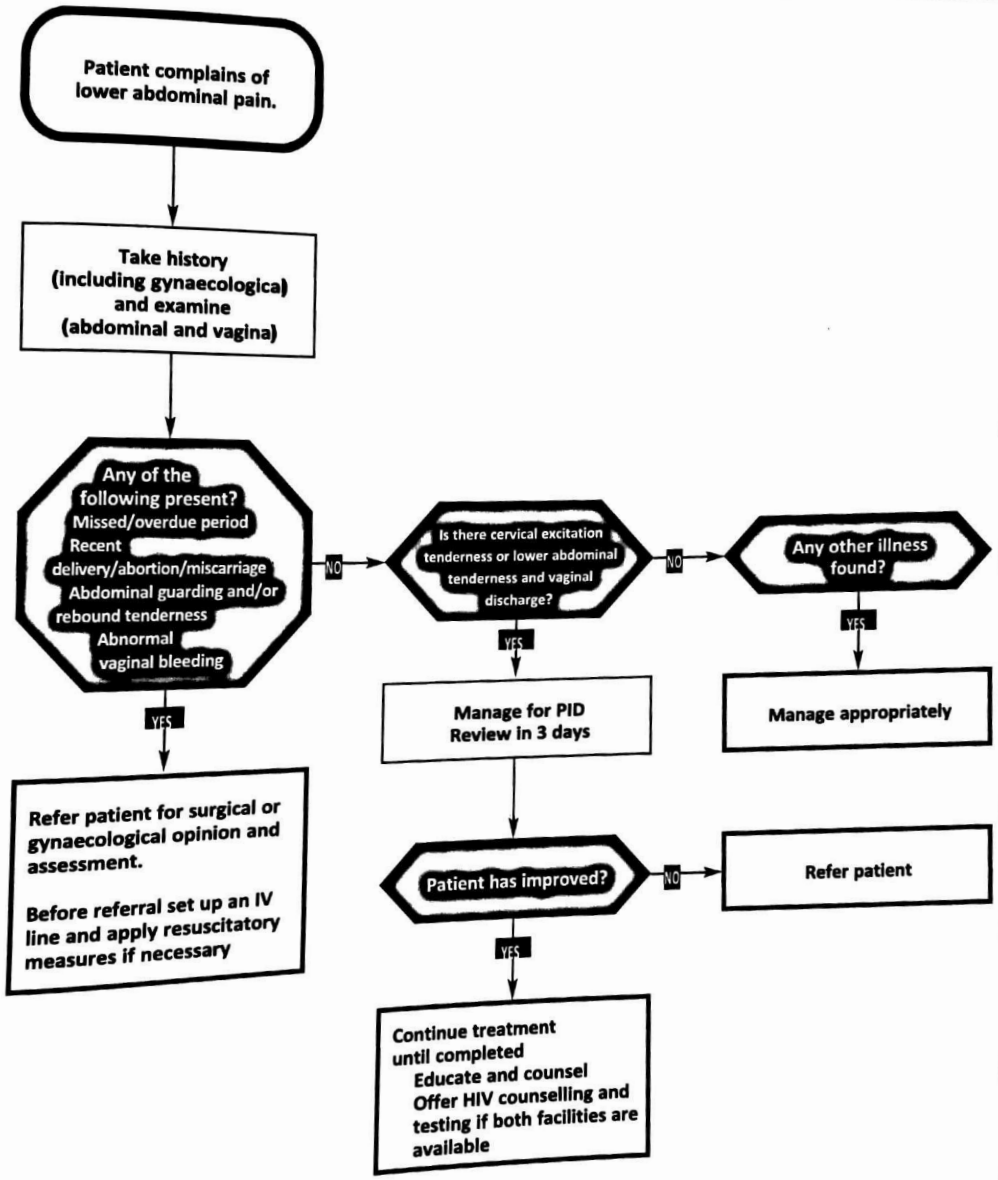


Notes:

1. KOH Test: 1 drop 10% KOH to reveal the amine odour (fishy)
2. Wet mount: smear on slide with 1 drop of saline and view at 400x

Risk factors need adaptation to local social and behavioural epidemiological situation.

FIGURE 9. LOWER ABDOMINAL PAIN



Recommended treatment for Pelvic inflammatory disease plus

- Ceftriaxone 250mg IM X 1 dose
- Doxycylin 100mg orally 12 hourly x 14 days
- Metronidazole 400mg 12 hourly x 14 days

Consider hospital admission for;

Severe illness, pregnancy & no improvement in 2-3 days

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TREATMENTOFSTI-ASSOCIATEDSYNDROMES

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NATIONAL STANDING ORDERS

FOR COMMUNITY HEALTH OFFICERS

Revised By:

COMMUNITY HEALTH PRACTITIONERS' REGISTRATION BOARD OF NIGERIA
IN COLLABORATION WITH NPHCDA

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2024